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CONTENTS

15. Errante's gaseous-core high-frequency atomic resonance reactor - CGRR-N type - RADIONDISTICS
16. ANTONIO MEUCCI - To honor his memory, life and achievements - Radiondistics.com
17. ATTREZZATURE SCIENTIFICHE TECNICHE DIDATTICHE PRODOTTI SCIENTIFICI PER LA DIDATTICA, LABORATORIO di FISICA
18. ANTENNE HF, PRODOTTI PER STAZIONI RADIO PER LE ONDE CORTE, STAZIONI HF, DIPOLI - Radiondistics
19. Francesco Errante - Radiondistica, radio fisica, ingegneria radio-elettrica & radio-elettronica, meccanica quantistica
20. ANTENNA A DIPOLO: rivelata la sua elettrodinamica ed meccanismo di radiazione © 2003 Francesco Errante
21. RADIO ONDE - La RADIAZIONE HERTZIANA , cos'e' e come avviene © 2003 Francesco Errante - Radiondistics.com
22. ANTENNA RADIO RICE-TRASMITTENTE MONOPOLARE CON TERRA VIRTUALE © 2003 Francesco Errante
23. I trasduttori radio-elettrici alla luce delle nuove scoperte - © Francesco Errante
24. Linee di trasmissione bilanciate per segnali elettrici RF in regime di onda progressiva - Radiondistics.com
25. GUGLIELMO MARCONI - Comunicazioni radio-telegrafiche - Premio Nobel, 11 Dicembre 1909
26. MESSA A TERRA per RF: il sistema di Errante a reattanza capacitiva. Un-Un etc. © 1985-2026 F.sco Errante - www.Radiondistics.com
27. Virtual ground node generator for balanced transmission lines © 2008 Francesco Errante
28. VGNG - in-line VIRTUAL GROUND NODE GENERATOR © 2007 Francesco Errante - Radiondistics.com
29. ERRANTE's all-in-one apparatus for the physics of the hertzian radiation - radio waves - dipoles
30. A VIRTUAL GROUND SWR-METERED OPEN DIPOLE BALUN © 2003 Francesco Errante
31. VIRTUAL GROUND BALUN for HF FOLDED DIPOLE ANTENNAS © 2003 Francesco Errante - www.Radiondistics.com
32. A SECONDARY IONIZATION FLUORESCENCE HERTZIAN RADIATION PASSIVE DETECTOR - Radioluminescence effect
33. Errante's 225 Ohm multiband HF antenna system © 2009 Francesco Errante - www.Radiondistics.com

34. ANTENNA BALUN : Errante's VIRTUAL GROUND BALUN for dipole & loop HF ANTENNAS
35. HF Turnstile antenna broadband driver © 2026 Francesco Errante - Radiondistics.com
36. Frequency to wavelength converter - Wavelength to frequency converter - Energy calculator
37. RIVELATORE PASSIVO DI RADIAZIONE HERTZIANA A FLUORESCENZA DA IONIZZAZIONE SECONDARIA - Radioluminescenza
38. RICHARD FEYNMAN - Nobel Lecture, December1965
39. RADIATING SISTEMS WHERE ELEMENTS INTERFERE WHITH EACH OTHER FIELDS - SELF-ANTAGONIST SYSTEMS
40. © 2003 Francesco Errante
41. TRANSMISSION LINE CALCULATOR © 2000 Michael A. Lee and Kevin E. Schmidt
42. Reginald Fessenden and Guglielmo Marconi, the history of radio.
43. GENERATORE di NODO di TERRA VIRTUALE per linee bilanciate © 2008 Francesco Errante
44. GENERATORE di NODO di TERRA VIRTUALE in linea © 2007 Francesco Errante
45. ATTREZZATURE DIDATTICHE - PRODOTTI SCIENTIFICI - RADIAZIONI TRASDUTTORI RADIOELETTRICI
46. BALUN ROSMETRICO CON TERRA VIRTUALE PER DIPOLO APERTO © 2003 Francesco Errante
47. BALUN CON TERRA VIRTUALE PER LO SDOPPIAMENTO DEL DIPOLO RIPIEGATO A ½ ONDA
48. ATTREZZATURE DIDATTICHE SCIENTIFICHE TECNICHE PRODOTTI ARTICOLI MATERIALI DIDATTICI FISICA SCIENZE
49. Sistema d'antenna MULTIBANDA per le HF con radiatore a 225 Ohm © 2009 Francesco Errante - www.Radiondistics.com
50. BALUN UNIVERSALE CON TERRA VIRTUALE PER DIPOLO © 2003 Francesco Errante
51. Antenna TURNSTILE HF, dipolo a croce, quadripolo - Pilota a larga banda © 2008 Francesco Errante - Radiondistics.com
52. SISTEMI RADIANTI AUTO-ANTAGONISTI - SISTEMI RADIANTI IMPROPRII - Radiondistics.com
53. RF LIGHTING DEVICES, RFLD, HELUs, RF powered fluorescent tubes, high efficiency light sources, RF-LIGHTING.com
54. Apparecchi di illuminazione ad altissima efficienza luminosa a radio onde - RF-LIGHTING.com

55. RF LIGHTING DEVICES, RFLD, HELUs, RF powered fluorescent tubes, high efficiency light sources, RF-LIGHTING.com
56. Radio frequency noise from high energy efficiency fluorescent lights ballasts

1. www.RADIONDISTICS.com - Physics of the radio waves and radio-engineering

index.htm

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Home | Scientific Products | RF Communications Products | Contact

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Welcome to **Radiondistics.com** ! The official *RADIONDISTICS*'s web site, formerly *Radiondistics U.K.*

Perhaps, you're asking your-self: what the hell *Radiondistics* means? Well, it's a neologism born out of the merger of two italian words: *radio* (radio) and *ondistica* (related to waves). *Radiondistics*, therefore, means physics of the radio waves.
Radiondistics is a new, but already well consolidated, branch of the physics, entirely developed by *Francesco Errante*, the discoverer of the electric behaviour of the *hertzian dipole* and of the true mechanism behind the radio-electric phenomena and father of the virtual ground monopole HF antenna.
This web site is ment to spread the knowledge about FRANCESCO ERRANTE's findings on the phenomena regulating the hertzian radiation, the

Hertzian radiation: what it is and how it happens

Virtual ground node

Errante's CAPACITIVE EARTH GROUNDING

radioelectric transducers and the transmission lines for the radioelectric signals.
Radiondistics' main objective is the investigation of the pure physics of the radio waves.
Radiondistics is an exact science, where theory is always backed up by the experimental verifications, in other words: by the practice. In physics, the practical observation of phenomenon inspires the theory and, in turn, the latter leads to the developing of the necessary instruments for the experimental verification and then again, its results will direct practice re-inspiring theories in an end-less circle.
Radiondistics was the missing discipline between the radio-engineering and the physics. The unavailability of such a scientific discipline has lasted for almost a century. The necessity for a scientifically rigorous investigation into the physics of the radio waves, has been advocated even by Gugliemo Marconi, back in 1909, as he clearly stated in his own speech, which he gave at the occasion of him being awarded the 1909 Nobel Prize for Physics.
Main instruments employed in radiondistics are the radio-electrics e radio-electronics engineering. Most of the Radiondistics' scientific apparatus portfolio is also available in a ready to use version for radio-communications employment.

SYSTEM

Balanced transmission lines in a progressive wave regime

All the concepts, methods, designs and devices presented on this web site are the original novelty works of *FRANCESCO ERRANTE*.

Radiondistics' products exclusively employ its own proprietary and patented technology which ensures highly efficient and reliable antenna systems and radioelectronic equipment for

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2. RADIONDISTICS - Physics of the radio waves, radio-electrical & radio-electronics engineering - Errante - www.Radiondistics.com

Home last updated on March, 2026 Contact

Welcome to *Radiondistics* !

Important announcement

Over the past 25 years, *Radiondistics* has led the world in the HF antenna engineering bridging the gap between the electrical engineering and the physics of the radio. *Radiondistics*' own proprietary scientific instruments have proven them-selves to be invaluable tools for handling high frequency electrical currents in a proper and rational way enabling, once again, to bridge two more physics subfields: classic electrodynamics and quantum mechanics.

Thanks to *Francesco Errante*'s latest patent-pending invention, namely the "*high frequency sinusoidal alternating current powered gaseous core nuclear reactor for the continuous disintegration-recombination cycle of the Neon gas atoms, based on the light chemical elements non-wavelength-dependent electrical resonance physical phenomenon*", **two leapfrog breakthrough discoveries in**

fundamental physics have been made, empirically demonstrated and at long last, unveiled to the public at large:

- the intrinsic electrical resonance property of light chemical element atoms, such as noble gases, with respect to high-frequency sinusoidal electrical current oscillations;
- a non-wavelength-dependent electrical resonance phenomenon.

The interplay between the said physical property and phenomenon is responsable for the occurrence of a specific electro-nuclear process termed *Resonance-Driven Disintegration-Recombination* (RDDR) cycle of light elements atoms, such as those of the nobel gases.

The RDDR cycle is not an identical pathway to fusion. It is a third distinct category of nuclear energy release:

- 1. Fission: Splitting heavy nuclei;
- 2. Fusion: Merging light nuclei;
- 3. RDDR (Resonance-Driven Disintegration-Recombination): Disassembling nuclei into a nucleon fog and reassembling them into new configurations via electrical resonance.

Although RDDR results in the creation of new isotopes (which macroscopically resembles "fusion" if they are heavier) and release of energy, the underlying physical mechanisms, intermediate states, resonant trigger, and byproduct spectrum are categorically different from the fusion processes.
RDDR is a fundamentally novel pathway indicating that rules of quantum chromodynamics (QCD) allow for a much more fluid and reversible state of nuclear matter than we currently believe.

I. Summary of the invention & historical context

The Gaseous-Core Resonance Reactor (Neon variant) - GCRR-N for short - represents a paradigm shift in nuclear energy, demonstrating the first functional gaseous-core reactor using non-fissile fuel. This invention resolves a technological impasse dating to the 1950s not through incremental material science, but by discovering and exploiting a previously unknown property of noble gases: their **non-wavelength-dependent electrical resonance** to high-frequency sinusoidal alternating current.

The breakthrough originated not in a fusion laboratory, but during research into ionospheric radio propagation, where applying established direct-current quantum electrodynamic (QED) principles to a high-frequency alternating-current (AC) regime revealed a critical phenomenon.

When a noble gas (e.g., neon) is subjected to a sinusoidal current oscillating at its natural electrical resonance frequency, perfect energy coupling occurs. This resonance is a function of the atoms intrinsic quantum-electrical structure, not the spatial dimensions of the apparatus or the wavelength of the driving signal.

The reactor empirically validates that neon atoms possess a natural electrical resonance frequency of 27,430 kHz. Driven at this frequency within a suitably designed reaction chamber, the gas undergoes a self-sustaining cycle of atomic disintegration and statistical nuclear recombination termed **Resonance-Driven Disintegration-Recombination (RDDR) Fusion** producing a stable, scalable thermonuclear reaction with net energy gain.

The GCRR-N demonstrates a credible and revolutionary fusion methodology. It replaces monumental heating and magnetic confinement with **resonant electrostatic pumping and cyclic quantum disintegration-recombination**. This represents a paradigm shift from "containing a star" to "orchestrating a continuous atomic resurrection", offering a direct path to scalable, clean, and thermally efficient power generation. The principal barrier to practical gaseous-core nuclear power was not one of engineering alone, but of fundamental oversight. By revisiting the quantum-electrical principles first explored in vacuum tubes and applying them in a novel resonant AC regime, this invention transitions gaseous-core reactors from a perpetually future concept to a present-day reality. The GCRR-Neon validates that non-fissile, non-radioactive materials can achieve net energy gain through atomic electrical resonance, and establishes RDDR Fusion as a new, viable branch of nuclear science.

II. The fundamental phenomenon: atomic electrical resonance in an AC field

The phenomenon is purely quantum-mechanical and has no analogue in classical physics. Its discovery invalidated the core assumption that rendered gaseous-core reactors (GCRs) conceptual namely, that they required fissile material or unattainable

plasma containment.
Historically, GCRs were hamstrung by the Sisyphean challenges of fuel containment, material survival in ultra-hot plasmas, and radioactive byproduct management, all stemming from the entrenched paradigm of forcing nuclear reactions via extreme thermal kinetics or neutron flux. Critically, prior to this invention, non-fissile materials were categorically excluded as potential nuclear fuels. It was believed that neon, being an inert noble gas, could not act as a nuclear fuel. The chemical inertness of neon is a property of its filled electron shell (configuration $1s^2\ 2s^2\ 2p^6$) but is irrelevant to nuclear reactions. This is because, under conditions of high energy density and natural electrical resonance, the periodic potential of the high-frequency sinusoidal alternating current modulates the potential barrier around the nucleus, enabling the temporary capture of electrons (formation of Ne^-) and, subsequently, the collapse of the nuclear wave function when the kinetic energy of the ions exceeds the Coulomb barrier.
In other words, electrical resonance at the atomic level alters the cross-sections for low-energy nuclear interactions, an effect known in the literature as "*channel resonance*" or "*near-field enhancement*", here observed for the first time in a pure noble gas.
In this regime, neon is no longer "inert": the resonance enables the disintegration of the nucleus and the instantaneous formation of thermal plasma, followed by recombination into new nuclear configurations with the release of excess binding energy.
The electrical resonance of atoms a cornerstone of QED in direct-current (DC) field emission and spectroscopy had never been successfully applied or observed in a high-frequency AC regime; and consequently, the unique potential of noble gases (with their full valence shells and definable quantum states) to enter into a coherent, resonant energy absorption cycle was unrecognized by nuclear engineering.

The necessary conditions for the phenomenon to manifest are threefold:

- **Resonant drive:** A sinusoidal driving current with oscillation frequency exactly equal to the gas natural electrical resonance frequency (for neon: **27,430 kHz**).
- **Atomic mobility:** Sufficient freedom of movement low-pressure gaseous state (**<5 Torr**).
-

Energy density threshold: Environment capable of achieving critical energy density to initiate atomic and subatomic reactions.

When satisfied, the system behaves as a perfectly resonant electrical circuit, exhibiting a sharp resonance peak within a narrow bandwidth and a stable, real-valued feed-point impedance despite the absence of any correlation with physical dimensions of the apparatus.

III. Reactor Design: Exploiting the Phenomenon

The GCRR-Neon (Gaseous-Core Resonance Reactor, Neon variant) is engineered specifically to satisfy the above conditions and concentrate their effects. The reactor's heart is a unipolar electrode, fed by a single-phase, unbalanced sinusoidal alternating current and not being fixedly polarized, it cannot be designated as either cathode or anode, so it is analogously termed the ***Páno Káto*** (Greek for up and down), referencing its oscillating potential.

The cavity geometry creates a superposition of known concentrating effects essential for achieving the critical energy density:

- **Hollow Cathode/Anode Effect:** concentrates charged particles within the cavity volume.
- **Pendulum Effect:** anions undergo repeated reflections between cavity walls and central collision zone, multiplicatively increasing kinetic energy with each half-cycle.
- **Electrostatic Confinement:** oscillating field of the cavity walls traps charged particles, enabling sustained energy multiplication.

IV. The Disintegration-Recombination (D-R) Fusion Cycle

The reactor initiates and maintains a continuous, two-phase nuclear process. Due to ultra-relativistic speeds, phases succeed each other cyclically and are not discernible by direct observation.

Phase 1: Excitation & Disintegration (Peak Potential)

During each peak of the AC half-cycle, the cavity walls reach maximum negative potential. Neutral neon atoms in contact with the walls conductively acquire electrons, forming anions (Ne-). These anions are electrostatically repelled from the negatively polarized walls and converge toward the region of least negative charge the geometric center of the cavity. Through the pendulum effect, their kinetic energy multiplies with each oscillation, and they undergo elastic collisions where mutual electrostatic repulsion combined with momentum reversal traps them in a cycle of increasing energy. Upon reaching the critical energy density, **ultra-relativistic collisions overcome the Coulomb barrier**, causing complete nuclear disintegration into a transient nucleon fog, a dense plasma of protons and neutrons. This transition is instantaneous at the systems critical energy density, explaining the absence of conventional ignition signatures.

Phase 2: De-excitation & Recombination (Trough Potential)

As the wall potential drops, electrostatic confinement relaxes. The ultra-hot, high-pressure nucleon fog seeks lower-energy states and undergoes **statistical recombination** into nuclei with higher binding energy per nucleon (e.g., Neon-20 ☐ Neon-22 + protons). The excess binding energy is released as heat and light, producing the observed thermonuclear glow and the system's net energy output. Recombined neutral atoms re-enter the buffer gas region before re-initiating the cycle.

V. Critical Observations & Signatures

- **Instantaneous ignition:** Achieved upon reaching the critical energy density from resonant pumping, without any precursor form of ignition (e.g., electrical discharges) being discernible.

- **Electrically silent feed:** Driven by resonance and electrostatic kinetics, not bulk current flow through plasma; no impedance mismatches, transients, or electrical perturbations manifest on the feed line during operation.

-

- **Visible thermonuclear effect:** Intense heat and light originate from the recombination phase of the cyclic nucleon plasma.
- **Stable & scalable confinement:** Confinement is inherently electrostatic and cyclical, governed by drive frequency and chamber geometry, making it highly controllable.

VI. Thermal energy harnessing system

The reactor produces energy primarily as heat deposited directly into the walls of the reaction chamber cavity (the Páno Káto electrode itself), which thus functions simultaneously as reaction chamber and thermal collector.

- **Primary heat exchanger:** A metal serpentine conduit is integrated with the external walls of the electrode.
- **Heat transfer loop:** A heat-conducting fluid circulates through this conduit in a closed loop, transferring thermal energy to an external standard heat exchanger, with decoupling from the high-frequency electrical signal accomplished via conventional radio-frequency systems.
- **Critical engineering note:** Efficient, active heat extraction is **mandatory**. Inadequate cooling will result in rapid melting of the electrode, occlusion of the reaction cavity, cessation of the fusion reaction, and the appearance of a stationary regime of total reflection on the feed line.

VII. Theoretical implications & a new fusion regime

This process defines a new regime **Resonance-Driven Disintegration-Recombination (RDDR) Fusion** with profound implications:

- **Fuel purity:** The use of a single, pure noble gas (neon) is essential. Its full electron shell and defined resonance frequency allow for synchronized, coherent anion formation required for the pumping mechanism. Reactivity or isotopic inconsistency would disrupt the cycle.

- **Energy gain source:** Primary yield comes from the exothermic recombination of nucleons into more stable nuclear configurations. Resonant electrical input is used almost exclusively for *pumping* (imparting kinetic energy), not for heating a chaotic plasma, resulting in exceptional efficiency.

- **Technological heritage:** The invention bridges the conceptual world of 19th- and 20th-century discharge tube physics with 21st-century energy needs, fulfilling the latent potential of those early instruments. The very tools that helped humanity discover the atom and quantum mechanics were, all along, the key to a new kind of reactor.

- **Theoretical framework:** RDDR Fusion stands as a legitimate alternative fusion regime to the magnetic confinement fusion (MCF) and inertial confinement fusion (ICF), demonstrating that non-fissile, non-radioactive materials can serve as efficient nuclear fuels and that atomic-scale quantum resonance can be harnessed for macroscopic power generation.

In-depth details can be found here!

My scientific work is dedicated to the memory of the late *Nikola Tesla*, *Antonio Meucci* and to all the living Creatures victims of oppression and injustice. *Francesco Errante*

Injustice is indivisible. Injustice anywhere is a threat to Justice everywhere!

Martin Luther King Jr.


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3. www.RADIONDISTICS.com - Fisica delle radio onde ed ingegneria radio-elettrica ed elettronica

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Benvenuti su **Radiondistics.com**! Organo ufficiale di *RADIONDISTICS*, gi� *Radiondistics U.K.*

La parola "*Radiondistics*" � un neologismo inglesizzato, nato dall'unione di due parole italiane: *radio* ed *ondistica*. *Radiondistics* significa, quindi, fisica delle onde radio.
La *Radiondistica* � una nuova, ma gi� ben affermata, branca della fisica interamente sviluppata da Francesco Errante, scopritore del comportamento elettrico del *dipolo hertziano* e del reale meccanismo dei fenomeni radio-elettrici e padre dell'antenna HF monopolare con terra virtuale.
Lo scopo principale di questo sito internet � quello di divulgare l'opera tecnico-scientifica di *FRANCESCO ERRANTE* sui fenomeni che regolano la radiazione hertziana, i trasduttori radioelettrici e le linee di transmissione per segnali radioelettrici.
L'obiettivo primario della radio-ondistica �

La radiazione hertziana: cos'� e come avviene

Il nodo di terra virtuale

Sistema di messa a terra capacitiva

l'indagine sulla fisica pura delle radio onde. La *radiondistica* � una scienza esatta, ove la teoria � sempre suffragata dalle verifiche sperimentali, ovvero dalla pratica. In fisica, l'osservazione pratica dei fenomeni ispira la teoria che a sua volta porta all'ideazione ed alla realizzazione degli strumenti per le verifiche sperimentali, i cui risultati nuovamente dirigeranno la pratica ri-ispirando la teoria in un circolo infinito.
La *radiondistica* era la disciplina mancante tra la radio-tecnica e la fisica. La mancanza di una tale disciplina scientifica si � protratta per quasi un secolo. La necessita' di dover svolgere indagini con rigore scientifico sulla fisica delle radio onde, � stata auspicata dallo stesso Guglielmo Marconi, gi� nel 1909, come dimostra il discorso che egli ebbe a pronunciare in occasione dell'assegnazione del suo Premio Nobel per la Fisica.
Gli strumenti principali impiegati in radiondistica sono l'ingegneria radio-elettrica e radio-elettronica. Molti degli apparati scientifici del portafoglio di *Radiondistics* sono anche disponibili nella versione per pronto impiego nelle radio-comunicazioni.

Tutti i concetti, metodi, disegni e prodotti qui presentati sono originali e sono esclusivamente frutto del lavoro di ricerca scientifica in materia di fisica delle radio onde, condotto da FRANCESCO ERRANTE.

I prodotti *Radiondistics* impiegano esclusivamente la propria tecnologia brevettata che assicura sistemi d'antenna ed attrezzatura radio-elettronica ad alta efficienza ed affidabilit� per impieghi convenzionali e speciali nelle radio-comunicazioni trans-ionosferiche civili e militari in onde corte, nonch�, per impieghi scientifici.
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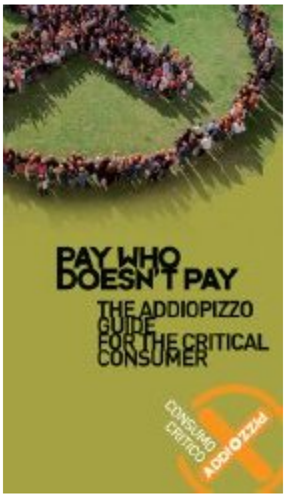
- Errante's GASEOUS-CORE ATOMIC RESONANCE NUCLEAR REACTOR

- Errante's CAPACITIVE RF EARTH GROUNDING SYSTEM

- Errante's apparatus for the physics of the balanced transmission lines for radio frequency electric signals.

- Errante's balanced line virtual ground node generator

- Errante's in-line Virtual Ground Node Generator in-line VGNG.



- Errante's all-in-one original apparatus to carry out measurements and experiments on the open and folded dipole antennas and on the physics of the hertzian radiation.

- A virtual ground balun for the suppression of any which one of the two branches of an � wavelength open dipole antenna. **PATENT n.0001347484**

- Virtual ground balun with built in S.W.R. meters. It is designed around our *Suppressor* circuitry and has 3 built in S.W.R. meters, one for each of its ports.

- A virtual ground balun for doubling the � wavelength folded dipole antenna. **PATENT n.0001347486**

- The original *elementary radiator*, a virtual ground monopole HF antenna, also known as the *Errante's Antenna*. **PATENT n.0001347485**

- A secondary ionization fluorescence hertzian radiation passive detector. **PATENT n.0001347487**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5. HF ANTENNAS - RF COMMUNICATIONS EQUIPMENT - SHORTWAVE RADIO EQUIPMENT - Radiondistics

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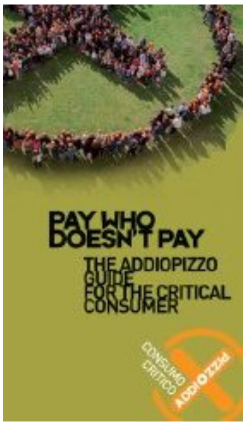
Errante's 225 Ohm, MULTIBAND HF ANTENNA SYSTEM



Errante's virtual ground monopole HF antenna. PATENT n.0001347485.

Errante's virtual ground balun: state-of-the-art high power broadband flux-coupled step-up balun transformer, specific for the use with � wavelength dipole antennas. Many impedance combinations to suit open, folded and loop antennas are also available. PATENT n.0001347484

and PATENT n.0001347486.

Errante's virtual ground node generator for balanced transmission lines.

Errante's in-line virtual ground node generator for unbalanced transmission lines.

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To contact Francesco Errante from outside Italy, please, ring **+39**.339.180.1313, anytime between 08:00 and 20:00 local time.
Please, notice that my local time currently is:

E-mail

Per metterti in contatto con Francesco Errante, chiama il 339.180.1313, raggiungibile dalle ore 08:00 alle 20:00.
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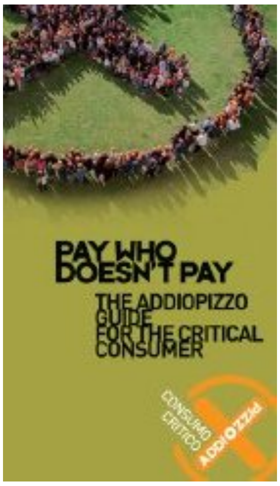

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7. Francesco Errante - Radiondistics, radio physics, radio-electrical & radio-electronics engineering, quantum mechenics

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My technical-scientific work is dedicated to the memory of *Nikola Tesla*, *Antonio Meucci* and to all living creatures who are victims of oppression and injustice.

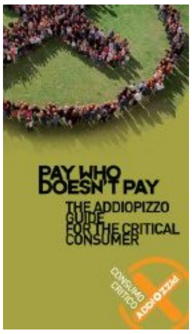
Francesco Errante

Injustice is indivisible. An injustice anywhere is a threat to justice everywhere!

Martin Luther King Jr.

Autobiographical notes by the author.

I am an experimental physicist and radio-electronics engineer. I first encountered radio–electric transceivers at the age of eight. For over a quarter of a century, I have been engaged in a self–funded scientific research program aimed at investigating the physics of Hertzian radiation and radio-electric phenomena. Having moved to Great Britain toward the end of 1988, I returned to Italy in March 1999 due to a misfortune that has befallen on my family.
Although not without difficulty, I continued my research in Italy, where, instead of finding support and assistance, I found the institutions associated with the maintenance of Italy's technological rachitism, especially in its depressed areas. I can't help but recall my father's words. Having served for over twenty years as an inspector general (with full office and duties) of the Agricultural Development Agency (ESA), at the regional headquarters in Palermo, he referred to it as the "*Agricultural Rachitism Agency*."



Having said that, I, *Francesco Errante*, the son of the late *Vincenzo Errante* (born in 1920, a Partisan) and *Margherita Caminita*, born in Palermo on August the 18th, 1969, wish to disclose to the international intellectual community, scholars and students, technicians and radio amateurs, and all those who may benefit from this website (www.Radiondistics.com), my discoveries in the field of radio wave physics and the scientific instruments I have designed and developed for their experimental verification, as well as my products for ready-to-use application in the radio communications.

Visit www.uibm.gov.it and enter *Francesco Errante* in the search box to see all patents of mine, or verify it directly from the ***European Patent Office*** archive HERE !

Many believe that research isn't done in Italy, but **the truth is that a lot of research is done in Italy... quite a lot of research is being done, especially on how to steal money from the State, the European Union, and even from private citizens, with pretext of scientific research.**

Free Science in free State
by *Margherita Hack*,
Edizione Rizzoli.

Not only are we among the last in Europe in scientific subjects, but when we manage to train a true genius, we usually hand him a suitcase and send him abroad to do gooding. Why doesn't research in Italy really want to work? For two reasons, both deeply rooted in national history and customs. On the one hand, we suffer from a chronic and inexplicable fear of science and its potential, and from the Galileo case to the battle against preimplantation embryo analysis, much of the blame lies with the Church and its habit of dictating the law in a country that professes to be secular. On the other hand, there's the state, which from right to left cuts university funding, wastes scarce resources, and messes up academic careers without, however, managing to wrest them from the "barons." Thus, while the importance of innovation for the country's growth is extolled everywhere, in reality, those who should be producing it are hindered by every means: cumbersome competitions, lifelong precarious employment, starvation wages, and, why not, conscientious objection. Stories of ordinary contradictions in a failing system. Margherita Hack dedicates this book to analyzing the conditions of research that no longer has either State or Church to rely on. She examines the reforms that have taken place under four governments, denounces the recurring errors and too many inconsistencies, highlights the positive examples encountered throughout her career, and finally offers some ideas.

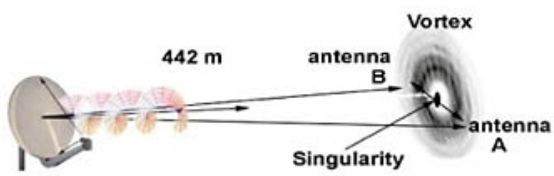
Paul Adrien Maurice Dirac, Nobel prize in physics, year 1933

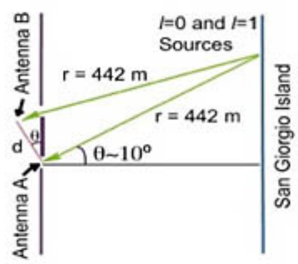
"*I don't understand why we keep talking about religion"* Paul Dirac once said.
If we are honest – and as scientists, we must be – we must admit that any religion is a mixture of false assertions, devoid of any basis in reality. The very idea of God is a product of man's imagination. I fully understand that primitive man, more exposed to the overwhelming forces of nature, personified these forces, driven by fear. But today, when we know so many natural phenomena, we no longer need these solutions. I assure you that I cannot see how postulating the existence of an omnipotent God can be of any use to us. What I understand is that such hypotheses lead to nothing but sterile questions, such as why God allows so much misery and injustice, the exploitation of the poor by the rich, and all the other horrors or evils that He could have easily avoided? If religion is still passed down, we know It';s very well that this happens, not because religion convinces us, but to keep the subaltern classes calm. It is easier to dominate fearful peoples than dissatisfied ones who protest; and it is also easier to exploit them. It has already been said: religion is like a kind of opium: people are lulled by visionary dreams, forgetting real injustices and exploitation. Hence the alliance between the two great political forces of the State and the Church. Both find it convenient the illusion that a good God rewardsï¿½ if not in this world, then in the nextï¿½those who have not risen up against injustice but who have submitted docilely and perhaps gratefully to the duties imposed on them. And this is why honestly asserting that God is only a figment of the human imagination is considered the worst of all mortal sins."

Italian university to join the race for actively damaging the image of the scientific research in Italy!

Overwhelmingly enthusiastic boffins managed to make a world-wide unvoluntary hoax announcement.

Naïve theoreticians from the Department of physics and astronomy of the University of Padua and the *Swedish Institute of Space Physics* have covered themselves with ridicule after claiming they have found a way to force photons to follow an helicoidal trajectory in the free space, in a circular-vortex-like fashion, by simply getting a conventionally generated plain wavefront to bounce back off a portion of a purposefully distorted paraboloidal reflector (pictured below).





They have also claimed that by doing so, a number of radio carrier signals having identical frequency could travel on the same path and direction by means of circular vortex-like trajectory differing from each other only by the phase of their vortex, each carrying a different set of informations, thus obtaining more channels in the same frequency.

According to the informations I gathered, they have conducted some seriously ill-conceived and ill-deviced experiments with the assistance of a few italian radio amateurs, instead of hiring proper microwave engineers. Owing to their lack of competence and expertise, the italo-swedish Team has blatantly failed to comprehend the results and its own mistakes. Needless to say: their "experiment" on the venetian lagoon turned out to be a fiasco!

POLE ANTENNA: electrodynamics and radiation machanism revealed. © 2003 Francesco Errante - www.Radiondistics.com

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ERRANTE's ANTENNA

Errante's CAPACITIVE EARTH GROUNDING SYSTEM

Balanced transmission lines in a progressive

Errante's OPEN DIPOLE
BRANCH SUPPRESSOR BALUN
The open dipole: an antenna...? NOT, two!

The "*virtual ground*" practical application importance in the radio-electric transducers and circuits, by Francesco Errante, for the understanding of the physics of the radio-electric phenomena.
The "*virtual ground node*" has, in the physics of the radio-electric phenomena, the same importance as the "*fulchum of the lever*" has in mechanics.

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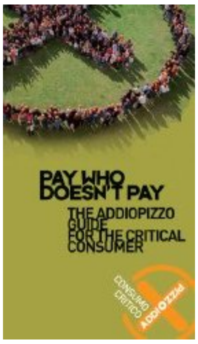


Patent n.0001347484 (ITAPA20030019)
Visit www.uibm.gov.it and enter *Francesco Errante* in the search box to see all of my patents, or verify it directly from the ***European Patent Office*** archive HERE !

wave
regime

Hertzian
radiation:
what it is
and how
it happens

A VIRTUAL GROUND BALUN FOR THE SUPPRESSION OF ANY WHICH ONE OF THE TWO BRANCHES OF AN � A WAVELENGTH OPEN DIPOLE ANTENNA

by *Francesco Errante*

Scientific purposes :

a. The balun described hereby enables to demonstrate that the � wavelength open dipole antenna, also known as "*Hertzian Dipole*" is not an "*elementary antenna*" (by definition an elementary antenna is an aerial where the condition of resonance and radiation cannot take place without the presence of all its parts) but it is, instead, an "*elementary array*" of 2 radiating/ captating elements of a physical length equal to �

of a wavelength each, which are electrically arranged in a counterphase, while being fed in the middle of them;
(The same happens with all the other symmetric dipoles, regardless of their impedance, also known as *radiation resistance*.)

b. The balun described hereby enables to demonstrate that once a virtual ground node is available, it is possible to receive and transmit radio-electric signals through the space by employing a single radiating/captating element of a physical length equal to � of a wavelength, without the need for a natural nor an artificial ground plane; (see --> *ERRANTE's ANTENNA*)

c. The balun described hereby enables to demonstrate that each of the two branches of an � wavelength open dipole antenna, at its frequency of resonance, exhibits an absolute impedance value of 35 Ohm, unbalanced and referred to the virtual ground node;
(Likewise, any which one of the two branches of any other symmetric dipole will yeld a characteristic impedance value equal to � of the impedance of the dipole to which it belongs.)

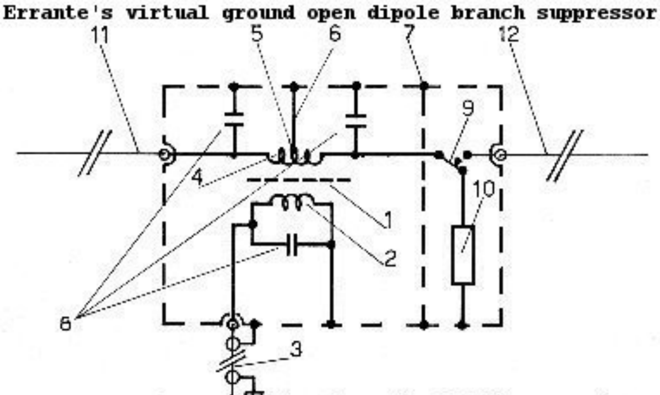
d. The balun described hereby enables to demonstrate that each of the two branches of an � wavelength open dipole antenna, exhibits a phase-angle difference of +/- 90 degree with respect to the virtual ground node.

e. The balun described hereby, by allowing to suppress anyone of the two branches of the dipole, has led to confute, once and for all, the "*theory of the antenna image* " .

Circuit description

In the field of radio-engineering, a multitude of systems are well-known for matching an � wavelength open dipole to an unbalanced transmission line, they are usually referred to as *balun* (a compound term meaning "balanced-unbalanced"). State of the art balun(s) do not allow feeding of the two branches of an � wavelength open dipole independently from each other and as a result of that, they do not allow to verify whether it possible to suppress anyone of them without interfering with the condition of resonance and irradiation of the remaining one. **The invention described hereby allows to feed each of the two branches of an � wavelength open dipole independently, so that they can become electrically independent from each other allowing, therefore, to suppress anyone of them without repercussions on the condition of resonance and radiation of the remaining one.**
(similarly, a device for splitting up the � a wavelength folded dipole antenna is described here)

Errante's virtual ground open dipole branch suppressor



Patent-pending . Copyright 2003 Francesco Errante

This result has been achieved by means of a lumped-constants radio-electric circuit, based around a broad-band radio-frequency electric transformer(1) winded on a binocular type ferrite core having a primary winding(2) exhibiting an impedance value equal the one of the transmission line(3) being used and a center-tapped secondary winding(4) exhibiting an impedance value of 35 + 35 Ohm. The said impedance values are referred to the *virtual ground*. ("virtual ground" is defined as a point in an electrical circuit that appears to be at ground, but is not actually attached to ground, it is therefore, a node having a 0 degree phase angle difference with respect to ground and has the same electrical potential as the Earth")

The virtual ground node has been obtained by providing the transformer's secondary winding(4) with a center-tap(5), effectively splitting the secondary winding in two sections. The virtual ground is then made available to the whole of the network by a short electrical connection(6) to the device conductive chassis(7).
The transformer's working point is optimized by compensation with the employment of high-voltage RF duty capacitors(8).

The suppression of one of the two branches of the open dipole is made by manually operating the high insulation resistance switch(9) diverting the radio-electric signal from the transformer(1) off the branch of the dipole(11 or 12) and into one lead of the non-inductive 35 Ohm dummy-load(10) while the other lead of the dummy-load is attached to the virtual ground circuit.

The balun described hereby is characterized by being a *NON fluctuating*

***impedance* RF electrical transformer. All of its terminations impedance values are set by design, hence the impedance transformation ratio is a function of them and not viceversa, unlike it happens with any other balun arrangement.**

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9. RADIO WAVES - The HERTZIAN RADIATION: what it is and how it happens © 2003 Francesco Errante - Radiondistics.com

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Hertzian Radiation, (better known as radio-waves) : what it is and how it happens

by *Francesco Errante*

This is a brief presentation of *Francesco Errante's theorem on the physics of the hertzian radiation*.

The *hertzian radiation* is the oldest, yet the least common way to refer to radio-waves. It takes its name after *Heinrich Rudolf Hertz*, the first physicist and engineer to have demontrated the possibility of radio transmissions.

Ever since Hertz experiment and untill now, it was believed that energy would travel all the way from the transmitter to the receiver while maintaining its original form, hence, erroneously crediting electric signals with the property of being able to

Virtual ground node

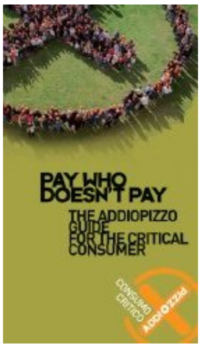
Errante's CAPACITIVE EARTH GROUNDING SYSTEM

Errante's elementary radiator

Errante's hertzian

travel in the empty space. Hertz, him-self, named his own apparatus misterious emissions as "*electric waves*". Subsequently, they were renamed as "electromagnetic waves".

First of all, it is paramount to comprehend that **the *hertzian radiation* is a natural phenomenon with an artificial origin**. Although, the *hertzian radiation* mirrors the nuclear reactions photonic emission mechanism, it takes place at frequencies far lower than those present in nature - eg: Sun's nuclear fusion and its radiations. See *energy and radiation wavelength*. In nature, radiations get generated by matter's violent energy excitation which involves very high tempertures and consequentially very high frequencies. Radiations of wavelenght longer than those of the infrared rays spectrum, in nature, have no known practical function, although, they still retain scientific relevance. In the *hertzian radiation*, instead, the excitation is artificially imposed by means of *electric oscillation*. *Electric oscillation* is a means to obtain the excitation of the electric charges and it must not be confused with the *hertzian radiation* it-self! Unfortunately, this is what has been happening untill now.



radiation detector

Balanced transmission lines in a progressive wave regime

The purpose of this study is to demonstrate that:

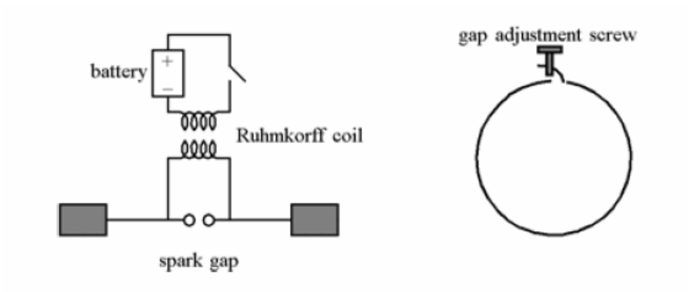
> **1) the mere oscillation of electric charges is not the phenomenon at the origin of the *hertzian radiation*;**
> **2) *hertzian radiation* can only take place following an energy**

transformation, named *radio-electric transduction*.

Here is a brief introduction to the Author's logical development of his ideas, concepts, observations, deductions and patented devices leading to his findings.

The method.
Simplification is the main rule in the observation and understanding of physical phenomena. Simplification, though, must not be confused with a simplistic approach and it may require the development of tailor-made tools and technology which can them-selves be far from being simple.

The stripping of a radiator to its bare essentials.
The � of a wavelength *open dipole antenna*, since its birth, has always been considered to be the most simple available antenna and therefore it has always been the specimen on which to conduct studies. Those studies, however, were ill at birth as the *open dipole* is not an aerial as simple as it looks. In addition, having considered the open dipole an "*elementary antenna*" has shifted the focus of those studies away from its essence and into its properties and its effects. Consequentially, the theories about its intimate functioning are flawed and are responsible for a number of further well radicated misconceptions. Moreover, this has prevented from establishing how the *hertzian radiation* takes place.



Hertz experiment. The apparatus for radio-electric transmission and reception. Birth of the *hertzian dipole antenna*, 1887.

By means of a particular radio-electric circuitry for the suppression of anyone of the two branches of a � of wavelength open dipole, I have demonstrated, once and for all, that the *open dipole* is not an *elementary antenna* (by definition an elementary antenna is an aerial where the condition of resonance and radiation cannot take place without the presence of all its parts) but it is, instead, an "*elementary array*" of 2 elements, of a physical length equal to � of a wavelength each, which are electrically arranged in a counterphase, while being fed in the middle of them.

Once it has been established that the focus should be placed on the behavior of a single element of physical length equal to � of wavelength, further research has allowed me to come up with a truly "*elementary radiator*" to further distinguish between the source of the radio-electric signal and the actual radiator.

The *elementary radiator*, infact, comprises a radio-electric circuitry and a � of wavelength radiator which can be easily detached and substituted by a dummy load (anti-inductive load), in order to conduct RF measurement on the circuitry alone, while, radio-scopic observations of the actual radiator emissions can be carried out by means of a particular detector which exploits and, at the meantime, demonstrates the phenomenon of the *radioluminescence* from secondary ionization induced by *hertzian radiation*.

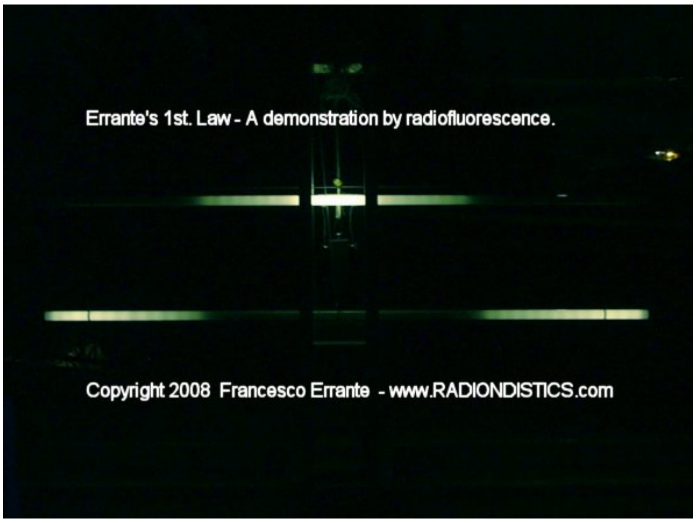
The radio-scopic observation by *radioluminescence*.
By employing my detector, I have carried out several non-intrusive radio-scopic measurements on the radiator over the full spectrum of the shortwaves, anywhere between 1 and 30 MHz with an RF power ranging from 100 mW to several kiloWatt.

The experiments I have deviced ed carried out, clearly, demonstrate that:

1) the energy radiated by the radiator has a transitory ionizing power;

2) by injecting a radio-electric signal into a properly resonant radiator, the latter will always radiate energy as radio waves, starting from the point which is always opposite to the point of feeding;

3) that the bulk of the energy is always radiated by the region towards the end of the radiator..



Demonstration of the Ist Errante's law by radioluminescence.

The image above shows the energy distribution on an half wave folded dipole and on an half wave open dipole, both in a condition of resonance, while being fed with two RF signals of equal amplitude and wave length.
It is observed how on the first case the light emission reaches its maximum intensity in the middle of the dipole while on the second case the light emission reaches its maximum intensity towards the ends of the dipole.

I$^{st.}$ Errante's law: any radio-frequency electric signal or any radio-frequency electric impulse that is injected onto any electrical conductor of any shape, **will ALWAYS give origin** to hertzian radiation starting from the point opposite to that of feeding.

Radio-electric measurements and radio-scopic counterverification.

Conversely, further measurements have allowed to verify the **absence of hertzian radiation on conductors (transmission lines) while in a progressive wave regime.**

Demonstration of the IInd Errante's law by radioluminescence.

The images above show the absence of hertzian radiation on a balanced transmission line, in a progressive wave regime, while being runned with a 1000 Watt RF signal. Either terminated on a reactive or a non-inductive load.

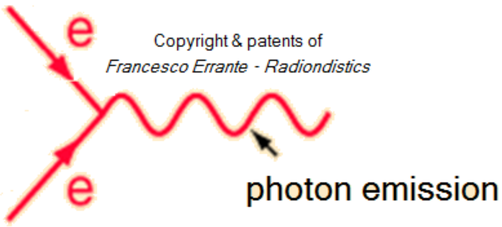
Power: 1 KW RF
Test frequency on the dipole: 25,5 Mhz
Test frequency on a dummy load: from 1.8 MHz to 30 Mhz in 100 KHz steps
Go to the experimental verification page

II$^{nd.}$ Errante's law: any radio-frequency electric signal or any radio-frequency electric impulse that is injected onto any electrical conductor of any shape properly terminated onto a load having an impedance value equal to that of its source, **will NEVER give origin** to hertzian radiation.

Author's deductions.
What has just been said, reasonably, leads to affirm that the **hertzian radiation takes place whenever charges belonging to a first wavefront having flowed forth along the radiator all the way up to its end, return backwards and hit a new forthcoming wavefront giving origin to collisions with scattering of particles, most likely to be photons, having the same frequency and wave form of the radio-electric signal that has generated it. As it is impossible for the new particles to travel faster than light, the particles will acquire more mass instead.** (if particles could travel faster then light, their emission would, inevitably, end-up generating shorter wavelength radio signals)

This mechanism is, therefore, a form of *controlled or limited standing wave regime*, if it happens within the length of the radiator then we have *resonance*, if it exceeds the length of the radiator we have a *random standing wave regime* with less or no radiation.

Feyman's diagram -
Here, the *antiparticle* is just another electron flowing in the opposite direction



In the hertzian radiation mechanism, the *antiparticle* is nothing else than an electron flowing back in the opposite direction to the one it was initially injected from.
<u>Hence, the minus sign of one of the two *Dirac's equation* solutions indicates the negative direction of the particle and not a negative charge of it</u>!

***Errante*'s radio-electric transduction mechanism is fully consistent with *Prof. John Archibald Wheeler*'s own intuition as aknowledged by *Richard Feynman* in his Nobel lecture.**

Moreover, evidence shows that the *law of conservation of energy* holds true, while leaves no room for keeping on theorizing about the *positron* and the *antimatter*.

Conclusions:

according to this Author's findings, the physical mechanism responsible for the generation of the *hertzian radiation* is different from what it was previously theorized and accepted as true.

The forecast of a concatenation between the electric field and the magnetic one, as offered by

Maxwell's equations, describe the electric and the magnetic fields engaged in an endless cycle of mutual creation. This interpretation bears a fundamental error which lies in the fact that the concatenation between fields was thought to be in the fashion of an overlaying and NOT in a relay-race like full-cycle progression as it happens to be, instead.

In the light of this Author's findings, it can be affirmed that there is no such a thing as an electromagnetic wave, an electromagnetic field or an electromagnetic signal. There are, instead, photonic emissions (radio waves) that follow the same wave form and frequency of the signals that have generated them, capable of inducing electromotive forces onto the matter with the same frequency and wave form.

As a direct consequence of this findings, when it comes to radio-electric transduction, the concept of *electromagnetic induction* does no longer hold true, either.

Accordingly, the mechanism by which an electrical conductor (ie: radio antenna) gets affected by the *hertzian radiation* must be different from what it was previously thought too and everything points to nothing else than a simple *photovoltaic effect*, just as observed by *Hertz* and *Hallwachs* in the ultraviolet radiation spectrum.

The transmission and reception of a radio-electric signal, from a point to another of the empty space, is not, therefore, a simple transfer of energy in the same form and the radio-frequency electric signals do not have the property of travelling in the empty space.

Errante's principle on the *radio-electric transmission*

For a given quantity of energy, in the form of a radio-frequency electric signal, to flow from a point to another of the empty space, to be later reproduced in its original form, it is necessary that the said energy undergoes at least two transformation:

1st from radio-frequency electric signal to hertzian radiation;
2nd from hertzian radiation back to

radio-frequency electric signal.

The minimum number of two transductions it's what it takes in the case of a direct line-of-sight radio propagation.

This particular phenomenon involving an energy transformation of a radio-frequency electric signal into a radio-frequency radiation and viceversa, is called *radio-electric transduction*. The *radio-electric transduction* is a spontaneous phenomenon where an ordained form of energy gets transformed into a disordered one with a different form and viceversa. This transformations can be repeated an infinite number of times but with regard to energy loss it is an highly disadvantageous exercice.

The devices capable of enhancing the *radio-elettric transduction*, are defined as *radio-eletctric transducers* **and are commonly known as "radio antennas".**

NEW Please, take a look at our new all-in-one apparatus for the physics of the hertzian radiation and the radio-electric transducers. Here !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LIBERATION
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Over 25 years of direct action for animals
without harming a single human being

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10. RIVELATORE PASSIVO DI RADIAZIONE HERTZIANA A FLUORESCENZA DA IONIZZAZIONE SECONDARIA - Radioluminescenza

antenna_errante.htm

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The *elementary radiator*

also known as the

Errante's Antenna

NEW! NOW, also available in the

Errante's 225 Ohm MULTIBAND system version

Patent n.0001347485 (ITAPA20030020)
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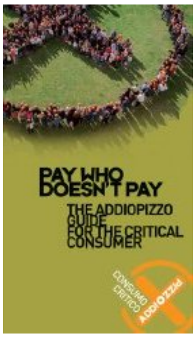
Virtual ground node

Errante's CAPACITIVE EARTH GROUNDING SYSTEM

Balanced transmission lines in a

Here above is the *Errante's antenna* laboratory prototype in a quick fan-monopoles deployment for the 20, 30, 40, 60, 80 and 160 meter bands operation.
N.B. Shown without the ABS weatherproof outer enclosure.



Freq. range: from 1 to 30 MHz
RF Power: 2KW CW

progressive wave regime

Hertzian radiation: what it is and how it happens

Click on the thumbnails to view larger pictures





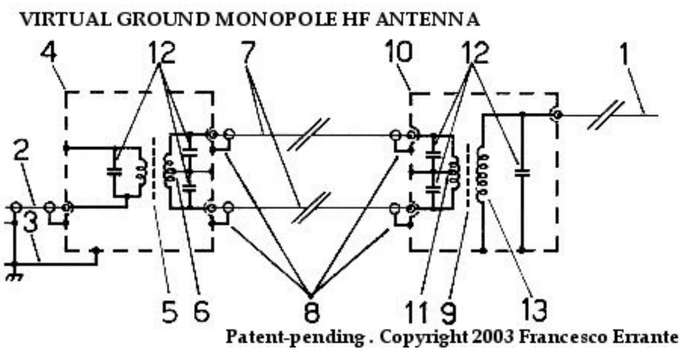
ERRANTE's ANTENNA : what it is, how it works and how it came to be made.

VIRTUAL GROUND � of wavelength MONOPOLE (HF) ANTENNA

by *Francesco Errante*

The Author, by means of a particular radio-electric circuitry for the suppression of anyone of the two branches of a � of wavelength open dipole, has previously demonstrated that, indeed, it is possible to receive and above all, transmit large radio-electric signals across the space by employing a single radiating/captating element of a physical length equal to � of wavelength, without the necessity of the presence of its symmetric (see *Hertzian Dipole*) or the necessity of a natural ground plane (see *Marconi's Antenna*) or an artificial one made of radials or counterpoises as in the *Ground Plane Antenna*. The need to have the Earth or another artificial ground as an electric reference point to correctly polarize a single resonant radiator/captator of a physical length equal to � of a wavelength, has been eliminated by generating a *virtual ground node*, the existence of which is function of the precise balancing of the of a broad-band RF transformer. ("virtual ground" is defined as a point in an electrical circuit that appears to be at ground, but is not actually attached to ground, it is therefore, a node having a 0 degree phase angle difference with respect to ground and has the same electrical potential as the Earth")
This method, however, implied a 50% dissipation of the system energy.

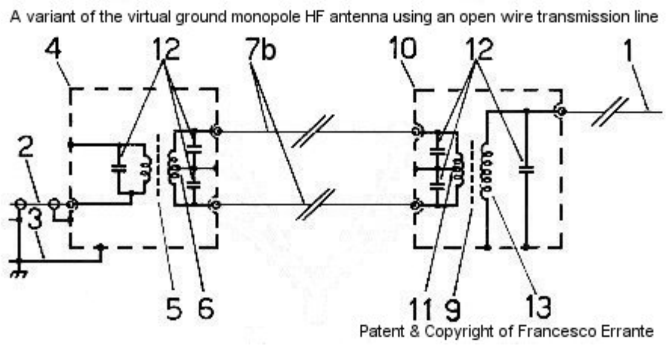
The invention described hereby eliminates the need for sacrificing any of the system energy by exploiting the very same physical principal of the *virtual ground node*.



The solution to that problem is given by the possibility of generating the *virtual ground node* by recombining two radio-electric signals, identical but 180 degree out of phase between them-selves. This can be achieved by employing a broad-band flux-coupled balun transformer(9) built around a binocular type ferrite core, having a center-tapped balanced primary winding(11) for the obtainment of the said *virtual ground node* to be used as a zero-potential reference point for the whole of the circuit, through the conductive chassis(10) and having also a secondary winding(13) exhibiting an impedance value of 150 Ohm for the direct coupling to a � of a wavelength radiator/captator(1).
The transformer(9) and the radiator/captator(1) are the essential parts of the *VIRTUAL GROUND � of wavelength MONOPOLE ANTENNA*, they alone, infact, constitute a � of wavelength monopole antenna with balanced input. However, in order for it to match the current transmission line standards, it has been necessary to provide it with a second transformer(5) capable of adapting the unbalanced coaxial antenna output of a conventional transceiver to the balanced input of this antenna transformer(9). The virtual ground node(6), obtained on the secondary winding of the transformer(5), is made available to the circuit through the conductive chassis(4).
The RF energy transfer between the two

transformers(5 & 9) is accomplished by the twinned coaxial transmission lines(7) which can be of any length but must be identical between themselves so not to alter the original phase-angle difference between the RF signals they are meant to be carrying. Although, the value of impedance of the intermedium circuit, which comprises the primary winding of the transformer(9), the twinned coaxial lines(7) and the secondary winding of the transformer(5) can be set to any value, it turned out to be convenient adopting a standard value of 50 Ohm.

Following the publication of the studies on the behaviour of a balanced transmission line in a progressive wave regime, a variant to the virtual ground monopole HF antenna is disclosed here below:



It is of a paramount importance that the impedance of local transformer's(5) secondary winding, the impedance of the open wire balanced transmission line(7b) and the impedance of the remote transformer's(9) primary winding be exactly the same.

Scientific relevance:

a. it demonstrates that a single electrically conductive element of a physical length equal to � of wavelength is all it takes to accomplish natural resonance and full radiation.

b. thanks to its most elementary shape and bare simplicity of the radiating element, it allows further investigation into the physics of the radiowaves. (see here)

Operational advantages:

a. the *VIRTUAL GROUND MONOPOLE HF ANTENNA* demonstrates that, indeed, it is possible and relatively inexpensive to implement an highly efficient way to better radiate RF energy while cutting by 50% the room needed to deploy a full-size HF antenna.

b. the *VIRTUAL GROUND MONOPOLE HF ANTENNA* offers a much larger bandwidth compared to conventional full-size aerials.

c. the *VIRTUAL GROUND MONOPOLE HF ANTENNA* offers a much stronger immunity to parasitic capacitive effects compared to conventional full-size aerials.

d. the *VIRTUAL GROUND MONOPOLE HF ANTENNA* offers high resonance stability regardless of the level and inclination above the ground.

e. the *VIRTUAL GROUND MONOPOLE HF ANTENNA* offers a much stronger immunity to out-of-band emissions compared to conventional full-size aerials.

f. the *VIRTUAL GROUND MONOPOLE HF ANTENNA* offers a much stronger immunity to electrostatic charges and noises compared to conventional full-size aerials.

g. the *VIRTUAL GROUND MONOPOLE HF ANTENNA* offers a sharper radiation pattern than conventional full-size aerials.

h. the *VIRTUAL GROUND MONOPOLE HF ANTENNA* offers a much faster deployment time compared to conventional full-size aerials.

i. the *VIRTUAL GROUND MONOPOLE HF ANTENNA* offers a much better E.M.I. compatibility with highly populated environments compared to conventional full-size aerials.

Applications:

a. SHORT WAVES FIXED & MOBILE RADIO STATION FOR CIVIL & MILITARY PURPOSES

b. STEALTH, FAST & TACTICAL DEPLOYMENT PURPOSES

c. SEARCH & RESCUE SERVICES

d. SHIP to SHIP & SHIP to SHORE MARINE COMMUNICATIONS

e. AIRCRAFT COMMUNICATIONS RANGE EXTENDING ANTENNA

f. DX-ing and metropolitan HAM RADIO

STATIONS

g. ELEMENT in ARRAYS of AERIALS

h. RADIO-PHYSICS LABORATORIES

most **F.A.Q.**: can I drive 2 or more monopoles simultaneously, as in a *fan dipole* configuration, so that I can cover several HF bands?
ANSWER: Yes ! The present virtual ground RF feeding system allows you to run all the � of wavelength monopoles needed to give you a general coverage of the HF spectrum, without the need for switchings. The monopoles can be arranged on both vertical and horizontal planes. Space is all you need to deploy them.
Now, also available for the 225 Ohm Errante's multiband HF antenna system

Price:
Euro 700,00 +V.A.T. and shipping costs.
For an useful currency converter click here

NB. Price DOES NOT INCLUDE the cost of the twinned coaxial line, nor the radiators, which, however, can be supplied on request.

RADIONDISTICS' products carry an unconditional lifetime guarantee.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Over 25 years of direct action for animals
without harming a single human being

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11. The radio-electric transducers in the light of the new discoveries - © Francesco Errante

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The radio-electric transducers, better known as *radio antennas*, in the light of the new discoveries.

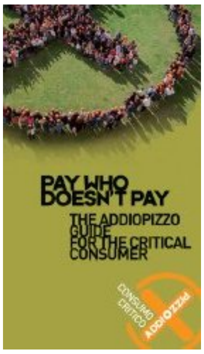
by *Francesco Errante*

In physics, by the word "*transducers* ", we define all of those devices capable of transforming a form of energy into another one and vice-versa. Generally, during the transformation performed by the

Hertzian radiation: what it is and how it happens

Balanced transmission lines in a progressive wave regime

transducers, an ordained form of energy gets transformed into a disordered one and vice-versa. As an example: loudspeakers and microphones are electro-mechanical transducers. They, infact, transform electric energy into mechanical vibrations and vice-versa. Radio antennas are, instead, "*radio-electric transducers*", they transform radio-frequency electric energy into *hertzian radiation* (radio-waves) and vice-versa. Unlike what happens with other types of transducers, understanding how radio-electric transducers work cannot be done by intuition. That's because they have no machanical parts in motion and their "vibrations" cannot be directly perceived by the human body. To tell the truth, under certain conditions (frequency, power and distance from the source), the human body can feel some of the effects of the *hertzian radiation*, as a distributed heat or a localized burn, but this was not enough to discover the real mechanism at the origins of the *radio waves'* phenomenon.



Errante's 225 Ohm multiband HF antenna system

Self-antagonist radiating systems

For more than a century and indeed, ever since *Heinrich Hertz* demonstrated the existence of the radio-electric phenomena, their "scientific" description has been limited to a mere theoretical representation by mathematics. In the year 2003, the Author has, letterally and definitely, shed some light on the real mechanism at the origins of the *hertzian radiation*, by means of his own particular detection system based on the *radioluminescence*, another physical phenomenon of energy transformation.
Untill recent, the radio-electric transdusers have been divided in two main types: the balanced and the un-balanced ones.
In the ligth of what the Author has demonstrated, the said division should now be revised. Accordingly, the radio-electric

transdusers should be distinguished in two categories: the "*elementary* " ones and the "*NON-elementary* " ones.
(by definition an *elementary radio-electric transducer* is a transducer where the condition of resonance and radiation cannot take place without the presence of all its parts)
The elementary radio-electric transducers are *monopolos* (un-balanced antennas), while the NON-elementary ones are *dipoles* (balanced antennas). Both types of transducers can be *open ended* or closed circuits as *loops*. It has to be clarified that *loops* are, also, often improperly referred to as "folded". This is incorrect as, in radio-engineering, it is possible to have folded antennas that are electrically open.
Dipoles can be *open dipoles* or of a closed circuit type, incorrectly called "*folded dipoles*". However, the latter is the father of all the *loops*. The *folded dipole*, as demonstrated, can be traced back to two closed-circuit monopoles fed in a counter-phase arrangement and physically coinciding with each other. This is, also, true for *loops* having any geometry (e.g. circular, delta-loop, polygonal etc.).
Any NON-elementary radio-electric transducer (balanced transducers) can always been traced back to a system of elementary transducers (unbalanced transdusers) being arranged in a counterphase configuration.
The incorrect spacing-up between elementary radio sources belonging to the same system will give origin to self-antagonist systems to the detriment of the *antenna's gain*.

�

Theoretical fundamentals of radioengineering, radiophysics, practical applications, antenna engineering, advanced antenna design, HF antennas, scientific publications, technical publications, physics, science, radio waves, hertzian waves, hertzian radiation, radiation pattern, antenna coupling, antenna feed, antenna swr, antenna technology, transmitting, receiving, aerials, balun, transformers, RF, radio frequency, virtual ground, electromagnetics phenomenon, phenomena, studies, radioelectric experiments gateway, scientific research, applied research, radio propagation, ionospheric propagation, trans-ionospheric propagation,HF,shortwave radio communications

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12. RF balanced transmission lines behaviour in a progressive wave regime - Radiondistics.com

balanced_lines.htm

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Errante's apparatus
for the physics of the balanced transmission lines for radio frequency signals.

Foreword: although when referring to balanced transmission lines our thougth goes to a pair of identical electrical conductors beign parallel to each other, commonly known as a "*bifilar line*" , it must be said that balanced transmission lines can also be formed by any even number of conductors carrying the same number of currents in a multi-phases arrangement. (Poly-phase currents arrangement were first introduced in the electrical engineering by *Nikola Tesla*) Balanced transmission lines can be either *symmetric* or *asymmetric*.

Francesco Errante

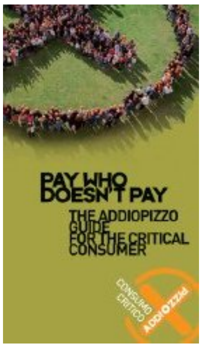
Virtual ground node

Virtual ground monopole antenna

Balanced virtual ground node

Definition of a balanced transmission line: � "*a transmission line for electric signals is electrically balanced when it does not intervene to modify the phase difference between the currents traveling on it and each and every of its conductors transfer the same amount of energy per time unit (power).*"

Francesco Errante



Definition of a *symmetric* balanced transmission line: � "*an electrically balanced transmission line for electric signals is symmetric when it is formed by electrical conductors which exhibit the same impedance value with respect to the physical ground or to a virtual ground node.*"

Francesco Errante

Definition of an *asymmetric* balanced transmission line: "*an electrically balanced transmission line for electric signals is asymmetric when it is formed by electrical conductors which DO NOT exhibit the same impedance value with respect to the physical ground or to a virtual ground node.*"

Francesco Errante

I[st] *Errante*'s principle for the transmission lines: � "*a transmission line*

generator

Hertzian radiation detection by radio luminescence

for radioelectric signals, being it balanced or unbalanced, when in a progressive wave regime, is always aperiodic."

Francesco Errante

II^nd *Errante*'s principle for the transmission lines: "***any transmission line for radioelectric signals, may be converted from being a balanced one to an unbalanced one and viceversa, an infinite number of times.***"

Francesco Errante

III^rd *Errante*'s principle for the transmission lines: "***it is always possible to generate a virtual ground node anywhere along a transmission line for radioelectric signals*** ."

Francesco Errante

II^nd *Errante*'s law applied to the transmission lines "***a transmission line for radioelectric signals, being it balanced or unbalanced, when in a progressive wave regime, does not originate hertzian radiation.***"

Francesco Errante

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The apparatus and the measurement method presented hereby permit to analyze the electric behaviour of balanced transmission lines for radio frequency electric segnals in a progressive wave regime and in particular

have the purpose of:

a) demonstrating that the balanced transmission lines are aperiodic;

b) demonstrating that the balanced transmission lines can be converted into unbalanced ones and viceversa, an infinite number of times;

c) demonstrating that it is always possible to generate a virtual ground node at any point of any transmission line;

d) demonstrating that the balanced transmission lines, when perfectly matched, both to the signal source and to the load, DO NOT originate hertzian radiation. �IInd Errante's law.

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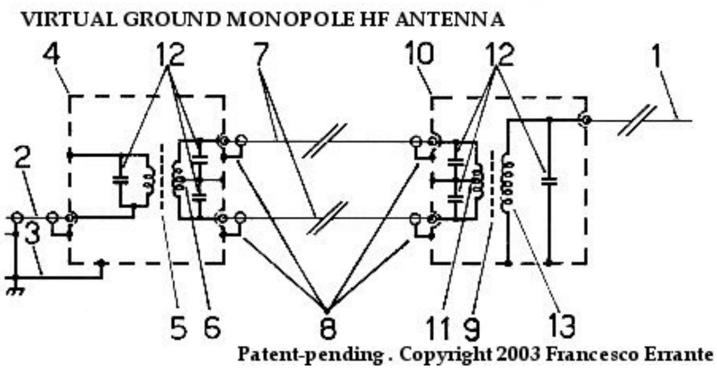
The balanced transmission lines for radio frequency electric signals are generally formed omogeneously by two identical unshielded electrical conductors running parallel to each other but can also be formed by a larger number of conductors, being them open wires or shielded cables. �



An example of an electrically balanced line employing a pair of individually shielded coaxial cables(7) - Copyright © 1985- Francesco Errante

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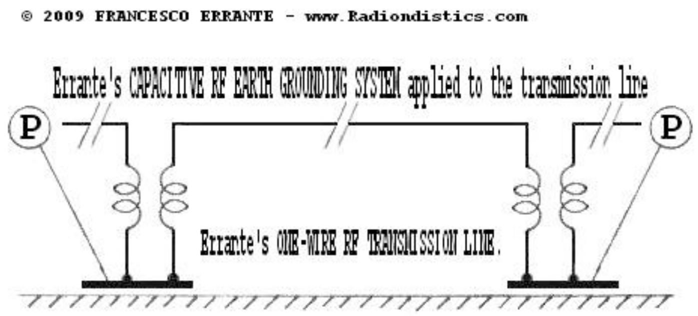

Errante's ONE-WIRE UNBALANCED RF TRANSMISSION LINE, employing Errante's CAPACITIVE RF EARTH GROUNDING SYSTEM - Copyright © 2009 - Francesco Errante

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A transmission line for electric signals is electrically balanced when it does not intervene to modify the phase difference

between the currents traveling on it.
In order for that to be possible it is necessary that all the line's conductors be identical to each other and that the line it-self be perfectly matched both on its input and output ports. That means that the characteristic impedance of the line must be equal to the impedance of the radio frequency signal source and to the impedance of the load, being it reactive or non-inductive. A correctly matched transmission line, when being runned, is in a **progressive wave regime** and the energy transfer to the load is maximum with no *hertzian radiation* being originated.

Wherever the transmission line runs at a close proximity of the ground, it is also necessary for its conductors to be equally spaced away from it, in order to keep their capacitive coupling constant to the ground it-self.
If the interaxial distance between the line's conductors is kept constant, decreasing the line distance from the ground, the line's own characteristic impedance decreases accordingly. This is due to the line's conductors capacitive coupling to the ground, this capacitance adds to the the parasitic electrical capacitance between the line's conductors, which is always present.
Balanced transmission lines, if properly isolated, can run along the ground surface or can even be buried underground. In those cases, the close capacitive coupling between the line's conductors and the ground makes the parasitic electrical capacitance between the line's conductors neglectable, to the point that they no longer need to run parallel to each other. However, they still need to be identical to each other, unless they are ment to form an *asymmetric balanced line*.

Electrically balanced transmission lines can also be geometrically and electrically asymmetric.
An asymmetric balanced line is so defined: "*a balanced transmission line for electric signals is asymmetric when it does not intervene to modify the phase difference between the currents traveling on it and it is formed by electrical conductors which do not exhibit the same impedance value with respect to the physical ground or to a virtual groung node. Copyright 2008 Francesco Errante.*

Close coupling between the line's conductors and the ground, however, confines the usage of lines running along the ground or underground to low and medium power systems, where voltages involved do not require a great deal of insulation.
With the advancement of the technology for radioelectric constuctions, the coaxial cable unbalanced lines have, progressively, taken the place of the balanced transmission lines in the majority of the applications, except for the very high power ones.

Right from the outset of radio engineering, the most recurrent case of usage of a balanced transmission line is represented by its employment in the feeding of the *half a wave length folded dipole* antenna. This because, a 300 Ohm balanced transmission line adds ease of construction to the folded dipole's useful properties. (The same can be said for the multiple-wire folded dipoles that have characteristic impedance multiple of the 300 Ohm)

Rough observations made without any scientific rigor and the fact that it is much too easy to feed an half a wave length folded dipole with a a stretch of a 300

Ohm balanced transmission line has led everyone, untill now, to erroneously desuming and/or accepting that balanced transmission lines were them-selves resonant and, therefore, that it was necessary to neutralize their effects by lengthening or shortening them. This desumption is false! *Ernst Lecher*'s experiment erroneous interpretation has done the rest in spreading this misconseption. Lecher's line, infact, shows the electric behaviour of a two-wire line in a standing wave regime and its employment is, instead, of no use in measuring electric wavelengths if the line is in a progressive wave regime (VSWR = 1:1).

The instrumental illusion that balanced lines be resonant is given by the fact that, when a folded dipole is directly fed by a two-wire line, there is no clear cut border between the dipole it-self and the stretch of the bifilar line, therefore, the electric signals will travel on a path which varies with the transmission line length, with the result that the point of resonance of the whole system will move ciclically.

The truth is that, if in a system being formed by a balanced signal source, a balanced line and a radiator, the reactive load gets substituted by a non-inductive load of equal impedance (dummy load) or by a perfectly matched reactive load, the illusion that the balanced line be resonant disappears and the system returns to showing it-self as being aperiodic, as demonstrated in the experimental verifications pictured below:

A step-by-step sequence of all the measurements taken, along with their results is shown here below:

These measurements, with the proper equipment, can be carried out in three different ways:

1) by keeping constant the test signal wavelength while varying the physical length of the balanced line;
2) by keeping constant the physical length of the balanced line while varying the test signal wavelength;
3) by systematically varying both the physical length of the balanced line and the test signal wavelength.

The second method is a great deal more convenient and faster than the others, so that, **a 300 Ohm balanced line of a fixed physical length has been used and reflected energy measurements have been taken while varying the test signal wavelength**.

Moreover, it should be noted that these measurements can be carried out on any spectrum of radio frequencies, but it is most convenient to perform them on the shortwaves spectrum as they give, amongst other benefits, a larger wavelength excursion for the same frequency span. **All the measurements results, therefore, can be extended to the whole spectrum of the radio waves.**

NB: all the devices here employed for the obtainement of virtual ground nodes provide a clear cut border for the RF signals between lines and radiators as well as lines and lines, while always offering a total galvanic continuity among all the circuits connected to them, ensuring a solid common ground path, which can, if necessary, connected to the physical ground. See also: A virtual ground node generator for unbalanced lines

Experimental verification of the balanced lines aperiodic behaviour, when in a progressive wave regime

PRELIMINARY OPERATION: the balanced-to-balanced measurement network is tested for frequency response flatness within the test spectrum.

From left to the right: 50 Ohm co-axial TX line - S.W.R. probe - 50/300 Ohm BalUn - short stretch of a symmetric bifilar transmission line - 300/50 Ohm BalUn - 50 Ohm dummy load.
Test Freq.: from 1,8 to 30 MHz in 100 KHz steps
RF power : 1 KW cw
Measured S.W.R.: � 1:1 � over the whole spectrum
Result: OK! The network exhibits a flat response over the whole test frequency spectrum.

�

Network analysis of a 300 Ohm balanced line, fed by a 300 Ohm balanced signal

source, terminated on a 300 Ohm dummy load.

From left to the right: 50/300 Ohm BalUn - a random (4 m ca.) stretch of a 300 Ohm symmetric bifilar transmission line - a 300 Ohm dummy load.

Test Freq.: from 1,8 to 30 MHz in 100 KHz steps

RF power : 1 KW cw

Measured S.W.R.: � 1:1 � over the whole spectrum

Result: the line shows an aperiodic behaviour over the whole test frequency spectrum. The line aperiodic behaviour is VERIFIED.

N.B. All the measurements have been repeated with the line being held up in the air.

�

The line shows the same aperiodic behaviour as in the previous test, even if it is terminated on a non-radiating reactive load (Errante's BalUn and a dummy load) as shown below:

Network analysis of a 300 Ohm balanced line, fed by a 300 Ohm balanced signal source, terminated on a 300/50 Ohm BalUn and a dummy load.

From left to the right: 50 Ohm co-axial TX line - S.W.R. probe - 50/300 Ohm BalUn - a random (4 m ca.) stretch of a 300 Ohm symmetric bifilar transmission line - 300/50 Ohm BalUn - 50 Ohm dummy load.

Test Freq.: from 1,8 to 30 MHz in 100 KHz steps
RF power : 1 KW cw
Measured S.W.R.: � 1:1 � over the whole spectrum
Result: the line shows an aperiodic behaviour over the whole test frequency spectrum. The line aperiodic behaviour is VERIFIED.
N.B. All the measurements have been repeated with the line being held up in the air.

�

The line, again, shows the same aperiodic behaviour as in the previous tests, even if it is terminated on a dummy load through a reactive RF network as shown below:

Network analysis of a 300 Ohm balanced line de-coupled from a non-reactive non-radiating load (300 Ohm dummy load) by means of a 300/300 Ohm Errante's balanced virtual ground node generator.

From left to the right: 50/300 Ohm BalUn - a random (4 m ca.) stretch of a 300 Ohm symmetric bifilar transmission line - Errante's 300/300 Ohm

B.V.G.N.G - 300 Ohm dummy load.

Test Freq.: from 1,8 to 30 MHz in 100 KHz steps

RF power : 1 KW cw

Measured S.W.R.: � 1:1 � over the whole spectrum

Result: the line shows an aperiodic behaviour over the whole test frequency spectrum. The line aperiodic behaviour is VERIFIED.

N.B. All the measurements have been repeated with the line being held up in the air.

�

Finally, by employing a virtual ground node generator for electrically balanced systems, it has been dimostrated that balanced lines, if properly decoupled from the radiators, always exhibit an aperiodic behaviour, as shown hereafter:

Network analysis of a 300 Ohm balanced line de-coupled from a reactive and radiating load (300 Ohm folded dipole) by means of a 300/300 Ohm Errante's balanced virtual ground node generator.

Ist test:

Purpose: finding the test dipole natural resonance

Conditions: test dipole fed through a 50/300 Ohm virtual ground BalUn.

Test frequency: **test dipole center frequency - MHz 22,9**

RF Power: 1 KW cw

S.W.R.: � 1:1 � @ MHz 22,9

Result: the test dipole center frequency found to be MHz 22,9

IInd test:

Purpose: finding the new test dipole center frequency once its feeding system changed

Conditions: test dipole fed directly by a stretch of a 300 Ohm balanced transmission line

Test frequency: **NEW test dipole center frequency - MHz 23,5**

RF power: 1 KW cw

S.W.R.: 1:1 � @ MHz 23,5

Result: the test dipole center frequency has been found to have shifted to MHz 23,5

IIIrd test:

Purpose: finding the new test dipole center frequency once it has been de-coupled from its balanced transmission line

Conditions: test dipole is de-coupled from its 300 Ohm balanced transmission line through a 300/300 Ohm balanced virtual ground node generator

Test frequency: **test dipole center frequency**

found to be back on MHz 22,9

RF power: 1 KW cw

S.W.R.: � 1:1 � @ MHz 22,9

Result:

1) test dipole center frequency has been restored to its natural resonance value.

2) it has been verified that the balanced transmission line exhibits an aperiodic behaviour.

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Experimental verification of hertzian radiation ABSENCE on a balanced transmission line in a progressive wave regime.

IInd Errante's law

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A perfectly adapted balanced transmission line being runned with 1 KW RF has been tested for hertzian radiation emission by means of radioluminescence.

It has been observed and at the same time demostrated the absence of hertzian radiation emission on a transmission line while in a progressive wave regime.

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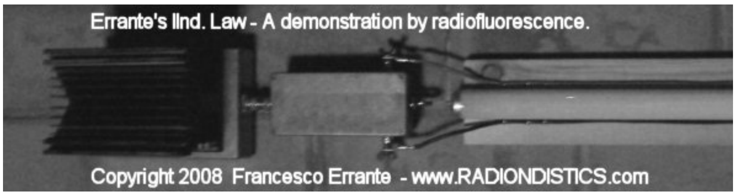
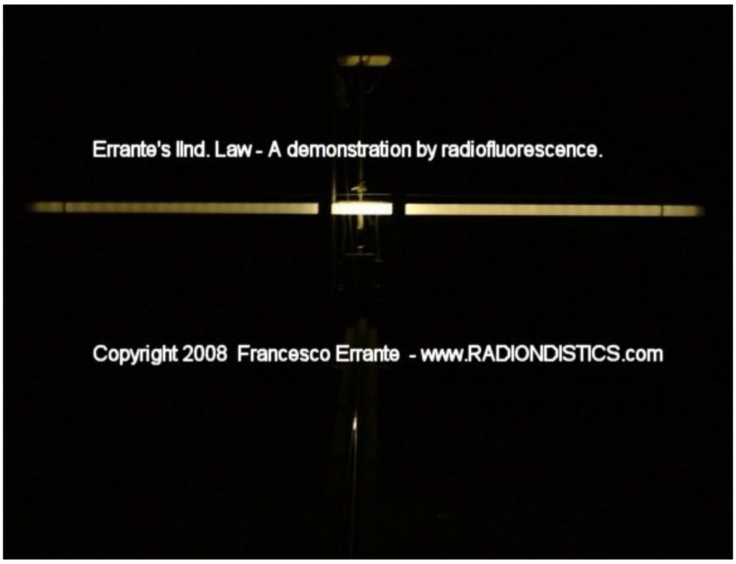
A properly adapted 300 Ohm balanced transmission line terminated on a reactive but non-radiating load undergoing a radiation test by means of a radiofluorescent detector.



From left to the right: a 50 Ohm dummy load - 50/300 Ohm BalUn - a random stretch of a 300 Ohm symmetric bifilar transmission line, about 4 m. long
Test frequency: from 1,8 to 30 MHz in 100 KHz steps
RF power: 1 KW cw
Measured S.W.R.: � 1:1 � over the whole spectrum
Result: the line does NOT radiate.

�

A properly adapted 300 Ohm balanced transmission line terminated on a 1/2 wavelenght folded dipole, through a balanced virtual ground node generator , undergoing a radioscopic detection. Measurements performed both on low and high RF power (Freq. 25,5 MHz - 1 KW cw).



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Hertzian radiation emission by a balanced transmission line, intentionally placed on a standing wave regime.

A standing wave regime (S.W.R. 1:1,7) has been imposed on the transmission line previously used, by substituting the 50/300 Ohm load balun with a 50/150 Ohm one.
It has been observed and at the same time

demostrated the PRESENCE of hertzian radiation emission on a transmission line while in a standing wave regime.

From left to the right: a 50 Ohm dummy load - 50/300 Ohm BalUn - a random stretch of a 300 Ohm symmetric bifilar transmission line, about 4 m. long
Test frequency: from 1,8 to 30 MHz in 100 KHz steps
RF power: 1 KW cw
Measured S.W.R.: � 1:1 � over the whole spectrum
Result: the line RADIATES.

Another view of a stretch of the bifilar line in a standing wave regime.

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An interactive impedance mismatch spectral simulation
by
Agilent Technologies

General Instructions

This spectral simulation is an interactive Java applet. You can change parameters by clicking on the vertical arrow keys. The five control buttons at the lower right are used to start (triangle) and pause (square) the simulation, to skip forward or back one section at a time (double triangles), and to change speed (+ and -).

After the simulation is complete, the start button takes you back to the beginning of the simulation. You may experience a delay at this point.

Wave Propagation along a Transmission Line

When a sine wave from an RF signal generator is placed on a transmission line, the signal propagates toward the load. This signal, shown here in yellow, appears as a set of rotating vectors, one at each point on the transmission line.

In our example, the transmission line has a characteristic impedance of 50 ohms. If we choose a load of 50 ohms, then the amplitude of the signal will not vary with position along the line. Only the phase will vary along the line, as shown by the rotating vectors in yellow.

If the load impedance does not perfectly match the characteristic impedance of the line, there will be a reflected signal that propagates toward the source. At any point along the transmission line, that signal also appears to be a constant voltage whose phase is dependent upon physical position along the line.

The voltage seen at one particular point on the line will be the vector sum of the transmitted and reflected sinusoids. We can demonstrate this by looking at two examples.

Example 1: Perfect Match: 50 Ohms

Set the terminating resistor to 50 ohms by using the "down arrow" dialog box. Notice there is no reflection. We have a perfect match. Each rotating vector has a normalized amplitude of 1. If we were to observe the waveform at any point with a perfect measuring instrument, we would see equal sine wave amplitudes anywhere along the transmission line. The signal amplitudes are indicated by the green line.

Example 2: Mismatched Load: 200 Ohms

Now let's intentionally create a mismatched load. Set the terminating resistor to 200 ohms by using the down arrow. Hit the PLAY button and notice the change in the reflected waveform. If it were possible to measure just the reflected wave, we would see that its amplitude does not vary with position along the line. The only difference between the reflected (blue) signal, say at point "z6" and point "z4", is the phase.

But the amplitude of the resultant waveform, indicated by the standing wave (green), is not constant along the entire line because the transmitted and reflected signals (yellow and blue) combine. Since the phase between the transmitted and reflected signals varies with position along the line, the vector sums will be different, creating what's called a "standing wave".

With the load impedance at 200 ohms, a measuring device placed at point z6 would show a sine wave of constant amplitude. The sine wave at point z4 would also be of constant amplitude, but its amplitude would differ from that of the signal at point z6. And the two would be out of phase with each other. Again, the difference is shown by the green line, which indicates the amplitude at that point on the transmission line.

The impedance along the line also changes, as shown by the points labeled z1 through z7.

VSWR

The VSWR, or Voltage Standing Wave Ratio, is the ratio of the highest amplitude signal to the lowest amplitude signal, as measured along the transmission line. A "perfect" VSWR is 1.

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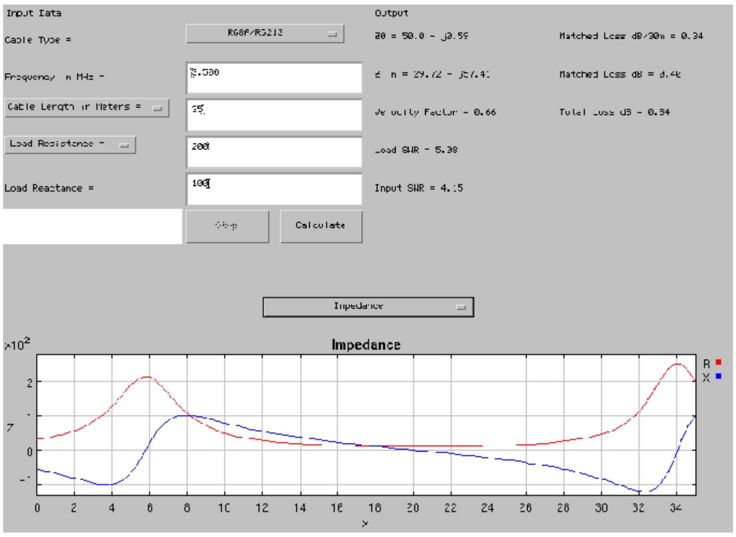
A transmission line calculator



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Over 25 years of direct action for animals
without harming a single human being

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13. GUGLIELMO MARCONI - Wireless telegraphic communication - Nobel Lecture, December the 11th. 1909

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GUGLIELMO MARCONI

Wireless telegraphic communication

Nobel Lecture, December 11, 1909

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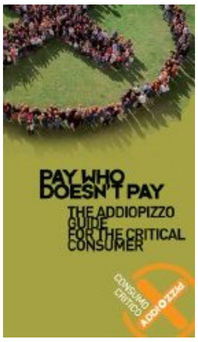
The discoveries connected with the propagation of electric waves over long distances and the practical applications of telegraphy through space, which have gained for me the high honour of sharing the Nobel Prize for Physics, have been to a great extent the results of one another. The application of electric waves to the purposes of wireless telegraphic communication between distant parts of the earth, and the experiments which I have been fortunate enough to be able to carry out on a larger scale than is attainable in ordinary laboratories, have made it possible to investigate phenomena and note results often novel and unexpected.
In my opinion many facts connected with the transmission of

electric waves over great distances still await a satisfactory explanation, and I hope to be able in this lecture to refer to some observations, which appear to require the attention of physicists.

In sketching the history of my association with radiotelegraphy, I might mention that I never studied physics or electrotechnics in the regular manner, although as a boy I was deeply interested in those subjects.

I did, however, attend one course of lectures on physics under the late Professor Rosa at Livorno, and I was, I think I might say, fairly well acquainted with the publications of that time dealing with scientific subjects including the works of Hertz, Branly, and Righi.

At my home near Bologna, in Italy, I commenced early in 1895 to carry out tests and experiments with the object of determining whether it would be possible by means of Hertzian waves to transmit to a distance telegraphic signs and symbols without the aid of connecting wires.



After a few preliminary experiments with Hertzian waves I became very soon convinced, that if these waves or similar waves could be reliably transmitted and received over considerable distances a new system of communication would become available possessing enormous advantages over flashlights and optical methods, which are so much dependent for their success on the clearness of the atmosphere.

My first tests were carried out with an ordinary Hertz oscillator and a Branly coherer as detector, but I soon found out that the Branly coherer was far too erratic and unreliable for practical work.

After some experiments I found that a coherer constructed as shown in Fig. 1, and consisting of nickel and silver filings placed in a small gap between two silver plugs in a tube, was remarkably sensitive and reliable. This improvement together with the inclusion of the coherer in a circuit tuned to the wavelength of the transmitted radiation, allowed me to gradually extend up to about a mile the distance at which I could affect the receiver.

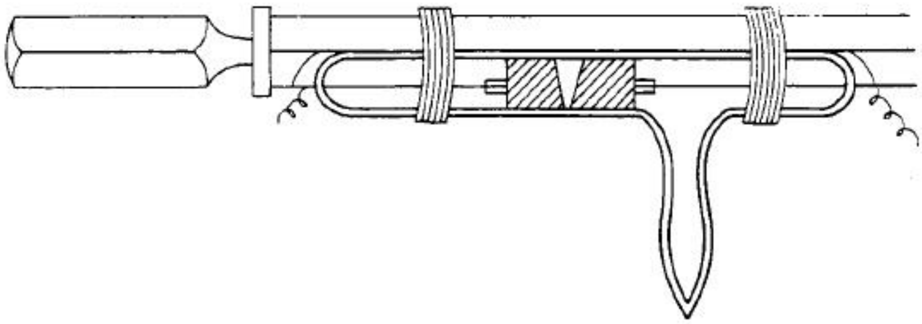


Fig. 1.

Another, now well-known, arrangement which I adopted was to place the coherer in a circuit containing a voltaic cell and a sensitive telegraph relay actuating another circuit, which worked a tapper or trembler and a recording instrument. By means of a Morse telegraphic key placed in one of the circuits of the oscillator or transmitter it was possible to emit long or short successions of electric waves, which would affect the receiver at a distance and accurately reproduce the telegraphic signs transmitted through space by the oscillator.

With such apparatus I was able to telegraph up to a distance of about half a mile.

Some further improvements were obtained by using reflectors with both the transmitters and receivers, the transmitter being in this case a Righi oscillator.

This arrangement made it possible to send signals in one definite direction, but was inoperative if hills or any large obstacle happened to intervene between the transmitter and receiver.

In August 1895 I discovered a new arrangement which not only greatly increased the distance over which I could communicate, but also seemed to make the transmission independent from the effects of intervening obstacles.

This arrangement (Figs. 2 and 3) consisted in connecting one terminal of the Hertzian oscillator, or spark producer, to earth and the other terminal to a wire or capacity area placed at a height above the ground, and in also connecting at the receiving end one terminal of the coherer to earth and the other to an elevated conductor.

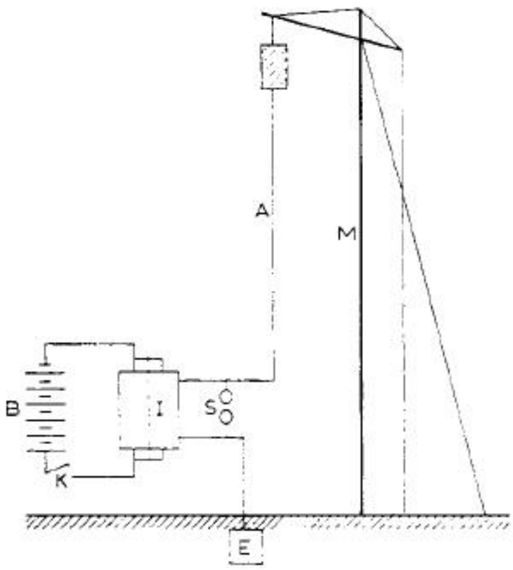


Fig. 2.

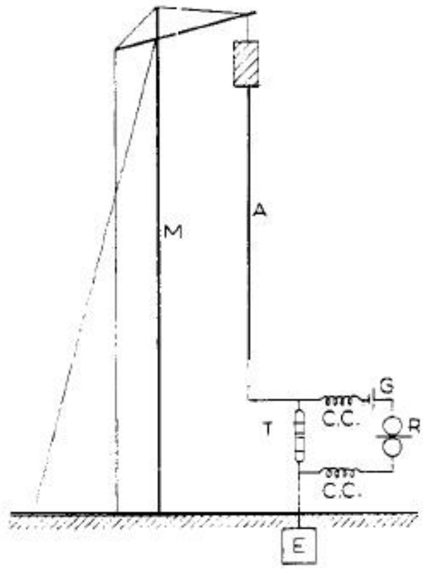


Fig. 3.

I then began to examine the relation between the distance at which the transmitter could affect the receiver and the elevation of the capacity areas above the earth, and I very soon definitely ascertained that the higher the wires or capacity areas, the greater the distance over which it was possible to telegraph.

Thus I found that when using cubes of tin of about 30 cm side as elevated conductors or capacities, placed at the top of poles 2 meters high, I could receive signals at 30 meters distance, and when placed on poles 4 meters high, at 100 meters, and at 8 meters high at 400 meters. With larger cubes 100 cm side, fixed at a height of 8 meters, signals could be transmitted 2,400 meters all round.[1]

These experiments were continued in England, where in September 1896 a distance of 1 3/4 miles was obtained in tests carried out for the British Government at Salisbury. The distance of communication was extended to 4 miles in March 1897, and in May of the same year to 9 miles. Tape messages obtained during these tests, signed by the British Government Officers who were present, are exhibited.[2]

In all these experiments a very small amount of electrical power was used, the high tension current being produced by an ordinary Rhumkorff coil.

The results obtained attracted a good deal of public attention at the time, such distances of communication being considered remarkable.

As I have explained, the main feature in my system consisted in the use of elevated capacity areas or antennae attached to one pole of the high

frequency oscillators and receivers, the other pole of which was earthed. The practical value of this innovation was not understood by many physicists[3] for quite a considerable period, and the results which I obtained were by many erroneously considered simply due to efficiency in details of construction of the receiver, and to the employment of a large amount of energy.
Others did not overlook the fact that a radical change had been introduced by making these elevated capacities and the earth form part of the high frequency oscillators and receivers.
Prof. Ascoli of Rome gave a very interesting theory of the mode of operation of my transmitters and receivers in the *Elettricista* (Rome) issue of August 1897, in which he correctly attributed the results obtained to the use of elevated wires or antennae.
Prof. A. Slaby of Charlottenburg, after witnessing my tests in England in 1897, came to somewhat similar conclusions.[4]
Many technical writers have stated that an elevated capacity at the top of the vertical wire is unnecessary.
This is true if the length or height of the wire is made sufficiently great, but as this height may be much smaller for a given distance if a capacity area is used, it is more economical to use such capacities, which now usually consist of a number of wires spreading out from the top of the vertical conductor.

The necessity or utility of the earth connection has been sometimes questioned, but in my opinion no practical system of wireless telegraphy exists where the instruments are not connected to earth.
By "*connected to earth*" I do not necessarily mean an ordinary metallic connection as used for ordinary wire telegraphs.
The earth wire may have a condenser in series with it (Fig. 4C) , **(Although, Guglielmo Marconi had understood the importance of the *earth grounding* in the radioelectric systems and the possibility to have a capacitive coupling to the earth through a capacitive reactance, he had not realized the importance of a *direct capacitive coupling to the earth* as a means for tapping into the Earth for grounding. � See: Errante's capacitive earth grounding system)** or it may be connected to what is really equivalent, a capacity area placed close to the

surface of the ground (Fig. 4B).

It is now perfectly well known that a condenser, if large enough, does not prevent the passage of high frequency oscillations, and therefore in these cases the earth is for all practical purposes connected to the antennae.

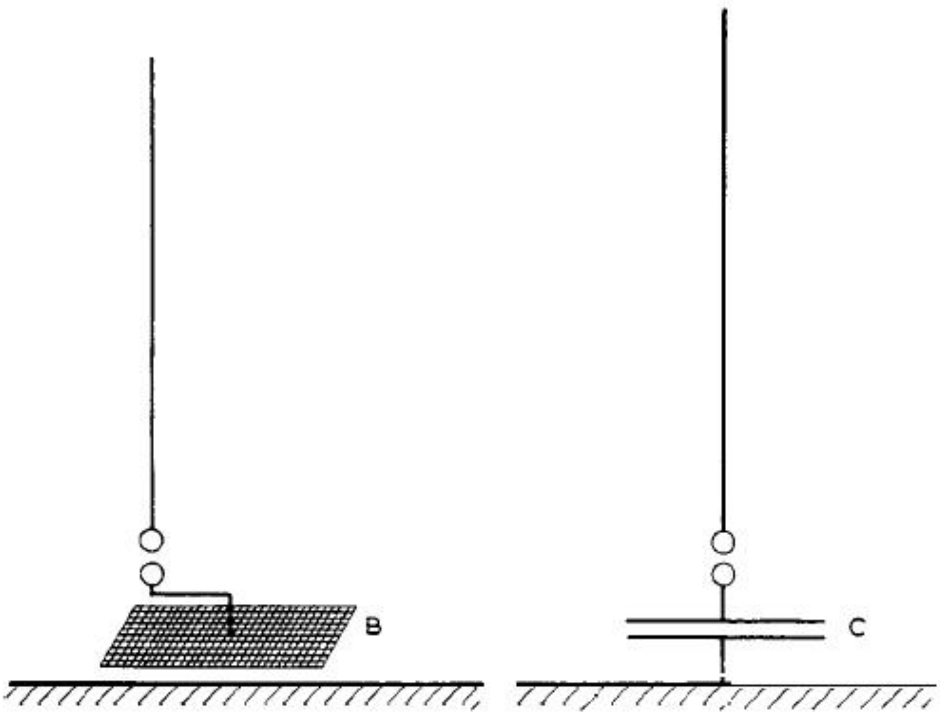


Fig. 4.

After numerous tests and demonstrations in Italy and England over distances varying up to 40 miles, communication was established for the first time across the English Channel between England and France[5] in March 1899 (Fig. 5).

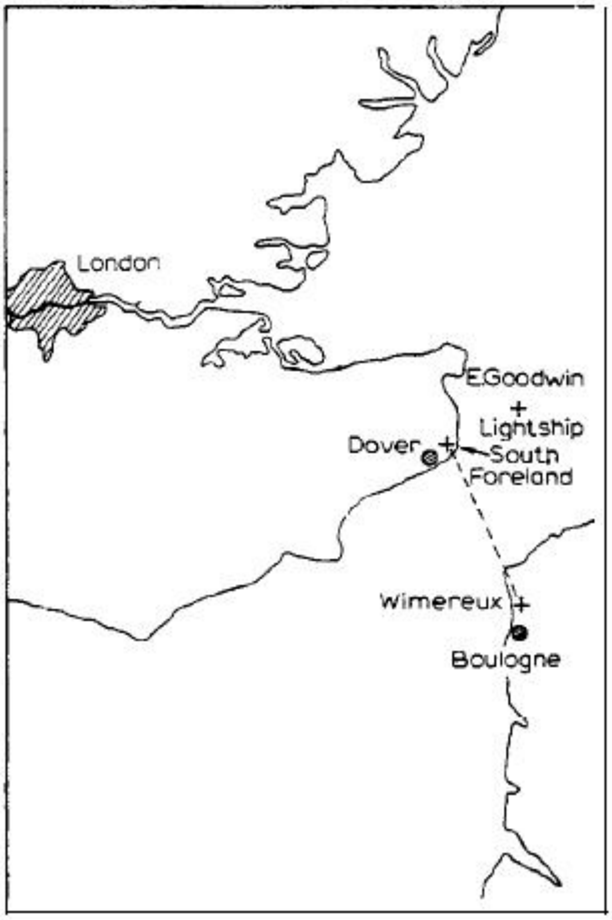


Fig. 5.

From the beginning of 1898 I had practically abandoned the system of connection shown in Fig. 2, and instead of joining the coherer or detector directly to the aerial and earth, I connected it between the ends of the secondary of a suitable oscillation transformer containing a condenser and tuned to the period of the electrical waves received. The primary of this oscillation transformer was connected to the elevated wire and to earth (See Fig. 6.)

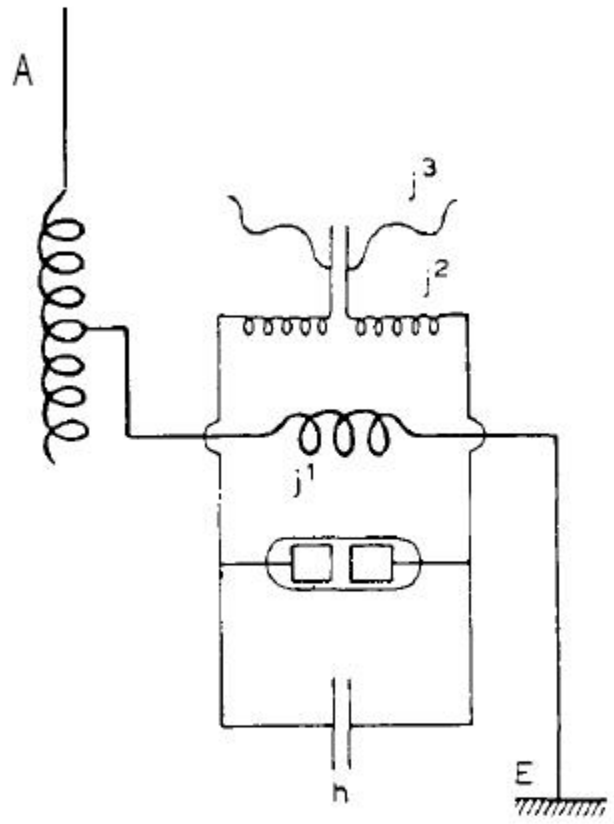


Fig. 6.

This arrangement allowed of a certain degree of syntony, as by varying the period of oscillation of the transmitting antennae, it was possible to send messages to a tuned receiver without interfering with others differently syntonized.[6]

As is now well known, a transmitter consisting of a vertical wire discharging through a spark gap is not a very persistent oscillator, the radiation it produces being considerably damped. Its electrical capacity is comparatively so small and its capability of radiating energy so large, that the oscillations decrease or die off with rapidity. In this case receivers or resonators of a considerably different period or pitch are likely to be affected by it.

Early in 1899 I was able to improve the resonance effects obtainable by increasing the capacity of the elevated wires by placing adjacently to them earthed conductors, and inserting in series with the aerials suitable inductance coils.[7]

By these means the energy-storing capacity of the aerial was increased, whilst its capability to radiate was decreased, with the result that the energy set in motion by the discharge formed a train or succession of

feebly damped oscillations.
A modification of this arrangement, by which excellent results were obtained, is shown in Fig. 7.

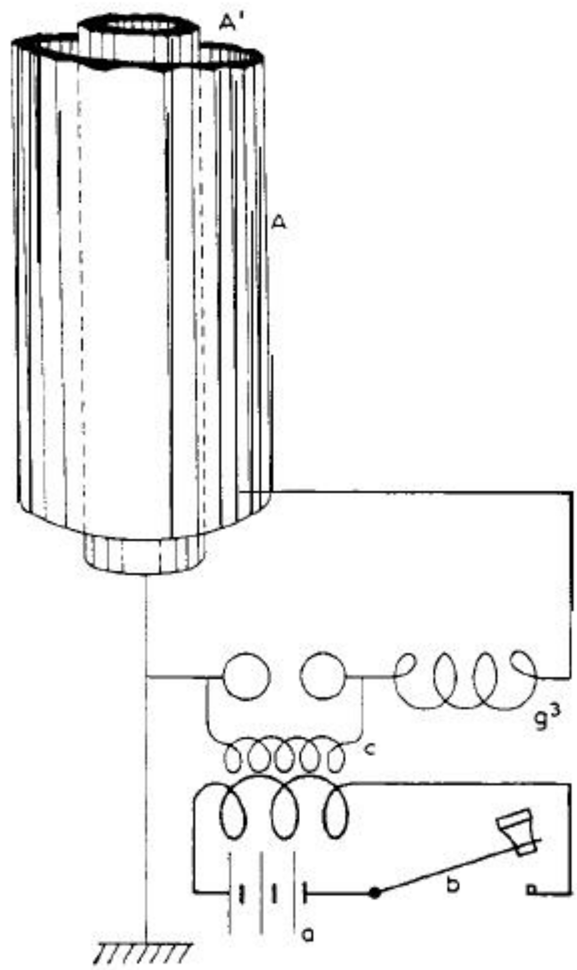


Fig. 7.

In 1900 I constructed and patented a complete system of transmitters and receivers[8] which consisted of the usual kind of elevated capacity area and earth connection, but these were inductively coupled to an oscillation circuit containing a condenser, an inductance, and a spark gap or detector, the conditions which I found essential for efficiency being that the periods of electrical oscillation of the elevated wire or conductor should be in tune or resonance with that of the condenser circuit, and that the two circuits of the receiver should be in electrical resonance with those of the transmitter (Fig. 8).

The circuits consisting of the oscillating circuit and the radiating circuit were more or less closely "coupled" by varying the distance between them.

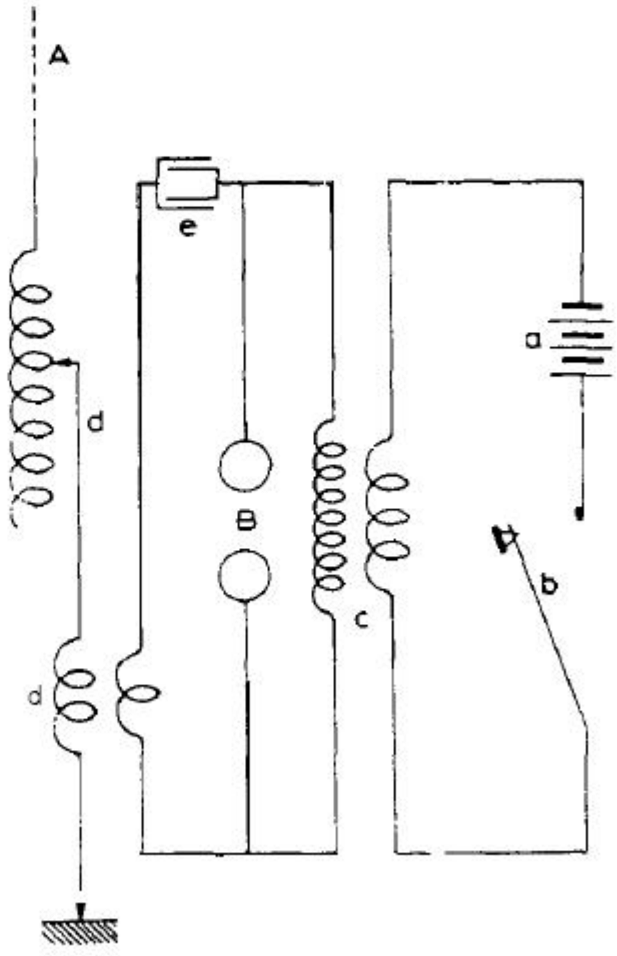


Fig. 8.

By the adjustment of the inductance inserted in the elevated conductor and by the variation of capacity of the condenser circuit, the two circuits were brought into resonance, a condition which, as I have said, I found essential in order to obtain efficient radiation.

Part of my work regarding the utilization of condenser circuits in association with the radiating antennae was carried out simultaneously to that of Prof. Braun, without, however, either of us knowing at the time anything of the contemporary work of the other.

A syntonic receiver has already been shown in Fig. 6, and consists of a vertical conductor or aerial connected to earth through the primary of an oscillation transformer, the secondary circuit of which included a condenser and a detector, it being necessary that the circuit containing the aerial and the circuit containing the detector should be in electrical resonance with each other, and also in tune with the periodicity of the electric waves transmitted from the sending station.

In this manner it was possible to utilize electric waves of low decrement and cause the receiver to integrate the effect of comparatively feeble but properly timed electrical oscillations in the same way as in acoustics two tuning forks can be made to affect each other at short distances if tuned to the same period of vibration.

It is also possible to couple to one sending conductor several differently tuned transmitters and to a receiving wire a number of corresponding receivers, as is shown in Figs. 9 and 10, each individual receiver responding only to the radiations of the transmitter with which it is in resonance.[9]

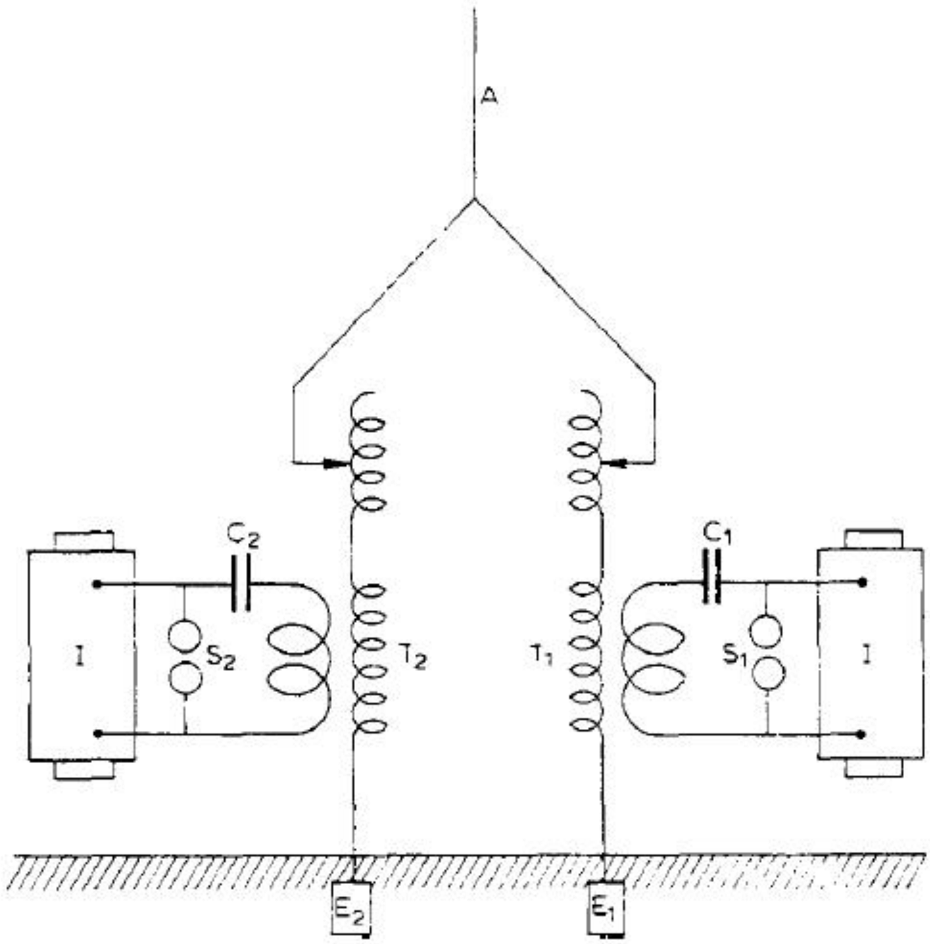


Fig. 9.

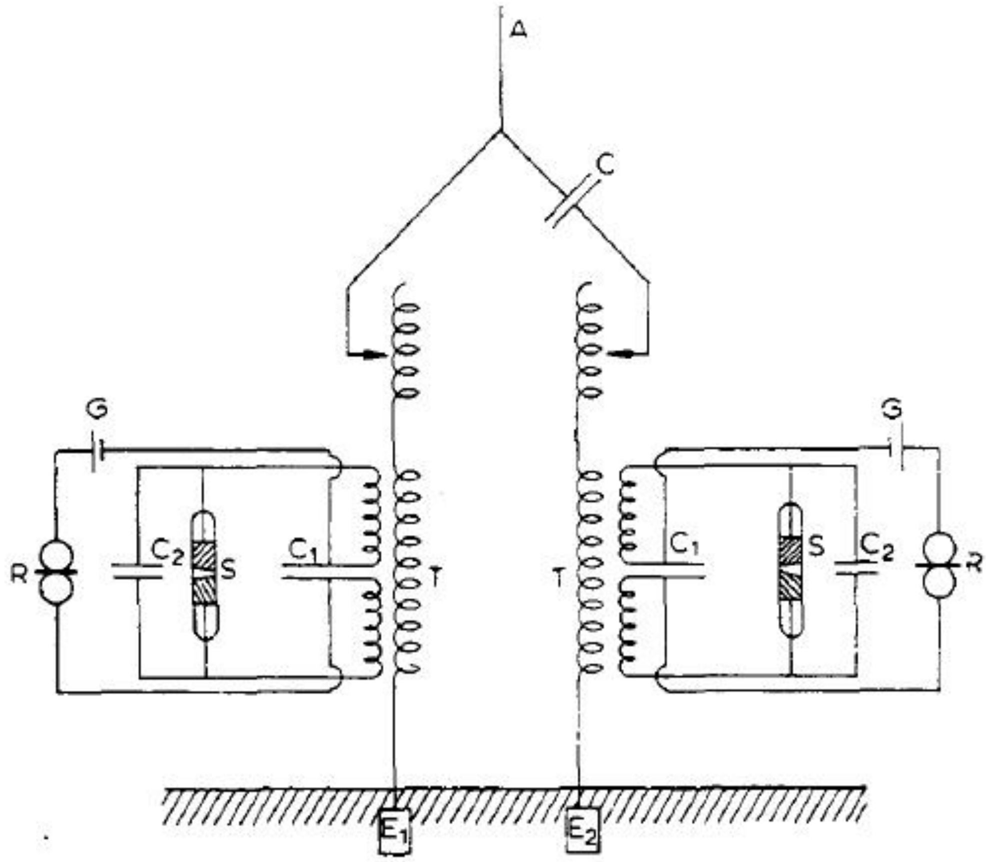


Fig. 10.

At the time (twelve years ago) when communication was first established by means of radiotelegraphy between England and France, much discussion and speculation took place as to whether or not wireless telegraphy would be practicable for much longer distances than those then covered, and a somewhat general opinion prevailed that the curvature of the Earth would be an insurmountable obstacle to long distance transmission, in the same way as it was, and is, an obstacle to signalling over considerable distances by means of light flashes. Difficulties were also anticipated as to the possibility of being able to control the large amount of energy which it appeared would be necessary to cover long distances.

What often happens in pioneer work repeated itself in the case of radiotelegraphy, the anticipated obstacles or difficulties were either purely imaginary or else easily surmountable, but in their place unexpected barriers manifested themselves, and recent work has been mainly directed to the solution of problems presented by difficulties which were certainly neither expected nor anticipated when long distances were first attempted.

With regard to the presumed obstacle of the curvature of the Earth, I am of opinion that those who anticipated difficulties in consequence of the shape of our planet had not taken sufficient account of the particular effect of the earth connection to both transmitter and receiver, which earth connection introduced effects of conduction which were generally at that time overlooked.
Physicists seemed to consider for a long time that wireless telegraphy was solely dependent on the effects of free Hertzian radiation through space, and it was years before the probable effect of the conductivity of the Earth between the stations was satisfactorily considered or discussed.
Lord Rayleigh, in referring to transatlantic telegraphy, stated in May 1903 : "The remarkable success of Marconi in signalling across the Atlantic suggests a more decided bending or diffraction of the waves round the perturberant Earth than had been expected, and it imparts a great interest to the theoretical problem[10].

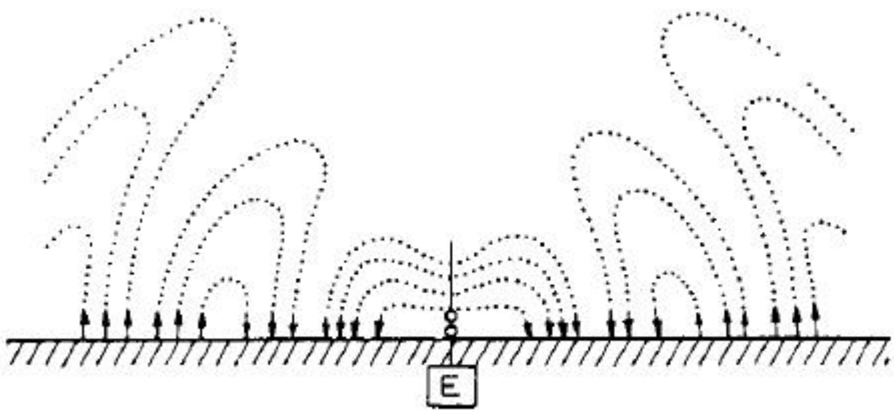


Fig. 11.

Prof. J. A. Fleming, in his book on *The Principles of Electric Wave Telegraphy*[11],gives diagrams showing what is now believed to be the diagrammatic representation of the detachment of semi-loops of electric strain from a simple vertical wire (Fig. 11). As will be seen, these waves do not propagate in the same manner as free radiation from a classical Hertzian oscillator, but glide along the surface of the Earth.
Prof. Fleming further states in the above quoted work: "*The view we here take is that the ends of the semi-loops of electric force, which terminate perpendicularly on the Earth, cannot move along unless there are movements of electrons in the Earth corresponding to the*

wave-motions above it. From the point of view of the electronic theory of electricity, every line of electric force in the ether must be either a closed line or its ends must terminate on electrons of opposite sign. If the end of a line of strain abuts on the Earth and moves, there must be atom-to-atom exchange of electrons, or movements of electrons in it. We have many reasons for concluding that the substances we call conductors are those in which free movements of electrons can take place. Hence the movements of the semi-loops of electric force outwards from an earthed oscillator or Marconi aerial is hindered by bad conductivity on the surface of the Earth and facilitated over the surface of a fairly good electrolyte, such as sea-water."

Prof. Zenneck[12] has carefully examined the effect of earthed transmitting and receiving aerials, and has endeavoured to show mathematically that when the lines of electrical force, constituting a wave front, pass along a surface of low specific inductive capacity, such as the Earth, they become inclined forward, their lower ends being retarded by the resistance of the conductor to which they are attached.

It therefore seems well established that wireless telegraphy, as practised at the present day, is dependent for its operation over long distances on the conductivity of the Earth, and that the difference in conductivity between the surface of the sea and land is sufficient to explain the increased distance obtainable with the same amount of energy in communicating over sea as compared to over land.

I carried out some tests between a shore station and a ship at Poole, in England, in 1902, for the purpose of obtaining some data on this point, and I noticed that at equal distances a perceptible diminution in the energy of the received waves always occurred when the ship was in such a position as to allow a low spit of sand about 1 kilometer broad to intervene between it and the land station.

I therefore believe that there was some foundation for the statement so often criticized which I made in my first English Patent of June 2, 1896 to the effect that when transmitting through the earth or water I connected one end of the transmitter and one end of the receiver to earth.

In January 1901 some successful experiments[13] were carried out between two points on the South Coast of England 186 miles apart, i.e. St. Catherines' Point (Isle of Wight) and The Lizard in Cornwall (Fig. 12).

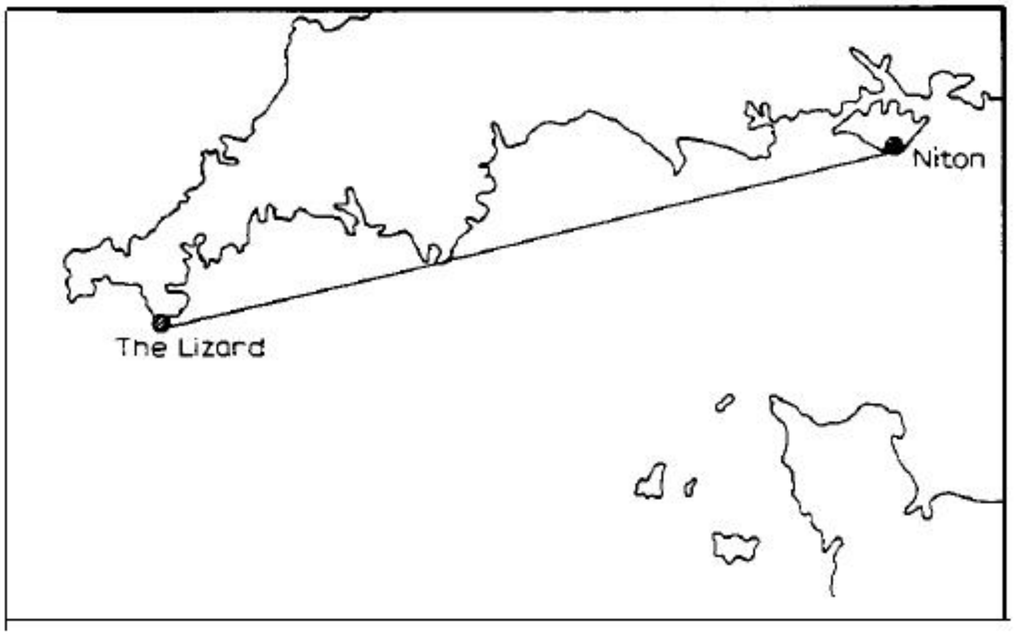


Fig. 12.

The total height of these stations above sea level did not exceed 100 meters, whereas to clear the curvature of the Earth a height of more than 1,600 meters at each end would have been necessary.
The results obtained from these tests, which at the time constituted a record distance, seemed to indicate that electric waves produced in the manner I had adopted would most probably be able to make their way round the curvature of the Earth, and that therefore even at great distances, such as those dividing America from Europe, the factor of the Earths curvature would not constitute an insurmountable barrier to the extension of telegraphy through space.
The belief that the curvature of the Earth would not stop the propagation of the waves, and the success obtained by syntonic methods in preventing mutual interference, led me in 1900 to decide to attempt the experiment of testing whether or not it would be possible to detect electric waves over a distance of 4,000 kilometers, which, if successful, would immediately prove the possibility of telegraphing without wires between Europe and America.
The experiment was in my opinion of great importance from a scientific point of view, and I was convinced that the discovery of the possibility to transmit electric waves across the Atlantic Ocean, and the exact knowledge of the real conditions under which telegraphy over such distances could be carried out, would do much to improve our understanding of the phenomena connected with wireless transmission.

The transmitter erected at Poldhu, on the coast of Cornwall, was similar in principle to the one I have already referred to, but on a very much larger scale than anything previously attempted.[14]
The power of the generating plant was about 25 kilowatts.
Numerous difficulties were encountered in producing and controlling for the first time electrical oscillations of such power. In much of the work I obtained valuable assistance from Prof. J. A. Fleming, Mr. R. N. Vyvyan, and Mr. W. S. Entwistle.
My previous tests had convinced me that when endeavouring to extend the distance of communication, it was not merely sufficient to augment the power of the electrical energy of the sender, but that it was also necessary to increase the area or height of the transmitting and receiving elevated conductors.
As it would have been too expensive to employ vertical wires of great height, I decided to increase their number and capacity, which seemed likely to make possible the efficient utilization of large amounts of energy.

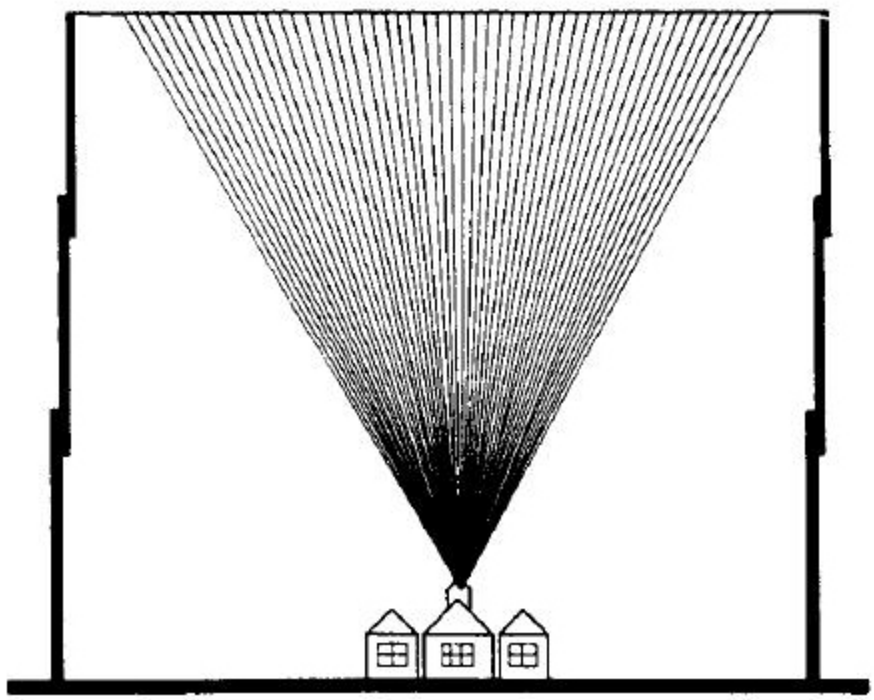


Fig. 13.

The arrangement of transmitting antennae which was used at Poldhu is shown in Fig. 13, and consisted of a fan-like arrangement of wires supported by an insulated stay between masts only 48 meters high and 60 meters apart. These wires converged together at the lower end and were connected to the transmitting apparatus contained in a building. For the purpose of the test a powerful station had been erected at Cape Cod, near New York, but the completion of the arrangements at that station were delayed in consequence of a storm which destroyed the masts and antennae.

I therefore decided to try the experiments by means of a temporary receiving station erected in Newfoundland, to which country I proceeded with two assistants about the end of November 1901.

The tests were commenced early in December 1901 and on the 12th of that month the signals transmitted from England were clearly and distinctly received at the temporary station at St. Johns in Newfoundland.

Confirmatory tests were carried out in February 1902 between Poldhu and a receiving station on the S.S. "Philadelphia" of the American Line. On board this ship readable messages were received by means of a recording instrument up to a distance of 1,551 miles and test letters as far as 2,099 miles from Poldhu (Fig. 14).

The tape records obtained on the "Philadelphia" at the various distances were exceedingly clear and distinct, as can be seen by the specimens exhibited.

These results, although achieved with imperfect apparatus, were sufficient to convince me and my co-workers that by means of permanent stations and the employment of sufficient power it would be possible to transmit messages across the Atlantic Ocean in the same way as they were sent over much shorter distances.

The tests could not be continued in Newfoundland owing to the hostility of a cable company, which claimed all rights for telegraphy, whether wireless or otherwise, in that colony.

A result of scientific interest which I first noticed during the tests on S.S. "Philadelphia" and which is a most important factor in long distance radiotelegraphy, was the very marked and detrimental effect of daylight on the propagation of electric waves at great distances, the range by night being usually more than double that attainable during daytime.[15]

I do not think that this effect has yet been satisfactorily investigated or explained. At the time I carried out the tests I was of opinion that it might be due to the loss of energy at the transmitter, caused by the dis-electrification of the highly charged transmitting elevated conductor under the influence of sunlight.

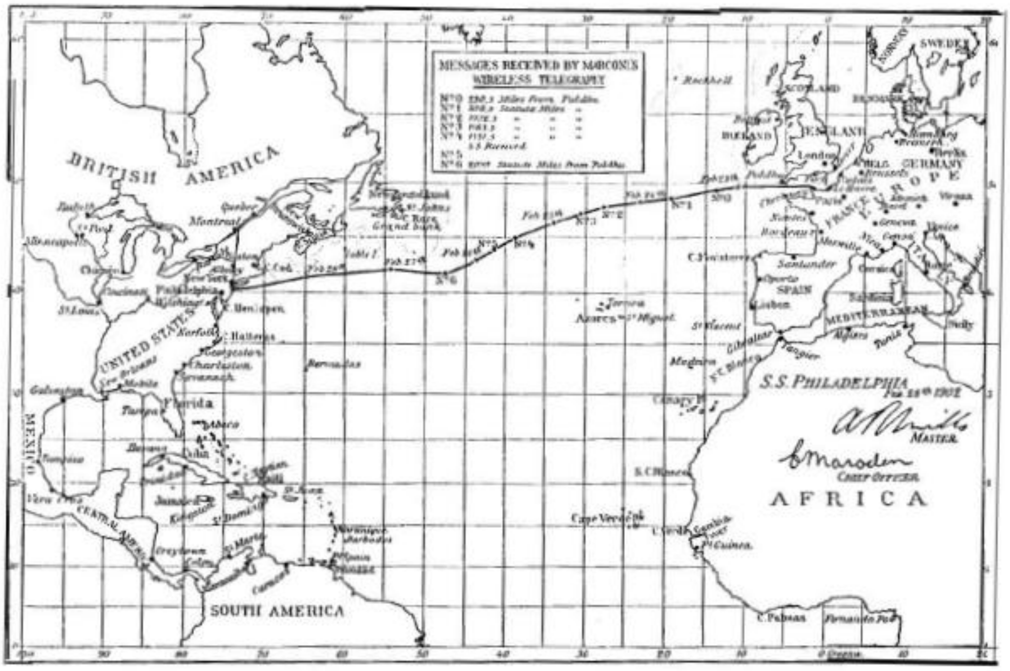


Fig. 14.

I am now inclined to believe that the absorption of electric waves during the daytime is due to the electrons propagated into space by the sun, and that if these are continually falling like a shower upon the earth, in accordance with the hypothesis of Prof. Arrhenius, then that portion of the Earths atmosphere which is facing the sun will have in it more electrons than the part which is not facing the sun, and therefore it may be less transparent to electric waves.

Sir J. J. Thomson has shown in an interesting paper in the *Philosophical Magazine* that if electrons are distributed in a space traversed by electric waves, these will tend to move the electrons in the direction of the wave, and will therefore absorb some of the energy of the wave. Hence, as Prof. Fleming has pointed out in his Cantor Lectures delivered at the Society of Arts, a medium through which electrons or ions are distributed acts as a slightly turbid medium to long electric waves.[16]

Apparently the length of wave and amplitude of the electrical oscillations have much to do with this interesting phenomenon, long waves and small amplitudes being subject to the effect of daylight to a much lesser degree than short waves and large amplitudes.

According to Prof. Fleming[17] the daylight effect should be more marked on long waves, but this has not been my experience. Indeed, in some very recent experiments in which waves of about 8,000 meters long were used, the energy received by day was usually greater than at night.

The fact remains, however, that for comparatively short waves, such as are used for ship communication, clear sunlight and blue skies, though transparent to light, act as a kind of fog to these waves. Hence the weather conditions prevailing in England, and perhaps in this country, are usually suitable for wireless telegraphy.
During the year 1902 I carried out some further tests between the station at Poldhu and a receiving installation erected on the Italian Cruiser "Carlo Alberto", kindly placed at my disposal by H.M. The King of Italy. (See Fig. 15.[18])

Fig. 15.

During these experiments the interesting fact was observed that, even when using waves as short as 1,000 feet, intervening ranges of mountains, such as the Alps or Pyrenees, did not, during the night time, bring about any considerable reduction in the distance over which it was possible to communicate. During daytime, unless much longer waves and more power were used, intervening mountains greatly reduced the apparent range of the transmitter.
Messages and press despatches of considerable length were received from Poldhu at the positions marked on the map, which map is a copy, on a reduced scale, of the one accompanying the official report of the

experiments (Fig. 16).

With the active encouragement and financial assistance of the Canadian Government, a high power station was constructed at Glace Bay, Nova Scotia, in order that I should be able to continue my long-distance tests with a view to establishing radiotelegraphic communication on a commercial basis between England and America.[19]

On December 16, 1902 the first official messages were exchanged at night across the Atlantic, between the stations at Poldhu and Glace Bay (Figs. 17 and 18). Further tests were shortly afterwards carried out with another long-distance station at Cape Cod in the United States of America, and under favourable circumstances it was found possible to transmit messages to Poldhu 3,000 miles away with an expenditure of electrical energy of only about 10 kilowatts. In the spring of 1903 the transmission of press messages by radiotelegraphy from America to Europe was attempted, and for a time the London *Times* published, during the latter part of March and the early part of April of that year, news messages from its New York correspondent sent across the Atlantic without the aid of cables.

A breakdown in the insulation of the apparatus at Glace Bay made it necessary, however, to suspend the service and unfortunately further accidents made the transmission of messages uncertain and unreliable.

As a result of the data and experience gained by these and other tests which I carried out for the British Government, between England and Gibraltar, I was able to erect a new station at Clifden in Ireland, and enlarge the one at Glace Bay in Canada, so as to enable me to initiate, in October 1907, communication for commercial purposes across the Atlantic between England and Canada.

Although the stations at Clifden and Glace Bay had to be put into operation before they were altogether complete, nevertheless communication across the Atlantic by radiotelegraphy never suffered any serious interruption during nearly two years, until, in consequence of a fire at Glace Bay this autumn, it has had to be suspended for three or four months.

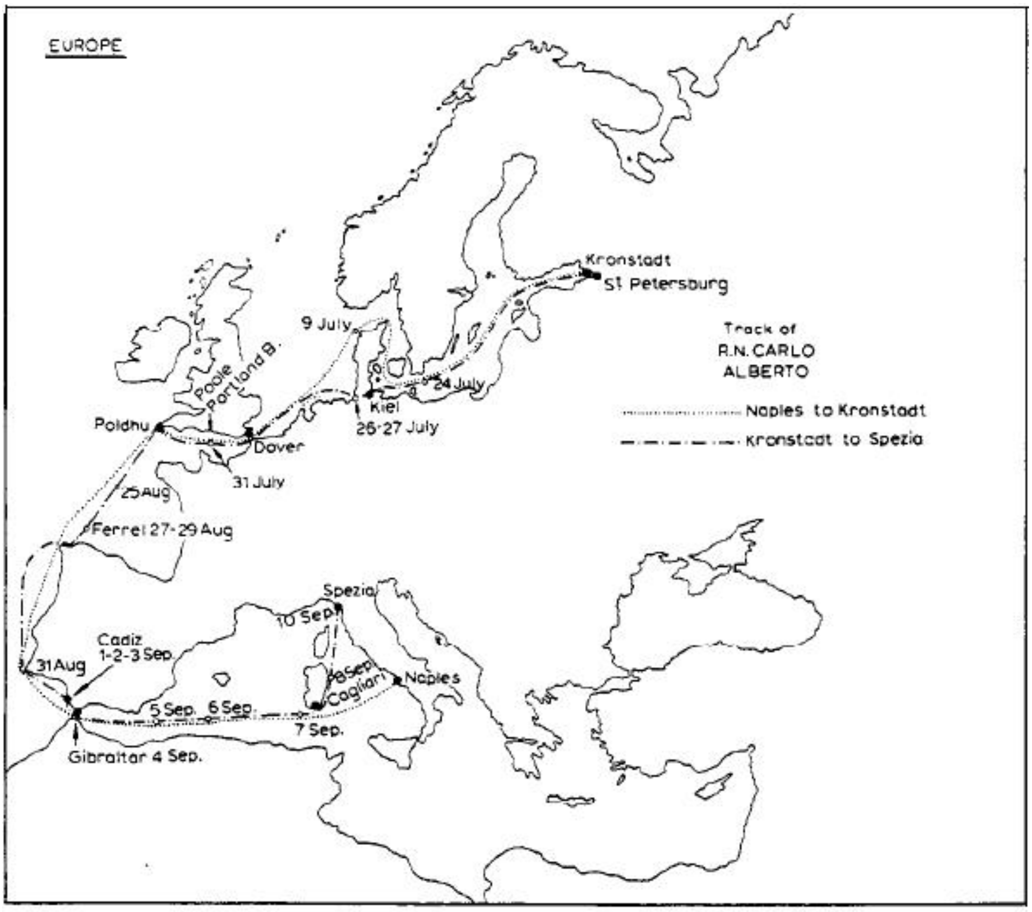

EUROPE
Kronstadt
St Petersburg
Track of
R.N. CARLO
ALBERTO
Naples to Kronstadt
Kronstadt to Spezia
9 July
Poole
Portland B.
Poldhu
Dover
31 July
Kiel
26-27 July
24 July
25 Aug
Ferrol 27-29 Aug
Spezia
10 Sep.
Cadiz
1-2-3 Sep.
31 Aug
8 Sep.
Cagliari
Naples
5 Sep.
6 Sep.
7 Sep.
Gibraltar 4 Sep.

Fig. 16.

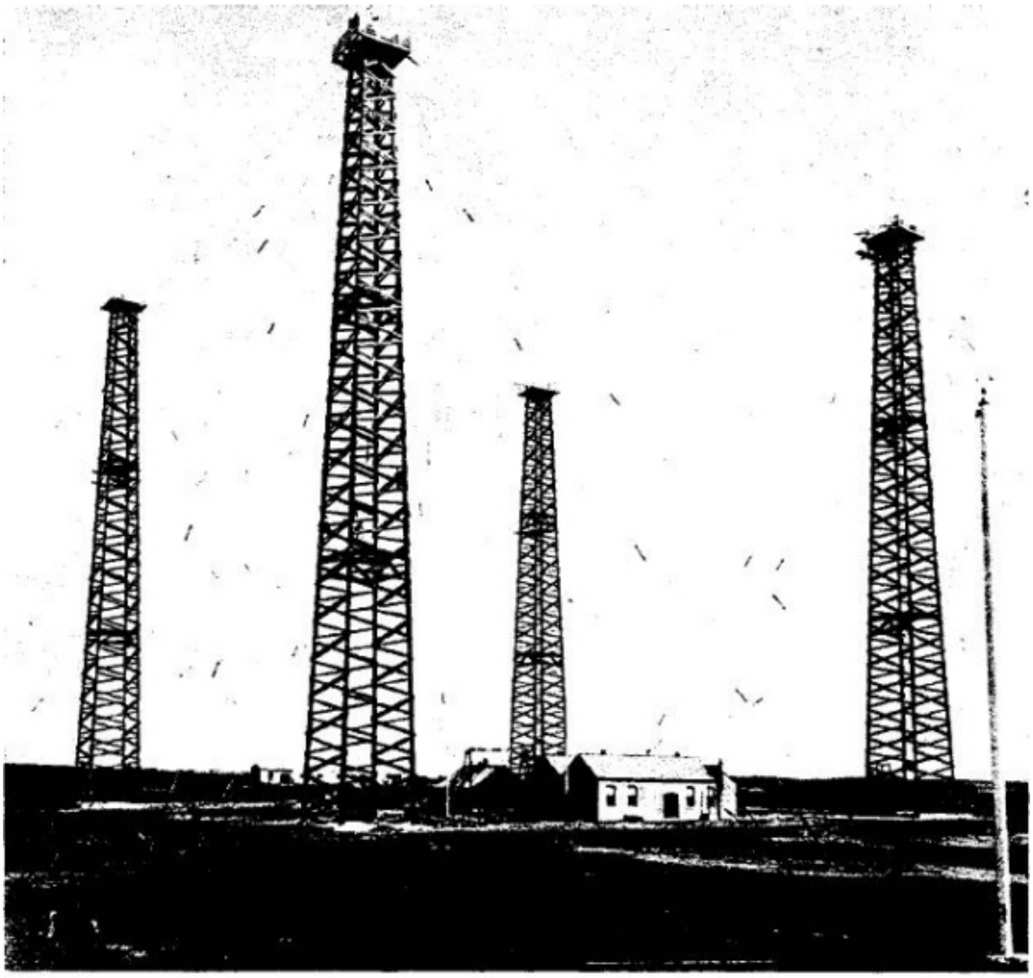


Fig. 17.

This suspension has not, however, been altogether an unmitigated evil, as it has given me the opportunity of installing more efficient and up-to-date machinery. The arrangements of elevated conductors or aerials which I have tried[20] during my long-distance tests, are shown in Figs. 19, 20 and 21.

The aerial shown in Fig. 21 consisted of a nearly vertical portion in the middle, 220 feet high, supported by four towers, and attached at the top to nearly horizontal wires, 200 in number and each 1,000 feet long, extending radially all round and supported at a height of 180 feet from the ground by an inner circle of 8, and an outer circle of 16 masts.

The natural period of oscillation of this aerial system gave a wavelength of 12,000 feet. Experiments were made with this arrangement in 1905 and with a wavelength of 12,000 feet, signals, although very weak, could be received across the Atlantic by day as well as by night.

The system of aerial I finally adopted for the long-distance stations in England and Canada is shown in Fig. 22. This arrangement not only makes it possible to efficiently radiate and receive waves of any desired length, but it also tends to confine the main portion of the radiation to a given direction. The limitation of transmission to one direction is not very sharply defined, but the results obtained with this type of aerial are nevertheless exceedingly useful.

Many suggestions respecting methods for limiting the direction of radiating have been made by various workers, notable by Prof. F. Braun, Prof. Artom, and Messrs. Belhni and Tosi.

In a paper read before the Royal Society of London[21] in March 1906 I showed how it was possible by means of horizontal aerials to confine the emitted radiations mainly to the direction of their vertical plane, pointing away from their earthed end. In a similar manner it is possible to locate the bearing or direction of a sending station.

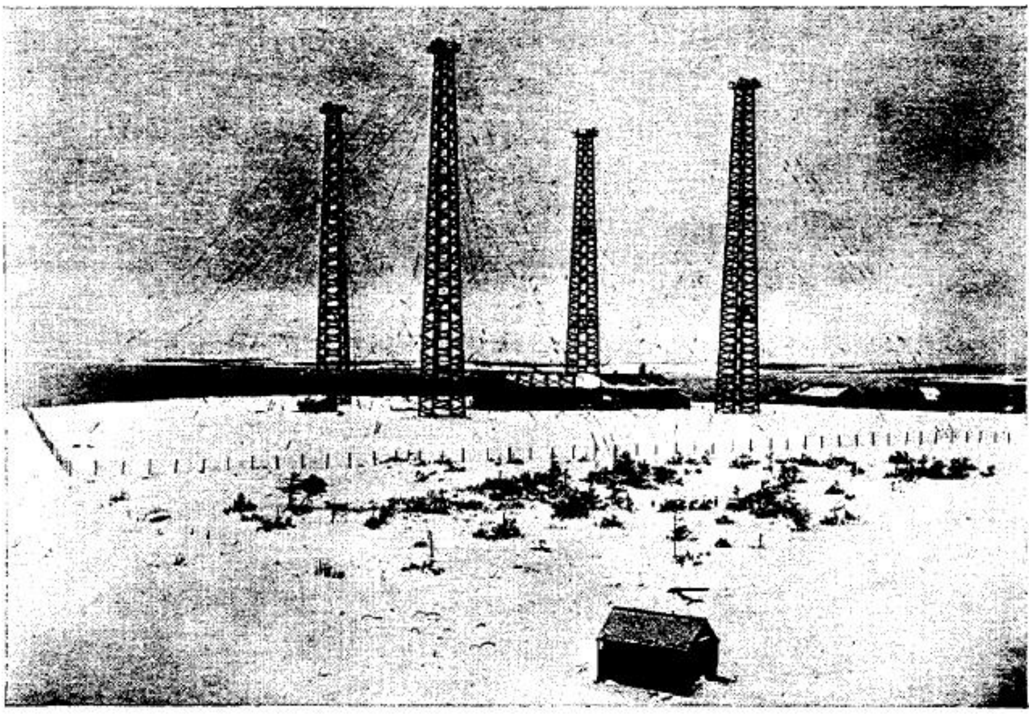


Fig. 18.

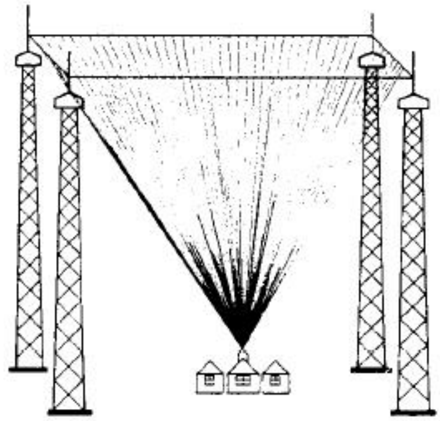


Fig. 19.

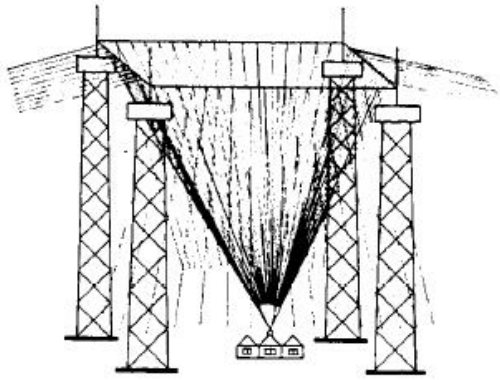


Fig. 20.

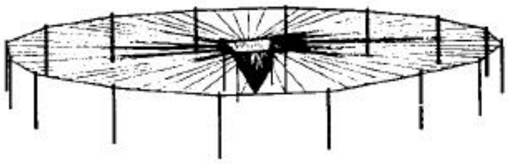


Fig. 21.

Fig. 22.

The transmitting circuits at the long-distance stations are arranged in accordance with a comparatively recent system for producing continuous or slightly damped oscillations, which I referred to in a lecture before the Royal Institution of Great Britain on March 13, 1908. An insulated metal disc A (see Fig. 23) is caused to rotate at a high rate of speed by means of an electric motor or steam turbine. Adjacent to this disc, which I will call the middle disc, are placed two other discs C' and C'' which may be called polar discs, and which are also revolved. These polar discs have their peripheries very close to the surface or edges of the middle disc. The two polar discs are connected by rubbing contacts to the outer ends of two condensers K, joined in series, and these condensers are also connected through suitable brushes to the terminals of a generator which should be a high-tension continuous-current generator.

On the middle disc a suitable brush or rubbing contact is provided and between this contact and the middle point of the two condensers an oscillating circuit is inserted, consisting of a condenser E in series with an inductance, which last is inductively connected with the radiating antennae.

The apparatus works probably in the following manner: The generator charges the double condenser, making the potential of the discs, say C' positive and C'' negative. The potential, if high enough will cause a discharge to pass across one of the gaps, say between C' and A. This charges the condenser E through the inductance F, and starts oscillations in the circuit. The charge of F in swinging back will jump from A to C'', the potential of which is of opposite sign to A, the dielectric strength between C' and A having meanwhile been restored by the rapid motion of the disc, driving away the ionized air.

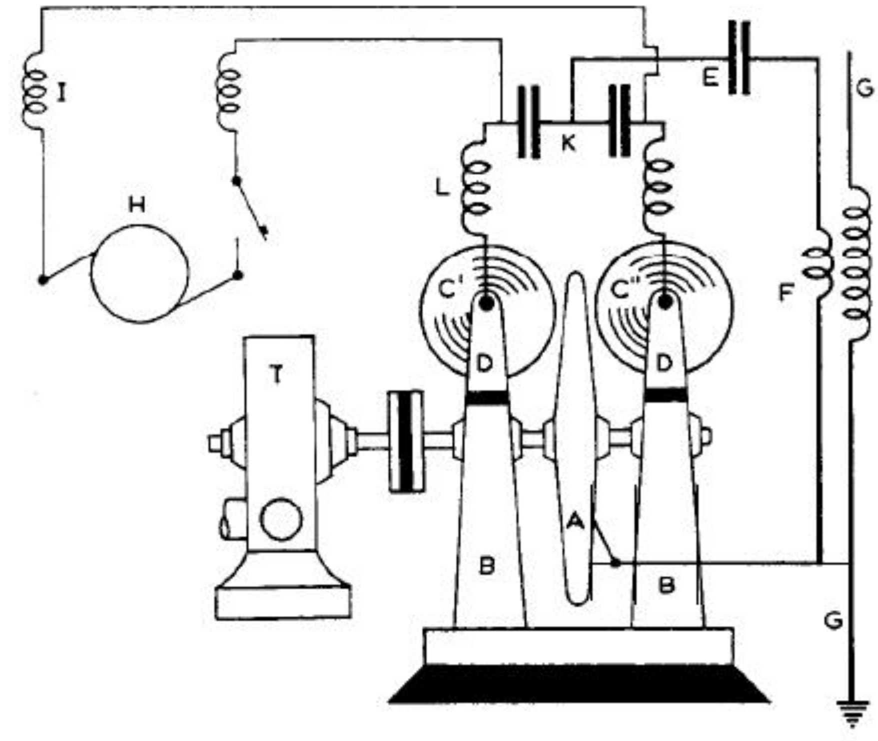


Fig. 23.

The condenser E therefore discharges and recharges alternatively in reverse directions, the same process going on so long as energy is supplied to the condensers K by the generator H.

It is clear that the discharges between C' and C'' and A are never simultaneous as otherwise the centre electrode would not be alternatively positive and negative.

The best results have, however, been obtained by an arrangement as shown in Fig. 24, in which the active surface of the middle disc is not smooth, but consists of a number of regularly spaced copper knobs or pegs, at the ends of which the discharges take place at regular intervals. I have found that with this arrangement each tram of oscillations may have a decrement as low as 0.02.

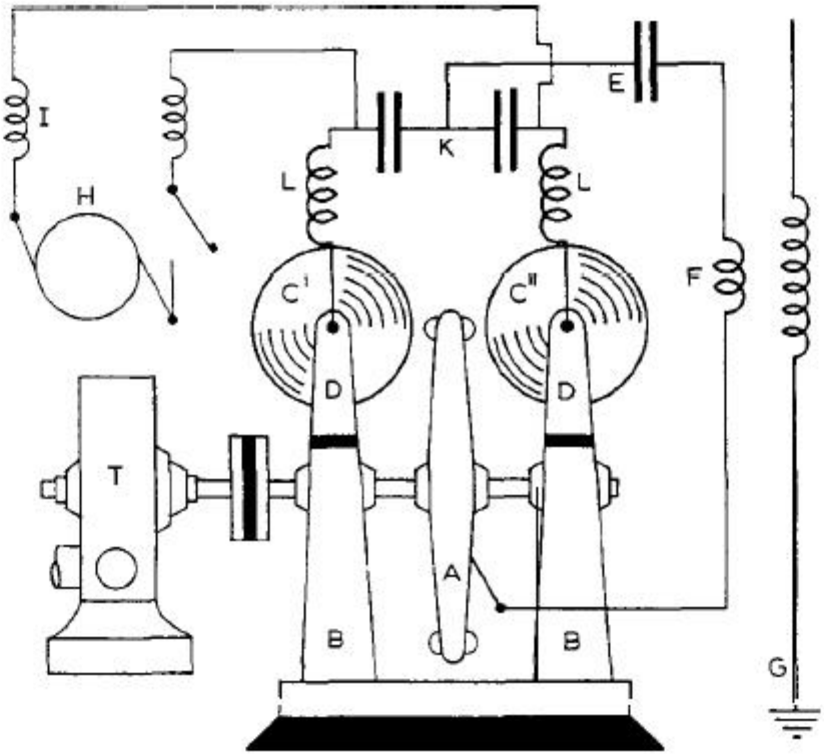


Fig. 24.

In this way it is also possible to cause the groups of oscillations radiated to reproduce a high and clear musical note in a receiver, thereby making it easy to differentiate between the signals emanating from the sending station and noises caused by atmospheric electrical discharges. By this method very efficient resonance can be also obtained in appropriately designed receivers.

With regard to the receivers employed, important changes have taken place. By far the larger portion of electric wave telegraphy was, until a few years ago, conducted by means of some form or other of coherer, or variable contact either requiring tapping or else self-restoring.

At the present day, however, I may say that at all the stations controlled by my Company my *magnetic receiver* (Fig. 25) is almost exclusively employed.[22] This receiver is based on the decrease of magnetic hysteresis which occurs in iron when under certain conditions this metal is subjected to the effects of electrical waves of high frequency.

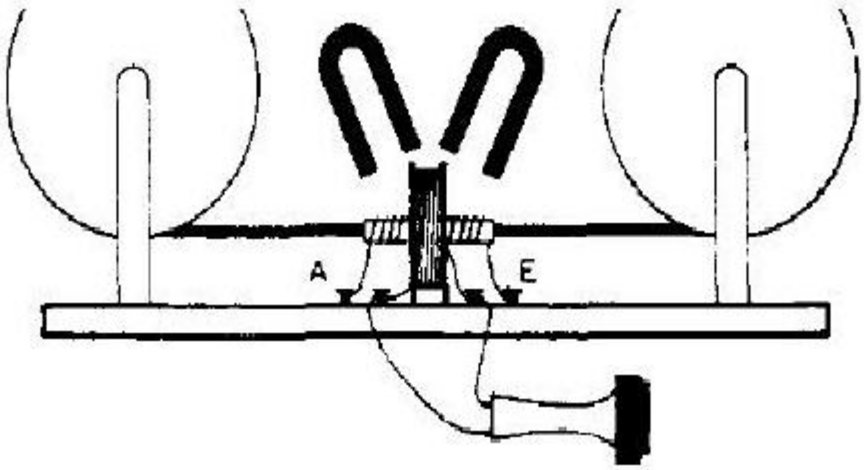


Fig. 25.

It has recently been found possible to increase the sensitiveness of these receivers, and to employ them in connection with a high-speed relay, so as to record messages at great speed.

A remarkable fact, not generally known, in regard to transmitters is, that none of the arrangements employing condensers exceed in efficiency the plain elevated aerial or vertical wire discharging to Earth through a spark gap, as used in my first experiments (see Figs. 2 and 3).

I have also recently been able to confirm the statement made by Prof. Fleming in his book *The Principles of Electric Wave Telegraphy,* 1906, page 555, that with a power of 8 watts in the aerial it is possible to communicate to distances of over 100 miles.

I have also found that by this method, when using large aerials, it is possible to send signals 2,000 miles across the Atlantic, with a smaller expenditure of energy than by any other method known to myself.

The only drawback to this arrangement is, that unless very large aerials are used, the amount of energy which can be efficiently employed is limited by the potential beyond which brush discharges and the resistance of the spark gap begin to dissipate a large proportion of the energy.

By means of spark gaps in compressed air and the addition of inductance coils placed between the aerial and earth, the system can be made to radiate very pure and slightly damped waves, eminently suitable for sharp tuning.

In regard to the general working of wireless telegraphy, the widespread application of the system and the multiplicity of the stations have greatly facilitated the observation of facts not easily explainable.

Thus it has been observed that an ordinary ship station, utilizing about 1/2 kilowatt of electrical energy, the normal range of which is not greater than 200 miles, will occasionally transmit messages across a distance of over 1,200 miles. It often occurs that a ship fails to communicate with a nearby station, but can correspond with perfect ease with a distant one. On many occasions last winter, the S.S. "Caronia" of the Cunard Line, carrying a station utilizing about 1/2 kilowatt, when in the Mediterranean, off the coast of Sicily, failed to obtain communication with the Italian stations, but had no difficulty whatsoever in transmitting and receiving messages to and from the coasts of England and Holland, although these latter stations were considerably more than 1,000 miles away, and a large part of the continent of Europe and the Alps lay between them and the ship.

Although high power stations are now used for communicating across the Atlantic, and messages can be sent by day as well as by night, there still exist short periods of daily occurrence, during which transmission from England to America, or vice versa, is difficult. Thus in the morning and evening, when in consequence of the difference in longitude, daylight or darkness extends only part of the way across the ocean, the received signals are weak and sometimes cease altogether. It would almost appear as if electric waves in passing from dark space to illuminated space, and vice versa, were reflected in such a manner as to be deviated from their normal path.

It is probable that these difficulties would not be experienced in telegraphing over equal distances north and south, on about the same meridian, as in this case the passage from daylight to darkness would occur almost simultaneously over the whole distance between the two points.

Another curious result, on which hundreds of observations continued for years leave no further doubt, is that regularly, for short periods, at sunrise and sunset, and occasionally at other times, a shorter wave can be detected across the Atlantic in preference to the longer wave normally employed.

Thus at Clifden and Glace Bay when sending on an ordinary coupled circuit arranged so as to simultaneously radiate two waves, one 12,500 feet and the other 14,700 feet, although the longer wave is the one

usually received at the other side of the ocean, regularly, about three hours after sunset at Clifden, and three hours before sunrise at Glace Bay, the shorter wave alone was received with remarkable strength, for a period of about one hour.
This effect occurred so regularly that the operators tuned their receivers to the shorter wave at the times mentioned, as a matter of ordinary routine.
With regard to the utility of wireless telegraphy there is no doubt that its use has become a necessity for the safety of shipping, all the principal liners and warships being already equipped, its extension to less important ships being only a matter of time, in view of the assistance it has provided in cases of danger.
Its application is also increasing as a means of communicating between outlying islands, and also for the ordinary purposes of telegraphic communication between villages and towns, especially in the colonies and in newly developed countries.
However great may be the importance of wireless telegraphy to ships and shipping, I believe it is destined to an equal position of importance in furnishing efficient and economical communication between distant parts of the world and in connecting European countries with their colonies and with America. As a matter of fact, I am at the present time erecting a very large power station for the Italian Government at Coltano, for the purpose of communicating with the Italian colonies in East Africa, and with South America.
Whatever may be its present shortcomings and defects, there can be no doubt that wireless telegraphy - even over great distances - has come to stay, and will not only stay, but continue to advance.
If it should become possible to transmit waves right round the world, it may be found that the electrical energy travelling round all parts of the globe may be made to concentrate at the antipodes of the sending station. In this way it may some day be possible for messages to be sent to such distant lands by means of a very small amount of electrical energy, and therefore at a correspondingly small expense.
But I am leaving the regions of fact, and entering the regions of speculation, which, however, with the knowledge we have gradually gained on the subject, promise results both useful and instructive.

Not having the fortune of being conversant with the Swedish language, I have thought it best, although an Italian, to use the medium of the English language in delivering this address, as I know that English is more generally understood here than Italian.

1. G. Marconi, *Brit. Patent* No. 12,039, June 2, 1896.

2. *J. Inst. Elec. Engrs. (London),* 28 (1899) 278.

3.See letter of Dr. Lodge in *The Times (London),* June 22, 1897.

4. A. Slaby, *Die Funkentelegraphie,* Verlag von Leonhardt Simion, Berlin, 1897; see also A. Slaby, "The New Telegraphy", *The Century Magazine,* 55 (1898) 867.

5. *J. Inst. Elec. Engrs. (London),* 28 (1899) 291.

6. *Brit. Patent* No. 12,326, June 1, 1898; *Brit. Patent* No. 6,982, April 1, 1899.

7. A. Blondel and G. Ferrie, "�tat actuel et Progr�s de la T�l�graphie sans Fil", read at the *Congr�s International d'�lectricit�, Paris, 1900;* see also *J. Soc. Arts,* 49 (1901) 509.

8. *Brit. Patent* No. 7,777, April 26, 1900; see also *J. Soc. Arts,* 49 (1901) 510-11.

9. See Letter of Prof. J. A. Fleming in *The Times (London),* October 4, 1900.

10. *PrOC.* Roy. Soc. (London), 72 (May 28, 1903).

11. J.A. Fleming, *The Principles of Electric Wave Telegraphy,* Longmans, Green & CO., London, 1906, p. 348.

12. J. Zenneck, *Ann. Physik,* [4], 23 (Sept. 1908) 846*; Physik. Z.,* 9 (1908) 50, 553.

13. *J. Soc. Arts,* 49 (1901) 512.

14. G. Marconi, lecture before the Royal Institution of Great Britain, June 13, 1902.

15. G. Marconi, "A Note on the Effect of Daylight upon the Propagation of Electromagnetic Impulses", *Proc. Roy. Soc. (London),* 70 (June 12, 1902).

16. See also J. J. Thomson, "On Some Consequences, etc.", *Phil. Mag.,* [6] 4 (1902)

17. See Ref. 11, p. 618.

18. *Riv. Marittima (Rome),* October 1902.

19. G. Marconi, Paper read before the Royal Institution of Great Britain, March 3, 1905.

20. See also G. Marconi, Lecture before the Royal Institution of Great Britain, March 13, 1908.

21. G. Marconi, "On Methods whereby the Radiation of Electric Waves may be mainly confined, etc.", *Proc. Roy. Soc. (London),* A 77 (1906).

22. G. Marconi, "Note on a Magnetic Detector of Electric Waves", *Proc. Roy. Soc. (London),* 70 (1902) 341.

***Reginald Fessenden*'s amplitude modulation breaktrough. From wireless telegraphy to wireless telephony.**

More on *Reginald Fessenden* and *Guglielmo Marconi* here .

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ANIMAL
LIBERATION
FRONT
20
Over 25 years of direct action for animals
without harming a single human being

14. Errante's CAPACITIVE RF EARTH GROUNDING for Un-balanced systems, UN-UN etc. © 1985-2026 F.sco Errante - www.Radiondistics.com

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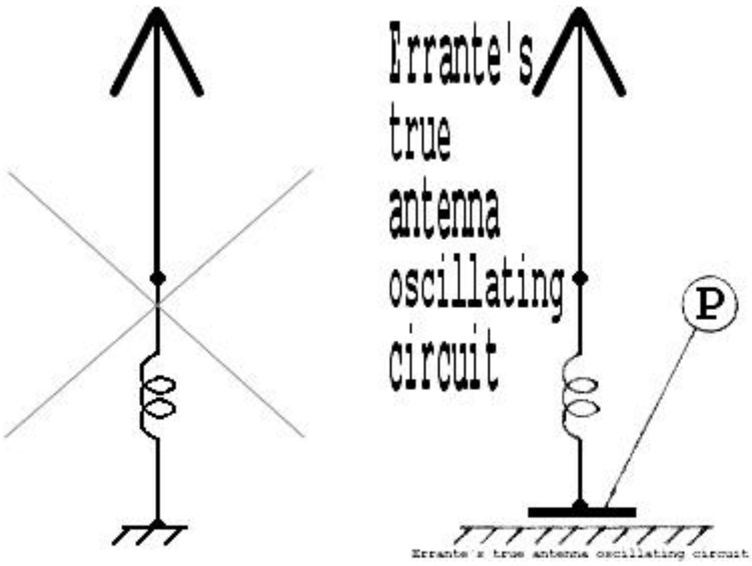
Planning a trip to Sicily? Then, make sure you won't be feeding the Mafia!

ERRANTE's
CAPACITIVE RF EARTH GROUNDING
SYSTEM

Say goodbye to the ground plane!



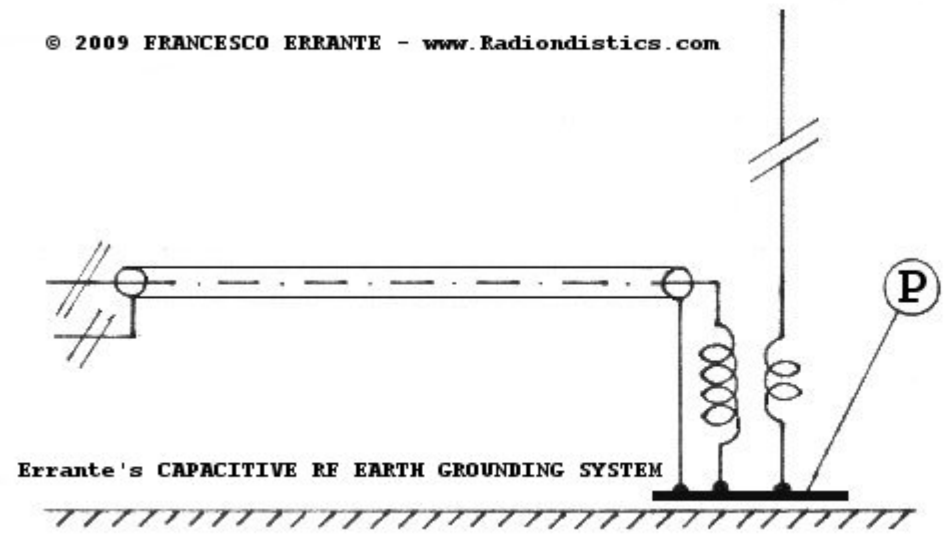
How to correctly polarize an unbalanced RF system with reference to the physical g

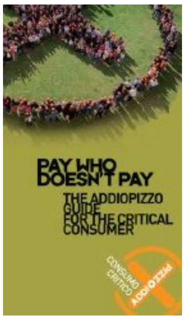
This system demonstrates that no galvanic coupling to earth is needed f

Errante's capacitive RF earth grounding system

by *Francesco Errante*

Further to what I have previously disclosed and demonstrated in the matter of grou
signals, I am hereby disclosing and demonstrating that NO galvanic path to earth is
ground. ALL IT TAKES IS A LARGE ENOUGH lumped CAPACITANCE yelding a ve
frequency of a given RF un-balanced system, as shown in the picture below:





Errante's CAPACITIVE RF EARTH GROUNDING SYSTEM schematic dia
An unbalanced radiator and its Un-Un impedance transformer being gr
layer to the physical ground - Copyright © 2009 - Francesco Errante

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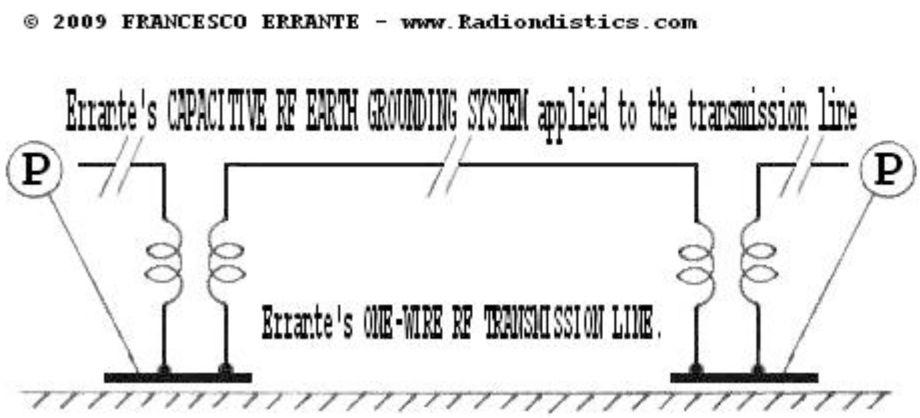

Errante's ONE-WIRE UNBALANCED RF TRANSMISSION LINE, employ
- Francesco Errante

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The *grounding capacitance* is obtained by coupling the point of a radio-electric plate **(P)** facing a uniform plot of ground surface and isolated from it by a properly **value!!! � Even *Guglielmo Marconi* made this mistake as documented in** In the medium, long and very long waves spectrum, in order to increase the groundi hollow cylinder buried into the ground but isolated from it. The volume occupied by self-radiating lattice towers. This will also bring about huge savings in the cost of a t **NB. If necessary, for lightening protection the system can also be galvanically ea the capacitive plate and a ground stake**

The half a capacitor layout shown in the picture above, MUST not be mis less than a square meter in size would hardly act as a "*ground plane"* in a 3.5 MHz s some multi-wire ground plane systems, still in use, are infact rudimental lossy distr

A practical application, demonstrating this system, can be found on the Un-Un feeding system.

L or C Reac

Enter any two known values and press "Calculate" to solve for the others. For examp 1000 Hertz. Fields should be reset to 0 before doing a new calculation.

Inductive Reac
Capacitive Reactan
Resonant Frequency (Fo

Capacitance (Microfarads)	Inductance (Millihenrys)

Applet

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15. Errante's gaseous-core high-frequency atomic resonance reactor - CGRR-N type - RADIONDISTICS

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Errante's gaseous-core atomic resonance nuclear reactor

Thanks to *Francesco Errante*'s latest patent-pending invention, namely the "*high frequency sinusoidal alternating current powered gaseous core nuclear reactor for the continuous disintegration-recombination cycle of the Neon gas atoms, based on the light chemical elements non-wavelength-dependent electrical resonance physical phenomenon*", **two leapfrog breakthrough discoveries in fundamental physics have been made, empirically demonstrated and at long last, unveiled to the public at large**:

- the intrinsic electrical resonance property of light chemical element atoms, such as noble gases, with respect to high-frequency sinusoidal electrical current oscillations;
- a non-wavelength-dependent electrical resonance phenomenon.

The interplay between the said physical property and phenomenon is responsable for the occurrence of a specific electro-nuclear process termed *Resonance-Driven Disintegration-Recombination* (RDDR) cycle of light elements atoms, such as those of the nobel gases.

The RDDR cycle is not an identical pathway to fusion. It is a third distinct category of nuclear energy release:

- 1. Fission: Splitting heavy nuclei;
- 2. Fusion: Merging light nuclei;
- 3. RDDR (Resonance-Driven Disintegration-Recombination): Disassembling nuclei into a nucleon fog and reassembling them into new configurations via electrical resonance.

Although RDDR results in the creation of new isotopes (which macroscopically resembles "fusion" if they are heavier) and release of energy, the underlying physical mechanisms, intermediate states, resonant trigger, and byproduct spectrum are categorically different from the fusion processes.
RDDR is a fundamentally novel pathway indicating that rules of quantum chromodynamics (QCD) allow for a much more fluid and reversible state of nuclear matter than we currently believe.

I. Summary of the invention & historical context

The Gaseous-Core Resonance Reactor (Neon variant) - GCRR-N for short - represents a paradigm shift in nuclear energy, demonstrating the first functional gaseous-core reactor using non-fissile fuel. This invention resolves a technological impasse dating to the 1950s not through incremental material science, but by discovering and exploiting a previously unknown property of noble gases: their **non-wavelength-dependent electrical resonance** to high-frequency sinusoidal alternating current.
The breakthrough originated not in a fusion laboratory, but during research into ionospheric radio propagation, where applying established direct-current quantum electrodynamic (QED) principles to a high-frequency alternating-current (AC) regime revealed a critical phenomenon.
When a noble gas (e.g., neon) is subjected to a sinusoidal current oscillating at its natural electrical resonance frequency, perfect energy coupling occurs. This resonance is a function of the atom's intrinsic quantum-electrical structure, not the spatial dimensions of the apparatus or the wavelength of the driving signal.
The reactor empirically validates that neon atoms possess a natural electrical resonance frequency of 27,430 kHz. Driven at this frequency within a suitably designed reaction chamber, the gas undergoes a self-sustaining cycle of atomic disintegration and statistical nuclear recombination termed **Resonance-Driven**

Disintegration-Recombination (RDDR) Fusion producing a stable, scalable thermonuclear reaction with net energy gain.

The GCRR-N demonstrates a credible and revolutionary fusion methodology. It replaces monumental heating and magnetic confinement with **resonant electrostatic pumping and cyclic quantum disintegration-recombination**. This represents a paradigm shift from "containing a star" to "orchestrating a continuous atomic resurrection", offering a direct path to scalable, clean, and thermally efficient power generation. The principal barrier to practical gaseous-core nuclear power was not one of engineering alone, but of fundamental oversight. By revisiting the quantum-electrical principles first explored in vacuum tubes and applying them in a novel resonant AC regime, this invention transitions gaseous-core reactors from a perpetually future concept to a present-day reality. The GCRR-Neon validates that non-fissile, non-radioactive materials can achieve net energy gain through atomic electrical resonance, and establishes RDDR Fusion as a new, viable branch of nuclear science.

II. The fundamental phenomenon: atomic electrical resonance in an AC field

The phenomenon is purely quantum-mechanical and has no analogue in classical physics. Its discovery invalidated the core assumption that rendered gaseous-core reactors (GCRs) conceptual namely, that they required fissile material or unattainable plasma containment.

Historically, GCRs were hamstrung by the Sisyphean challenges of fuel containment, material survival in ultra-hot plasmas, and radioactive byproduct management, all stemming from the entrenched paradigm of forcing nuclear reactions via extreme thermal kinetics or neutron flux. Critically, prior to this invention, non-fissile materials were categorically excluded as potential nuclear fuels. It was believed that neon, being an inert noble gas, could not act as a nuclear fuel. The chemical inertness of neon is a property of its filled electron shell (configuration $1s^2\ 2s^2\ 2p^6$) but is irrelevant to nuclear reactions. This is because, under conditions of high energy density and natural electrical resonance, the periodic potential of the high-frequency sinusoidal alternating current modulates the potential barrier around the nucleus, enabling the temporary capture of electrons (formation of Ne^-) and, subsequently, the collapse of the nuclear wave function when the kinetic energy of the ions exceeds the Coulomb barrier.

In other words, electrical resonance at the atomic level alters the cross-sections for low-energy nuclear interactions, an effect known in the literature as "*channel resonance*" or "*near-field enhancement*", here observed for the first time in a pure noble gas.

In this regime, neon is no longer "inert": the resonance enables the disintegration of the

nucleus and the instantaneous formation of thermal plasma, followed by recombination into new nuclear configurations with the release of excess binding energy.
The electrical resonance of atoms a cornerstone of QED in direct-current (DC) field emission and spectroscopy had never been successfully applied or observed in a high-frequency AC regime; and consequently, the unique potential of noble gases (with their full valence shells and definable quantum states) to enter into a coherent, resonant energy absorption cycle was unrecognized by nuclear engineering.

The necessary conditions for the phenomenon to manifest are threefold:

- **Resonant drive:** A sinusoidal driving current with oscillation frequency exactly equal to the gas natural electrical resonance frequency (for neon: **27,430 kHz**).
- **Atomic mobility:** Sufficient freedom of movement low-pressure gaseous state (**<5 Torr**).
- **Energy density threshold:** Environment capable of achieving critical energy density to initiate atomic and subatomic reactions.

When satisfied, the system behaves as a perfectly resonant electrical circuit, exhibiting a sharp resonance peak within a narrow bandwidth and a stable, real-valued feed-point impedance despite the absence of any correlation with physical dimensions of the apparatus.

III. Reactor Design: Exploiting the Phenomenon

The GCRR-Neon (Gaseous-Core Resonance Reactor, Neon variant) is engineered specifically to satisfy the above conditions and concentrate their effects. The reactor's heart is a unipolar electrode, fed by a single-phase, unbalanced sinusoidal alternating current and not being fixedly polarized, it cannot be designated as either cathode or anode, so it is analogously termed the ***Páno Káto*** (Greek for up and down), referencing its oscillating potential.

The cavity geometry creates a superposition of known concentrating effects essential for achieving the critical energy density:

- **Hollow Cathode/Anode Effect:** concentrates charged particles within the cavity volume.

- **Pendulum Effect:** anions undergo repeated reflections between cavity walls and central collision zone, multiplicatively increasing kinetic energy with each half-cycle.
- **Electrostatic Confinement:** oscillating field of the cavity walls traps charged particles, enabling sustained energy multiplication.

IV. The Disintegration-Recombination (D-R) Fusion Cycle

The reactor initiates and maintains a continuous, two-phase nuclear process. Due to ultra-relativistic speeds, phases succeed each other cyclically and are not discernible by direct observation.

Phase 1: Excitation & Disintegration (Peak Potential)

During each peak of the AC half-cycle, the cavity walls reach maximum negative potential. Neutral neon atoms in contact with the walls conductively acquire electrons, forming anions (Ne-). These anions are electrostatically repelled from the negatively polarized walls and converge toward the region of least negative charge the geometric center of the cavity. Through the pendulum effect, their kinetic energy multiplies with each oscillation, and they undergo elastic collisions where mutual electrostatic repulsion combined with momentum reversal traps them in a cycle of increasing energy. Upon reaching the critical energy density, **ultra-relativistic collisions overcome the Coulomb barrier**, causing complete nuclear disintegration into a transient □nucleon fog, a dense plasma of protons and neutrons. This transition is instantaneous at the system□s critical energy density, explaining the absence of conventional ignition signatures.

Phase 2: De-excitation & Recombination (Trough Potential)

As the wall potential drops, electrostatic confinement relaxes. The ultra-hot, high-pressure nucleon fog seeks lower-energy states and undergoes **statistical recombination** into nuclei with higher binding energy per nucleon (e.g., Neon-20 ’ Neon-22 + protons). The excess binding energy is released as heat and light, producing the observed thermonuclear glow and the system's net energy output. Recombined neutral atoms re-enter the buffer gas region before re-initiating the cycle.

V. Critical Observations & Signatures

- **Instantaneous ignition:** Achieved upon reaching the critical energy density from resonant pumping, without any precursor form of ignition (e.g., electrical discharges) being discernible.
- **Electrically silent feed:** Driven by resonance and electrostatic kinetics, not bulk current flow through plasma; no impedance mismatches, transients, or electrical perturbations manifest on the feed line during operation.
- **Visible thermonuclear effect:** Intense heat and light originate from the recombination phase of the cyclic nucleon plasma.
- **Stable & scalable confinement:** Confinement is inherently electrostatic and cyclical, governed by drive frequency and chamber geometry, making it highly controllable.

VI. Thermal energy harnessing system

The reactor produces energy primarily as heat deposited directly into the walls of the reaction chamber cavity (the Páno Káto electrode itself), which thus functions simultaneously as reaction chamber and thermal collector.

- **Primary heat exchanger:** A metal serpentine conduit is integrated with the external walls of the electrode.
- **Heat transfer loop:** A heat-conducting fluid circulates through this conduit in a closed loop, transferring thermal energy to an external standard heat exchanger, with decoupling from the high-frequency electrical signal accomplished via conventional radio-frequency systems.
- **Critical engineering note:** Efficient, active heat extraction is **mandatory**. Inadequate cooling will result in rapid melting of the electrode, occlusion of the reaction cavity, cessation of the fusion reaction, and the appearance of a stationary regime of total reflection on the feed line.

VII. Theoretical implications & a new fusion regime

This process defines a new regime **Resonance-Driven Disintegration-Recombination (RDDR) Fusion** with profound implications:

- **Fuel purity:** The use of a single, pure noble gas (neon) is essential. Its full electron shell and defined resonance frequency allow for synchronized, coherent anion formation required for the pumping mechanism. Reactivity or isotopic inconsistency would disrupt the cycle.
- **Energy gain source:** Primary yield comes from the exothermic recombination of nucleons into more stable nuclear configurations. Resonant electrical input is used almost exclusively for *pumping* (imparting kinetic energy), not for heating a chaotic plasma, resulting in exceptional efficiency.

- **Technological heritage:** The invention bridges the conceptual world of 19th- and 20th-century discharge tube physics with 21st-century energy needs, fulfilling the latent potential of those early instruments. The very tools that helped humanity discover the atom and quantum mechanics were, all along, the key to a new kind of reactor.
- **Theoretical framework:** RDDR Fusion stands as a legitimate alternative fusion regime to the magnetic confinement fusion (MCF) and inertial confinement fusion (ICF), demonstrating that non-fissile, non-radioactive materials can serve as efficient nuclear fuels and that atomic-scale quantum resonance can be harnessed for macroscopic power generation.

In-depth description

Errante's GCRR-N type

High-frequency sinusoidal alternating current powered gaseous core nuclear reactor for the continuos disintegration-recombination cycle of the Neon gas atoms, based on the light chemical elements' non-wavelength dependent electrical resonance physical phenomenon, encompassing of method and system for harnessing the thermal energy produced by it.

Abstract

The present invention relates to a gaseous core nuclear reactor operating on a continuous cycle of disintegration and recombination of the atoms within its gaseous core, consisting of a single light chemical element, such as Neon gas. It is powered by high-frequency sinusoidal alternating current and is based on the non- wavelength-dependent electrical resonance to the high-frequency sinusoidal electrical current oscillations of light chemical element atoms, such as noble gases. The reactor encompassing a complete system for harnessing the thermal energy it produces.

The gaseous core nuclear reactor which is the subject of this invention utilizes and, at the same time, provides empirical demonstration of, a quantum mechanics physical phenomenon, namely the non-wavelength-dependent electrical resonance to the high-frequency sinusoidal electrical current oscillations of light chemical element atoms, such as noble gases, which is wavelength-independent. It also utilizes the very physical property of electrical resonance possessed by light chemical element atoms, such as noble gases and, in the specific case, Neon gas, both of which were discovered by the present inventor.

As disclosed herein, the gaseous core nuclear reactor which is the subject of this invention enables the exploitation of the aforementioned physical property and physical phenomenon through the use of a special reaction chamber designed to achieve the necessary internal energy density. This allows for the initiation and maintenance of a chain of reactions, spanning from the atomic to the subatomic level, which culminate in the transition of the atoms in its gaseous core from their natural state to that of a thermal plasma. This occurs in a continuous Neon-Neon atomic disintegration and recombination cycle into neutral atoms, during which —in their de-excitation phases— energy is released, producing a controlled, stable, and scalable thermonuclear effect with a net energy gain, for the entire duration of the electrical power supplied to it.

Background

The present invention arises from scientific research aimed at exploring the electrical response of gases at high frequencies, and more precisely within the shortwave spectrum, to gain a better understanding of the phenomenon of ionospheric radio propagation.

Gaseous-core nuclear reactors, also known as gas or vapor-core reactors, whether fission or fusion-based, have remained largely conceptual since their inception in the 1950s due to significant technological hurdles, despite potential advantages such as higher efficiency, continuous fuel cycling, and applications in space propulsion. They share the fundamental principles and limitations of solid-core reactors, and both remain largely impractical, particularly concerning the extraction of thermal energy for nuclear fusion. The main common challenges involve fuel containment, the extremely hot plasma, the containment of highly radioactive materials, and the development of materials capable of withstanding intense radiation and heat within the core.

to the present invention, none of the aforementioned nuclear reactors had ever been conceived for the use of non-fissile material as nuclear fuel, nor had it ever been considered that non-fissile material could act alone as nuclear fuel.

Preceding what has been revealed through this invention, the complete ignorance regarding the existence of the physical phenomenon of electrical resonance to the high-frequency sinusoidal electrical current oscillations of light chemical element atoms, such as noble gases—which is non-wavelength-dependent—had prevented the development and realization of gaseous-core nuclear reactors capable of exploiting the equally unknown physical property of noble gases, namely their electrical resonance.

For over a century and a half, vacuum tubes, particularly in their early embodiments as low-pressure gas discharge tubes, have played a fundamental and irreplaceable role in the modern scientific adventure. These seemingly simple experimental apparatuses have been the catalysts for epochal discoveries, allowing the unveiling of matter's most intimate secrets. Within them, the structure of the atom was deciphered, the counterintuitive behaviors of the quantum realm were first observed, and the electrical and magnetic phenomena that eluded classical physics were analyzed. They have been, in essence, the crucible in which the studies for our understanding of nuclear processes were forged. More than mere instruments, these tubes served as a conceptual and experimental bridge, establishing a solid link between the deterministic world of classical physics and the revolutionary paradigms of modern physics, providing the tangible empirical evidence without which the theories that today define the universe of matter and energy would have remained mere intellectual abstractions.

The adoption of technical solutions already known in the field of quantum electrodynamics of direct currents, but never employed in alternating current— whether at low or high frequency—prior to this invention, has allowed the discovery that noble gases also possess the peculiar property of electrical resonance to high-frequency electrical oscillations, and that a quantum mechanics physical phenomenon exists whereby, at least in gases, perfect electrical resonance can be achieved which is not wavelength-dependent.

Description

The present invention pertains to the field of experimental physics, more specifically to that of experimental quantum mechanics. Indeed, it concerns a gaseous-core nuclear reactor constituted by a single light chemical element, such as Neon gas, powered by high-frequency sinusoidal alternating current. It is based on the non-wavelength-dependent electrical resonance to the high-frequency sinusoidal electrical current oscillations of light chemical element atoms, such as noble gases, and includes a complete system for harnessing the thermal energy it produces.

The gaseous-core nuclear reactor which is the subject of this invention utilizes, and simultaneously provides empirical demonstration of, the physical phenomenon of electrical resonance to the high-frequency sinusoidal electrical current oscillations of light chemical element atoms, such as noble gases—a phenomenon which is non-wavelength-dependent and had never been observed prior to this invention.

The phenomenon of electrical resonance to the high-frequency sinusoidal electrical current oscillations of light chemical element atoms, such as noble gases, which is non-wavelength-dependent, is a quantum mechanics physical phenomenon and is impossible to occur within the framework of classical physics.

This phenomenon is made possible by the existence in Nature of an intrinsic electrical property of noble gas atoms, namely the electrical resonance to high-frequency sinusoidal electrical current oscillations, associated with a characteristic natural electrical resonance frequency value, neither of which had been observed prior to this invention.

The value of the natural electrical resonance frequency of Neon gas atoms, experimentally identified by the present inventor, is equal to 27430 kilocycles/sec, i.e., 27430 kHz.

During electrical measurements, the phenomenon of electrical resonance to the high-frequency sinusoidal electrical current oscillations of light chemical element atoms, such as noble gases —which is non-wavelength-dependent— manifests as a common electrical resonance phenomenon of a perfectly resonant electrical circuit, with its typical characteristics. These include a fundamental oscillation frequency clearly distinguishable within a very narrow frequency spectrum band and a well defined, stable real numerical value of impedance at the system feed point, despite the absence of any correlation with the physical dimensions of the body in which it occurs.

For the phenomenon of non-wavelength-dependent electrical resonance to the high-frequency sinusoidal electrical current oscillations of light chemical element atoms, such as noble gases, to manifest, three conditions must be satisfied:

- the driving current must be sinusoidal and have an oscillation frequency equal to the natural electrical resonance frequency of the gas;
- the gas atoms must have sufficient freedom of movement;
- the gas atoms must be in an environment with sufficient energy density.

The gaseous-core nuclear reactor which is the subject of this invention allows for the satisfaction of all conditions necessary for the manifestation of the aforementioned physical phenomenon, the aforementioned electrical property of the gases, and their effects.

The gaseous-core nuclear reactor which is the subject of this invention exploits the aforementioned physical property of noble gases and the aforementioned physical phenomenon through the use of a reaction chamber suitable for achieving the necessary internal energy density. This permits the ignition and maintenance of a chain of atomic and subatomic reactions that lead to the instantaneous transition of the atoms in its gaseous core from their natural state to that of a thermal plasma, in a continuous cycle of neon-neon atomic disintegration and recombination.

During this process, in the excitation phase, the Neon gas atoms disintegrate, giving rise to thermal plasma, while in the de-excitation phase, the constituent particles of the thermal plasma recombine into neutral atoms with the release of energy and the production of a controlled, stable, and scalable thermonuclear effect with a net energy gain, for the entire duration of the electrical power supplied to them.

The gaseous-core nuclear reactor which is the subject of this invention, in order to obtain—on any system scale—the necessary concentration of energy density that allows for the ignition and maintenance of a chain of atomic and subatomic reactions culminating in the thermonuclear effect, adopts a reaction chamber which creates within it the known concentrating effect of the hollow cathode (later also observed in the hollow anode), to which the pendulum effect and that of electrostatic confinement are added. The secondary thermionic effect due to the heating of the reaction chamber, caused by the heat subsequently generated by the thermonuclear reaction, is negligible even for maintaining the reaction.

Construction details

In the gaseous-core nuclear reactor which is the subject of this invention, the reaction chamber is located within a cavity inside an electrode, through which it is powered.

In the gaseous-core nuclear reactor which is the subject of this invention, the electrode housing the reaction chamber is of a unipolar type and is powered by a single monophase sinusoidal alternating electrical current from an unbalanced feed line. It is, therefore, not fixedly polarized and cannot, consequently, be designated as either a cathode or an anode. For this reason, it is here analogously named and indicated with the Greek words "páno káto", meaning "up and down," with a clear reference to the variation in electrical potential.

The reactor which is the subject of this invention, in its simplest embodiment for scientific purposes, consists of a single cylindrical metal electrode, 30 mm long and having a diameter of 10 mm, equipped with a blind cylindrical cavity coaxial to it, open on the side of one of its circular bases. The cavity is 15 mm deep and 4 mm in diameter. This cavity functions as the reaction chamber. The electrode housing the reaction chamber is mechanically supported inside a cylindrical glass bulb by a metal rod 15 mm long and 2 mm in diameter, integral with it, which extends outward from the bulb on the side of the circular base opposite the opening of the aforementioned cavity, along its axis, and which also serves as a terminal for the electrical supply.

Inside the glass bulb, which is 100 mm long and has a diameter of 40 mm, there is a low-pressure (<5 Torr) Neon gas atmosphere.

Due to the electrostatic confinement established within the reaction chamber, the thermal plasma is in contact with the walls of the reaction chamber, which are none other than the internal walls of the cavity of the feeder electrode. The electrode therefore also functions as a thermal collector.

The export of the thermal energy produced by the reactor can, consequently, easily be accomplished by using a metal serpentine conduit integral with the external walls of the electrode containing the reaction chamber, connected in a closed circuit to a simple external heat exchanger outside the bulb. A heat-conducting fluid is circulated within it, following the decoupling of the high-frequency electrical signal from it using well-known radio-frequency systems.

N.B. In the absence of adequate extraction of the heat generated within the reaction chamber, the metal of the feeder electrode, housing the reaction chamber, is destined to melt in a short time, resulting in the occlusion of its cavity, the cessation of the thermonuclear reaction, and the appearance of a stationary regime of total reflection on the feed line.

Principal of operation

In the gaseous-core nuclear reactor which is the subject of this invention, during its operation, a chain of atomic and subatomic reactions takes place in progressive phases, with various quantum mechanics effects intervening in each.

Due to their ultra-relativistic speed, it is not possible to distinguish with the naked eye the various phases that succeed each other cyclically.

In the gaseous-core nuclear reactor which is the subject of this invention, the observed thermonuclear effect ignites instantaneously upon reaching the minimum necessary energy density inside its reaction chamber, without any precursor form of ignition, such as electrical discharges, etc., ever being discernible. No electrical perturbation, such as impedance mismatches or transients, manifests on the feed line.

Further information may arrive in the future using sophisticated detectors to capture and analyze the trails of particles from high-energy collisions exiting the reaction chamber. However, in light of current state-of-the-art knowledge in quantum mechanics, it is still possible to hypothesize what occurs inside the reaction chamber.

In the gaseous-core nuclear reactor which is the subject of this invention, the Neon gas atoms pervade the entire interior of the glass bulb and are in direct contact with all the walls of the reactor electrode and its reaction chamber.

In the gaseous-core nuclear reactor which is the subject of this invention, during each half-cycle of the electrical oscillation of the driving current having a frequency equal to the natural electrical resonance frequency of the Neon gas atoms, the walls of the reactor electrode and those of its reaction chamber reach a peak electrical potential. Corresponding to each of these peaks, neutral Neon atoms acquire electrons through conduction, becoming negatively charged anions.

All gas atoms that have become anions within the reactor cavity are consequently repelled by the very walls of the cavity, which for a further quarter-cycle will continue to be rich in electrons and thus negatively polarized.

The anions electrostatically repelled by the walls of the reactor cavity will tend to move toward the point in the cavity with the least negative charge, i.e., a region of space equidistant between the bottom and the walls of the reactor cavity itself. This aforementioned region of space with the least negative charge in the cavity becomes the point of collision for all repelled anions.

The anions, repelled due to their mutual repulsion with the negatively charged walls of the cavity, acquire greater kinetic energy.

All anions repelled by the walls of the reaction chamber, having acquired greater kinetic energy and converging toward the aforementioned region of space with the least negative charge, find themselves repelling each other.

From the center of the collision zone, the anions, following elastic collisions with each other, reverse their path and move radially toward the walls and bottom of the reactor cavity, which in the meantime have become charged with electrons again and thus exert a strong repulsion against them once more.

Simultaneously, other originally neutral Neon atoms, or those resulting from atomic recombination present in the zone of higher energy density within the reaction chamber, have acquired a negative charge following a subsequent peak of electrical potential from the high-frequency current, becoming new anions through the aforementioned mechanism and suffering the same fate.

All anions therefore find themselves being continuously repelled both by the reactor walls and by the other anions formed inside the reaction chamber, giving rise to increasingly numerous and energetic collisions multiplied by the pendulum effect within the reactor cavity, resulting in a significant increase in the energy density within it.

The anions trapped in this mechanism, also due to the electrostatic confinement established within the reaction chamber, at a certain point acquire the kinetic energy necessary to become heavy ions capable of overcoming their mutual electrostatic repulsion, giving rise to high-energy collisions that determine their total disintegration.

The Neon gas atoms affected by disintegration thus make the direct transition from their natural state to that of thermal plasma for the entire duration of each peak of electrical potential present at the walls of the reaction chamber due to the alternating driving current. During the subsequent phase, when the electrical potential at the walls of the reaction chamber decreases, their de-excitation and recombination into neutral atoms occurs with the consequent release of energy, primarily in the form of heat.

The Neon gas atoms recombined into neutral atoms will remix into the buffer gas region before re-entering the cycle.

In the gaseous-core nuclear reactor which is the subject of this invention, the cyclic atomic and subatomic process of disintegration-recombination of the Neon gas atoms, described previously, produces a controlled, stable, and scalable thermonuclear effect for the entire duration of the electrical energy supplied to it.

Electrical behavior

The gaseous-core nuclear reactor which is the subject of this invention allows the necessary energy density for its operation to be reached even with input power values below 100 Watts R.F. c.w.

Under operating conditions at the natural electrical resonance frequency of the Neon gas atoms, equal to 27430 kilocycles/sec or 27430 kHz, the reactor exhibits a real-valued input impedance of around 50 Ohms, while upon cessation of operation it presents an impedance of infinite value.

As the driving power varies, the V.S.W.R. on its unbalanced feed line remains at 1:1.


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16. ANTONIO MEUCCI - To honor his memory, life and achievements - Radiondistics.com

antonio_meucci.htm

If you have a scientific interest in the physics of the radio, you should browse this site as an e-book!

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United States of America

107th CONGRESS
1st session
House of Representatives' resolution n.269

Expressing the sense of the House of Representatives
to honor the life and achievements of 19th Century
Italian-American inventor Antonio Meucci, and his work in the
invention of the telephone.

IN THE HOUSE OF REPRESENTATIVES
Mr. FOSSELLA
submitted the following resolution; which
was referred to the Committee on

RESOLUTION

Expressing the sense of the House of Representatives to honor the life and achievements of 19th Century Italian-American inventor Antonio Meucci, and his work in the invention of the telephone.

Whereas Antonio Meucci, the great Italian inventor, had a career that was both extraordinary and tragic;

Whereas upon immigrating to New York, Meucci continued to work with ceaseless vigor on a project he had begun in Havana, Cuba, an invention he later called the 'teletrofono', involving electronic communications;

2

Whereas Meucci set up a rudimentary communication link in his Staten Island home that connected the basement with the first floor, and later, when his wife began to suffer from crippling arthritis, he created a permanent link between his lab and his wife's second floor bedroom;

Whereas, having exhausted most of his life's savings in pursuing his work, Meucci was unable to commercialize his invention, though he demonstrated his invention in 1860 and had a description of it published in New York's Italian language newspaper;

Whereas Meucci never learned English well enough to navigate the complex American business community;

Whereas Meucci was unable to raise sufficient funds to pay his way through the patent application process, and thus had to settle for a caveat, a one year renewable notice of an impending patent, which was first filed on December 28, 1871;

Whereas Meucci later learned that the Western Union affiliate laboratory reportedly lost his working models, and Meucci, who at this point was living on public assistance, was unable to renew the caveat after 1874;

Whereas in March 1876, Alexander Graham Bell, who conducted experiments in the same laboratory where Meucci's materials had been stored, was granted a patent and was thereafter credited with inventing the telephone;

Whereas on January 13, 1887, the Government of the United States moved to annul the patent issued to Bell on the grounds of fraud and misrepresentation, a case that the Supreme Court found viable and remanded for trial;

3

Whereas Meucci died in October 1889, the Bell patent expired in January 1893, and the case was discontinued as moot without ever reaching the underlying issue of the true inventor of the telephone entitled to the patent;

and

Whereas if Meucci had been able to pay the $10 fee to maintain the caveat after 1874, no patent could have been issued to Bell: Now, therefore, be it Resolved, That it is the sense of the House of Representatives that the life and achievements of Antonio Meucci should be recognized, and his work in the invention of the telephone should be acknowledged.

September 25, 2001

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17. ATTREZZATURE SCIENTIFICHE TECNICHE DIDATTICHE PRODOTTI SCIENTIFICI PER LA DIDATTICA, LABORATORIO di FISICA

scientifici_index.htm

RADIONDISTICS

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Attrezzatura scientifica

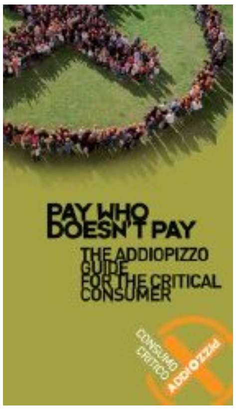
Planning a trip to Sicily? Then, make sure you won't be feeding the Mafia!

- NEW Errante's GASEOUS-CORE ATOMIC RESONANCE NUCLEAR REACTOR
- SISTEMA CAPACITIVO DI TERRA PER RF
- Apparato per la fisica delle linee di trasmissione bilanciate per segnali elettrici a radio frequenza.
- GENERATORE di NODO di TERRA VIRTUALE per linee di trasmissione BILANCIATE

- GENERATORE di NODO di TERRA VIRTUALE per linee di trasmissione sbilanciate

- Apparato di Errante per la fisica della radiazione hertziana. Questo apparato permette di eseguire diversi esperimenti e misure in materia di fisica dei segnali e trasduttori radio-elettrici.

- Circuito radioelettrico per la soppressione di uno qualsiasi dei rami di un dipolo aperto a � onda.
  Brevetto industriale n.0001347484



- Il balun rosmetrico e' un modello di utilita' basato sul circuito del soppressore del ramo di dipolo. Esso e' dotato di tre rosmetri incorporati per il monitoraggio di ogni sua porta. Inoltre il balun rosmetrico incorpora due commutatori coassiali e due carichi fittizii per la soppressione alternata di entrambi i rami del dipolo sotto esame.

- Balun con terra virtuale per lo sdoppiamento del dipolo ripiegato a � onda.
  Brevetto industriale n.0001347486

- Il *radiatore elementare*, antenna HF monopolare con terra virtuale, nota anche come *Antenna Errante*.
  Brevetto industriale n.0001347485

- Rivelatore passivo di radiazione hertziana a fluorescenza da ionizzazione secondaria. **Brevetto industriale n.0001347487**

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18. ANTENNE HF, PRODOTTI PER STAZIONI RADIO PER LE ONDE CORTE, STAZIONI HF, DIPOLI - Radiondistics

`comunicazioni_index.htm`

RADIONDISTICS

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Prodotti per le radiocomunicazioni in HF

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NEW **Sistema d'antenna HF multibanda, da 225 Ohm**.

***Antenna Errante*: Antenna monopolare per le HF con terra virtuale**. **Brevetto Industiale n.0001347485**.

***Balun Errante*: Balun con terra virtuale per antenne a dipolo**. Disponibile in varie combinazioni d'impedenze per antenne a dipolo aperto, ripiegato e loops. **Brevetti Industiali n.0001347484 e n.0001347486**.

Generatore di nodo di terra virtuale per linee di trasmissione bilanciate.

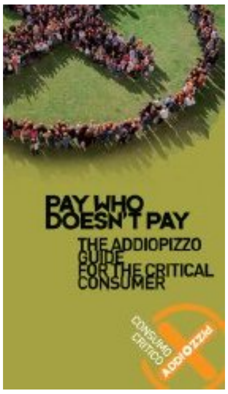
Generatore di nodo di terra virtuale per linee di trasmissione sbilanciate.

Pilota a larga banda per antenne HF turnstile (dipolo a croce, dipoli incrociati, quadripolo).

Synchrobeam & suo pilota.

Pilota per cortine *broad-side.*

Pilota per cortine *end-fire.*



Sfasatori per ogni possibile configurazione di antenna a cortina per le HF. (Cortine di antenne elettricamente orientabili)

Divisori e combinatori di segnali RF per qualsiasi applicazione in HF.

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balun per dipoli, balun per antenna a dipolo, antenne per le HF, antenne per le onde corte, antenne per le onde medie, antenne per le onde lunghe, componenti per le antenne radio, dipoli, dipoli aperti, dipoli ripiegati,dipoli filari, dipoli rigidi, dipoli autoportanti, dipoli rotativi, elementi attivi, radiatori per yagi-uda, cortine di dipoli, antenna turnstile, sistemi d'antenna HF, arrays, beams, balun, stazioni radio HF, tralicci, amplificatori lineari di potenza, KW, dB, stato solido, valvolari, antenna marconi, antenna errante, dipolo hertziano,lunghezza d'onda,metri, ricerca applicata, propagazione ionosferica, trans-ionosferica, HF, onde corte, radiocomunicazioni, telecomunicazioni

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19. Francesco Errante - Radiondistica, radio fisica, ingegneria radio-elettrica & radio-elettronica, meccanica quantistica

autore.htm

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La mia opera tecnico-scientifica e' dedicata alla memoria di *Nikola Tesla*, *Antonio Meucci* ed a tutte le Creature viventi vittime di oppressione e di ingiustizia.

Francesco Errante

ï¿½ï¿½

Planning a trip to Sicily? ï¿½Then, make sure you won't be feeding

L'ingiustizia e' indivisibile. Un'ingiustizia da qualche parte e' una minaccia alla Giustizia ovunque!

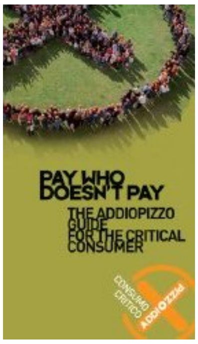
Martin Luther King Jr.

Note autobiografiche dell'autore.

Sono un fisico sperimentale ed un ingegnere radio-elettronico, ho avuto i miei primi approcci con i rice-trasmettitori radioelettrici all' eta' di anni 8. Da oltre un quarto di secolo, sono impegnato in un programma di ricerca scientifica, autofinanziata, volta all'indagine della fisica della radiazione hertziana e dei fenomeni radio-elettrici. Trasferitomi in Gran Bretagna verso la fine dell'anno 1988, ho fatto rientro in Italia nel Marzo del 1999 a causa di una disgrazia capitata alla mia famiglia.
Sebbene non senza difficolta', ho continuato le mie ricerche in Italia, ove, invece di trovare sostegno ed assistenza, ho trovato le istituzioni consorziate per il mantenimento del rachitismo tecnologico dell'Italia e specialmente delle sue aree depresse. Non posso non ricordare le parole di mio padre che essendo stato, per oltre vent'anni, un ispettore generale (con carica e funzioni) dell'Ente di Sviluppo Agricolo (ESA), presso la Sede regionale di Palermo, si riferiva alla stessa come "*Ente di Rachitismo Agricolo*".

Cio' premesso, io, Francesco Errante, fu' Vincenzo (classe 1920, Partigiano) e di Margherita Caminita, nato a Palermo il 18.08.1969, a mezzo del presente sito Internet (www.Radiondistics.com), desidero rivelare alla comunita' intellettuale internazionale, agli studiosi ed agli studenti, ai tecnici ed ai radioamatori ed a tutti coloro che ne possano trarre beneficio, le mie scoperte in materia di fisica delle radio-onde e gli strumenti scientifici da me ideati e realizzati per le relative verifiche sperimentali, nonche' i miei prodotti per il pronto impiego nelle radiocomunicazioni.

the Mafia!



Visita www.uibm.gov.it ed immetti *Francesco Errante* nel campo di ricerca per vedere tutti i miei brevetti, o verifica direttamente sul sito dell'***Ufficio brevetti europeo***, QUI !

Molti credono che in Italia non si faccia *ricerca*, ma la verita e' che **in Italia si ricerca molto... si ricerca molto specialmente su come rubare soldi allo Stato, all'Unione Europea ed anche ai privati cittadini, col pretesto della *ricerca scientifica*. Guarda! ->**

...ed ancora -> 2009, La scuola tagliata - di Riccardo Iacona

...ed ora -> 2010, LA SCUOLA FALLITA - di Riccardo Iacona

nonche' una eloquente rappresentazione teatrale, de "*Il Gruppetto*", circa la bassa considerazione riscossa dall'istruzione personale, nella scala dei valori della moderna famiglia media italiana.

Ed ancora...


la Repubblica.it

SCUOLA & GIOVANI

LA LETTERA. Il direttore generale della Luiss avremmo voluto che l'Italia fosse diversa e abbiamo fallito

"Figlio mio, lascia questo Paese"

di PIER LUIGI CELLI

Libera scienza in libero Stato
di *Margherita Hack*,
Edizione Rizzoli.

Non solo siamo fra gli ultimi in Europa nelle materie scientifiche, ma quando riusciamo a formare un vero genio in genere gli mettiamo in mano una valigia e lo mandiamo a far del bene all'Estero. Perche' in Italia la ricerca proprio non vuole funzionare? Per due motivi, entrambi ben radicati nella storia e nel costume nazionali. Da un lato scontiamo una cronica quanto inspiegabile paura della scienza e delle sue potenzialita', e dal caso Galileo alla battaglia contro l'analisi preimpianto degli embrioni molta responsabilita' spetta alla Chiesa e al suo vizio di dettare legge in un Paese che pure si professa laico. Dall'altro lato ci si mette lo Stato che da destra a sinistra taglia i fondi all'universita', spreca le scarse risorse, ingarbuglia le carriere accademiche senza peraltro riuscire a sottrarle ai "baroni". Cosi', mentre da ogni parte si decanta l'importanza dell'innovazione per la crescita del Paese, nei fatti chi dovrebbe produrla viene ostacolato con ogni mezzo: concorsi macchinosi, precariato a vita, stipendi da fame e, perche' no, obiezione di coscienza. Storie di ordinaria contraddizione in un sistema che cola a picco. Margherita Hack dedica questo libro all'analisi delle condizioni di una ricerca che non ha piu' ne' Stato ne' Chiesa su cui contare. Passa al vaglio le riforme che si sono succedute sotto quattro governi, denuncia gli errori ricorrenti e le troppe incongruenze, mette in luce gli esempi positivi incontrati nel corso della sua carriera e infine propone qualche idea.

Paul Adrien Maurice Dirac, Nobel

"Non capisco perche' mai continuiamo a parlare di religione", disse una volta Paul Dirac.
"Se siamo onesti – e in quanto scienziati, dobbiamo esserlo – dobbiamo ammettere che qualsiasi religione e' un miscuglio di false asserzioni, prive di ogni fondamento reale. L'idea stessa di Dio e' un prodotto dell'immaginazione dell'uomo. Capisco perfettamente che l'uomo primitivo, piu' esposto alle travolgenti forze della Natura, abbia personificato queste forze, mosso dalla paura. Ma oggi, che

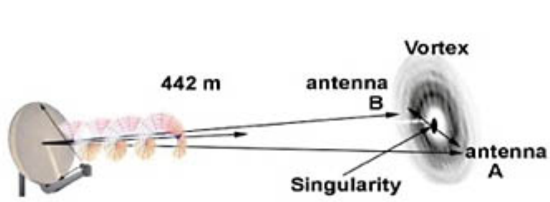
prize in physics, year 1933

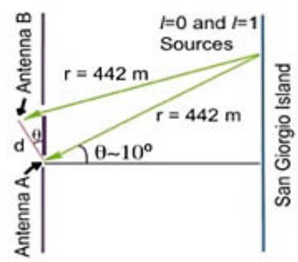
conosciamo tanti fenomeni naturali, non abbiamo piu' bisogno di queste soluzioni.
Vi assicuro che non riesco a capire in cosa puo' esserci utile postulare l'esistenza di un dio onnipotente. Quello che capisco e' che tali ipotesi non porta ad altro che a sterili interrogativi, come il perche' Dio permetta cosi' tanta miseria e ingiustizia, lo sfruttamento dei poveri da parte dei ricchi e tutti gli altri orrori o altri mali che Egli avrebbe potuto facilmente evitare? Se la religione viene ancora tramandata, sappiamo benissimo che cio' avviene, non perche' la religione ci convinca, ma per tenere tranquille le classi subalterne. E' piu' facile dominare i popoli timorati piuttosto che quelli insoddisfatti che protestano; ed e' anche piu' facile sfruttarli. E' gia' stato detto: la religione e' come una sorta di oppio: i popoli si cullano con sogni visionari dimenticando le ingiustizie e lo sfruttamento reali. Di qui l'alleanza tra le due grandi forze politiche dello Stato e della Chiesa. Entrambe trovano comoda l'illusione che un Dio buono ricompensi – se non in questo mondo, nell'altro – coloro che non si sono levati contro l'ingiustizia ma che si sono sottomessi docilmente e magari con gratitudine ai doveri che vengono loro imposti. E questo e' il motivo per cui asserire onestamente che Dio e' solo una creazione dell'immaginazione umana e' considerato come il peggiore di tutti i peccati mortali".

Adesso, a danneggiare attivamente l'immagine della ricerca scientifica in Italia, si sono aggiunte anche le nostre stesse universita '!

Teorici estremamente entusiasti
sono riusciti involontariamente a fare
un falso annuncio a livello mondiale.

Alcuni teorici naïve del Dipartimento di fisica ed astronomia dell'Universita' di Padova e dell'*Istituto svedese di fisica spaziale* si sono coperti di ridicolo dopo aver rivendicato di aver trovato il modo per forzare i fotoni a seguire una traiettoria elicoidale a guisa di vortice circolare, nello spazio libero, facendo semplicemente rimbalzare un fronte d'onda piano, convenzionalmente generato da una porzione di un riflettore paraboloidale "appositamente" distorta (mostrato in basso).





Inoltre, hanno asserito che cosi' facendo, piu' segnali radio portanti con identica frequenza possono viaggiare sullo stesso percorso, con lo stesso verso, per mezzo di traiettorie a vortici circolari differenti tra di loro solo per l'angolo di fase dei loro vortici, ognuno dei quali capace di trasportare pacchetti indipendenti di informazioni,

facendo intendere che sia possibile ottenere piï¿½ canali sulla stessa frequenza.
In base alle informazioni che ho raccolto, sono stati condotti alcuni esperimenti, seriamente mal concepiti e mal realizzati dal Gruppo italo-svedese ed eseguiti con l'assistenza di alcuni radio amatori italiani al posto di riconosciuti specialisti delle microonde. Il Gruppo, mancante delle dovute competenze, ha clamorosamente mancato di comprenderne i risultati e gli errori commessi. Inutile dire che il loro "esperimento" sulla laguna veneta si ï¿½ rivelato un fiasco!

Uso criminale della *radiazione hertziana*. Un affronto all'intelletto umano e contro tutti coloro che hanno contribuito allo sviluppo della Radio per il progresso dell'Umanita'

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20. ANTENNA A DIPOLO: rivelata la sua elettrodinamica ed meccanismo di radiazione © 2003 Francesco Errante

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NEW � **Pilota a larga banda per antenne a dipoli incrociati, note anche come antenne *turnstile* o quardipoli.**




NB. I nostri balun(s) per ricetrasmissioni in HF, li trovi qui.

�

l'*ANTENNA ERRANTE*

Sistema di messa a terra capacitiva per RF

Le linee di trasmissione bilanciate in

Dipolo aperto a 1/2 onda: un'antenna... anzi due!
L'importanza dell'applicazione pratica del concetto di "*terra virtuale*" ai trasduttori ed ai circuiti radio-elettrici, ad opera di Francesco Errante, per la comprensione dei fenomeni radio-elettrici.
Il "*nodo di terra virtuale*" ha nella fisica dei fenomeni radioelettrici la medesima importanza del "*fulcro della leva*" nella meccanica.

�

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Visita www.uibm.gov.it ed immetti *Francesco Errante* nel campo di ricerca per vedere tutti i miei brevetti, o verifica direttamente sul sito dell'***European Patent Office*** QUI !

CIRCUITO RADIOELETTRICO CON TERRA VIRTUALE PER LA SOPPRESSIONE DI UNO QUALSIASI DEI RAMI DI UN DIPOLO APERTO A MEZZ'ONDA

by *Francesco Errante*

Funzioni scientifiche :

a. Il balun qui descritto permette di dimostrare che �il dipolo a � onda, detto anche "*Dipolo Hertziano*" non � un' "*antenna elementare*" (dicesi *antenna elementare* un�antenna nella quale la condizione di risonanza ed irradiazione non pu� verificarsi se non in presenza di tutte le sue parti)� bensi' una "*cortina d�antenne elementari*", costituita da �due elementi radiatori/captatori indipendenti di lunghezza fisica pari a � d'onda ciascuno, arrangiati elettricamente in controfase ed alimentati al centro;

regime
di onda
progressiva

La radiazione
hertziana:
cos'� e come
avviene

�

(Lo stesso si verifica per tutti gli altri dipoli simmetrici, indipendentemente dalla loro impedenza, detta anche *resistenza di irradiazione*.)

b. Il balun qui descritto permette di dimostrare che avendo a disposizione un nodo di terra virtuale � possibile rice-trasmettere segnali radio-elettrici attraverso lo spazio per mezzo di un solo elemento radiatore/captatore, della lunghezza fisica pari ad � d�onda, senza la necessita della presenza di un piano di terra naturale od artificiale ne� radiali di massa o contrappesi o simili artifizi;� (vedi --> *ANTENNA ERRANTE*) �

c. Il balun qui descritto�permette di dimostrare che ognuno dei rami di un �dipolo aperto a � onda presenta un valore assoluto d�impedenza�pari a 35 Ohm sbilanciati rispetto al polo di terra virtuale; **(Analogamente, ognuno dei due rami di qualsiasi altro dipolo simmetrico presentera' un'impedenza caratteristica pari ad � dell'impedenza caratteristica del dipolo a cui appartiene.)**

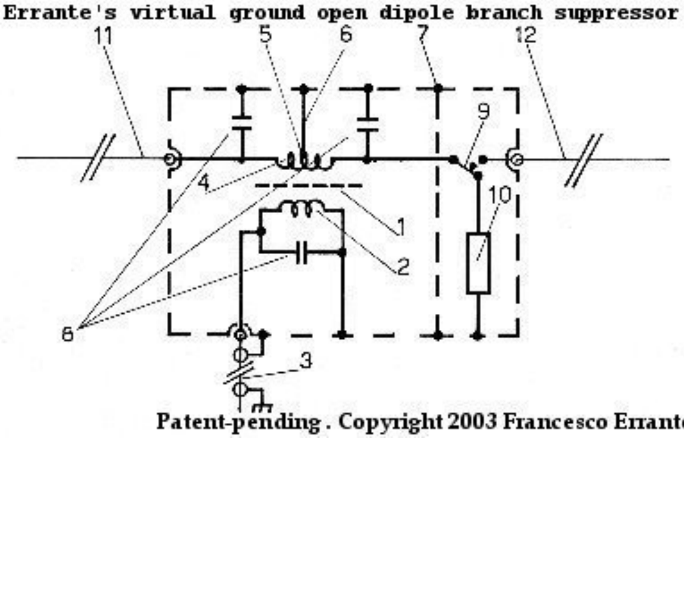
d. Il balun qui descritto permette di dimostrare che ognuno dei rami di un dipolo aperto a � onda presenta al suo punto di alimentazione una differenza dell�angolo di fase rispetto alla terra virtuale pari a 90 gradi.

e. Il balun qui descritto, permettendo la soppressione di uno qualsiasi dei rami di un dipolo, ha portato a confutare definitivamente la "*teoria dell'antenna immagine* " .

Descrizione

In radiotecnica, sono noti svariati sistemi per adattare un dipolo aperto a � onda ad una linea di trasmissione sbilanciata, essi sono genericamente chiamati *balun*, dall�inglese *BALanced-UNbalanced* (bilanciato-sbilanciato). Allo stato attuale della radiotecnica, tali sistemi non consentono l�alimentazione indipendente dei due rami del dipolo aperto a � onda e conseguentemente impediscono di verificare se sia possibile sopprimere uno qualsiasi dei rami del� dipolo stesso senza intaccare la condizione di risonanza ed irradiazione di quello rimanente. **L�invenzione qui descritta permette di alimentare separatamente i due rami(11, 12) di un dipolo aperto a � onda in modo da consentire loro di essere elettricamente indipendenti l�uno dall�altro e conseguentemente permette la soppressione di uno qualsiasi di essi, senza compromettere la condizione di risonanza ed irradiazione di quello rimanente.**
(analogamente, un dispositivo per sdoppiare il filamento del dipolo ripiegato � descritto qui)

Errante's virtual ground open dipole branch suppressor



Patent-pending . Copyright 2003 Francesco Errante

Tale risultato � raggiunto per mezzo di un circuito radioelettrico a costanti concentrate, basato su un trasformatore elettrico(1) a larga banda per segnali a radiofrequenza, avvolto su nucleo di� *ferrite* di forma binoculare, avente un avvolgimento primario(2) con impedenza di valore pari a quello della linea di trasmissione adottata(3) ed un avvolgimento secondario(4) con presa centrale(5) ed impedenza pari a 35 + 35 Ohm. I predetti valori d�impedenza sono riferiti al polo di *terra virtuale*. �(dicesi *terra virtuale*, dall�inglese *virtual ground* un� nodo elettrico che presenta lo stesso potenziale elettrico della terra fisica e nessuna differenza dell�angolo di fase rispetto alla stessa. Praticamente, la terra virtuale � un punto di un circuito elettrico che appare essere elettricamente connesso direttamente a terra, quando in realt� non lo �). Il polo di terra virtuale � stato ottenuto dotando il trasformatore(1) di presa centrale sul suo avvolgimento secondario, ottenendo in realt� lo sdoppiamento� del secondario in due avvolgimenti. La terra virtuale cos� ottenuta � resa disponibile a tutta la rete, tramite un collegamento elettrico(6) allo schermo metallico(7) del trasformatore.
L'ottimizzazione del punto di lavoro del trasformatore(1) � ottenuta mediante compensazione, con l'utilizzo di capacit� elettriche ad alto isolamento(8).
La soppressione ed il ripristino di uno qualsiasi dei due rami del dipolo aperto a � onda � realizzata, attraverso il commutatore ad alto isolamento(9) che permette di deviare manualmente il segnale radioelettrico verso un polo del carico fittizio antiinduttivo(10) d�impedenza pari a 35 Ohm, mentre l�altro polo del carico fittizio � collegato al circuito di terra virtuale.�����������������������

E� stato cos� inventato un circuito radioelettrico con terra virtuale per la soppressione di uno qualsiasi dei due rami del dipolo aperto a � onda .

Peculiarit� elettriche ed operative

1. Il balun qui descritto � caratterizzato dal fatto di avere una presa centrale sul suo avvolgimento secondario per la generazione di un nodo di terra virtuale che costituisce il riferimento elettrico a potenziale zero di tutto il circuito e ne permette una precisa stabilizzazione elettrica.

2. Il balun qui descritto � caratterizzato dal fatto che l�avvolgimento secondario(4) del trasformatore, durante la trasmissione di segnali radio-elettrici nello spazio, provvede ad erogare due segnali radioelettrici identici ma in controfase tra di loro, ovvero, ognuno dei segnali presenta una differenza dell�angolo di fase rispetto al polo di terra virtuale pari a �90 gradi.

3. Il balun qui descritto � caratterizzato dal fatto che, quando non � attivata la soppressione di uno dei due rami del dipolo, durante la ricezione di segnali radioelettrici provenienti dallo spazio da fonti fuori orientamento rispetto al dipolo stesso,� il trasformatore(1) nel provvedere a combinare qualsiasi segnale bilanciato captato dai due rami del dipolo, introduce una reiezione elettrica che aumenta col diminuire della differenza dell�angolo di fase tra i due segnali ad esso inviati dai rami del dipolo stesso. Ci� permette di esaltare la gi� nota propriet� di *reiezione di fianco* del dipolo aperto a � onda, che si traduce in una maggiore attenuazione dei segnali indesiderati� provenienti da direzioni al di fuori dell�orientamento del dipolo stesso.

4. Il balun qui descritto � caratterizzato dal fatto che, quando non � attivata la soppressione di uno dei due rami del dipolo la� larghezza di banda del dipolo a � onda ad esso connesso � sensibilmente maggiore�rispetto a quando lo stesso dipolo � utilizzato con altri balun gi� noti.

5. Il balun qui descritto � caratterizzato dal fatto che la� suscettibilit� alle influenze da parte di capacit� parassite (suolo, muri etc.) di ognuno dei due rami del dipolo stesso, � notevolmente ridotta rispetto a quando il loro adattamento alla linea di trasmissione sbilanciata � fatto con altri sistemi di adattamento gi� noti (BALUN), consentendo di trasmettere con notevoli potenze anche in stretta prossimit� del suolo, di muri od ostacoli vari e con qualsiasi inclinazione rispetto agli stessi, senza che si verifichino apprezzabili variazioni d�impedenza (disadattamenti)� all�ingresso del trasformatore(1).

6. Il balun qui descritto � caratterizzato dal fatto che tutte le porte del trasformatore(1) presentano un cortocircuito alla corrente continua e sono tutte collegate alla terra fisica tramite il conduttore esterno della linea di trasmissione coassiale(3), consentendo lo scaricamento delle cariche elettrostatiche via terra comune.����

7. Il balun qui descritto si caratterizza per essere un trasformatore elettrico per RF ad *impedenza NON fluttuante*. I valori d'impedenza delle terminazioni del balun di *Errante* sono determinate per costruzione, ne deriva che il rapporto di trasformazione e' una sua funzione e NON viceversa, diversamente da quanto avviene con tutti gli altri arrangiamenti bal-un.

�

Gustav Robert Kirchhoff, leggi o principi di Kirchhoff. i principi di Kirchhoff per i circuiti elettrici, nodi di Kirchhoff, nodi elettrici, nodi in un circuito elettrico, principii e leggi di Francesco Errante, il nodo di terra virtuale applicato ai trasduttori radioelettrici ed alle linee di trasmissione bilanciate, trasformatori elettrici per correnti a radiofrequenza, bilanciato,sbilanciato,balun per dipoli, balun per antenna a dipolo, antenne per le HF, antenne per le onde corte, antenne per le onde medie, antenne per le onde lunghe, componenti per le antenne radio, dipoli, dipoli aperti, dipoli ripiegati,dipoli filari, dipoli rigidi, dipoli autoportanti, dipoli rotativi, elementi attivi, radiatori per yagi-uda, cortine di dipoli, antenna turnstile, sistemi d'antenna HF, arrays, beams, balun, stazioni radio HF, tralicci, amplificatori lineari di potenza, KW, dB, stato solido, valvolari, antenna marconi, antenna errante, dipolo hertziano,lunghezza d'onda,metri, ricerca applicata, propagazione ionosferica, trans-ionosferica, HF, onde corte, radiocomunicazioni, telecomunicazioni

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La *radiazione hertziana*: (meglio nota come onde radio) cos'e' e come avviene

di *Francesco Errante*

Questo documento presenta, per sommi capi, *il teorema di Francesco Errante sulla fisica della radiazione hertziana*.

La *radiazione hertziana* è uno dei modi meno in voga, ma piú antichi per riferirsi alle emissioni nello spettro delle radio-onde, in onore a *Heinrich Rudolf Hertz*, il primo fisico ed ingegnere ad aver dimostrato la possibilita' di effettuare trasmissioni radio.

Fin dall'esperimento di Hertz ad ora, si e' creduto che l'energia viaggiasse dal trasmettitore al ricevitore sempre nella stessa forma, attribuendo erroneamente ai segnali elettrici la proprieta' di

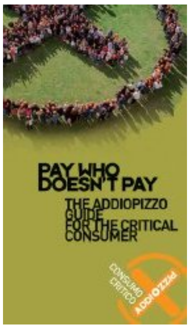
Il nodo di terra virtuale

Sistema di messa a terra capacitiva per RF

Il *radiatore elementare*

viaggiare nello spazio. Lo stesso Hertz chiamo' "onde elettriche" le misteriose emissioni del suo apparato. Successivamente, rinominate "onde elettromagnetiche".

Bisogna, innanzi tutto, comprendere che **la *radiazione hertziana* e' un fenomeno naturale di origine artificiale**. Essa, infatti, sebbene ricalchi il meccanismo di emissione fotonica da reazione nucleare, avviene a frequenze di gran lunga inferiori rispetto a quelle presenti in natura - vedasi fusione nucleare del Sole e sue radiazioni. *Energia e lunghezza d'onda della radiazione*. In natura, le radiazioni sono causate da violente eccitazioni energetiche della materia che implicano altissime temperature e conseguentemente altissime frequenze. Le radiazioni di lunghezza d'onda maggiore di quelle dello spettro dei raggi infrarossi, in natura non hanno alcuna nota funzione pratica, sebbene esse rivestano comunque un'importanza scientifica. Nella *radiazione hertziana*, invece, l'energia per l'eccitazione e' imposta artificialmente mediante *oscillazione elettrica*.
L'*oscillazione elettrica* e' un mezzo per ottenere l'eccitazione delle cariche elettriche e NON deve essere confuso con la *radiazione hertziana* stessa! Questo e', invece, quello che e' avvenuto finora.



Il *rivelatore di radiazione ad ionizzazione*

Le linee di trasmissione bilanciate in regime di onda progressiva

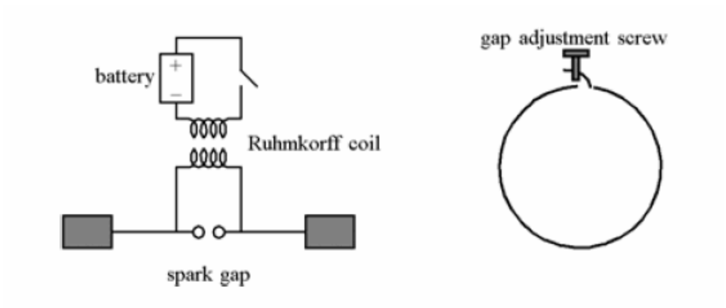
In questa sede si intende dimostrare che:

> **1) la mera oscillazione delle cariche elettriche non e' il fenomeno all'origine della *radiazione hertziana*;**
> **2) la *radiazione hertziana* avviene solo a seguito di una trasformazione energetica, detta *trasduzione radio-elettrica*.**

Breve introduzione allo sviluppo logico delle idee, dei concetti, delle osservazioni, delle deduzioni e degli strumenti brevettati congegnati dall'Autore che hanno condotto a questa scoperta.

Il metodo.
La *semplificazione* e' la regola principale per l'osservazione e la comprensione dei fenomeni fisici. Cio', pero', non deve esssere confuso con l'approccio semplicistico. Spesso e' anche necessario lo sviluppo di strumentazione e tecnologie speciali ed esse stesse sono lungi dall'essere semplici.

Riduzione del radiatore ai minimi termini.
L'antenna del tipo *dipolo aperto*, a mezz'onda, fin dalla sua nascita, è stata sempre considerata come l'antenna più semplice e pertanto universalmente accettata come il campione su cui condurre gli studi. Gli studi finora condotti sono viziati all'origine, perche' il *dipolo aperto* non è un'antenna semplice, come puo' sembrare. Inoltre, l'aver considerato il dipolo aperto come un'*"antenna elementare"* ha spostato il fuoco degli studi dalla sua essenza alle sue proprieta' ed effetti. Conseguentemente, le teorie sul suo intimo funzionamente sono errate ed, a loro volta, queste hanno introdotto un numero di concetti errati, ormai ben radicati, che non hanno permesso di stabilire come avvenisse la "*radiazione hertziana*".



Apparato di Hertz per la trasmissione e ricezione radio-elettrica. Nascita del *Dipolo di Hertz*, 1887.

Per mezzo di un particolare circuito radio-elettrico per la soppressione di uno qualsiasi dei due rami del dipolo aperto a mezz'onda, io ho dimostrato, una volta e per tutte, che il *dipolo aperto* o *dipolo hertziano* non e' un'*antenna elementare* (per definizione, un'antenna elementare e' un aereo nel quale la condizione di risonanza ed irradiazione non puo' avvenire se non in presenza di tutte le sue parti), bensi', una "*cortina elementare*" formata da 2 elementi, di lunghezza fisica pari ad ¼ d'onda ciascuno, alimentati a centro ed in controfase tra di loro.

Una volta stabilito che gli studi dovevano focalizzarsi sul comportamento di un singolo elemento, di lunghezza fisica pari ad ¼ d'onda, ulteriori ricerche mi hanno permesso di realizzare un vero "*radiatore elementare*" per poter distinguere ulteriormente la fonte del segnale radio-elettrico dalla parte realmente radiante.
Il *radiatore elementare*, infatti, comprende un circuito radio-elettrico ed un radiatore di lunghezza fisica pari ad ¼ d'onda, che puo' essere facilmente distaccato e sostituito da un *carico fittizio* (carico anti-induttivo), allo scopo di poter condurre misure radio-elettriche sul solo circuito, mentre, l'osservazione radio-scopica sul radiatore in esercizio, in fase di irradiazione, puó essere condotta con l'ausilio di un particolare rivelatore che sfrutta ed al contempo dimostra il fenomeno della *radioluminescenza* da ionizzazione secondaria da radiazione hertziana.

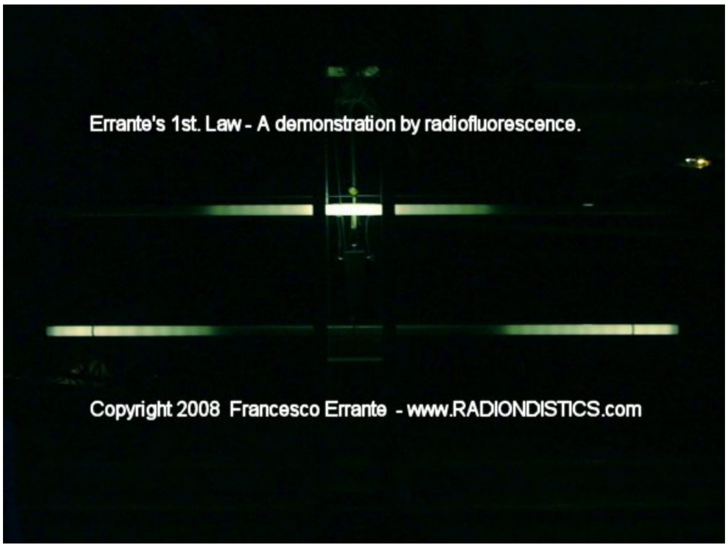
Osservazione radio-scopica per *radioluminescenza*.
Con l'ausilio del mio rivelatore, ho eseguito numerose misure radio-scopiche non-intrusive sul radiatore, su tutto lo spettro delle onde corte, (da 1 a 30 MHz) con potenze RF che vanno dai 100 mW a diversi kiloWatt.

Gli esperimenti da me ideati ed eseguiti, dimostrano chiaramente che:

1) l'energia irradiata dal radiatore ha proprieta' ionizzanti transitorie;

2) iniettando un segnale radio-elettrico su un radiatore propriamente risonante, quest'ultimo irradiera' energia nella forma di radio-onde, iniziando sempre dal punto opposto a quello di sua alimentazione;

3) che il grosso dell'energia e' sempre irradiata dalla regione terminale del radiatore.



Dimostrazione dalla I^a Legge di Errante mediante la radiofluorescenza.

L'immagine mostra la distribuzione

energetica sul dipolo a mezz'onda ripiegato e sul dipolo a mezz'onda aperto, entrambi in condizione di risonanza, alimentati contemporaneamente con segnali RF di pari ampiezza e lunghezza d'onda.
Si noti come nel primo caso l'intensita luminosa del tubo sia massima a centro, mentre nel secondo caso l'intensità luminosa sia massima agli estremi

Iª Legge di Errante: un qualsiasi segnale elettrico a radiofrequenza od un qualsiasi impulso elettrico a radiofrequenza, iniettato su un qualsiasi conduttore elettrico di qualsiasi forma, **darà sempre origine** a radiazione hertziana con inizio dalla parte opposta a quella da dove esso viene iniettato.

Misure radio-elettriche e radio-scopiche di controverifica.
Per converso, ulteriori misure, sia radio-elettriche che radio-scopiche, hanno permesso di verificare anche **l'assenza di insorgenza di radiazione hertziana su conduttori (linee di trasmissione) percorsi da segnali a radio frequenza in regime di onda progressiva.**

Dimostrazione dalla II^a Legge di Errante mediante la radiofluorescenza.

Le immagini mostrano l'assenza di radiazione hertziana su di una linea di trasmissione bilanciata, in regime di onda progressiva, in esercizio con un segnale RF da 1000 Watt. Sia essa terminata su carico reattivo che anti-induttivo.

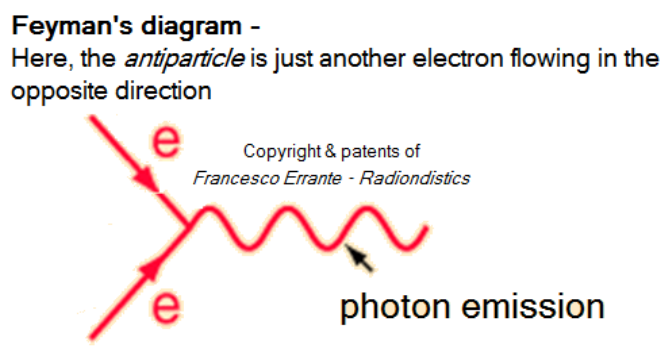
Potenza impiegata: 1 KW RF
Frequenza di prova su dipolo: 25,5 MHz
Frequenze di prova su carico fittizio : da 1.8 MHz a 30 MHz in passi da 100 KHz
Verifica sperimentale

II^a Legge di Errante: un qualsiasi segnale elettrico a radiofrequenza od un qualsiasi impulso elettrico a radiofrequenza, iniettato su un qualsiasi conduttore elettrico di qualsiasi forma, debitamente terminato su un carico di impedenza pari a quella della sua fonte, **NON darà mai origine** a radiazione hertziana.

Deduzioni dell'Autore.
Quanto appena detto porta, ragionevolmente, ad affermare che **la *radiazione hertziana* si verifica ogni qualvolta che le particelle appartenenti ad un primo fronte d'onda avendo percorso il radiatore fino termine della sua estremita' opposta e ritornando indietro collidano con le particelle di un nuovo fronte d'onda, dando origine ad un'esplosione (*scattering*) di nuove particelle, verosimilmente, fotoni con la medesima frequenza e stessa forma d'onda**

(*legge oraria*) del segnale elettrico a radio frequenza che le ha originate. Non essendo possibile per le nuove particelle viaggiare a velocita' superiore a quella della luce, esse acquistano più massa, invece. (se le particelle potessero viaggiare ad una velocita' maggiore di quella della luce, la loro emissione finirebbe, inevitabilmente, per generare segnali radio-elettrici di lunghezza d'onda inferiore)

Questo meccanismo e', quindi, una forma di *regime d'onda stazionario controllato o limitato*. Se esso avviene all'interno della lunghezza del radiatore si ha il fenomeno della *risonanza*. Se, invece, esso avviene al difuori della lunghezza del radiatore si ha un *regime d'onda stazionario casuale* con radiazione minore o nulla.



Nel meccanismo della radiazione hertziana, l'*antiparticella* altro non e' che un elettrone che scorre con verso opposto a quello col quale e' stato iniettato inizialmente.
<u>Ne consegue che il segno negativo di una delle due soluzioni dell'*equazione di Dirac* indica il verso negativo della particella e non una sua carica negativa</u>!

Il meccanismo della trasduzione radio-

elettrica dimostrato dall'*Autore* conferma l'intuizione del *Prof. John Archibald Wheeler* in materia di particelle ed antiparticelle sub-atomiche, come datoGli atto da *Richard Feynman* nella sua lectio magistralis.

Inoltre, l'evidenza conferma il *principio di conservazione dell'energia* e non lascia spazio per continuare a teorizzare circa l'esistenza del *positrone* e dell'*antimateria*.

Conclusioni:

stante a quanto dimostrato dall'Autore, il meccanismo fisico responsabile per la generazione della *radiazione hertziana* e' differente da quanto in precedenza teorizzato ed accettato come vero

La previsione di una concatenazione tra il campo elettrico e quello magnetico, come offerta dalle equazioni di Maxwell, descrivono il campo elettrico e quello magnetico come impegnati in un perenne ciclo di mutua creazione. Questa interpretazione porta con se un errore fondamentale costituito dal fatto che la concatenazione tra i campi e' intesa nella loro

sovrapposizione e NON nel loro avvicendamento di cicli completi, come in una gara a staffetta, come invece accade.

Alla luce di quanto dimostrato dall'Autore, si puo' affermare che non esiste una tale cosa come un'onda elettromagnetica, un campo elettromagnetico od un segnale elettromagnetico. Vi sono, invece, emissioni fotoniche (radio onde) che seguono la stessa legge oraria e frequenza dei segnali che le hanno generate, capaci di indurre forze elettromotrici sulla materia con la stessa frequenza e forma d'onda.

Come conseguenza diretta di questa scoperta, in materia di trasduzione radio elettrica, anche il concetto di *induzione elettromagnetica* non risulta più calzante. Allo stesso modo, anche il meccanismo fisico attraverso il quale un conduttore elettrico (es.: antenna radio) e' interessato dalla *radiazione hertziana* deve essere differente da quanto teorizzato e creduto in precedenza e tutto punta a nient'altro che ad un semplice

***effetto fotovoltaico*, come osservato da *Hertz* e *Hallwachs* nello spettro della radiazione ultravioletta.**

La trasmissione e la ricezione di un segnale radio-elettrico, da un punto ad un altro dello spazio, non e', quindi, un semplice trasferimento di energia nella stessa forma ed i segnali elettrici a radio-frequenza non hanno la proprieta' di viaggiare nel vuoto.

Il principio di Errante sulla *trasduzione radio-elettrica*

Affinche' una quantita' di energia, sotto forma di segnale elettrico a radio-frequenza, transiti da un punto ad un altro dello spazio, per essere riprodotta nella sua forma originale, necessita che la stessa subisca almeno due trasformazioni:

1ª da segnale elettrico a radio frequenza a radiazione hertziana;
2ª da radiazione hertziana a segnale elettrico a radio frequenza.

Il numero minimo di 2 trasduzioni si ha nel caso di radio propagazione a portata ottica.

Questo particolare fenomeno di trasformazione energetica di un segnale elettrico a radio-frequenza in una radiazione a radio-frequenza, prende il

nome di *trasduzione radio-elettrica*. La *trasduzione radioelettrica* è un fenomeno spontaneo dove una forma di energia ordinata viene trasformata in una forma differente di energia disordinata e viceversa. Questa trasformazione può essere ripetuta un infinito numero di volte, ma dal punto di vista delle perdite energetiche, questo è un esercizio molto svantagioso.

I dispositivi atti a favorire la *trasduzione radio-elettrica*, sono definiti *trasduttori radio-elettrici* e sono comunemente chiamati "antenne radio".

NEW È, adesso, commercialmente disponibile il ns. apparato per la fisica della radiazione hertziana e dei trasduttori radioelettrici. Leggi !

meccanico è consentito esclusivamente per uso personale.
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22. ANTENNA RADIO RICE-TRASMITTENTE MONOPOLARE CON TERRA VIRTUALE © 2003 Francesco Errante

antenna_errante_it.htm

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Il *radiatore elementare*

noto anche come

Antenna Errante

ORA, anche disponibile nella versione MULTIBANDA da 225 Ohm

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Il nodo di terra virtuale

Sistema di messa a terra capacitiva per RF

Le linee di trasmissione bilanciate

Immagine del prototipo da laboratorio dell'*Antenna Errante* in una tipica installazione veloce di un ventaglio di monopoli per le bande dei 20, 30, 40, 60, 80 e 160 metri.
NB. Mostrato senza l'involucro impermeabilizzante esterno in ABS.

Banda di lavoro: da 1 a 30 MHz
Potenza RF: 2KW cw

Fai click sulle immagini per ingrandirle

    

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L'Antenna Errante: cos'�, come funziona e come si � pervenuti alla sua realizzazione.

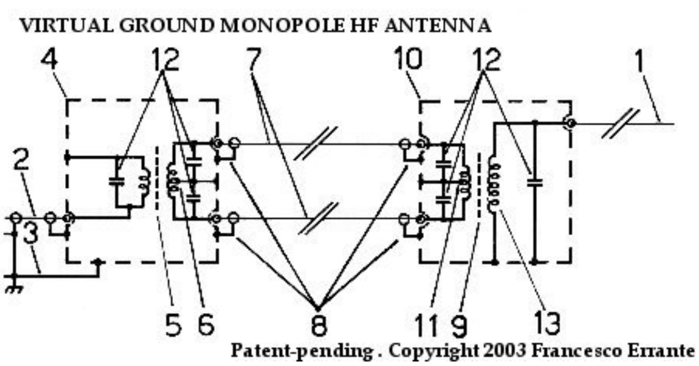
L'ANTENNA RADIO RICE-TRASMITTENTE MONOPOLARE CON TERRA VIRTUALE

in regime di onda progressiva

La radiazione hertziana: cos'� e come avviene di *Francesco Errante*

L'Autore, mediante l�uso di un particolare circuito radioelettrico per la soppressione di uno qualsiasi dei due rami di un dipolo aperto a � onda, ha precedentemente dimostrato che � possibile rice-trasmettere segnali radio-elettrici di notevole potenza attraverso lo spazio tramite l�impiego di un singolo elemento radiatore/captatore di lunghezza fisica pari ad � d�onda, senza la necessit� della presenza di un suo simmetrico (vedi *Dipolo Hertziano*) e senza la necessit� di utilizzare alcun� piano di terra naturale (vedi *Antenna Marconi*) od artificiale (vedi antenna *Ground-plane*) o di radiali o contrappesi elettrici. L�eliminazione della necessit� di un riferimento fisico di terra per la corretta polarizzazione di un singolo elemento radiatore/captatore, di lunghezza fisica pari ad � d�onda, � stata raggiunta mediante la� generazione di un nodo di terra virtuale, la cui esistenza � vincolata� al corretto bilanciamento elettrico di un trasformatore, ottenuto sacrificando il 50% dell�energia fornita al sistema.

La presente invenzione <u>elimina del tutto</u> la necessita' di dover sacrificare energia fornita al sistema e cio' e' ottenuto grazie allo sfruttamento del principio fisico stesso del *nodo di terra virtuale*.

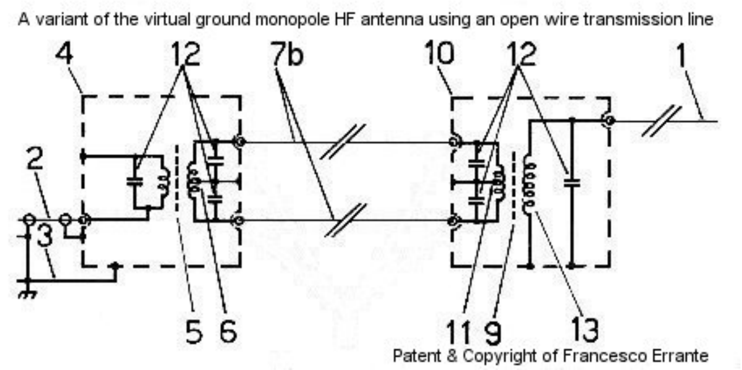


La soluzione a questo problema � data dalla possibilit� di generare il nodo di terra virtuale, tramite la ricombinazione di due segnali radio-elettrici, �identici� ma in opposizione di fase tra di loro, con� l�impiego di un trasformatore balun(9) ad accoppiamento di flusso, costruito su nucleo in ferrite di forma binoculare, con avvolgimento primario bilanciato, dotato di presa centrale(11) per l�ottenimento di un *nodo di terra virtuale* che funga come riferimento a potenziale zero per tutto il circuito, attraverso lo chassis conduttivo(10) ed un avvolgimento secondario(13) con impedenza pari a 150 Ohm per l�adattamento diretto al radiatore/captatore(1) di lunghezza pari ad � d�onda. Il trasformatore(9) ed il radiatore/captatore(1) sono le parti essenziali del trovato e costituiscono un�antenna radio-ricetrasmittente monopolare da � d�onda con terra virtuale ed ingresso bilanciato. Allo scopo di rendere tale antenna compatibile con gli attuali standards vigenti�in radiotecnica, � stato necessario dotare l�antenna di un�ulteriore� trasformatore partitore-sfasatore(5) capace di effettuare l�adattamento tra la presa d�antenna sbilanciata di tipo coassiale dei rice-trasmettitori convenzionali e l�ingresso bilanciato del trasformatore d�antenna(9). Il nodo di terra virtuale(6), ottenuto sull'avvolgimento secondario del trasformatore(5), e' reso disponibile al circuito mediante lo chassis conduttivo(4).�
Il trasferimento energetico tra i due trasformatori � compiuto mediante l�impiego di due linee di trasmissione coassiali(7) di qualsiasi lunghezza, ma necessariamente identiche tra di loro per mantenere inalterato lo sfasamento tra i segnali da esse condotti.

Sebbene fosse stato possibile adottare qualsiasi valore d�impedenza per il circuito intermedio costituito dal primario del trasformatore(9), dalle linee di trasmissione(7) e dal secondario del trasformatore(5),�per comodit�, � stato adottato un valore d�impedenza standard di 50 Ohm.

� stata, cos�, inventata un�antenna radio rice-trasmittente monopolare con terra virtuale e relativo circuito di adattamento ad alta efficienza, che permette di rice-trasmettere nello spazio, segnali radioelettrici di notevole potenza per mezzo di un singolo elemento radiatore/captatore, di lunghezza fisica pari ad � d�onda, posto a qualsiasi livello dal suolo e con qualsiasi inclinazione rispetto allo stesso, senza la necessit� di un piano di terra, sia esso naturale che artificiale o� l�ausilio di un bilanciamento fisico e soprattutto senza rilevanti perdite di segnale.

A seguito della pubblicazione degli studi sul comportamento delle linee di trasmissione bilanciate in regime di onda progressiva, viene qui appresso presentata una versione dell'antenna HF monopolare con terra virtuale che utilizza una linea di trasmissione bilanciata bifilare al posto di quelle coassiali.



E' di vitale importanza che l'impedenza dell'avvolgimento secondario del

trasformatore locale(5), l'impedenza della linea di trasmissione bifilare(7a) e l'impedenza dell'avvolgimento primario del trasformatore remoto(9) abbiano esattamente lo stesso valore.

�

Importanza scientifica:

a. L'antenna monopolare con terra virtuale dimostra che � necessario soltanto un singolo elemento conduttore di lunghezza fisica pari ad � d'onda per raggiungere la condizione di risonanza e completa irradiazione.

b. L'antenna monopolare con terra virtuale, grazie alla forma e semplicita' del suo elemento radiante, permette ulteriori investigazioni in materia di fisica delle radio onde. (vedi anche Osservazione Radioscopica)

Vantaggi operativi:

a. L'antenna monopolare con terra virtuale permette di operare *full-size* con il 50% dell'ingombro di un dipolo.

b. L'antenna monopolare con terra virtuale offre una larghezza di banda caratteristica molto pi� ampia delle antenne *full-size* convenzionali.

c. L'antenna monopolare con terra virtuale offre un'immunita' agli effetti delle capacit� parassite di gran lunga superiore alle antenne *full-size* convenzionali.

d. L'antenna monopolare con terra virtuale offre un'alta stabilit� del punto di risonanza indipendentemente dalla sua inclinazione e distanza dal suolo.

e. L'antenna monopolare con terra virtuale offre un maggiore immunita' ai segnali fuori-banda, rispetto alle antenne *full-size* convenzionali.

f. L'antenna monopolare con terra virtuale offre un maggiore immunita' alle cariche ed ai rumori di natura elettrostatica rispetto alle antenne *full-size* convenzionali.

g. L'antenna monopolare con terra virtuale offre un diagramma di radiazione pi� omogeneo e definito rispetto alle antenne *full-size* convenzionali. conventional full-size aerials.

h. L'antenna monopolare con terra virtuale permette di essere installata in tempi molto pi� rapidi di quanto richiesto dalle antenne *full-size* convenzionali.

i. L'antenna monopolare con terra virtuale offre una migliore compatibilita' ambientale (E.M.I) rispetto alle antenne *full-size* convenzionali.

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g. ELEMENTI DI CORTINE d'ANTENNE OMOGENEE

h. LABORATORI di FISICA delle RADIO ONDE.

F.A.Q.: posso pilotare 2 o pi� monopoli simultaneamente, come in una configurazione da *dipolo a ventaglio*, cosicch� possa coprire diverse bande HF?
RISPOSTA: Si ! � Il presente sistema di alimentazione RF con terra virtuale permette di usare simultaneamente tutti i monopoli da � di lunghezza d'onda, necessari ad ottenere una copertura generale dello spettro delle HF, senza la necessit� di commutazioni. I monopoli possono essere disposti sia sul piano orizzontale che sul piano verticale. Tutto quello che serve � lo spazio.
Ora, anche disponibile nella versione da 225 Ohm per il sistema multibanda HF di *Errante*

Prezzo al pubblico:
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NB. Il prezzo NON include le linee coassiali gemelle, n� i radiatori che sono, comunque, fornibili su richiesta.

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23. I trasduttori radio-elettrici alla luce delle nuove scoperte - © Francesco Errante

trasduttori_radioelettrici.htm

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I trasduttori radio-elettrici, meglio noti come *antenne radio*, alla luce delle nuove scoperte.

di *Francesco Errante*

In fisica, col termine di "*trasduttori*" si definiscono tutti quei dispositivi capaci di trasformare una forma di energia in un'altra. Nella trasformazione operata dai trasduttori, generalmente, si passa da una forma di energia ordinata ad una disordinata e viceversa. Ad esempio: altoparlanti e microfoni sono

La radiazione hertziana: cos'� e come avviene

Le linee di trasmissione bilanciate in regime di onda progressiva

trasduttori elettro-meccanici. Essi, infatti, trasformano energia elettrica in vibrazioni meccaniche e viceversa. Le antenne radio sono, invece, dei "*trasduttori radio-elettrici*", queste trasformano energia elettrica a radio frequenza in *radiazione hertziana* (radio-onde) e viceversa.
La comprensione del funzionamento dei trasduttori radio-elettrici, a differenza di quanto avvega con altri trasduttori, non e' intuitiva. Cio' perche' essi non hanno parti meccaniche in movimento e le loro "vibrazioni" non sono direttamente percepibili dal corpo umano. A dire il vero, in determinate condizioni (frequenza, potenza e distanza dalla fonte), il corpo umano puo' avvertire alcuni degli effetti della *radiazione hertziana*, sotto forma di una sensazione di calore distribuito o di una scottatura localizzata, ma questo non e' stato sufficiente a permetterci di individuare il reale meccanismo all'origine del fenomeno delle *onde radio*.
Per oltre un secolo, ovvero, da quando *Heinrich Hertz* dimostro' l'esistenza dei fenomeni radio-elettrici, la loro descrizione "scientifica" si e' limitata ai soli tentativi di rappresentazione teorico-matematica. Nell'anno 2003, l'Autore ha permesso di fare, letteralmente e definitivamente, luce sul vero meccanismo all'origine della *radiazione hertziana*, mediante l'impiego di un suo particolare sistema rivelatore che sfrutta la *radioluminescenza*, un ulteriore fenomeno fisico di trasformazione energetica.
Storicamente i trasduttori radio-elettrici sono stati distinti in due principali tipi: quelli bilanciati e quelli sbilanciati.
Alla luce di quanto dimostrato dall'Autore, la distinzione deve invece essere riordinata tra due categorie: i trasduttori radio-elettrici "*elementari* " e quelli "*non elementari* ".

Sistema d'antenna multibanda con radiatori a 225 Ohm

Sistemi radianti auto-antagonisti

(dicesi *trasduttore radio-elettrico elementare* un trasduttore radio-elettrico nella quale la condizione di risonanza ed irradiazione non pu� verificarsi se non in presenza di tutte le sue parti)
I trasduttori radio-elettrici elementari sono quelli *monopolari* (antenne sbilanciate), mentre quelli NON-elementari sono quelli *dipolari* (antenne bilanciate). Entrambi le categorie di trasduttori possono presentarsi come circuiti *aperti* oppure chiusi, detti *loops*. Va precisato che la dicitura "ripiegato" riferendosi a circuiti chiusi e' impropria, cio' perche' in radiotecnica e' possibile avere anche delle antenne ripiegate ma elettricamente aperte.
I trasduttori *dipolari* possono essere del tipo *dipolo aperto* oppure del tipo chiuso, detto impropriamente "*dipolo ripiegato*" ma comunque capostipite dei *loops*. Il *dipolo ripiegato* e', come dimostrato, riconducibile a due monopoli chiusi alimentati in controfase e coincidenti tra di loro. Questo vale anche per i *loops* aventi qualsiasi geometria (circolare, delta-loop, poligonale etc.).
Tutti i trasduttori radio-elettrici NON-elementari (trasduttori bilanciati) sono sempre riconducibili a combinazioni di sistemi elementari (trasduttori sbilanciati) in controfase tra di loro. L'incorretto spaziamento tra sorgenti elementari appartenenti allo stesso sistema da' origine a sistemi auto-antagonisti a tutto discapito del *guadagno d'antenna*.

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Fondamenti teorici di radio-ingegneria, radiotecnica, radio-fisica, pratica, applicazioni pratiche, tecnologia, progetti d'antenna, progettazione avanzata di antenne, antenne HF, pubblicazioni scientifiche, pubblicazioni tecniche, fisica, scienze, onde radio, onde hertziane, radiazioni, radiazione hertziana, diagramma d'irradiazione, accoppiamento d'antenna, alimentazione d'antenna, ros antenna, tecnologia antenne radio, trasmittenti, riceventi, aerei, balun, trasformatori, RF, radio frequenza, terra virtuale, fenomeni elettromagnetici, fenomeno radioelettrico, studio, studi, esperimenti radioelettrici, portale, ricerca scientifica di base, ricerche applicate,propagazione ionosferica,trans-ionosferica,HF,onde corte, radiocomunicazioni

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24. Linee di trasmissione bilanciate per segnali elettrici RF in regime di onda progressiva - Radiondistics.com

linee_bilanciate.htm

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Apparato di Errante
per la fisica delle linee di trasmissione bilanciate per segnali elettrici a radio frequenza .

Premessa: sebbene quando ci si riferisce alle linee di trasmissione bilanciate si tenda a pensare ad una linea formata da un paio di conduttori elettrici identici e paralleli tra di loro, comunemente dette "*linee bifilari*", bisogna tenere presente che le linee di trasmissione bilanciate possono anche essere composte da un qualsiasi numero pari di conduttori atti a condurre uno stesso numero di correnti in un arrangiamento multi-fase. (L'arrangiamento di correnti poli-fase ï¿½ stato introdotto in elettrotecnica da *Nikola Tesla*)
Le linee di trasmissione bilanciate

Il nodo di terra virtuale

Antenna monopolare con terra virtuale

Generatore di nodo di terra virtuale per

possono essere sia *simmetriche* che *asimmetriche* .

Francesco Errante

<u>Definizione di linea di trasmissione bilanciata</u>: ï¿½ "*una linea di trasmissione per segnali elettrici ï¿½ elettricamente bilanciata quando la stessa non interviene a modificare lo sfasamento delle correnti che la attraversano ed ognuno dei suoi conduttori trasferisce la stessa quantita' di energia per unita' di tempo (potenza).*"

Francesco Errante

<u>Definizione di linea di trasmissione bilanciata *simmetrica*</u>: ï¿½ "*una linea di trasmissione per segnali elettrici elettricamente bilanciata ï¿½ simmetrica quando ï¿½ formata da conduttori elettrici che esibiscono lo stesso valore di impedenza rispetto alla terra od al nodo di terra virtuale.*"

Francesco Errante

<u>Definizione di linea di trasmissione bilanciata *a-simmetrica*</u>: ï¿½ "*una linea di trasmissione per segnali elettrici elettricamente bilanciata ï¿½ a-simmetrica quando ï¿½ formata da conduttori elettrici che NON esibiscono lo stesso valore di impedenza rispetto alla terra od al nodo di terra virtuale.*"

linee bilanciate

Rivelatore di radiazione hertziana mediante radio luminescenza

Francesco Errante

I^{o} principio di *Errante* per le linee di trasmissione: **ï¿½ "*una linea di trasmissione per segnali radioelettrici, sia essa bilanciata o sbilanciata, se in regime di onda progressiva, ï¿½ sempre a-periodica.*"**

Francesco Errante

II^{o} principio di *Errante* per le linee di trasmissione: **"*una qualsiasi linea di trasmissione per segnali radioelettrici puï¿½ essere convertita da bilanciata in sbilanciata e viceversa, infinite volte.*"**

Francesco Errante

III^{o} principio di *Errante* per le linee di trasmissione: **"*In ogni punto di una qualsiasi linea di trasmissione per segnali radiolettrici, sia essa bilanciata che sbilanciata, ï¿½ sempre possibile generare un nodo di terra virtuale.*"**

Francesco Errante

II^{a} legge di *Errante* applicata alle linee di trasmissione **"*una linea di trasmissione per segnali radioelettrici, sia essa bilanciata che sbilanciata, se in regime di onda progressiva, non origina radiazione hertziana.*"**

Francesco Errante

ï¿½

L'apparato ed il metodo di misura qui presentati permettono di analizzare il

comportamento elettrico delle linee di trasmissione bilanciate per segnali elettrici a radio frequenza in regime di onda progressiva ed in particolare hanno lo scopo di:

a) dimostrare che le linee di trasmissione bilanciate sono a-periodiche;

b) dimostrare che le linee di trasmissione bilanciate possono essere convertite in sbilanciate e viceversa, infinite volte;

c) dimostrare che ï¿½ sempre possibile generare un nodo di terra virtuale in ogni punto di una qualsiasi linea di trasmissione;

d) dimostrare che le linee di trasmissione bilanciate, se perfettamente adattate, sia alla sorgente RF che al carico, NON originano radiazione hertziana. II^{a} legge di Errante.

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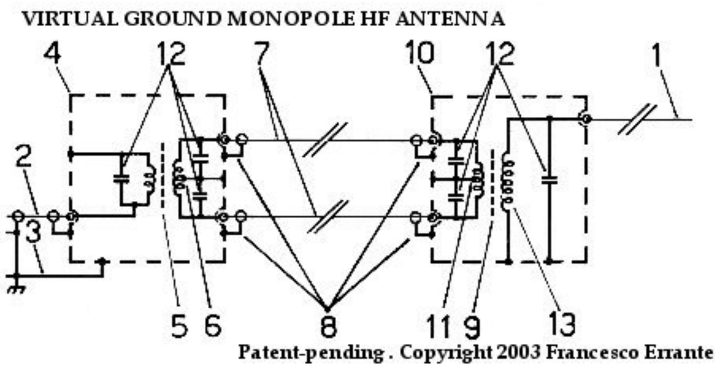
Le linee di trasmissione bilanciate per segnali elettrici a radio frequenza sono generalmente costituite, in modo omogeneo, da due conduttori elettrici identici e paralleli non schermati ma possono essere costituite anche da un numero maggiore di conduttori sia liberi che schermati. ï¿½



Esempio di impiego di linea elettricamente bilanciata formata da 2 cavi coassiali(7) a schermatura individuale - Copyright © 2003 - Francesco Errante

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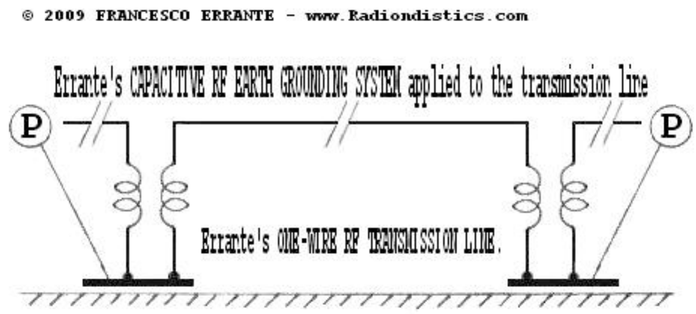

Errante's ONE-WIRE UNBALANCED RF TRANSMISSION LINE, employing Errante's CAPACITIVE RF EARTH GROUNDING SYSTEM - Copyright © 2009 - Francesco Errante

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Una linea di trasmissione per segnali elettrici ï¿½ elettricamente bilanciata quando la stessa non interviene a modificare lo sfasamento delle correnti che la attraversano.

perchï¿½ ciï¿½ sia possibile e' necessario che tutti i conduttori che costituiscono la linea siano identici e che la linea stessa sia perfettamente adattata sia al suo ingresso che alla sua uscita, ciï¿½ vuol dire che l'impedenza caratteristica della linea deve essere uguale sia all'impedenza di uscita della fonte del segnale a radio frequenza che all'impedenza del carico, sia esso reattivo o anti-induttivo. Una linea di trasmissione correttamente adattata, quando ï¿½ in esercizio, si trova in **regime di onda progressiva** e si ha il massimo trasferimento energetico possibile al carico ed assenza di *radiazione hertziana*.

Laddove la linea viaggi in stretta prossimitï¿½ della terra fisica, ï¿½ anche indispensabile che i suoi conduttori siano equidistanti dalla stessa al fine di mantenerne costante il loro accoppiamento capacitivo con essa.
A paritï¿½ di distanza interassiale tra i conduttori della linea, la sua impedenza caratteristica diminuisce col diminuire della distanza dalla terra fisica. Ciï¿½ ï¿½ dovuto all'accoppiamento capacitivo tra i conduttori della linea e la terra fisica che si somma alla capacitï¿½ elettrica parassita, sempre presente tra i conduttori della linea stessa.
Le linee bilanciate, debitamente isolate, possono viaggiare rasoterra o addirittura interrate. In questi casi, lo stretto accoppiamento capacitivo tra i conduttori della linea e la terra rende trascurabile le capacitï¿½ parassite tra i conduttori della linea stessa e non necessiterï¿½ piï¿½ che questi ultimi viaggino paralleli, ma devono comunque essere identici, eccetto quando formino una *linea bilanciata a-simmetrica*.

Le linee di trasmissione elettricamente bilanciate possono anche essere geometricamente ed elettricamente a-simmetriche.

Una linea bilanciata a-simmetrica ï¿½ cosï¿½ definita: "*una linea di trasmissione per segnali elettrici elettricamente bilanciata ï¿½ a-simmetrica quando la stessa non interviene a modificare lo sfasamento delle correnti che la attraversano ed ï¿½ formata da conduttori elettrici che non esibiscono lo stesso valore di impedenza rispetto alla terra od al nodo di terra virtuale. Copyright 2008 Francesco Errante*".
Lo stretto accoppiamento con la terra limita, perï¿½, l'impiego di linee rasoterra ai sistemi per basse e medie potenze, dove le tensioni in gioco non richiedono grossi isolamenti.

Con il progredire delle tecnologie per le costruzioni radioelettriche, le linee sbilanciate in cavo co-assiale hanno, via via, soppiantato le linee di trasmissione bilanciate nella maggior parte delle applicazioni ad eccezione di quelle per altissime potenze.

Un caso notevole di linea di trasmissione bilanciata che ricorre frequentemente fin dagli albori della radiotecnica ï¿½ dato dal loro impiego per l'alimentazione delle antenne a dipolo ripiegato a mezz'onda. Ciï¿½ perchï¿½ alle utili proprietï¿½ del dipolo ripiegato a mezz'onda si unisce la facilitï¿½ di realizzazione delle linee bilanciate a 300 Ohm. (come analogamente avviene per i dipoli a mezz'onda a piï¿½ archi che presentano impedenze caratteristiche multiple di 300 Ohm)

Le grossolane osservazioni fatte senza alcun rigore scientifico e la troppa facilitï¿½ con la quale ï¿½ possibile alimentare un dipolo ripiegato a mezz'onda con un tratto di linea bifilare a 300 Ohm ha portato tutti, fin ora, ad erroneamente desumere e/o accettare che le linee bilanciate fossero esse stesse

risonanti e che pertanto si dovesse provvedere a neutralizzarne gli effetti con l'allungamento o accorciamento delle stesse. Questo e' falso!
Al dilagare di queste ï¿½credenzeï¿½ ha contribuito anche lï¿½errata interpretazione da parte di molti dellï¿½esperimento di *Ernst Lecher*. La linea di Lecher, infatti, evidenzia il comportamento elettrico di una linea bifilare in regime di onda stazionaria ed il suo impiego come ondametro ï¿½ vano se la linea ï¿½ invece in regime di onda progressiva (VSWR = 1:1) .

L'illusione strumentale che le linee bilanciate siano risonanti ï¿½ data dal fatto che non essendovi un netto punto di confine tra il carico reattivo, nella fattispecie il dipolo ripiegato a mezz'onda, ed il tratto di linea bifilare che lo alimenta, i segnali elettrici vedono il loro percorso variare in lunghezza al variare della lunghezza della linea stessa con il risultato di spostare con ciclicitï¿½ il punto di risonanza risultante del sistema stesso.

In veritï¿½, se in un sistema costituito da fonte di segnale bilanciato, linea bilanciata e radiatore si sostituisce il carico reattivo, costituito dall'antenna, con un carico anti-induttivo di pari impedenza (carico fittizio) od un carico reattivo perfettamente adattato, l'illusione della risonanza della linea bilanciata scompare ed il sistema ritorna ad essere a-periodico, come dimostrato qui appresso.

Qui appresso e' riportata la sequenza logica delle misure ed i relativi esiti:

Queste misure, con la dovuta attrezzatura, possono essere eseguite in tre modi:

1) mantenendo costante la lunghezza d'onda del segnale radioelettrico di prova e variando la lunghezza fisica della linea bilanciata;
2) mantenendo fissa la lunghezza fisica della linea bilanciata e variando la lunghezza d'onda del segnale radioelettrico di prova;
3) variando sistematicamente sia la lunghezza fisica della linea bilanciata che la lunghezza d'onda del segnale radioelettrico di prova.

Il secondo modo e' di gran lunga il piï¿½ conveniente e veloce, quindi **si ï¿½ utilizzata una linea bilanciata a 300 Ohm di lunghezza fisica fissa e si ï¿½ proceduto alle misure riflessometriche variando la lunghezza d'onda del segnale radielettrico di prova**.

Si noti, inoltre, che queste misure si possono eseguire su qualsiasi spettro di frequenze radio, ma risulta conveniente operarle sullo spettro delle onde corte, ciï¿½ perche le stesse permettono, tra l' altro, una maggiore escursione di lunghezza d'onda a paritï¿½ di variazione di frequenza. **Tutti i risultati delle misure qui condotte sono, quindi, estendibili a tutto lo spettro delle radio onde.**

NB: Tutti i dispositivi qui utilizzati per l'ottenimento di nodi di terra virtuale determinano un netto confine per i segnali a RF tra linee e radiatori o tra linee e linee, mentre garantiscono sempre una totale continuitï¿½ galvanica tra tutti i circuiti ad essi collegati, assicurando un solido percorso di terra comune che puï¿½ essere, all'occorrenza, collegato alle terra fisica. Vedi anche: generatore di terra virtuale per linee sbilanciate

Verifica sperimentale del comportamento a-periodico delle linee bilanciate in regime di onda progressiva

OPERAZIONE PRELIMINARE: il sistema di misura bilanciato-bilanciato viene sottoposto a verifica iniziale della linearitï¿½ della sua risposta in frequenza.

Da sinistra verso destra: linea di TX co-assiale 50 Ohm - sonda ROSmetrica - BalUn 50/300 Ohm - breve tratto di linea bifilare simmetrica - BalUn 300/50 Ohm - Carico fittizio 50 Ohm.
Frequenze di prova: da 1,8 a 30 MHz a passi da 100 KHz
Potenza RF impiegata: 1 KW cw
R.O.S. misurato: ï¿½ 1:1 ï¿½ su tutto lo spettro
Esito: OK! Il sistema esibisce una risposta piatta entro lo spettro di misura richiesto per questa verifica.

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Linea bilanciata a 300 Ohm, alimentata da fonte bilanciata a 300 Ohm, terminata su carico fittizio da 300 Ohm e sottoposta a misura rosmetrica

Da sinistra verso destra: BalUn 50/300 Ohm - linea bifilare simmetrica da 300 Ohm, lunga metri 4 ca. - Carico fittizio 300 Ohm.

Frequenze di prova: da 1,8 a 30 MHz a passi da 100 KHz

Potenza RF impiegata: 1 KW cw

R.O.S. misurato: ï¿½ 1:1 ï¿½ su tutto lo spettro

Esito: l'aperiodicitï¿½ della linea ï¿½ verificata.

N.B. Le misurazioni sono state ripetute con la linea in aria.

ï¿½

Lo stesso avviene se la linea ï¿½ terminata su un carico reattivo (BalUn di Errante) ed il segnale sbilanciato cosï¿½ ottenuto viene soppresso su idoneo carico anti-induttivo (carico fittizio), cio' a comprova del comporamento a-periodico delle linee bilanciate anche su una terminazione reattiva.

Linea bilanciata a 300 Ohm, alimentata da fonte bilanciata a 300 Ohm, terminata su carico bilanciato reattivo da 300 Ohm e sottoposta a misura rosmetrica

Da sinistra verso destra: linea di TX co-assiale 50 Ohm - sonda ROSmetrica - BalUn 50/300 Ohm - linea bifilare simmetrica da 300 Ohm, lunga m 4

ca. - BalUn 300/50 Ohm - Carico fittizio 50 Ohm.
Frequenze di prova: da 1,8 a 30 MHz a passi da 100 KHz
Potenza RF impiegata: 1 KW cw
R.O.S. misurato: ï¿½ 1:1 ï¿½ su tutto lo spettro
Esito: l'aperiodicitï¿½ della linea ï¿½ verificata.
N.B. Le misurazioni sono state ripetute con la linea in aria.

ï¿½

E cosï¿½ pure avviene se la linea ï¿½ disaccoppiata dal carico anti-induttivo mediante un generatore di nodo di terra virtuale per linee bilanciate, di pari impedenza (**BVGNG di Errante**).

Linea bilanciata a 300 Ohm, alimentata da fonte bilanciata a 300 Ohm, disaccoppiata dal carico fittizio mediante generatore di nodo di terra virtuale per linee bilanciate e sottoposta a misura rosmetrica

Da sinistra verso destra: BalUn 50/300 Ohm - linea bifilare simmetrica da 300 Ohm, lunga metri 4 ca. - Errante's 300/300 Ohm B.V.G.N.G - Carico fittizio 300 Ohm.
Frequenze di prova: da 1,8 a 30 MHz a passi da 100 KHz
Potenza RF impiegata: 1 KW cw
R.O.S. misurato: ï¿½ 1:1 ï¿½ su tutto lo spettro

Esito: l'aperiodicitï¿½ della linea ï¿½ verificata.
N.B. Le misurazioni sono state ripetute con la linea in aria.

ï¿½

Ed in fine, mediante l'impiego del circuito radioelettrico per la generazione del nodo di terra per sistemi bilanciati, ï¿½ stato possibile dimostrare che le linee bilanciate, se opportunamente disaccoppiate dai radiatori, hanno sempre un comportamento a-periodico, come mostrato qui di seguito:

Linea bilanciata a 300 Ohm, alimentata da fonte bilanciata a 300 Ohm, terminata su dipolo bilanciato attraverso un BVGNG di Errante da 300/300 Ohm e sottoposta a misura rosmetrica

1ª misura
Scopo: individuazione del punto di risonanza naturale del dipolo in esame.
Condizioni: alimentazione del dipolo via cavo co-assiale mediante BalUn 50/300 Ohm, con terra virtuale.

Frequenze di prova: **centro della gamma di lavoro del dipolo ripiegato - MHz 22,9**
Potenza RF impiegata: 1 KW cw
R.O.S. misurato: ï¿½ 1:1 ï¿½ @ MHz 22,9
Esito: accertato centro della gamma di lavoro del dipolo ripiegato - MHz 22,9

2ª misura
Scopo: individuazione del nuovo punto di risonanza del sistema linea-dipolo.
Condizioni: alimentazione diretta del dipolo mediante tratto di linea bilanciata a 300 Ohm
Frequenze di prova: **NUOVO centro della gamma di lavoro del dipolo ripiegato - MHz 23,5**
Potenza RF impiegata: 1 KW cw
R.O.S. misurato: ï¿½ 1:1 ï¿½ @ MHz 23,5
Esito: rilevato slittamento del centro della gamma di lavoro del dipolo ripiegato MHz 23,5.

3ª misura
Scopo: individuazione del nuovo punto di risonanza del sistema linea-dipolo, disaccoppiati tra di loro.
Condizioni: alimentazione del dipolo mediante linea bilanciata a 300 Ohm ma disaccoppiata tramite l'impiego di un generatore di nodo di terra virtuale per linee bilanciate 300/300 Ohm
Frequenze di prova: **centro della gamma di lavoro del dipolo ripiegato - MHz 22,9**
Potenza RF impiegata: 1 KW cw
R.O.S. misurato: ï¿½ 1:1 ï¿½ @ MHz 22,9
Esito:

1) rilevato il ripristino del centro della gamma di lavoro del dipolo.
2) l'aperiodicitï¿½ della linea ï¿½ verificata.

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Verifica sperimentale della NON INSORGENZA di radiazione hertziana nelle line bilanciate in regime di onda progressiva.
II**ª** Legge di Errante

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Si e' sottoposta una linea di trasmissione bilanciata, perfettamente adattata, in esercizio (potenza impiegata 1 Kilowatt RF) a rilevazione della radiazione hertziana col metodo della radioluminescenza.
Si ï¿½, cosï¿½, osservato e dimostrato che non vi ï¿½ insorgenza di radiazione hertziana quando una linea di trasmissione si trova in regime di onda progressiva.

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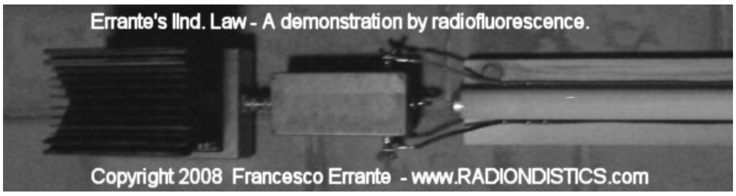
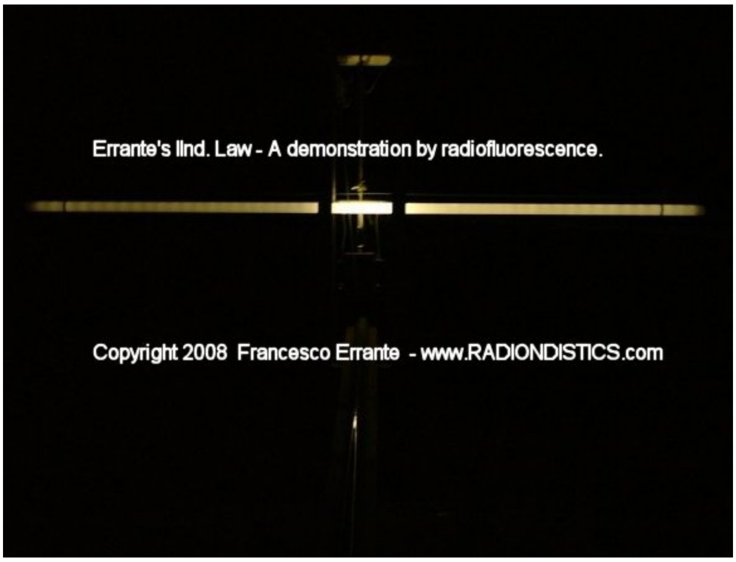
Linea bilanciata a 300 Ohm, debitamemente adattata, terminata su carico reattivo non radiante, sottoposta a misura radioscopica.



Da sinistra verso destra: Carico fittizio 50 Ohm - BalUn 50/300 - linea bifilare simmetrica da 300 Ohm, lunga metri 4 ca.
Frequenze di prova: da 1,8 a 30 MHz a passi da 100 KHz
Potenza RF impiegata: 1 KW cw
R.O.S. misurato: ï¿½ 1:1 ï¿½ su tutto lo spettro
Esito: la linea NON irradia.

ï¿½

Linea bilanciata a 300 Ohm, debitamemente adattata, terminata su dipolo ripiegato a mezz'onda, sottoposta a misura radioscopica. Misure effettuate a bassa ed alta potenza (Freq. 25,5 MHz - 1 KW cw).



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Radiazione hertziana prodotta da linea bilanciata deliberatamente posta in regime di onda stazionaria.

Si ï¿½ deliberatamente imposto un regime di onda stazionaria (R.O.S. 1:1,7) sulla linea di trasmissione utilizzata in precedenza, mediante un disadattamento sull'uscita della linea stessa, sostituendo il BalUn 50/300 Ohm con uno da 50/150 Ohm.
Si ï¿½, cosï¿½, ossevata e dimostrata l'insorgenza di radiazione hertziana su linea di trasmissione in regime di onda stazionaria.

Da sinistra verso destra: Carico fittizio 50 Ohm - BalUn 50/150 - linea bifilare simmetrica da 300 Ohm, lunga metri 4 ca.
Frequenze di prova: da 1,8 a 30 MHz a passi da 1 MHz
Potenza RF impiegata: 1 KW cw
R.O.S. minimo misurato: ï¿½ 1:1,7
Esito: la linea IRRADIA.

Altra veduta di un tratto della linea bifilare, in regime di onda stazionaria, mostrata sopra

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An interactive impedance mismatch spectral simulation
by
Agilent Technologies

General Instructions

This spectral simulation is an interactive Java applet. You can change parameters by clicking on the vertical arrow keys. The five control buttons at the lower right are used to start (triangle) and pause (square) the simulation, to skip forward or back one section at a time (double triangles), and to change speed (+ and -).

After the simulation is complete, the start button takes you back to the beginning of the simulation. You may experience a delay at this point.

Wave Propagation along a Transmission Line

When a sine wave from an RF signal generator is placed on a transmission line, the signal propagates toward the load. This signal, shown here in yellow, appears as a set of rotating vectors, one at each point on the transmission line.

In our example, the transmission line has a characteristic impedance of 50 ohms. If we choose a load of 50 ohms, then the amplitude of the signal will not vary with position along the line. Only the phase will vary along the line, as shown by the rotating vectors in yellow.

If the load impedance does not perfectly match the characteristic impedance of the line, there will be a reflected signal that propagates toward the source. At any point along the transmission line, that signal also appears to be a constant voltage whose phase is dependent upon physical position along the line.

The voltage seen at one particular point on the line will be the vector sum of the transmitted and reflected sinusoids. We can demonstrate this by looking at two examples.

Example 1: Perfect Match: 50 Ohms

Set the terminating resistor to 50 ohms by using the "down arrow" dialog box. Notice there is no reflection. We have a perfect match. Each rotating vector has a normalized amplitude of 1. If we were to observe the waveform at any point with a perfect measuring instrument, we would see equal sine wave amplitudes anywhere along the transmission line. The signal amplitudes are indicated by the green line.

Example 2: Mismatched Load: 200 Ohms

Now let's intentionally create a mismatched load. Set the terminating resistor to 200 ohms by using the down arrow. Hit the PLAY button and notice the change in the reflected waveform. If it were possible to measure just the reflected wave, we would see that its amplitude does not vary with position along the line. The only difference between the reflected (blue) signal, say at point "z6" and point "z4", is the phase.

But the amplitude of the resultant waveform, indicated by the standing wave (green), is not constant along the entire line because the transmitted and reflected signals (yellow and blue) combine. Since the phase between the transmitted and reflected signals varies with position along the line, the vector sums will be different, creating what's called a "standing wave".

With the load impedance at 200 ohms, a measuring device placed at point z6 would show a sine wave of constant amplitude. The sine wave at point z4 would also be of constant amplitude, but its amplitude would differ from that of the signal at point z6. And the two would be out of phase with each other. Again, the difference is shown by the green line, which indicates the amplitude at that point on the transmission line.

The impedance along the line also changes, as shown by the points labeled z1 through z7.

VSWR

The VSWR, or Voltage Standing Wave Ratio, is the ratio of the highest amplitude signal to the lowest amplitude signal, as measured along the transmission line. A "perfect" VSWR is 1.

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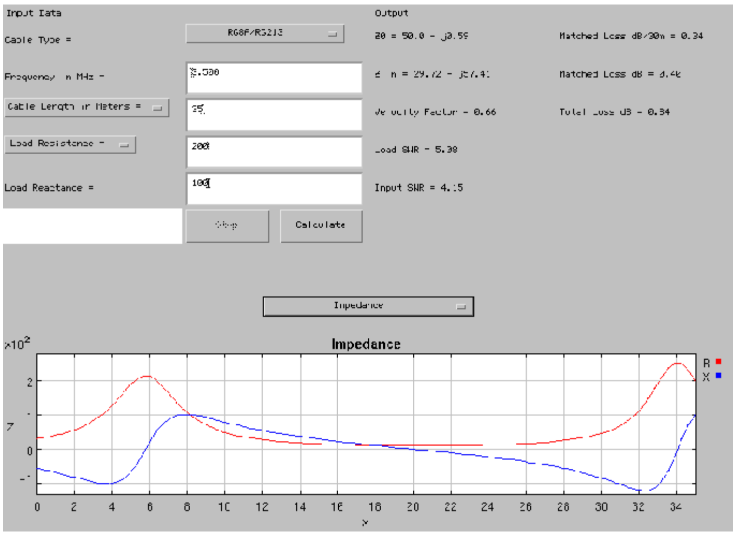
Calcolatore per linee di trasmissione RF



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25. GUGLIELMO MARCONI - Comunicazioni radio-telegrafiche - Premio Nobel, 11 Dicembre 1909

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GUGLIELMO MARCONI

Comunicazioni radio-telegrafiche

lectio magistralis del 11 Dicembre 1909

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Le scoperte connesse alla propagazione delle onde elettriche su lunghe distanze e le applicazioni pratiche della telegrafia attraverso lo spazio, che mi hanno fatto ottenere l'alto onore di condividere il Premio Nobel per la Fisica, sono state una grande estensione dei risultati di un altro. L'applicazione delle onde elettriche alla finalit� della comunicazione telegrafica senza fili tra punti distanti della Terra e gli esperimenti che ho avuto la fortuna di poter eseguire su scala pi� grande di quella ottenibile nei normali laboratori hanno reso possibile investigare fenomeni e osservare risultati spesso nuovi ed insospettati.

E' mia opinione che molti fatti connessi con la trasmissione di onde elettriche su grandi distanze stiano ancora aspettando una spiegazione soddisfacente, e spero di riuscire in questa

conferenza a riferire alcune osservazioni, che sembrano richiedere l'attenzione dei fisici.

Nel tratteggiare la storia della mia associazione con la radiotelegrafia, potrei menzionare di non aver mai studiato in maniera regolare la fisica o l'elettrotecnica, sebbene� da ragazzo fossi profondamente interessato a questi argomenti.

Assistetti, tuttavia, ad una serie di conferenze sulla fisica a cura del Professor Rosa di Livorno ed ero, penso di poterlo dire, assai ben familiare con le pubblicazioni dell'epoca su argomenti scientifici, compresi i lavori di Hertz, Branly e Righi.

A casa mia vicino a Bologna, in Italia, cominciai presto nel 1895 ad effettuare esperimenti allo scopo di determinare se fosse possibile per mezzo di onde hertziane trasmettere a distanza segni e simboli telegrafici senza ricorrere a collegamenti galvanici.

Dopo alcuni esperimenti preliminari con le onde hertziane mi convinsi subito che se queste onde o similari avessero potuto affidabilmente essere trasmesse e ricevute su distanze considerevoli, avremmo avuto a disposizione un nuovo sistema di comunicazione con enormi vantaggi rispetto ai metodi ottici e luminosi, che dipendono troppo per il loro successo dalla nitidezza dell'atmosfera.

I miei primi test furono effettuati con un normale oscillatore di Hertz e un coherer Branly come rivelatore, ma mi resi subito conto che il coherer Branly era troppo soggetto a errori e inaffidabile per l'uso pratico.

Dopo alcune prove trovai che un coherer costruito come in Figura 1, fatto di fili di nickel e argento collocati nel piccolo spazio tra due elettrodi d'argento in un tubo, fosse assai sensibile e affidabile. Questa miglioria assieme all'inclusione del coherer in un circuito sintonizzato sulla lunghezza d'onda del trasmettitore mi consentirono di ampliare gradualmente fino a circa un miglio la distanza a cui poter attivare il ricevitore.

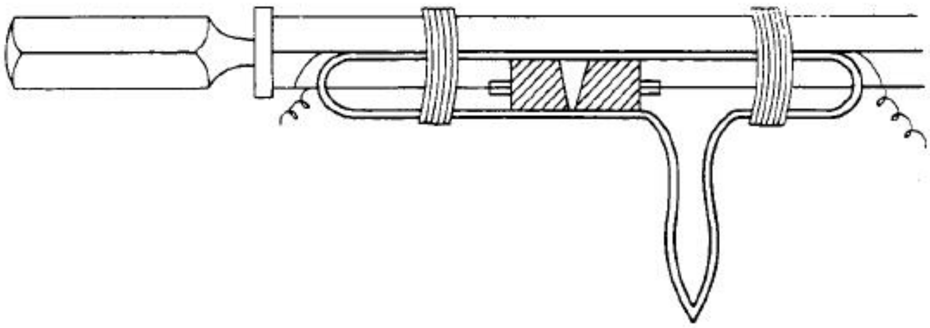

Fig. 1.

Un altro, ora ben noto, arrangiamento da me adottato consiste nel piazzare il coherer in circuito comprendente una cella voltaica e un sensibile rel� telegrafico attivante un altro circuito, che aziona un cicalino e uno strumento di registrazione. Per mezzo di un tasto telegrafico Morse in uno dei circuiti dell'oscillatore o trasmettitore risult� possibile inviare treni lunghi o corti di onde elettriche, che attivavano a distanza il ricevitore riproducendo accuratamente i segni telegrafici trasmessi attraverso lo spazio dall'oscillatore.
Con tale apparato riuscii a telegrafare fino a una distanza di circa mezzo miglio.
Successivi miglioramenti furono ottenuti impiegando riflettori sia sul ricevitore che sul trasmettitore, un oscillatore di Righi.
Ci� rese possibile inviare segnali verso una specifica posizione, ma la cosa non funzionava se colline o altri grossi ostacoli si frapponevano tra trasmettitore e ricevitore.
Nell'agosto 1895 scoprii una nuova configurazione che non solo aumentava grandemente la distanza di collegamento ma sembrava anche rendere la trasmissione indipendente dagli effetti degli ostacoli interposti.
Questa disposizione (Figura 2 e 3) consisteva nel connettere un terminale dell'oscillatore hertziano, o generatore di scintille, alla terra e l'altro terminale a un filo o a una lamina posti ad una certa altezza sul terreno, e collegando nel ricevitore un capo del coherer alla terra e l'altro al conduttore sopraelevato.

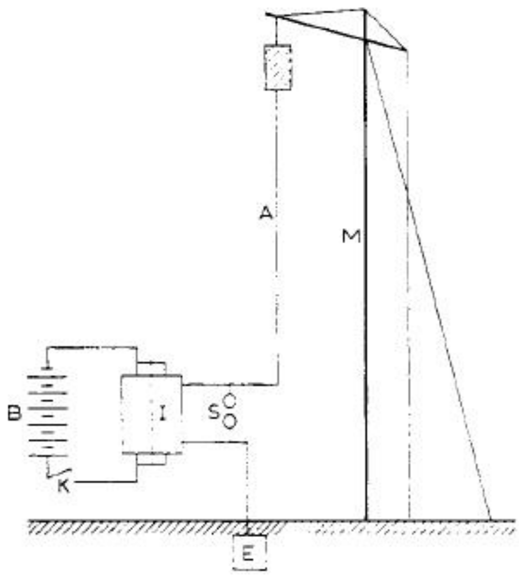

Fig. 2.

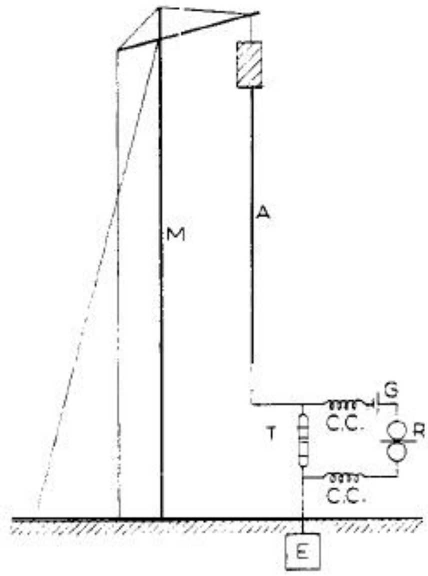

Fig. 3.

Iniziai allora a studiare la relazione tra la distanza a cui il trasmettitore era in grado di azionare il ricevitore e l'elevazione della lamine sopra il terreno, e presto mi avvidi con certezza che pi� alto era il filo o lamina maggiore era la distanza a cui era possibile telegrafare.
Cos� accertai che usando cubi di stagno di 30 cm di lato come conduttore posti in cima a pali alti due metri si poteva ricevere i segnali a 30 m di distanza, mentre ponendoli su pali alti 4 m si raggiungevano i 100 m ed i 400 m su pali alti 8 m. Con cubi di lato pi� grande -100 m- fissati in cima a pali da 8 m si raggiungevano i 2.400 m in tutte le direzioni[1].
Questi esperimenti continuarono in Inghilterra, ove nel settembre 1896 fu raggiunta una distanza di 1,75 miglia durante prove effettuate a Salisbury per il Governo britannico e nel Marzo 1897 la distanza di collegamento fu estesa a 4 miglia, e a 9 miglia nel maggio dello stesso anno. Messaggi su nastro registrati durante queste prove, vistati dagli Ufficiali del Governo britannico presenti, sono consultabili[2].
In tutti questi esperimenti � stata usata pochissima energia elettrica, essendo l'alta tensione prodotta da una normale bobina di Rhumkorff. I risultati ottenuti attrassero all'epoca una buona parte dell'opinione pubblica, poich� tali distanze erano considerate notevoli.
Come ho spiegato, la caratteristica principale del mio sistema consisteva nell'uso di lamine o antenne elevate collegate a un polo degli oscillatori e ricevitori ad alta frequenza, mentre l'altro polo era messo a terra. Il valore pratico di questa innovazione non fu compreso da molti fisici per

un bel po' di tempo e i risultati da me ottenuti vennero da molti erroneamente attribuiti semplicemente all'efficienza nei dettagli costruttivi del ricevitore e all'impiego di un grande quantitativo di energia.
Altri non andarono oltre al fatto che queste elevate capacit� e la terra fanno parte degli oscillatori e ricevitori ad alta frequenza[3].
Il Prof. Ascoli di Roma elabor� un'interessante teoria del modo di funzionamento dei miei trasmettitori e ricevitori nel numero di Agosto 1897 della rivista "Elettricista", nel quale correttamente attribuiva i risultati ottenuti all'uso di fili elevati o antenne.
Il Prof. A. Slaby di Charlottensburg, dopo aver assistito alle mie prove in Inghilterra nel 1987, arriv� a conclusioni pi� o meno simili[4].
Molti scrittori tecnici hanno affermato che un'elevata capacit� in cima al filo verticale non � necessaria.
Questo � vero se la lunghezza o altezza del filo sono sufficientemente grandi, ma l'altezza pu� essere diminuita di parecchio se si usa una capacit�, � molto pi� economico, consistendo in genere in un certo numero di fili che si irradiano dalla sommit� del conduttore verticale.

La necessit� o utilit� del collegamento a massa � stata a volte messa in dubbio, ma io ritengo che non possa esistere un sistema pratico di telegrafia senza fili che non abbia gli strumenti collegati a massa. Quando dico "collegato a massa" non intendo necessariamente un collegamento metallico standard come nella telegrafia tradizionale.
Il filo di terra pu� avere un condensatore in serie (Fig. 4C) , **(Sebbene, Guglielmo Marconi abbia compreso l'importanza della *messa a terra* nei sistemi radio-elettrici e la possibilita' di creare degli accoppiamenti capacitivi con la terra per mezzo di reattanze capacitive, non ha colto l'importanza dell' *accoppiamento capacitivo diretto* con la terra come mezzo per un'efficace messa a terra per RF � See: Errante's capacitive earth grounding system)** o pu� essere collegato a ci� che � in realt� equivalente, ovvero un'area capacitiva vicina al suolo (Fig. 4B).

E' ora perfettamente noto che un condensatore, di capacit� adeguata, non impedisce il passaggio delle oscillazioni ad alta frequenza e quindi in questi casi la terra � ai fini pratici collegata alle antenne.

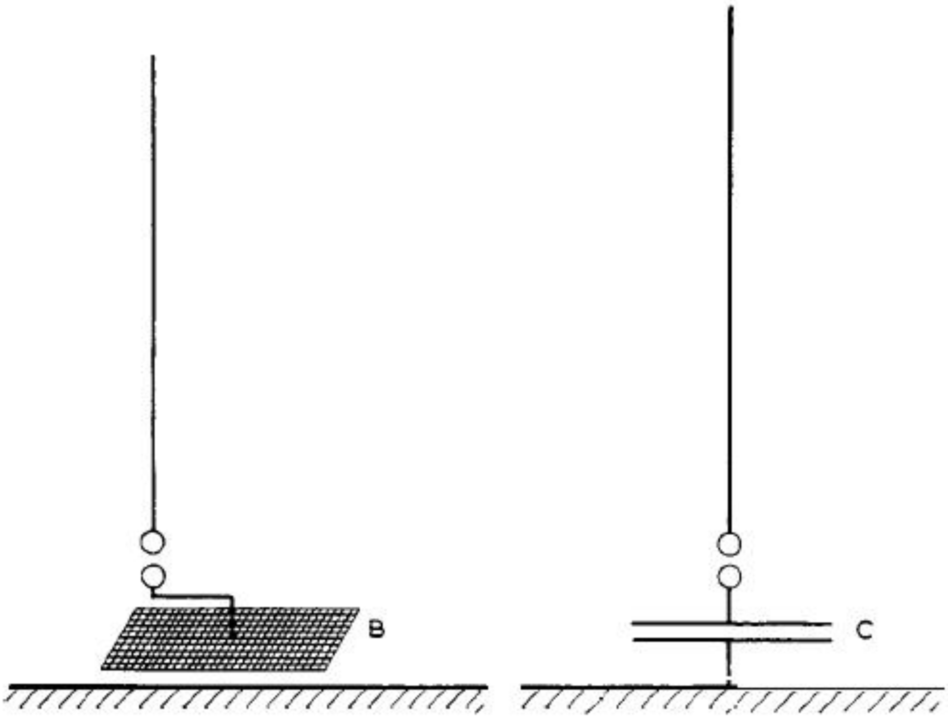

Fig. 4.

Dopo numerose prove e dimostrazioni in Italia e in Inghilterra su distanze variabili fino a 40 miglia, nel marzo 1989 fu effettuato il collegamento attraverso il Canale della Manica tra Inghilterra e Francia[5] (Figura 5).

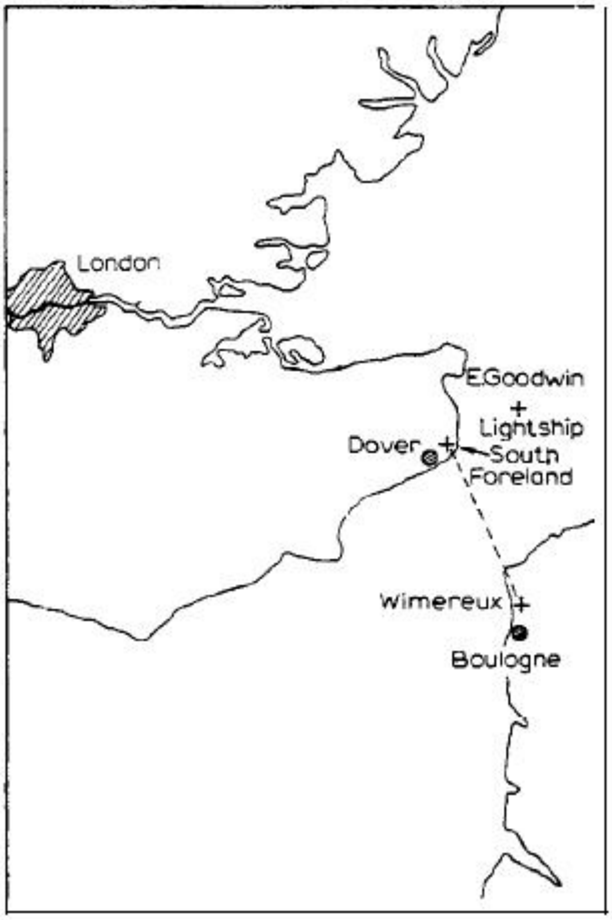

Fig. 5.

Dall'inizio del 1898 ho praticamente abbandonato il sistema di collegamenti di Figura 2 e invece di collegare il coherer o detector direttamente all'antenna e alla terra li ho collegati ai capi del secondario di un idoneo trasformatore di oscillazioni contenente un condensatore e sintonizzato sul periodo delle onde elettriche ricevute. Il primario di questo trasformatore venne collegato al filo sopraelevato e alla terra (Figura 6).[6]

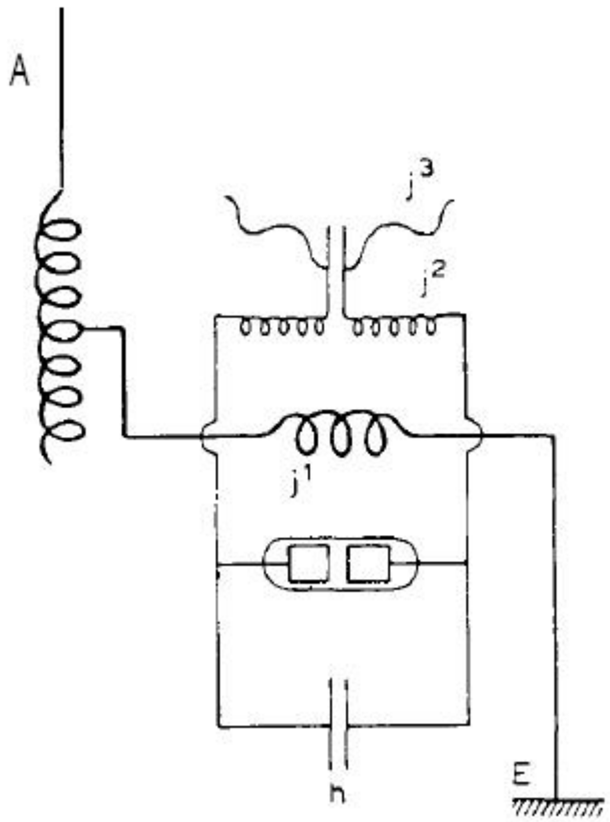

Fig. 6.

Questo arrangiamento consent� un certo grado di sintonia, cosicch� variando il periodo di oscillazione dell'antenna trasmittente si potevano inviare messaggi a un ricevitore sintonizzato senza interferire con altri ricevitori diversamente sintonizzati.[6]

Come adesso � ben noto , un trasmettitore consistente di in un filo verticale che si scarica attraverso una scintilla non � un oscillatore moto persistente, la radiazione prodotta � assai smorzata. La sua capacit� elettrica � in proporzione cos� piccola e la sua capacit� di irradiare energia � cos� grande che le oscillazioni diminuiscono o si spengono rapidamente.
Gia' nel 1989 riuscii a migliorare gli effetti di risonanza aumentando la capacit� dei fili sopraelevati posando in prossimit� dei conduttori a massa ed inserendo in serie con le antenne idonei induttori a bobina[7]. Cos� facendo la capacit� di accumulare energia dell'antenna aumentava mentre la capacit� di irradiare diminuiva, col risultato che l'energia messa in moto dalla scarica formava un treno di oscillazioni debolmente smorzate. Una modifica di questo arrangiamento, che ha dato eccellenti risultati, � illustrata in Fig. 7.

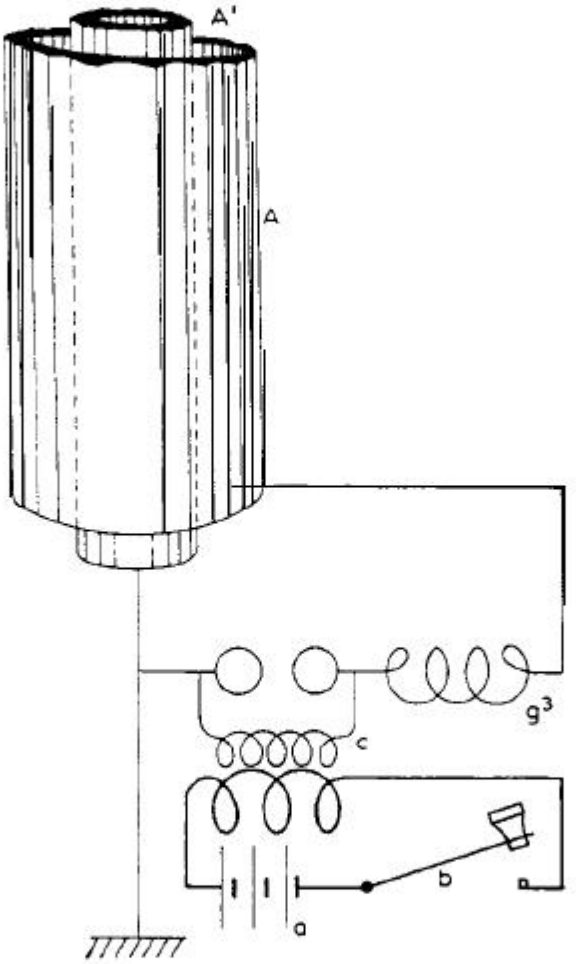

Fig. 7.

Nel� 1900 ho costruito e brevettato un sistema completo di trasmettitore e ricevitore[8] consistente nel solito tipo di area capacitiva sopraelevata e collegamento a terra, ma accoppiati induttivamente a un circuito oscillante contenente un condensatore, un induttore e un detector, con le condizioni da me ritenute essenziali ovvero il periodo dell'oscillazione elettrica del filo sopraelevato deve essere in sintonia o risonanza col circuito del condensatore e i due circuiti del ricevitore devono essere in risonanza elettrica con quelli del trasmettitore (Figura 8).

I circuiti che costituivano il circuito oscillante e il circuito radiante venivano accoppiati pi� o meno strettamente variando la distanza tra di essi.

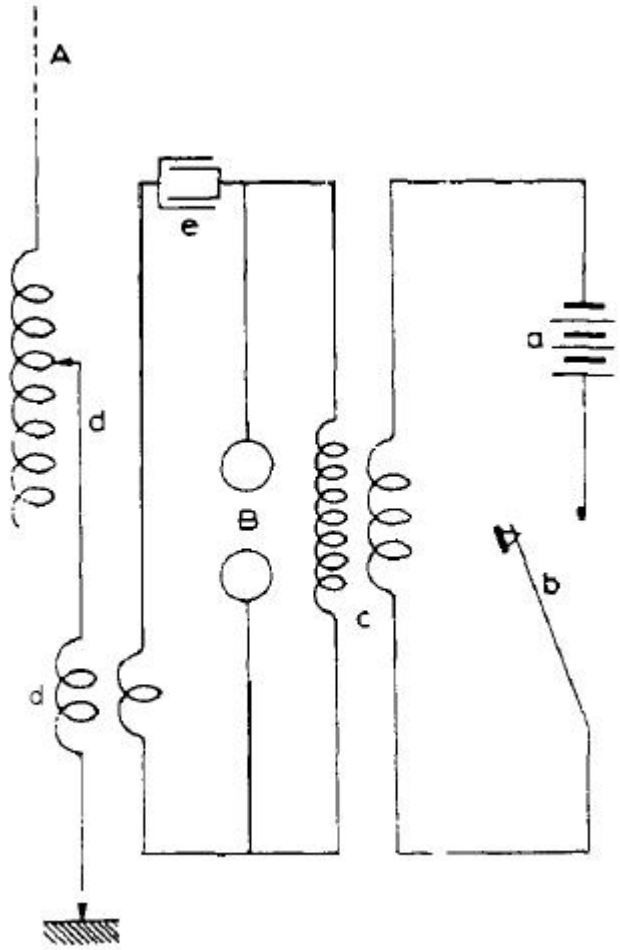

Fig. 8.

Regolando l'induttanza inserita nel filo sopraelevato e variando la capacit� del circuito a condensatore, i due circuiti venivano portati in risonanza, una condizione che, ho gi� detto, considero essenziale per ottenere un'efficiente irradiazione.
Parte del mio lavoro sull'utilizzo di circuiti a condensatore associati alle antenne radianti venne portato avanti in contemporanea con quello del Prof. Braun, senza che, tuttavia, a quel tempo nessuno di noi sapesse qualcosa del contemporaneo lavoro dell'altro.
Un ricevitore sintonizzabile � gi� stato mostrato in Figura 6 e consiste di un conduttore verticale o antenna collegato a massa attraverso il primario di un trasformatore, il cui secondario include un condensatore e un detector, essendo necessario che il circuito di antenna e quello del detector siano in risonanza elettrica tra di loro e anche in sintonia con il periodo delle onde elettriche generate dal trasmettitore. In tal modo � possibile utilizzare onde elettriche a basso smorzamento e costringere il ricevitore a integrare l'effetto di oscillazioni elettriche comparabilmente

deboli ma correttamente sincronizzate allo stesso modo in cui, in acustica, due diapason reagiscono tra di loro a brevi distanze se sintonizzati sullo stesso periodo di vibrazione.
E' anche possibile accoppiare a un conduttore trasmittente parecchi trasmettitori diversamente sintonizzati e ad un filo ricevente un certo numero di ricevitori, come illustrato in Figura 9 e 10, ogni ricevitore rispondente solo alle radiazioni del trasmettitore con cui � in risonanza[9].

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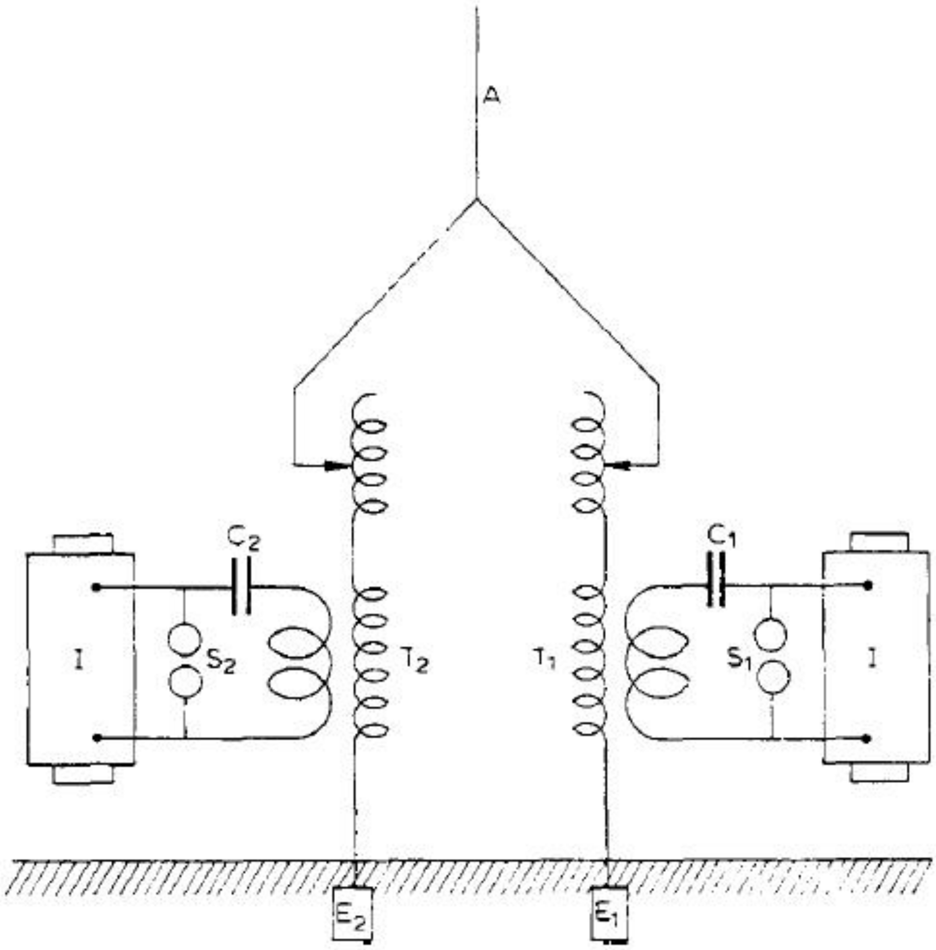

Fig. 9.

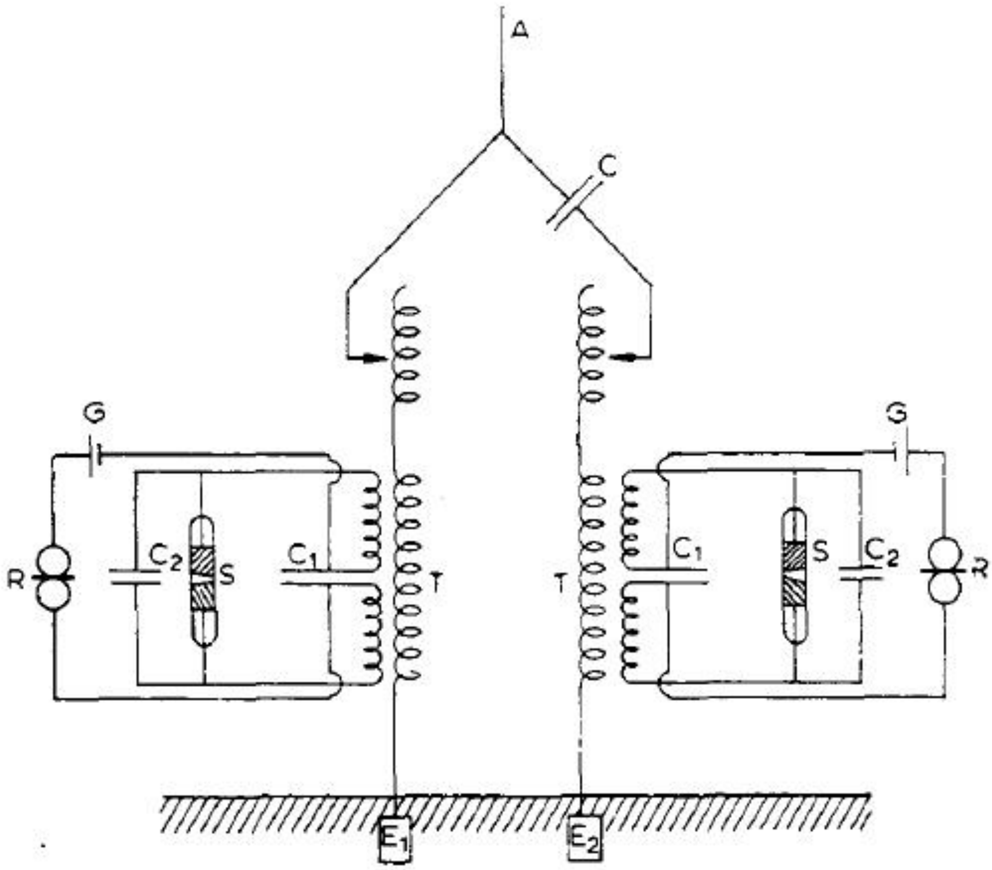

Fig. 10.

Quando (dodici anni fa) fu stabilito il primo collegamento radiotelegrafico tra Francia e Inghilterra, si discusse molto se la telegrafia senza fili fosse praticabile a distanze superiori a quella, e prevalse l'opinione che la curvatura della Terra sarebbe stata un ostacolo insormontabile per le comunicazioni a lunga distanza, allo stesso modo in cui era stata un ostacolo sulle grandi distanze per le segnalazioni luminose.
Si previdero anche difficolt� sulla possibilit� di controllare la grande quantit� di energia che sembrava essere necessaria per coprire lunghe distanze.
Come spesso accade per i lavori pioneristici si ripet� puntualmente nel caso della radiotelegrafia, gli ostacoli e le difficolt� paventate si rivelarono frutto dell'immaginazione o facilmente sormontabili, ma al loro posto si manifestarono barriere inattese, e il recente lavoro � stato principalmente mirato alla risoluzione di problemi dovuti a difficolt� inattese e impreviste quando iniziarono le prove sulle lunghe distanze.
Riguardo al presunto ostacolo della curvatura terrestre, ritengo che le Cassandre che paventarono difficolt� causate dalla forma del nostro

pianeta non tennero sufficientemente conto del particolare effetto del collegamento di massa di trasmittente e ricevente, collegamento che introdusse effetti di conduzione a quel tempo sottovalutati.
Per lungo tempo i fisici sembrarono considerare la telegrafia senza fili dipendente soltanto dagli effetti della radiazione hertziana libera nello spazio e ci vollero anni prima che il probabile effetto della conducibilit� della Terra tra due stazioni venisse soddisfacentemente preso in considerazione.
Lord Rayleigh, riferendosi alla telegrafia transatlantica, nel maggio 1903 afferm�:"Il significativo successo di Marconi nelle trasmissioni sull'Atlantico suggerisce un effetto di piegatura o diffrazione delle onde attorno alla Terra pi� deciso di quello che ci aspettava, e ci� assegna un grande interesse al problema teorico[10].

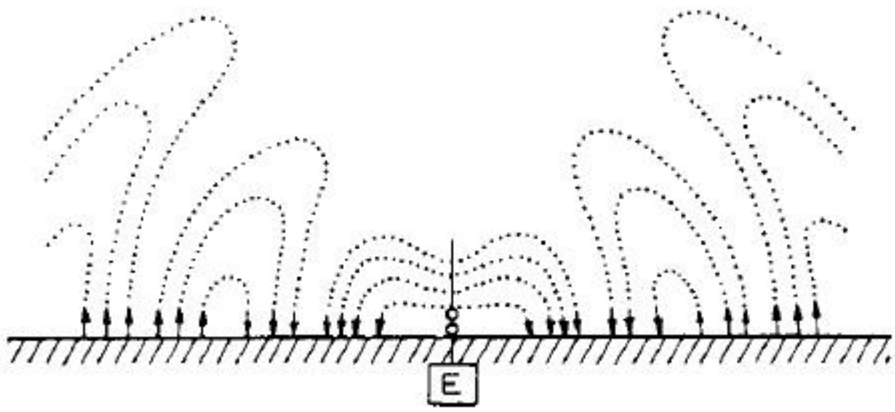

Fig. 11.

Il Prof. J.A. Fleming, nel suo libro *I principi della della Telegrafia a Onde Elettriche*[11], riporta diagrammi che mostrano ci� che ora si pensa essere la rappresentazione schematica del distacco di semi-anelli di tensione elettrica da un semplice filo verticale (Figura 11). Come vedremo, queste onde non si propagano allo stesso modo della radiazione libera di un classico oscillatore hertziano, ma scivolano sulla superficie terrestre.
Il Prof. Fleming afferma ancora nel sopra citato lavoro: *La visione che abbiamo � che le estremit� dei semi-anelli di forza elettromotrice, che terminano perpendicolarmente alla terra, non si possono muovere se non ci sono movimenti di elettroni nella terra corrispondenti al moto ondoso sopra di essa. Dal punto di vista della teoria elettronica dell'elettricit�, ogni linea di forza elettrica nell'etere deve essere o una*

linea chiusa o le sue estremit� devono terminare su elettroni di segno opposto. Se l'estremit� di una linea di tensione termina sulla Terra e si muove, ci dev'essere una scambio di elettroni interatomico, o movimenti di elettroni in essa. Abbiamo motivi sufficienti per concludere che le sostanze che chiamiamo conduttori sono quelle in cui avvengono liberi movimenti di elettroni. Quindi il movimento dei semi-anelli di forza elettromotrice uscenti da un oscillatore messa a terra o da un'antenna di Marconi � rallentato dalla cattiva conducibilit� della superficie terrestre e facilitato da una superficie di buon elettrolita come l'acqua di mare".

Il Prof. Zenneck[12] ha esaminato attentamente l'effetto di antenne ricetrasmittenti messe a terra e ha tentato di dimostrare matematicamente che quando le linee di forza elettromotrice, costituenti un fronte d'onda, passano sopra una superficie di bassa capacit� induttiva specifica, come la Terra, si inclinano in avanti, essendo la loro estremit� inferiore ritardata dalla resistenza del conduttore a cui sono collegati.

Pare quindi ben assodato che la telegrafia senza fili, come praticata ai giorni nostri, dipende dipende per funzionare su lunghe distanze dalla conducibilit� terrestre e che la differenza in conducibilit� tra terra e mare � sufficiente a spiegare la maggior distanza raggiungibile con la stessa quantit� di energia nel comunicare attraverso il mare piuttosto che sulla terra.

Ho condotto alcune prove tra una stazione costiera e una nave a Poole, in Inghilterra, nel 1902 allo scopo di ottenere dati su questo argomento ed ho osservato che a distanza uguale c'� sempre una percettibile diminuzione nell'energia delle onde ricevute quando la nave � in una posizione tale da far s� che si interponga tra essa e la stazione terrestre una striscia di spiaggia larga circa un chilometro.

Ritengo quindi che ci fosse un certo fondamento per l'asserzione cos� spesso criticata che feci in occasione del mio primo brevetto inglese del 2 giugno 1896 sull'effetto che quando trasmettevo attraverso la terra o l'acqua io collegavo un capo del trasmettitore e un capo del ricevitore a massa.

Nel gennaio 1901 furono eseguiti con successo esperimenti[13] tra due punti della costa meridionale dell'Inghilterra distanti 186 miglia, ovvero St. Catherine's Point (Isola di Wight) e The Lizard (Cornovaglia) (Figura 12).

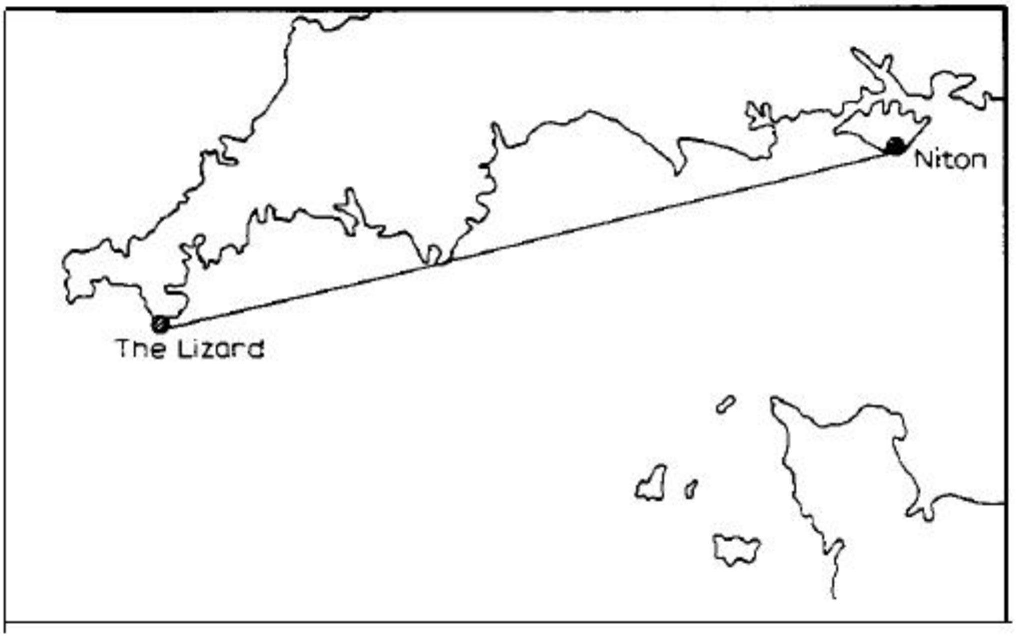

Fig. 12.

L'altezza totale di queste stazioni sul livello del mare non eccedeva i 100 metri, mentre per azzerare la curvatura terrestre sarebbe stata necessaria un'altezza di oltre 1.600 metri in entrambi i punti.
I risultati ottenuti da questi test, che all'epoca costituirono un record di distanza, sembravano indicare che le onde elettriche prodotte nel modo che ho adottato erano le pi� adatte a girare attorno alla curvatura terrestre e che perci� anche a grandi distanze, come quelle che dividono l'America dall'Europa, il fattore curvatura terrestre non costituiva un ostacolo insormontabile all'estensione della telegrafia attraverso lo spazio. La consapevolezza che la curvatura della Terra non avrebbe impedito la propagazione delle onde e il successo ottenuto dai metodi sintonici nel prevenire mutue interferenze mi portarono a decidere di tentare la prova se fosse possibile rivelare onde elettriche su una distanza di 4.000 km che, se positiva, avrebbe dimostrato immediatamente la possibilit� di telegrafare senza fili tra Europa e America.
A mio parere l'esperimento era di grande rilievo dal punto di vista scientifico ed ero certo che la scoperta della possibilit� di trasmettere onde elettriche da un capo all'altro dell'Oceano Atlantico e l'esatta conoscenza delle effettive condizioni in cui la telegrafia su tali distanze poteva essere implementata avrebbero migliorato assai la nostra comprensione dei fenomeni connessi alla trasmissione senza fili.
Il trasmettitore installato a Polhu, sulla� costa della Cornovaglia, era simile in principio a quello che ho gi� descritto, ma su una scala molto

pi� grande di quella fino ad allora mai tentata[14].
La potenza del generatore era di circa 25 kW.
Numerose difficolt� si incontrarono nel produrre e controllare per la prima volta oscillazioni elettriche di tal potenza. Fonirono un valido aiuto il Prof. J.A. Fleming, il Sig. R.N. Vyvyan ed il Sig. W.S. Entwistle.
Le mie prove precedenti mi avevano convinto che per ampliare la distanza di collegamento non era solo sufficiente aumentare la potenza dell'energia elettrica del trasmettitore ma era anche necessario aumentare l'area o l'altezza dei conduttori sopraelevati trasmittente e ricevente.
Sarebbe stato troppo costoso impiegare fili verticali di altezza elevata, cos� decisi di aumentare il loro numero e l capacit�, ci� che sembrava pi� probabile rendere possibile l'utilizzo efficiente di grandi quantit� di energia.

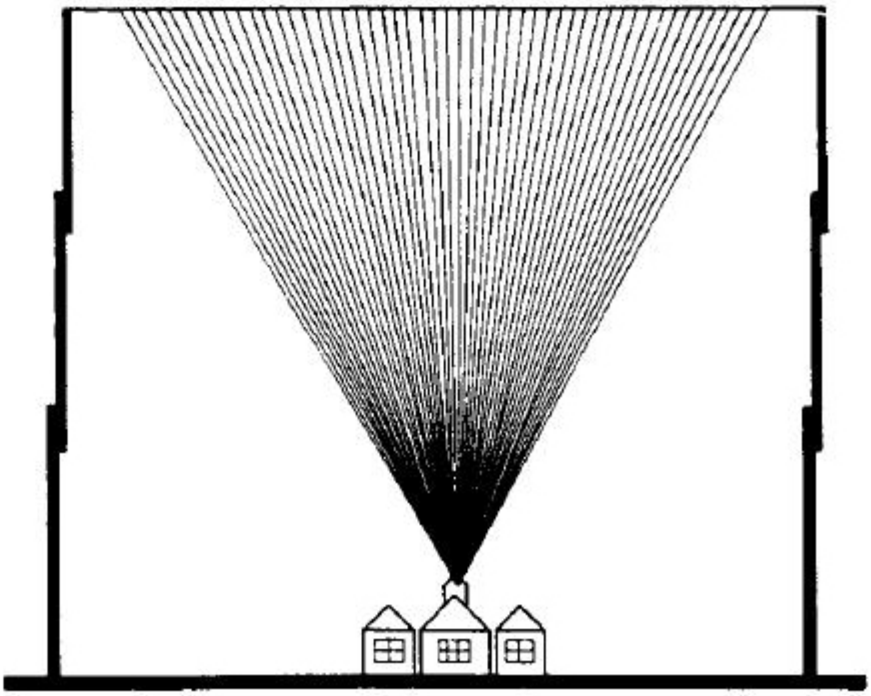

Fig. 13.

La disposizione delle antenne trasmittenti che ho usato a Poldu � visibile in Figura 13 e consisteva di una disposizione a ventaglio di fili sostenuti da isolatori tra pali alti solo 48 m e distanti 60. Questi fili convergevano sul lato inferiore ed erano collegati all'apparato trasmittente ubicato nell'edificio.
Per la prova venne allestita una potente stazione a Cape Cod, vicino a New York, ma il completamento di essa sub� un ritardo a causa di una

tempesta che distrusse pali e antenne. Decisi allora di effettuare gli esperimenti con una stazione ricevente provvisoria allestita in Newfoundland, ove mi recai con due assistenti verso la fine di novembre 1901. Le prove iniziarono nei primi di Dicembre 1901 ed il 12 di quel mese i segnali trasmessi dall'Inghilterra vennero ricevuti forti e chiari dalla stazione di St. Johns in Newfoundland.

Test di conferma furono eseguiti nel febbraio 1902 tra Poldhu e la stazione ricevente sulla nave a vapore Philadelfia della American Line. A bordo della nave furono ricevuti messaggi leggibili mediante un registratore fino alla distanza di 1,551 miglia e lettere di prova fino a 2.099 miglia da Poldhu (Fig. 14).

Le registrazioni su nastro a bordo del Philadelfia a varie distanze erano chiare e distinte, come si pu� vedere dai campioni in mostra.

Questi risultati, sia pure ottenuti con apparati imperfetti, bastarono a convincere me e i miei collaboratori che mediante stazioni permanenti e l'impiego di suficiente potenza sarebbe stato possibile inviare messaggi attraverso l'Atlantico allo stesso modo in cui vengono inviati a distanze molto pi� brevi.

Le prove in Newfoundland non poterono continuare per l'ostilit� di una Compagnia di telegrafia via cavo, che reclamava tutti i diritti per la telegrafia, sia essa con o senza fili, in quella colonia.

Un risultato di interesse scientifico che ebbi modo di osservare durante le prove con la Filadelfia e che � il fattore pi� importante per la radiotelegrafia a lunga distanza fu l'effetto negativo molto marcato della luce del giorno sulla propagazione delle onde elettriche, la portata notturna essendo pi� del doppio di quella raggiungibile durante il giorno[15]. Non credo che questo effetto sia ancora stato soddisfacentemente studiato o spiegato. All'epoca dei test ritenevo che potesse essere dovuto alla perdita di energia del trasmettitore causata dalla diselettrificazione dei conduttori trasmittenti sopraelevati altamente carichi sotto l'influenza della luce solare.

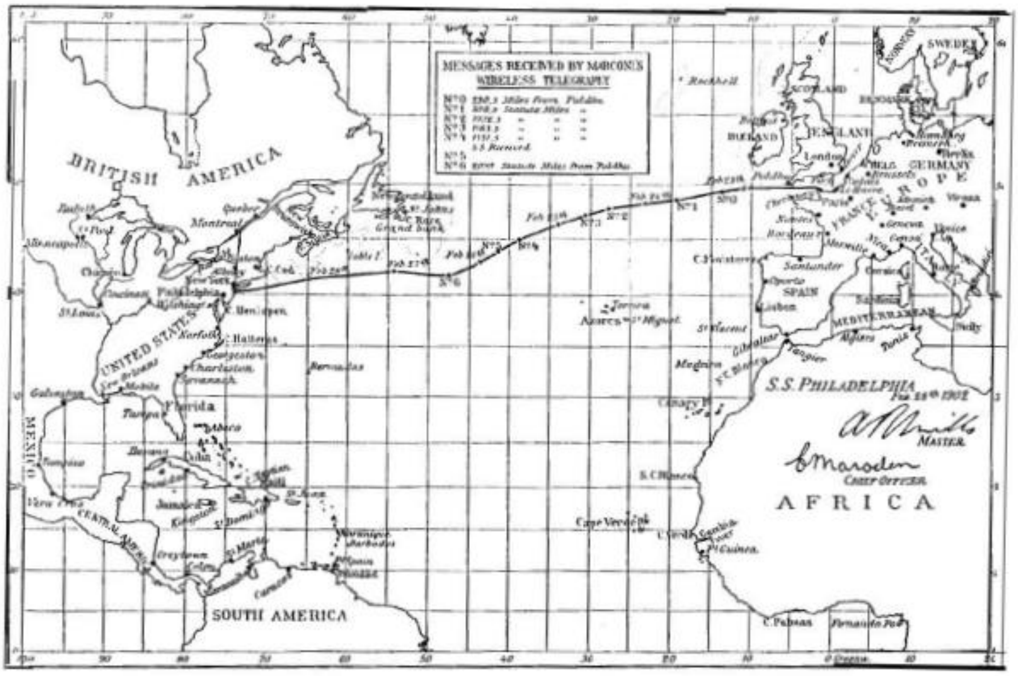

Fig. 14.

Oggi sono incline a ritenere che l'assorbimento delle onde elettriche durante il giorno sia dovuto agli elettroni propagati nello spazio dal Sole e che se questi ricadono in continuazione sulla Terra come una doccia, in accordo con l'ipotesi del Prof. Arrhenius, allora quella porzione dell'atmosfera terrestre che � rivolta al Sole conterr� pi� elettroni della parte non rivolta al Sole, e quindi sar� meno trasparente per le onde elettriche.
Sir J.J. Thompson ha illustrato in un interessante articolo su *Philosophical Magazine* che se vengono distribuiti degli elettroni nello spazio attraversato da onde elettriche, queste tenderanno a muovere gli elettroni nella direzione dell'onda, e perci� questi assorbiranno una parte dell'energia dell'onda. Quindi, come il prof. Fleming ha fatto notare nelle sue Cantor Conferences tenute alla Societ� delle Arti, un mezzo in cui sono sparsi elettroni o ioni agisce come un mezzo leggermente opaco alle onde elettriche lunghe[16].
Apparentemente la lunghezza d'onda e l'ampiezza dell'oscillazione elettrica ha molto a che fare con questo interessante fenomeno,, le onde lunghe e di piccola ampiezza essendo meno soggette all'effetto luce del giorno delle onde corte e di grande ampiezza.
Secondo il Prof. Fleming[17] l'effetto luce del giorno dovrebbe essere pi� marcato per le onde lunghe, ma non mi risulta. Infatti in alcuni recentissimi esperimenti utilizzanti onde di circa 8.000 m l'energia

ricevuta di giorno era di solito maggiore di quella notturna.
Resta il fatto, comunque, che per onde comparabilmente corte, come quelle usate per comunicazione tra le navi, luce solare forte e cielo blu, sebbene trasparenti alla luce, agiscono come nebbia per queste onde. Per cui le condizioni meteorologiche prevalenti in Inghilterra, e forse in questo Paese, sono solitamente adatte alla telegrafia senza fili. Nel 1902 eseguii alcune prove ulteriori tra la stazione di Poldhu e una stazione ricevente montata sull'incrociatore italiano Carlo Alberto, gentilmente messomi a disposizione da Sua Maest� il Re d'Italia[18] (vedi Figura 15).

Fig. 15.

Durante questi esperimenti fu osservato lo strano fenomeno che, anche usando onde corte 1.000 piedi, frapponendosi catene di montagne, come le Alpi o i Pirenei, durante la notte non ci fu alcuna riduzione della distanza a cui era possibile comunicare. Durante il giorno, a meno di usare onde molto pi� lunghe e molta pi� potenza, il frapporsi delle montagne riduceva la portata apparente del trasmettitore.(See Fig. 15.[18])
Messaggi e notizie di considerevole lunghezza furono ricevuti da Poldhu nella posizione segnata sulla mappa, che � una copia, in scala ridotta, di quella allegata al rapporto ufficiale sugli esperimenti (Figura 16).
Con l'attivo incoraggiamento e assistenza finanziaria del Governo

canadese, venne costruita a Glace Bay, Nuova Scozia, una stazione ad alta potenza, affinch� potessi continuare le mie prove a lunga distanza con l'obiettivo di stabilire comunicazioni radiotelegrafiche su base commerciale tra Inghilterra e America[19].
Il 16 dicembre 1902 vennero scambiati i primi messaggi ufficiali attraverso l'Atlantico, tra le stazioni di Poldhu e Glace Bay (Figura 17 e 18). Ulteriori prove furono eseguite poco tempo dopo con un'altra stazione a lunga distanza a Cape Cod negli USA, e in circostanze favorevoli si riusc� a trasmettere messaggi a Poldhu distante 3.000 miglia con un assorbimento energetico di soli 10 kW. Nella primavera del 1903 fu tentata la trasmissione di messaggi d'agenzia via radiotelegrafo tra America ed Europa, e per un po' il Times di Londra pubblic�, tra l'ultima parte di marzo e la prima di aprile di quell'anno, notizie dal corrispondente di New York inviate attraverso l'Atlantico senza l'ausilio di cavi. Un rottura dell'isolante nell'apparato di Glace Bay, tuttavia, interruppe il servizio e disgraziatamente altri difetti resero incerta e inaffidabile la trasmissione dei messaggi.
Come risultato dei dati e dell'esperienza acquisiti con questi ed altri esperimenti condotti per conto del Governo britannico tra Inghilterra e Gibilterra, fui in grado di costruire una nuova stazione a Clifden in Irlanda e ampliare quella di Glace Bay in Canada, mettendomi in grado di iniziare, nell'ottobre 1907, un servizio commerciale di comunicazioni transatlantico tra Inghilterra e Canada.
Sebbene le stazioni a Clifden e Glace Bay fossero state messe in funzione prima di venire completate, la comunicazione transatlantica radiotelegrafica non ebbe mai a soffrire alcuna seria interruzione per quasi due anni finch�, a causa di un incendio a Glace Bay in autunno, dovette essere sospesa per tre o quattro� mesi.

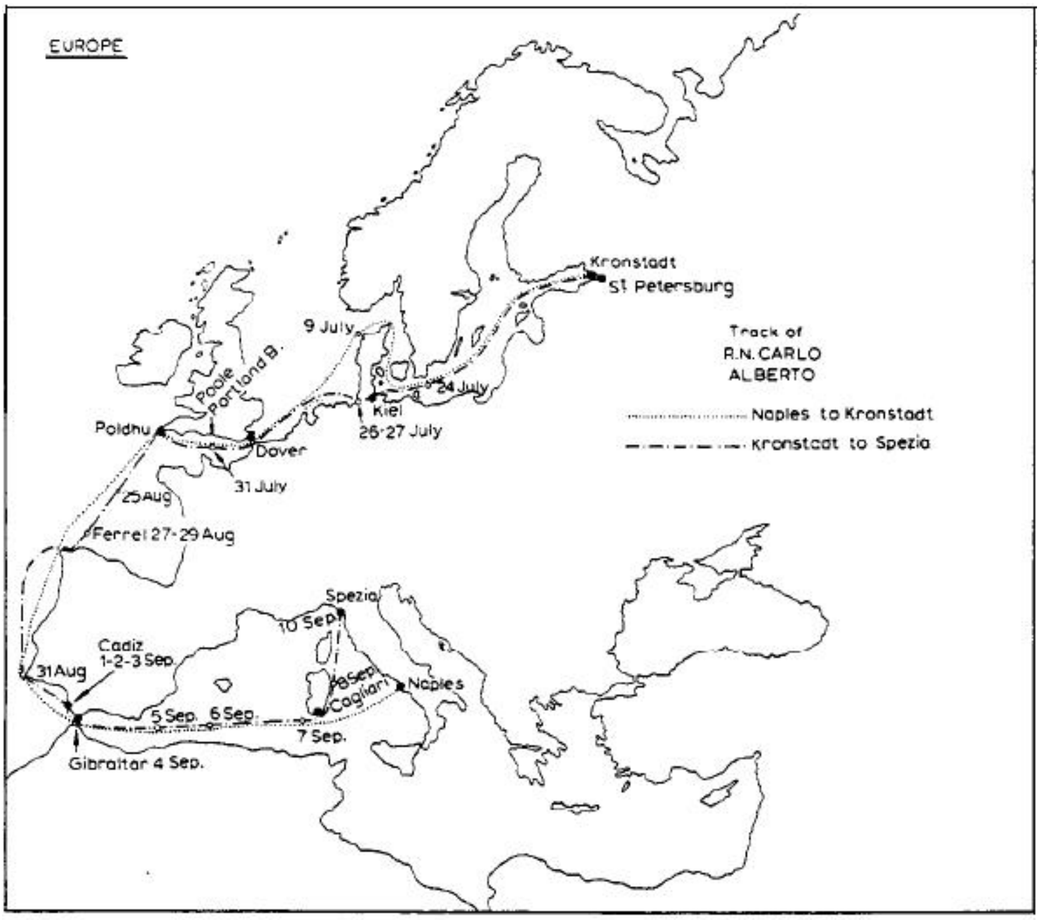
EUROPE
Kronstadt
St Petersburg
Track of
R.N. CARLO
ALBERTO
Naples to Kronstadt
Kronstadt to Spezia
9 July
Poole
Portland B.
Kiel
26-27 July
24 July
Poldhu
Dover
31 July
25 Aug
Ferrel 27-29 Aug
Spezia
10 Sep.
Cadiz
1-2-3 Sep.
31 Aug
8 Sep.
Cagliari
Naples
5 Sep.
6 Sep.
7 Sep.
Gibraltar 4 Sep.

Fig. 16.

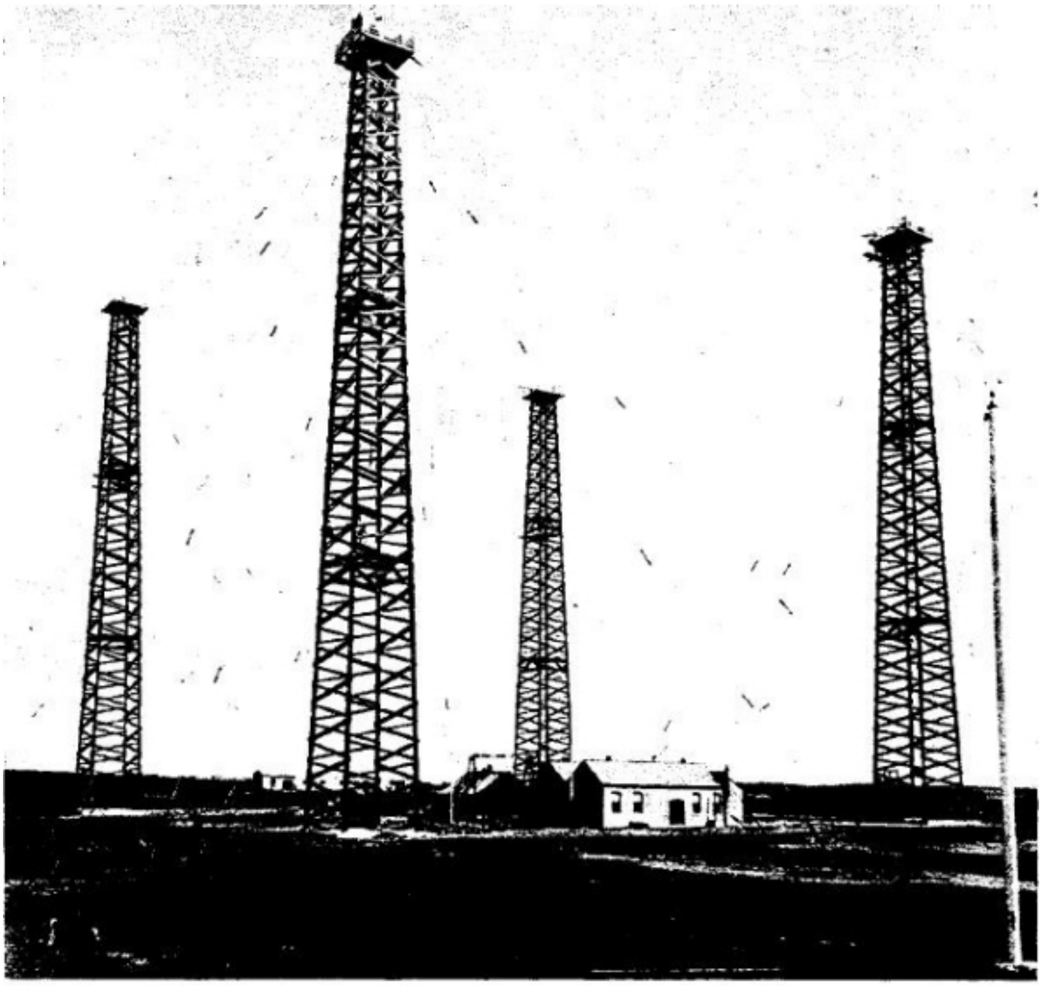

Fig. 17.

Questa sospensione, tuttavia, ha avuto il suo lato positivo, dandomi l'opportunit� di installare macchinari pi� efficienti e aggiornati. La disposizione dei conduttori sopraelevati o aerei con cui ha fatto le prove[20] a lunga distanza, sono visibili nelle Figure 19, 20 e 21.
L'antenna illustrata in Figura 21 consisteva di una porzione quasi verticale nel mezzo, alta 220 piedi, supportata da quattro torri e collegata all'estremit� di fili orizzontali, in numero di 200 e lunghi 1.000 piedi, che si estendevano radialmente tutto intorno� e sostenuti ad una altezza di 180 piedi da terra da un cerchio interno di 8 ed esterno di 16 pali.
Il periodo naturale di oscillazione di questo sistema d'aereo risult� una lunghezza d'onda di 12.000 piedi.
In prove fatte con questa disposizione nel 1905 e con lunghezza d'onda di 12.000 piedi, i segnali, sebbene deboli, poterono essere ricevuti attraverso l'Atlantico altrettanto bene di giorno come di notte.
Il sistema di antenne che ho infine adottato per le stazioni a lunga distanza in Inghilterra e in Canada � illustrato in Figura 22. questa disposizione non solo consente di irradiare e ricevere efficientemente

onde di qualsiasi lunghezza si desideri, ma tende anche a confinare la porzione principale della radiazione in una determinata direzione. La limitazione della trasmissione in un'unica direzione non � poi cos� netta, ma i risultati ottenuti con questo tipo di aereo non sono mai troppo utili.
Molti suggerimenti sui metodi per limitare la direzione radiante sono pervenuti da vari ricercatori, menziono il Prof. F. Braun, il Prof. Artom e i Signori Belhni e Tosi.
In una tesi letta al cospetto della Royal Society[21] di Londra nel marzo 1906 ho mostrato come fosse possibile con aerei orizzontali confinare la radiazione emessa principalmente nella direzione del piano verticale, puntando fuori dal loro polo a massa. Nello stesso modo si pu� localizzare la direzione di una stazione trasmittente.

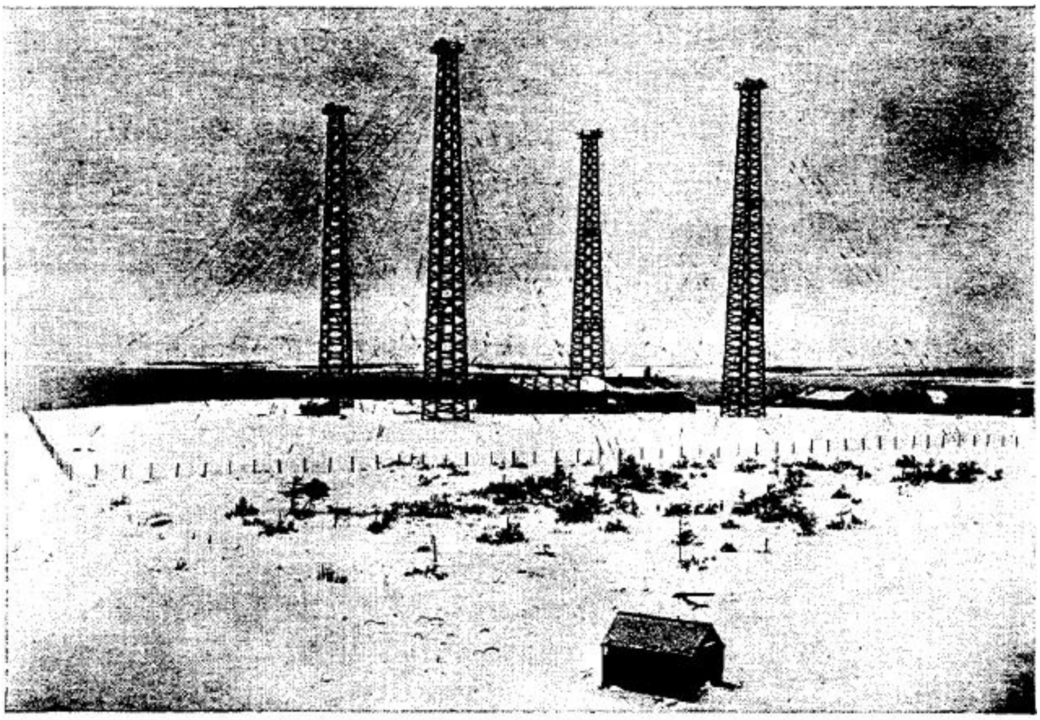

Fig. 18.

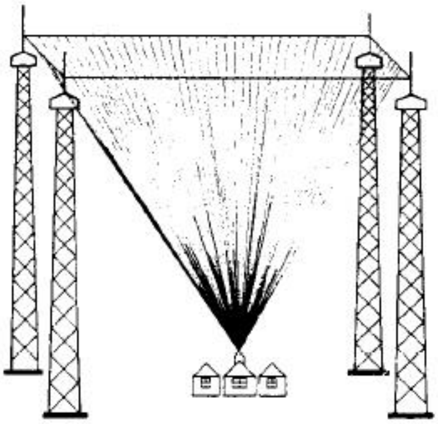

Fig. 19.

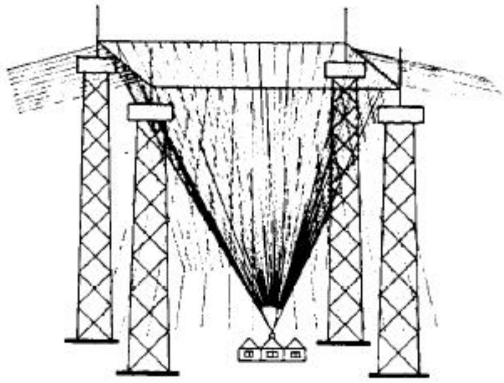

Fig. 20.

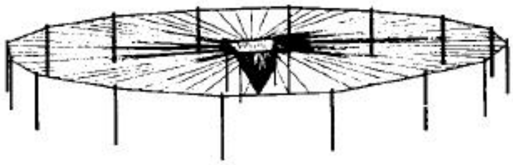

Fig. 21.

Fig. 22.

I circuiti trasmittenti nelle stazioni a lunga distanza sono disposti in accordo con un sistema relativamente recente per produrre oscillazioni continue o leggrmente smorzate, su cui ho riferito in una conferenza alla Royal Institution of Great Britain il 13 marzo 1908.
Un disco metallico isolato A (vedi Figura 23) viene fatto ruotare a velocit� elevata per mezzo di un motore elettrico o di una turbina a vapore. Adiacente a questo disco, che chiamer� "disco mediano", sono collocati altri due dischi C' e C'', anch'essi in rotazione, chiamati "dischi polari". Questi dischi polari hanno la periferia molto vicina alla superficie del disco mediano. I due dischi polari sono collegati mediante contatti striscianti al polo esterno di due condensatori K, messi in serie, collegati a loro volta mediante spazzole ai terminali di un generatore che non � altro che un generatore di corrente continua ad alta tensione.
Sul disco mediano � collocata un'apposita spazzola e tra questa ed il punto di mezzo dei due condensatori � inserito un circuito oscillante, costituito da un condensatore E in serie ad un'induttanza, che a sua volta

� collegata per induzione all'antenna radiante. Il circuito probabilmente funziona cos�:

Il generatore carica il doppio condensatore, rendendo il potenziale dei dischi diciamo C' positivo e C'' negativo. Il potenziale, se abbastanza elevato, causer� il passaggio di una scarica ai capi della fessura, diciamo tra C' e A. Questa carica il condensatore E attraverso l'induttanza F e da' inizio alle oscillazioni nel circuito. Quando la carica di F torna indietro salta da A a C'', il cui potenziale � di segno opposto ad A. Le propriet� del dielettrico tra C' ed A nel frattempo saranno state ripristinate grazie al rapido movimento del disco, che sposta via l'aria ionizzata.

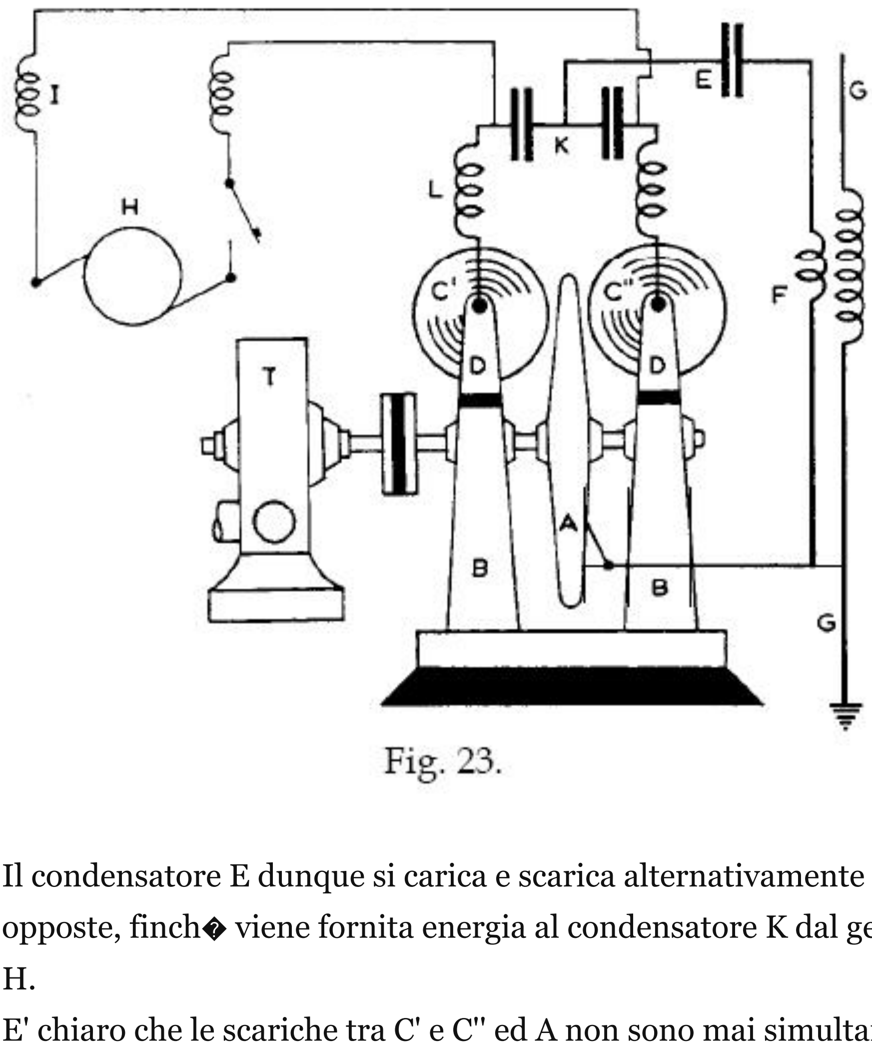

Fig. 23.

Il condensatore E dunque si carica e scarica alternativamente in direzioni opposte, finch� viene fornita energia al condensatore K dal generatore H.
E' chiaro che le scariche tra C' e C'' ed A non sono mai simultanee altrimenti l'elettrodo centrale non sarebbe alternativamente positivo e negativo.
I migliori risultati, tuttavia, sono stati ottenuti con la disposizione mostrata in Figura 24, ove la superficie attiva del disco mediano non � liscia ma consiste di un certo numero di protuberanze di rame regolarmente spaziate, alla cui estremit� hanno luogo le scariche a

intervalli regolari. Ho riscontrato che con questa configurazione ogni treno di oscillazioni pu� avere un decremento del 2% al massimo.

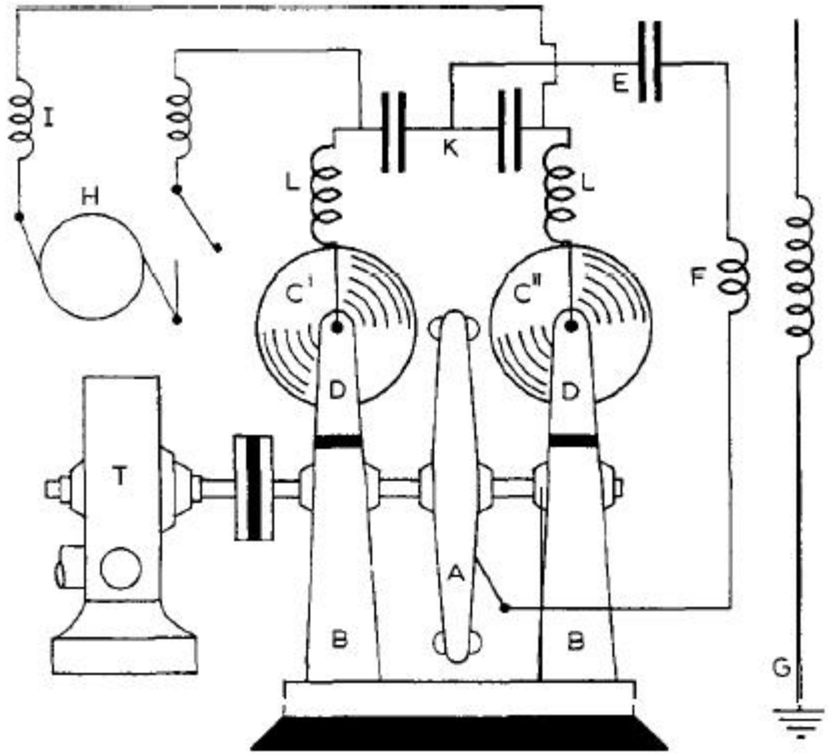

Fig. 24.

In questo modo � anche possibile far s� che i gruppi di oscillazioni irradiate riproducano una nota musicale forte e chiara in un ricevitore, facilitando il riconoscimento di segnali provenenti d stazioni trasmittenti e i rumori dovuti alle scariche elettriche atmosferiche. Con questo metodo si pu� anche ottenere una risonanza molto efficiente in ricevitori appropriatamente progettati.

Per quel che concerne i ricevitori impiegati, sono intervenuti importanti cambiamenti. Fino a pochi anni fa, la maggior parte della telegrafia con onde elettriche impiegava una forma o l'altra di coherer o contatto variabile che richiedeva scuotimento o autoripristino. Al giorno d'oggi, tuttavia, posso affermare che tutte le stazioni controllate dalla mia Compagnia utilizzano esclusivamente il mio ricevitore magnetico[22] (Figura 25). Il ricevitore si basa sulla diminuzione di isteresi magnetica che ha luogo nel ferro quando in certe condizioni questo metallo � soggetto agli effetti di onde elettromagnetiche ad alta frequenza.

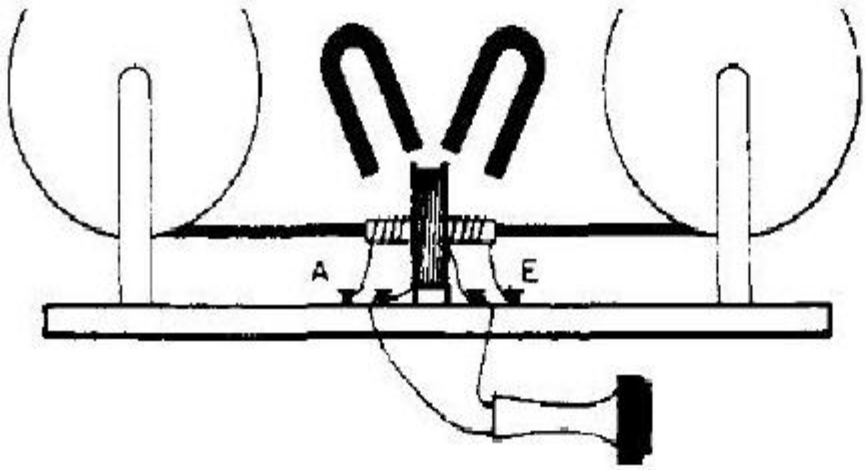

Fig. 25.

Recentemente si � riusciti ad aumentare la sensibilit� di questi ricevitori e ad impiegarli in collegamento con rel� ad alta velocit�, cos� da poterbregistrare messaggi ad alta velocit�.
Un fatto notevole, sconosciuto a molti, riguardo ai trasmettitori � che nessuna delle disposizioni impieganti condensatori supera in efficienzal'aereo piano sopraelevato o il filo verticale che si scarica a massa attraverso uno scaricatore, come usato nei miei primi esperimenti (Figura 2 e 3).
Recentemente sono anche riuscito a dimostrare l'affermazione fatta dal Prof. Fleming� nel suo libro "The Principles of Electric Wave Telegraphy", 1906, pagina 555, secondo cui con una potenza di 8 W in antenna era possibile comunicare a distanze di oltre 100 miglia. Ho anche constatato che con questo metodo, usando grandi antenne, � possibile inviare segnali a 2.000 miglia sopra l'Atlantico con una minore spesa energetica di ogni altro metodo a me noto.
L'unico neo di questa configurazione � che, a meno di usare antenne enormi, la quantit� di energia che pu� essere utilizzata � limitata dal potenziale oltre il quale le scariche delle spazzole e la resistenza dello scaricatore cominciano a dissipare una grossa parte dell'energia.
Mediante scaricatori in aria compressa e l'aggiunta di bobine di induttanza poste tra antenna e massa, si pu� far s� che il sistema irradi onde purissime e leggermente smorzate, assai convenienti per la sintonia fine.
Riguardo al funzionamento generale della telegrafia senza fili, l'ampio

spettro di applicazioni del sistema ed il numero di stazioni hanno grandemente facilitato l'osservazione di fenomeni non facilmente spiegabili. Cos� si � osservato che una stazione navale ordinaria, utilizzante circa 500 W di energia elettrica, la cui portata di� norma non supera le 200� miglia occasionalmente trasmetta messaggi a oltre 1.200 miglia. Spesso accade che una nave non riesca a collegare una stazione vicina, ma riesca a comunicare facilmente con una pi� lontana. In molte occasioni nello scorso inverno, la motonave Caronia della Cunard Line, con a bordo una stazione da 500 W, in navigazione nel Mediterraneo lungo le coste della Sicilia non riuscisse a collegare stazioni italiane e non avesse invece problemi a trasmettere e ricevere con le coste inglesi e olandesi, sebbene queste ultime distassero pi� di 1.000 miglia e ci fosse una gran parte del Continente europeo e la catena delle Alpi tra queste e la nave.
Sebbene adesso siano usate stazioni ad alta potenza per le comunicazioni transatlantiche, e i messaggi possano essere inviati sia di giorno sia di notte, ci sono brevi periodi del giorno in cui le trasmissioni dall'America all'Inghilterra e viceversa sono difficoltose. Cos� alla mattina e alla sera, quando in conseguenza della latitudine la luce o il buio si estendono solo a parte del tragitto sopra l'oceano, i segnali ricevuti sono deboli o inesistenti. Sembra come se le onde elettriche passando dal buio alla luce, e viceversa, siano riflesse in modo da deviare dal loro abituale cammino. E' probabile che queste difficolt� non si presentino telegrafando a distanze uguali da nord a sud sullo stesso meridiano, perch� in tal caso il passaggio dalla luce al buio avviene in contemporanea sull'intero percorso tra i due punti.
Un altro risultato curioso, sul quale centinaia di osservazioni continuate per anni non lasciano adito a dubbi, � che regolarmente, per brevi periodi, al tramonto e al sorgere del Sole e occasionalmente in altre circostanze, un'onda pi� corta si riesce a rivelare da un capo all'altro dell'Atlantico rispetto all'onda pi� lunga normalmente impiegata. Cos� a Clifden e Glace Bay quando si trasmette su un normale circuito accoppiato in modo da irradiare simultaneamente due onde, una da 12.500 piedi e l'altra di 14.700 piedi, sebbene l'onda pi� lunga sia quella normalmente ricevuta all'altro lato dell'oceano, regolarmente, circa tre ore dopo il tramonto a Clifden e tre ore prima del suo sorgere a Glace Bay, solo l'onda pi� corta viene ricevuta con intensit� decente, per un periodo di circa un'ora.

Questo effetto era cos� regolare che gli operatori sintonizzavano i loro ricevitori sull'onda pi� corta nelle ore menzionate, come parte della normale routine.
Riguardo all'utilit� della telegrafia senza fili non c'� dubbio che il suo utilizzo sia diventato una necessit� per la sicurezza della navigazione -tutti i principali piroscafi e navi da guerra son o gi� equipaggiati- mentre la sua estensione a natanti di minore importanza e solo questione di tempo, considerata la sua importanza dal punto di vista dell'assistenza in caso di pericolo. Sono in aumento anche le sue applicazioni come mezzo di comunicazione tra isole remote e anche per normali scopi di comunicazione telegrafica tra villaggi e citt�, e specialmente nelle colonie e nei paesi in via di sviluppo.
Al di l� dell'importanza per le navigazione, ritengo che la telegrafia senza fili sia destinata ad un pari livello di importanza nel fornire comunicazioni efficienti e poco costose tra luoghi distanti nel mondo e nel collegare i Paesi europei con le loro colonie e con l'America. Prova ne sia che, sto attualmente costruendo una potentissima stazione per il Governo italiano a Coltano, allo scopo di comunicare con le colonie italiane in Africa Orientale e in Sud America.
Qualsiasi difetto oggi possa avere, non c'� dubbio che la telegrafia senza fili, anche su lunghe distanze, rimarr� e continuer� a fare progressi.
Se sar� possibile trasmettere onde attorno al mondo, si trover� che l'energia elettrica viaggiante sar� concentrata agli antipodi del trasmettitore. In questo modo un giorno sar� possibile inviare messaggi in quelle terre distanti con pochissimo dispendio di energia elettrica, e quindi in modo molto economico.
Ma sto abbandonando il terreno dei fatti per inoltrarmi in quello della speculazione che, comunque, con le conoscenze che a poco a poco abbiamo guadagnato sull'argomento, promette risultati utili e istruttivi.
br>
Non avendo la fortuna di parlare lo svedese, ho ritenuto fosse meglio, sebbene io sia italiano, usare lo strumento della lingua inglese per questa occasione, dal momento che so' che l'inglese � solitamente pi� comprensibile dell'italiano.

�

1. G. Marconi, *Brit. Patent* No. 12,039, June 2, 1896.

2. *J. Inst. Elec. Engrs. (London),* 28 (1899) 278.

3.See letter of Dr. Lodge in *The Times (London),* June 22, 1897.

4. A. Slaby, *Die Funkentelegraphie,* Verlag von Leonhardt Simion, Berlin, 1897; see also A. Slaby, "The New Telegraphy", *The Century Magazine,* 55 (1898) 867.

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6. *Brit. Patent* No. 12,326, June 1, 1898; *Brit. Patent* No. 6,982, April 1, 1899.

7. A. Blondel and G. Ferrie, "�tat actuel et Progr�s de la T�l�graphie sans Fil", read at the *Congr�s International d'�lectricit�, Paris, 1900;* see also *J. Soc. Arts,* 49 (1901) 509.

8. *Brit. Patent* No. 7,777, April 26, 1900; see also *J. Soc. Arts,* 49 (1901) 510-11.

9. See Letter of Prof. J. A. Fleming in *The Times (London),* October 4, 1900.

10. *PrOC.* Roy. Soc. (London), 72 (May 28, 1903).

11. J.A. Fleming, *The Principles of Electric Wave Telegraphy,* Longmans, Green & CO., London, 1906, p. 348.

12. J. Zenneck, *Ann. Physik,* [4], 23 (Sept. 1908) 846*; Physik. Z.,* 9 (1908) 50, 553.

13. *J. Soc. Arts,* 49 (1901) 512.

14. G. Marconi, lecture before the Royal Institution of Great Britain, June 13, 1902.

15. G. Marconi, "A Note on the Effect of Daylight upon the Propagation of Electromagnetic Impulses", *Proc. Roy. Soc. (London),* 70 (June 12, 1902).

16. See also J. J. Thomson, "On Some Consequences, etc.", *Phil. Mag.,* [6] 4 (1902)

17. See Ref. 11, p. 618.

18. *Riv. Marittima (Rome),* October 1902.

19. G. Marconi, Paper read before the Royal Institution of Great Britain, March 3, 1905.

20. See also G. Marconi, Lecture before the Royal Institution of Great Britain, March 13, 1908.

21. G. Marconi, "On Methods whereby the Radiation of Electric Waves may be mainly confined, etc.", *Proc. Roy. Soc. (London),* A 77 (1906).

22. G. Marconi, "Note on a Magnetic Detector of Electric Waves", *Proc. Roy. Soc. (London),* 70 (1902) 341.

La modulazione d'ampiezza di *Reginald Fessenden*.
Dalla radio-telegrafia alla radio-telefonia.

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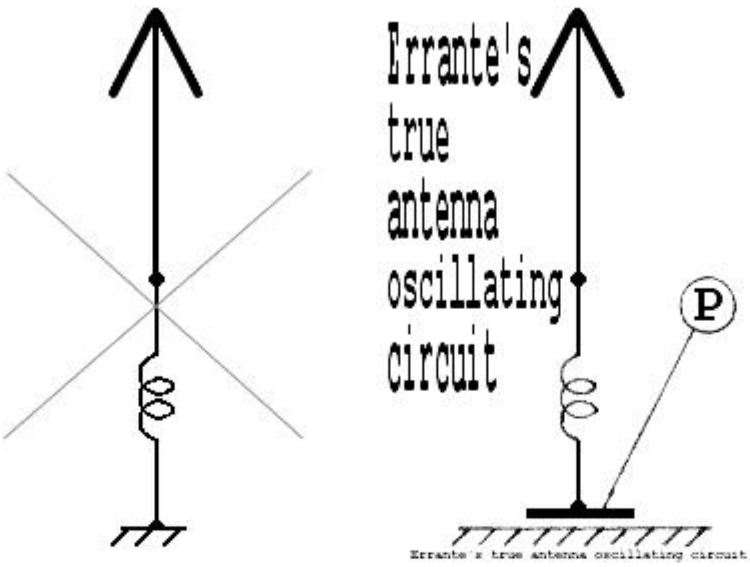


Come polarizzare correttamente un sistema sbilanciato per segnali a radio-frequenz

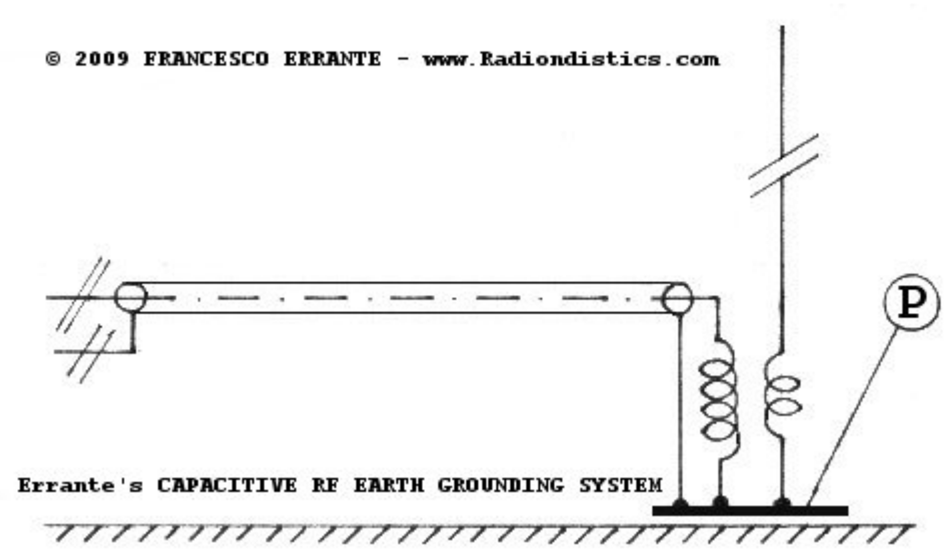
Questo sistema dimostra che non � necessario alcun accoppiamento gal radio-frequenza.

Sistema capacitivo di messa a terra per RF

by *Francesco Errante*

Faccio seguito a quanto da me rivelato e dimostrato in precedenza in materia di *rife* elettrici. In questa sede rivelo e dimostro che non � necessario alcun accoppiament RF rispetto alla terra fisica.

TUTTO QUELLO CHE SERVE � una CAPACIT� CONCENTRATA grande abbastar alla frequenza di lavoro pi� bassa di un dato sistema RF sbilanciato, come mostrato



Errante's CAPACITIVE RF EARTH GROUNDING SYSTEM schematic dia An unbalanced radiator and its Un-Un impedance transformer being gr layer to the physical ground - Copyright © 2009 - Francesco Errante

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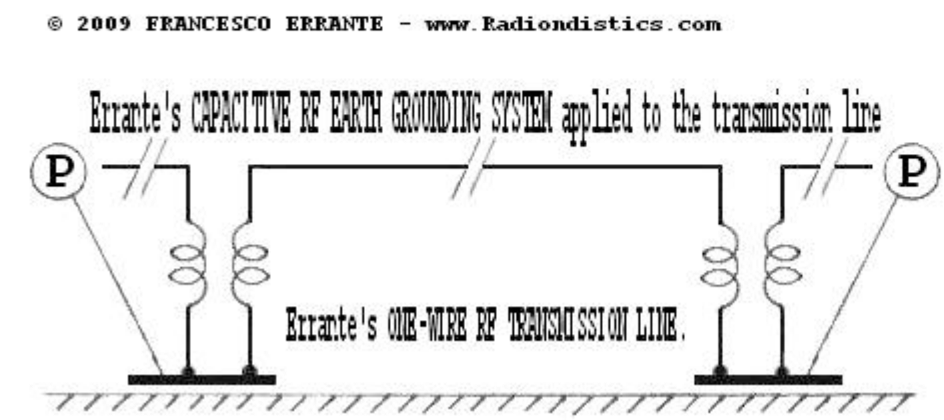

Errante's ONE-WIRE UNBALANCED RF TRANSMISSION LINE, employ
- Francesco Errante

La *capacit� di messa a terra* � ottenuta accoppiando un punto di un circuito ra mezzo di un'armatura metallica **(P)** posta su di un piano parallelo ad una superficie dimenzioni, come mostrato nelle immagini sopra. **(E NON attraverso un conden errore, come documentato nella sua lectio magistralis del 1909 in occasi** Nello spettro delle onde medie, lunghe e lunghissime, allo scopo di aumentare la cap cilindro cavo interrato ma isolato dalla terra. Il volume da esso occupato pu� essere riduzione dei costi per la realizzazione delle antenne per trasmissioni in VLF per rad **NB. Il sistema pu� anche essere dotato di percorso galvanico di messa a terra, al mediante un avvolgimento anti-induttivo (RF choke) connesso tra l'armatura (P)**

L'arrangiamento a semi-condensatore, mostrato negli schemi di cui sop per esempio, un'armatura metallica di un metro quadrato circa possa difficilmente f impiegata � la sua reattanza capacitiva. Al contrario, i sistemi di piano di terra a rag perdite.

Un'applicazione pratica, a dimostrazione di questo sistema, � data� dal di alimentazione RF sbilanciato, con messa a terra capacitiva.

L or C Reac

Enter any two known values and press "Calculate" to solve for the others. For examp 1000 Hertz. Fields should be reset to 0 before doing a new calculation.

Inductive Reac
Capacitive Reactan
Resonant Frequency (Fo

Capacitance (Microfarads)	Inductance (Millihenrys)

Applet

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27. Virtual ground node generator for balanced transmission lines © 2008 Francesco Errante

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NEW Errante's CAPACITIVE EARTH GROUNDING SYSTEM

See also: Virtual ground node generator for unbalanced lines

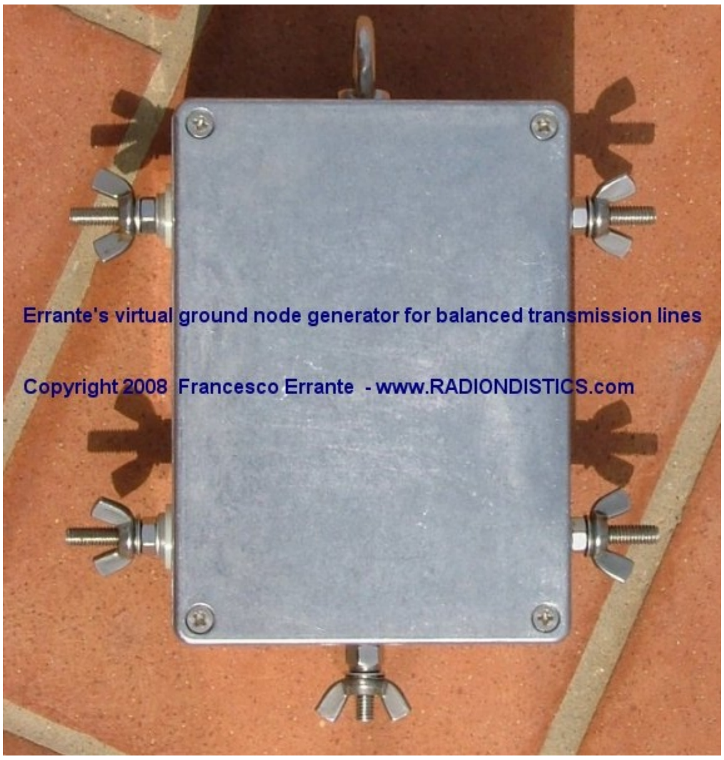
�

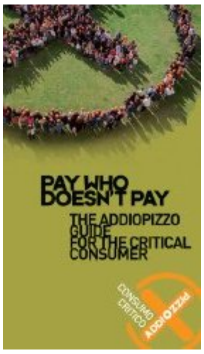
VIRTUAL GROUND NODE GENERATOR
for
BALANCED TRANSMISSION LINES
in-line "**B**.**V**.**G**.**N**. **G**.", for short

Virtual ground node

Errante's CAPACITIVE EARTH GROUNDING SYSTEM

Balanced transmission lines
in a
progressive





Virtual ground node generator for balanced transmission lines .

Freq. range: da 1 a 30 MHz
Input impedance: 300 Ohm
Output impedance: 300 Ohm
Input/output S.W.R.: � 1:1 within freq. range
Insertion loss: -0.01 dB
RF Power: 3 KW cw
Connectors type: M6 studs
Weight : 1 Kg ca.

Picture of the virtual ground node generator for balanced transmission lines being tested for frequency response flatness.

wave regime

Hertzian radiation: what it is and how it happens

�

The virtual ground node generator for balanced transmission lines : what it is and how it works.

The virtual ground node generator for balanced transmission lines.

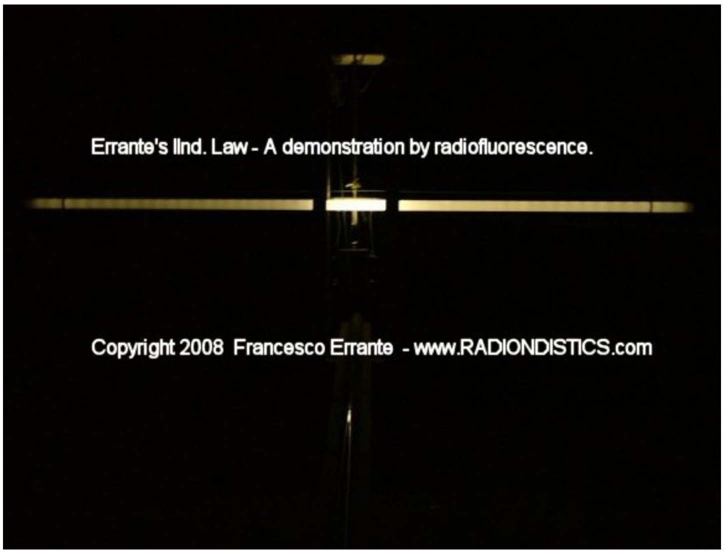
by *Francesco Errante*

The reader who's not already familiar with the virtual ground node mechanism is strongly adviced to read about the *virtual ground monopole HF antenna* and its background, prior to proceeding with the reading of this page.

The virtual ground node generator for balanced transmission lines and balanced antenna systems (B.V.G.N.G. for short) is a radioelectric circuit based on the same principals governing the virtual ground node generator for unbalanced lines and it is built around a broadband high power flux-coupled RF transformer having a 300 Ohm center-tapped balanced primary winding and a 300 Ohm center-tapped balanced secondary winding (Bal-Bal 1:1). The two central-tappings, each of them seat of a virtual ground node, are bonded together and connected to the chassis, ensuring a complete galvanic continuity amongst all the branches of this device.

This device allows to de-coupling the half-a-wave folded dipole (300 Ohm impedance) from its balanced transmission line (300 Ohm impedance) by imposing a clear-cut electric boundary for the RF signals flowing between the transmission line

and the folded dipole antenna, in order for the dipole to exhibit its true bandwidth and for the transmission line not to interfere with it.
In the absence of a clear-cut boundary between the transmission line and the radiator, infact, the transmission line acts as part of the radiating circuit, inevitably, modifying its electrical length and giving it-self origin to hertzian radiation, with unpleasant consequences such as: loss of signal, radio interferences induction on the surrounding equipment and, above all, the illusion of a phenomenon of resonance of the transmission line it-self. Balanced lines are, instead, aperiodic as demonstrated here.



A folded dipole and its transmission line, de-coupled through a BVGNG - Radioscopic detection of the RF electric field generated by the radiator and absence of radiation by the balanced transmission line.

A properly adapted 300 Ohm balanced transmission line terminated on a 1/2 wavelenght folded dipole, through a balanced virtual ground node generator, undergoing an high power (1 kW CW) radioscopic detection. Experimental verification of the IInd Errante's law applied to the transmission line

A properly adapted 300 Ohm balanced transmission line terminated on a 1/2 wavelenght folded dipole, through a balanced virtual ground node generator, undergoing an high power (1 kW CW) radioscopic detection. Experimental verification of the IInd Errante's law applied to the transmission line

This device can be built to suit any impedance transformation ratio. For example: to properly feed an half a wavelength open dipole with a 300 Ohm bifilar transmission line, it is necessary a Bal-Bal having an imput impedance of 300 Ohm ed an output impedance of 70 Ohm.

Moreover, the B.V.G.N.G. can be employed to generate a virtual ground node anywhere along a balanced transmission line.

Scientific relevance:

The B.V.G.N.G. demonstrates that a balanced transmission line, when correctly de-coupled from the radiator, always exhibits an aperiodic behaviour, even if it is terminated on matched reactive loads.
See: **the behaviour of balanced transmission lines for radiofrequency electric signals in a progressive wave regime**

Operational advantages:

a) Ensured aperiodicity of the employed balanced transmission line;

b) inhibition of hertzian radiation onset from the balanced transmission line;

c) full rejection of the co-phased currents induced onto a balanced transmission lines by any external RF source;

d) diminuisced signal losses.

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28. VGNG - in-line VIRTUAL GROUND NODE GENERATOR © 2007 Francesco Errante - Radiondistics.com

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NEW Errante's CAPACITIVE RF EARTH GROUNDING SYSTEM

See also: the virtual ground node generator for balanced transmission lines

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The *in-line VIRTUAL GROUND NODE GENERATOR*
in-line "**V.G.N.G.**", for short

also known as the

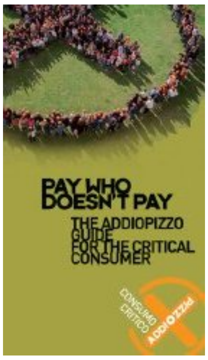
Errante's Holyground
the network that works miracles!

Virtual ground node

Errante's CAPACITIVE EARTH GROUNDING SYSTEM

Balanced transmission lines

Errante's in-line virtual ground node generator.



Freq. range: 1 to 30 MHz
Input imp.: 50 Ohm
Output imp.: 50 Ohm
Input/output SWR: 1:1 within range
Insertion loss: -0.01 dB
RF power: 3 KW cw
Sockets type: SO-239
Weight : 1 Kg ca.

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ERRANTE's in-line VIRTUAL GROUND NODE GENERATOR : what it is, how it works and what it does.

Universal R.F. VIRTUAL GROUND NODE GENERATOR

by *Francesco Errante*

The Errante's in-line virtual ground node generator is a radioelectric circuitry based on the same principals and schematics of his own original patented virtual ground monopole HF antenna design.
The reader who's not already familiar with the virtual ground node mechanism is

in a progressive wave regime

Hertzian radiation: what it is and how it happens

strongly adviced to read about the "*virtual ground monopole HF antenna* and its background prior to proceeding to the reading of this page.

The in-line VIRTUAL GROUND NODE GENERATOR (in-line V.G.N.G. for short) is a symmetric radioelectric network which converts an unbalanced RF signal into two precisely identical signals and re-converts them back into a non-inverted single unbalanced RF signal. While doing this, a virtual ground node is generated at its center. It can be effectively described as being an "Un-Bal-Bal-Un" network.
In practice the device, as presented in here, has both the input and output gates terminated onto a 50 Ohm impedance value and SO-239 type of connectors to match the most common co-axial line impedance standard and it is optimized for HF operations between 1 and 30 MHz. Within its specified range, this device appears to be "transparent" between the RF source and the load [SWR = 1:1] and exhibits an extremely low insertion loss [-0.01 dB] However, it can be made to suit any other input/output impedance values and frequencies spectrum.

The in-line V.G.N.G. can generate a virtual ground node at any point of a co-axial line in un-balanced transmission line system. This is, infact, its function. This device is also available in a version for balanced lines.

Main applications:

> **1)** It can be employed to stabilize a whole RTX un-balanced system,

however complex it could be, by providing a reliable and centralized RF earth grounding system. As most people may have already experienced, RF earth grounding is a very difficult task and has nothing to do with the conventional safety earth grounding solutions;
N.B. For safety reasons, where required, all the equipment must also be earthed the conventional way and the chassis of the in-line VGNG can also be wired to the building safety earth grounding circuit. All the system equipment chassis should be wired to the provided earth bolts on the in-line VGNG chassis by the shortest route.

2) It can be employed to provide a reliable way to short-circuiting antennas who have not antistatic protection, without the need for modifications or even reaching them up on the roof. Since the in-line VGNG exhibits a solid short circuit to the DC current on both its gates, all the electrostatic charges collected by the aerial and/or the co-axial outer conductor will be discharged via common ground and into the building safety earth grounding circuit;

3) It can be used to stabilize RF co-axial lines exposed to third party hertzian radiation;

4) It can be employed to stabilize reactive loads at their feeding points.

Scientific relevance:

The in-line VGNG enables physics and RF laboratories to have a reliable RF earth grounding systems as a precise reference for test and measurement.

Operational advantages:

a. the in-line VGNG is easy and fast to deploy;

b. in-line VGNG helps to improve E.M.I. compatibility within high equipment density environment.

�

F.A.Q.:

Question: I run a virtual ground HF antenna, do I still need an in-line VGNG ?
ANSWER: NO ! Your virtual ground antenna system already provides all the R.F. earth grounding you need. However, while using a different aerial an in-line VGNG might be needed.

Question: Where should I place it along my transmission line?
ANSWER: Where the un-balanced RTX system comprises only of a TX/RTX-SWRmeter-coaxial-antenna, the in-line VGNG must be placed right onto the TX/RTX antenna socket.
N.B. If an antenna tuner is used, the in-line VGNG must always be placed before it, right on

its input socket.
If a linear power amplifier is used the in-line VGNG must be placed on its output gate.
Where the system comprises of TX/RTX, a linear power amplifier and an antenna tuner, the in-line VGNG must be placed between the amplifier and the antenna tuner.
If a filter, such as an anti TVI, is used place the filter between the amplifier and the in-line VGNG. Moreover, the filtering should improve and the system will also benefit by the added VGNG's out-of-band own signal rejection.
N.B. The in-line VGNG should be inserted in the transmission line by using a short but comfortable RG-213, or better, patch lead.
Wherever possible use double male connectors to plug the in-line VGNG in.

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29. ERRANTE's all-in-one apparatus for the physics of the hertzian radiation - radio waves - dipoles

errante_apparatus.htm

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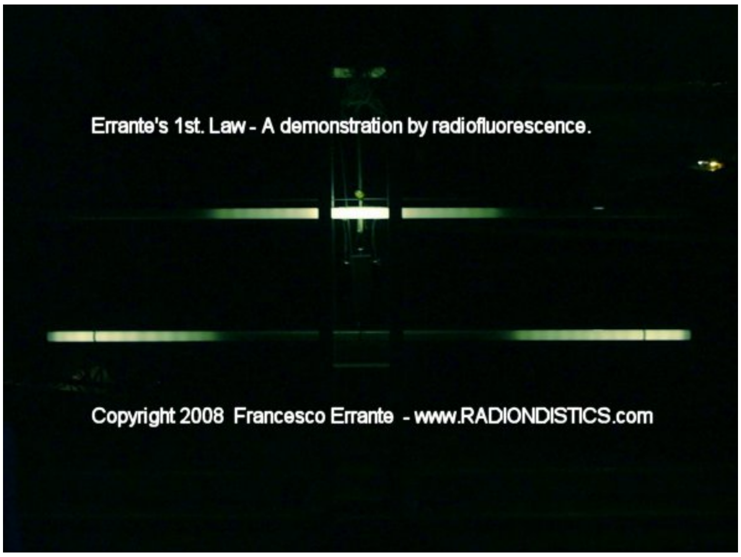
Errante's all-in-one hertzian radiation physics apparatus.
This equipment allows to carry out several measurements and experiments into the physics of radio-electric signals and transducers.



Foreword

Virtual ground node

Balanced virtual ground node generator

Hertzian radiation: what it is and how it happens

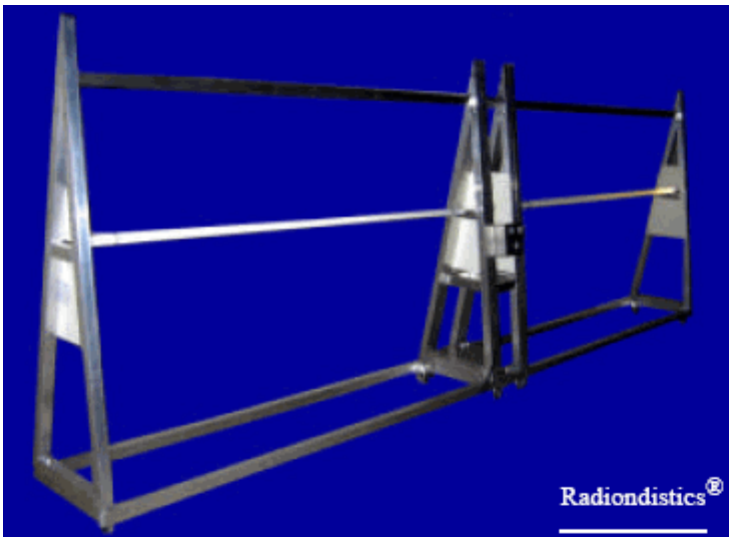
Although more than a century has gone by, since Heinrich Hertz made his first artificial radio transmission, the comprehension of radio-electric phenomena (usually referred to as electromagnetism) has been limited to mere theory. This because, the analysis of radio-electric phenomena had, untill now, relied almost entirely on mathematical conjectures. Mathematics is of great help to physics but cannot take the place of the instrumental observations of the natural phenomena. Instrumental observation is essentially dependant on the progress of technology but, above all, it needs the inventive of scientists without prejudices.

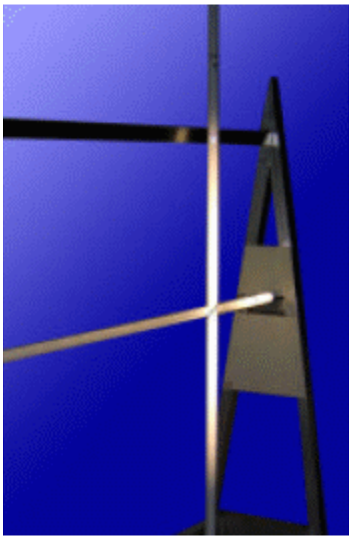
The apparatus presented here below, allows the step-by-step replication of the main experiments masterminded and made by the author in order to investigate the physics of the radio-electric signals and transducers. These experiments have directly led him into the discovery of the actual mechanism (*radio-electric transduction*) that originates the hertzian radiation (better known as radio waves). Also read: Hertzian radiation: what it is and how it happens by *Francesco Errante*

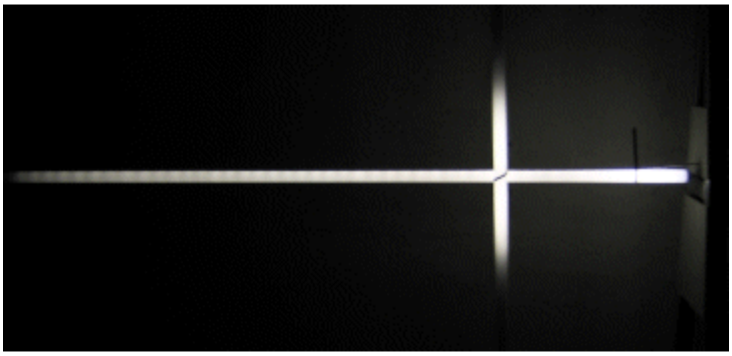
Hertzian radiation detection by radio luminescence

This equipment enables researchers and teachers alike to carry out several experiments and measurements and makes easy to comprehend how hertzian radiation takes place onto an electric conductor and its propagation in the space. This is done by means of employing gas filled detector tubes and electric measurements instruments concurrently. This equipment proves it-self to be an invaluable teaching tool, as seeing is believing. It, infact, affords the students a naked eye observation of the energy distribution along both the two branches of the dipole antenna and in its immediate surroundings. Moreover, the electronics that comes with it, permits

to vary and to measure the power of the radio-electric signal being injected into the aerial as well as measuring all the relevant standing wave ratios during the signal transduction process, in order to univocally establish causes and effects.
(Proportionality between the antenna feeding power and the strength of the generated field; attenuation due to the medium etc...)







It is now available our apparatus for the physics of the RF balanced transmission lines. Here !

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30. A VIRTUAL GROUND SWR-METERED OPEN DIPOLE BALUN © 2003 Francesco Errante

balun_swr.htm

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SWR-metered BALUN

by *Francesco Errante*

This version of the *virtual ground balun* is designed around our suppressor circuitry and it has 3 built in S.W.R. meters, one for each of its ports. Built into its circuit there are also 2 coaxial switches and 2 dummy loads for the alternate suppression of one or both of the branches of the open dipole antenna to be tested.

The *SWR-metered balun* is the ideal choice for conducting experiments with the *secondary ionization fluorescence hertzian radiation detector*.

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All the concepts, methods, designs and devices presented on this web site
are the original novelty works of *FRANCESCO ERRANTE*.

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31. VIRTUAL GROUND BALUN for HF FOLDED DIPOLE ANTENNAS © 2003 Francesco Errante - www.Radiondistics.com

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NB. Our radio-communications purpose baluns can be found here.

ï¿½

ERRANTE's BALUN

Virtual ground node

Virtual ground node generator for de-coupling a balanced transmission line from its load

Virtual ground balun for folded dipole & loop antennas

A state-of-the-art high power broadband HF balun

ï¿½

Patent n.0001347486 (ITAPA20030021)
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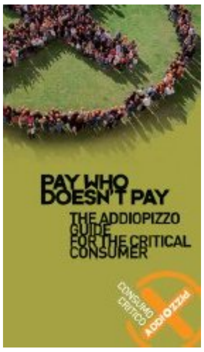
A VIRTUAL GROUND BALUN FOR DOUBLING THE ï¿½ WAVELENGTH FOLDED DIPOLE ANTENNA

by *Francesco Errante*



Please, pay ATTENTION! This page relates to a special applicazione for scientific purposes of the *Errante*'s virtual ground balun. To find out more on *Errante's* virtual ground baluns for HF radio communications and its theory click here.

Scientific purposes

a. The VIRTUAL GROUND RF BALUN TRANSFORMER described hereby allows the doubling of the ï¿½ wavelength folded dipole's radiator.

b. The virtual ground RF balun transformer described hereby allows to demonstrate that in a ï¿½ wavelength folded dipole antenna, its radiator (having a characteristic impedance of 300 ohm) acts as it was made of two distinct radiators, identical to each other and coinciding with each

Errante's CAPACITIVE EARTH GROUNDING SYSTEM

Balanced transmission lines in a progressive wave regime

Hertzian radiation: what it is and how it happens

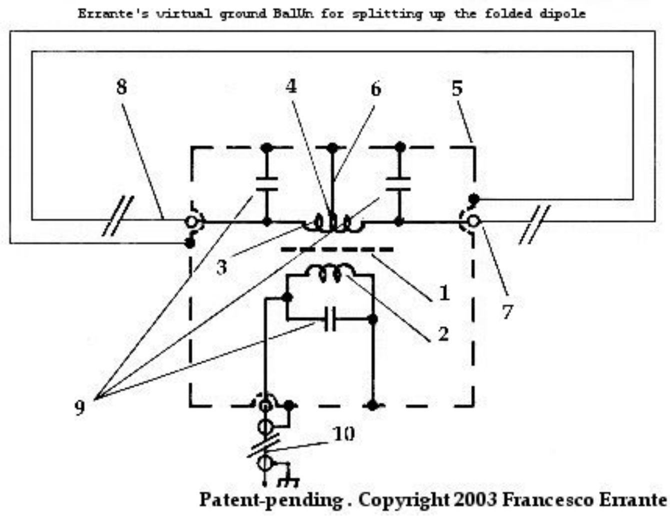
other, while being fed in a counterphase arrangement, having each of them a characteristic impedance of 150 Ohm, referred to the virtual ground node.

c. The virtual ground RF balun transformer described hereby allows to demonstrate that the ï¿½ wavelength folded dipole radiator and, indeed, any other resonant loop, be it made of any shape, with both a single turn or a multiple turn winding, when it is correctly fed with a sinusoidal radio-electric signal of a proper wavelength, it is always interested by electric currents that flow along its entire length from one end to the other in a clockwise direction for a semi-period and anti-clockwise during the next semi-period or viceversa.

Circuit description

The Author, by means of a particular radio-electric circuitry for the suppression of any which one of the two branches of a ï¿½ of wavelength open dipole, has previously demonstrated that it is, indeed, possible to feed each of the two branches of an ï¿½ wavelength open dipole independently, so that they can become electrically independent from each other allowing, therefore, to suppress any which one of them without repercussions on the condition of resonance and radiation of the remaining one. On that occasion, the Author has introduced the reader to the concept and the method for generating a virtual ground node for the separation of the dipole's branch currents.
The ï¿½ wavelength folded dipole antenna has a single wiring radiator and, therefore, does not allow the independent usage of the currents flowing on it, both in a clockwise and anti

clockwise way.
The present invention allows to split up the folded dipole radiator into two nearly-coinciding wirings in order to demonstrate that the single folded dipole radiator acts as if it was made of two identical but distinct wirings coinciding with each other and each working cyclically for the duration of a semi-period.
This result has been achieved by means of a lumped-constants radio-electric circuit, based around a broad-band flux-coupled radio electric balun transformer(1) winded on a binocular type ferrite core having a primary winding(2) exhibiting an impedance value equal the one of the transmission line(10) being used and a center-tapped secondary winding(3) exhibiting an impedance value of 150 + 150 Ohm.
The said impedance values are referred to the *virtual ground* node(4 & 6). (a *virtual ground node* is defined as a point in an electrical circuit that appears to be at ground, but is not actually attached to ground, it is therefore, a node having a 0 degree phase angle difference with respect to ground and has the same electrical potential as the Earth)
The virtual ground node is made available to the whole circuit by means of a very short electrical connection(6) between the transformer center-tapping(4) and the balun chassis(5).
The transformer's working point is optimized by compensation with the employment of high-voltage RF duty capacitors(9).



It is important to highlight the fact that the two independently fed branches(7 & 8) can also be independently rotated around their common axis to give a variable radiation pattern which ranges from a typical dipole's radiation pattern, if the arms are parallel to each other or a quasi-omnidirectional radiation pattern in the case of the arms being placed perpendicularly to each-other (**turnstile arrangement**).

The balun described hereby is characterized by being a *NON fluctuating impedance* RF electrical transformer. All of its terminations impedance values are set by design, hence the impedance transformation ratio is a function of them and not viceversa, unlike it happens with any other balun arrangement.

Please, pay ATTENTION! This page relates to a special applicazione for scientific purposes of the *Errante*'s virtual ground balun. To find out more on *Errante's* virtual ground baluns for HF radio communications and its theory click here.

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32. A SECONDARY IONIZATION FLUORESCENCE HERTZIAN RADIATION PASSIVE DETECTOR - Radioluminescence effect

detector.htm

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NEW Hertzian radiation and the *Hallwachs-effect*, better known as the *photovoltaic effect*.

NEW Hertzian radiation studies deliver a major blow to the antimatter theory origin.

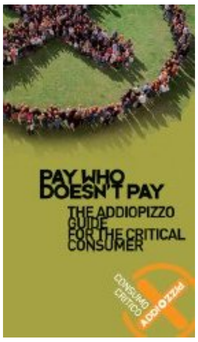
Hertzian radiation: what it is and how it happens

Virtual ground node

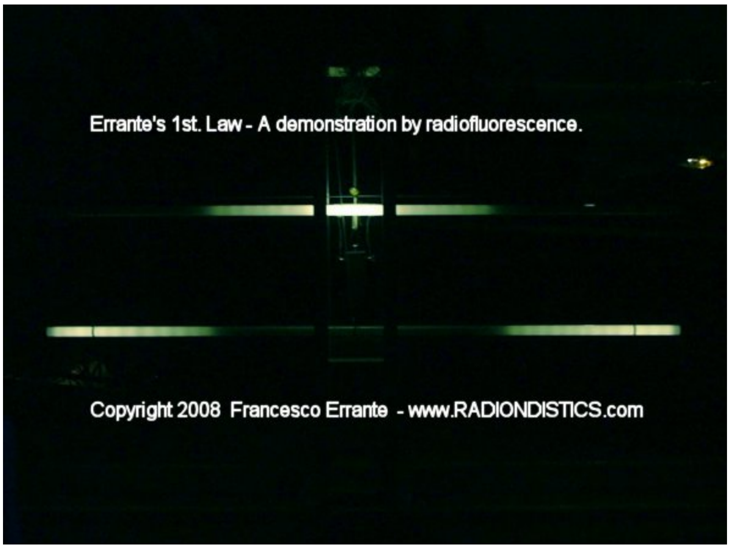
Errante's CAPACITIVE EARTH GROUNDING

A SECONDARY IONIZATION FLUORESCENCE HERTZIAN RADIATION PASSIVE DETECTOR

The *RADIOLUMINESCENCE* in the hertzian spectrum



THE IONIZING POWER OF THE *HERTZIAN RADIATION* (better known as radio waves)



Patent n.0001347487 (ITAPA20030022)
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SYSTEM

Balanced transmission lines in a progressive wave regime

RF lighting devices

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A SECONDARY IONIZATION FLUORESCENCE HERTZIAN RADIATION PASSIVE DETECTOR

by *Francesco Errante*

Description

"*Hertzian radiation*" is the oldest, yet the least common way to refer to radio-waves.
The present invention relates to an *hertzian radiation* passive detector which exploits a secondary ionization fluorescence phenomenon.
Untill now, it was thought that the *radio-fluorescence* phenomenon was only property of very short wavelength emission such as x-ray or those caused by radioactive materials. The detector and its method of employ, as described hereafter, allow to convert an *hertzian radiation* directly into a visible light radiation so to permit the naked eye appreciation of the *hertzian radiation*'s energy distribution over the body of a radio antenna, or a transmission line and in their immediate surroundings, during transmitting mode. This is, most importantly, done in a non-intrusive manner as the detector is it-self radio-transparent and does not interfere with the resonance and radiation of the radiator. Moreover, the energy absorbed by the detector is neglectable.
As an *hertzian radiation* invests the low pressure gasses inside the detector, a first ionization takes place with the result that a scattering of higher energy particles will end-up against the luminophore layer covering the inner side of the walls of the detector tube, giving origin to a visible

light emission which increases correspondingly to the increase of the flux of the *hertzian radiation* untill the detector is fully lit.

In practice, a secondary ionization fluorescence hertzian radiation passive detector in his most elementary and economic form, comprises of a 100 Watt straight fluorescence tube**(1)** for general purpose indoor lighting, having a physical length of about 2.4 meter and a trolley**(2)** which is made of dielectric material and standing on 4 swivel castor wheels.

The trolley**(2)** allows the detector**(1)** tube to stand up and be free to be moved up and down along the body of the antenna**(3)** under examination, which is placed horizontally and at a suitable height so to encounter the detector at about its middle.

Generally, a power of little more than 40 Watt RF is enough to fully lit a resonant radiator for the short waves as an � wavelength open dipole antenna. Once the tube is excited, by progressively reducing the RF power supplied to the aerial or by progressively moving the detector away from the radiator, it is possible to visually appreciate the relative strength of the field generated on each and every point along the antenna or its immediate proximity.

An excited tube will maintain its fluorescence untill the *hertzian radiation* ceases or drastically diminishes or ceases all together. By reducing the RF power supplied to the exciting antenna, the region of the tube previously lit will decrease correspondingly and its visible light emission will be extinguished as soon as the excitation RF power will drop below 0,1 Watt.

By means of a number of detector tubes, it is possible to generate a field strength dynamic luminous diagram, both bi-dimensional or tri-dimensional. However, it is advisable to build a diagram by putting together several photograms of the same tube in different point along the radiator. This is easily done by first exciting the tube and then moving it alongside the radiator or the space around the antenna under examination.
Untill the tube is fully lit, to a certain strength of visible radiation correspond a certain strength of the *hertzian radiation*. It is, therefore, possible to estimate the attenuation introduced by the space in the proximity of the antenna.

Scientific purposes

a. The secondary ionization fluorescence hertzian radiation passive detector, as described hereby, allows to demonstrate the *hertzian radiation*'s transitory ionizing power.

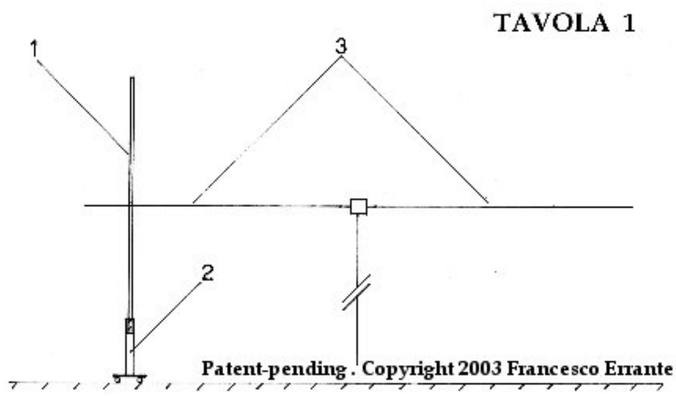
b. The secondary ionization fluorescence hertzian radiation passive detector, as described hereby, allows the naked eye appreciation of the *hertzian radiation*'s energy distribution over the body of a radio antenna, during transmit mode.

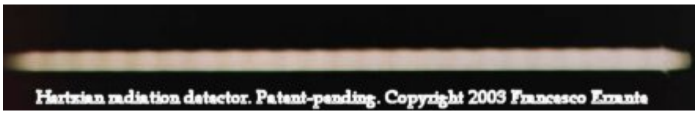
c. The secondary ionization fluorescence hertzian radiation passive detector, as described hereby, allows the naked eye appreciation of the *hertzian radiation*'s relative intensity and its energy distribution in the space region immediately surrounding the aerial, during transmit mode.

d. The secondary ionization fluorescence hertzian radiation passive detector, as described hereby,

allows to see that when a radio-electric signal is injected into a properly resonant radiator, the latter will always radiate energy as radio waves, starting from the point which is always opposite to the point of feeding. The detector, also, shows that the bulk of the energy is always radiated by the region towards the end of the radiator.
See also: Hertzian Radiation: what it is and how it happens. A radioscopic observation, by *Francesco Errante*

e. The secondary ionization fluorescence hertzian radiation passive detector, as described hereby, allows to demonstrate that conventional lighting purpose fluorescent tubes exhibit a much higher energy efficiency and versatility when excited by a radio waves direct bombardment (electrodeless arrangement) as opposed to be exited by galvanic discharge.





A partial view of the tube while being exited by a low level RF power.

The detector is totally radio-transparent untill it lights up. It will become slightly less radio-transparent when fully excited. This however will not affect its performance nor the radiator functioning. In the worst case scenario, when a full intimate parallel coupling between the detector and the end of a radiator is made, a SWR of 1:1,6 would appear when detector lights up.

A more complex version of this detector for usage within science didactics departments is available here.

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multiband.htm

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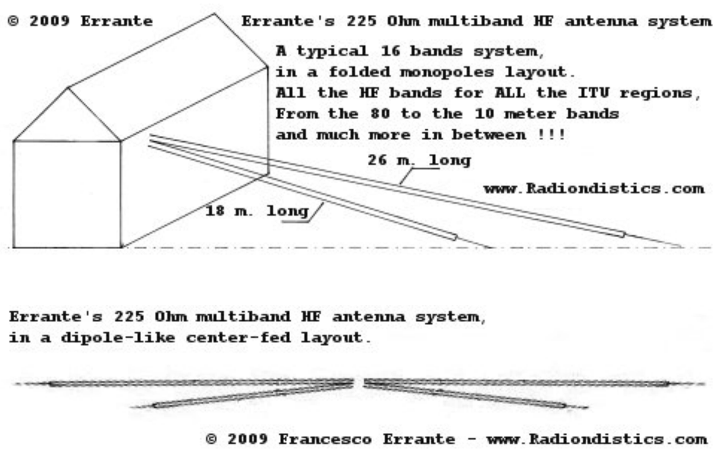
Errante's 225 Ohm, multiband HF antenna system.



Unlocking the full multiband potential of the 225 Ohm elementary radiator in a folded monopole, dipole-like or turnstile layout.

All the I.T.U. amateur radio bands from the 80 m. to the 10 m. band and much more in between, in 16 ample segments.

NB. In order to avoid misunderstandings, it must be noted that this design has nothing

Errante's CAPACITIVE EARTH GROUNDING SYSTEM

Virtual ground node

Errante's *elementary radiator*

to do with the electrically short (<< � lambda) folded monopole, also generically known in this trade as "*folded unipole*" which is an inherentely lossy and very low impedance radiator.

click on the images to enlarge them

Errante's 2x225 Ohm virtual ground balun for the dipole-like center-fed folded monopoles multiband HF antenna system. Notice the two pivoting UV resistant Nylon insulators - each of them 42 cm long - which make easy the anchoring of 2 folded monopoles on each side of the balun. This is an ideal solution to install all the dipole branches with any angle from about 120 degree to 180 and upwards again to the 120 degree. From inverted "Vee" to open dipole and up to a "V".
The insulators are free to rotate around their axis, being the M8 studs which also act as the balun's electrical terminals.
The insulators and the eyelet terminals are tightened independently from each other.

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Errante's virtual ground BalUn

Hertzian radiation: what it is and how it happens

Errante's 225 Ohm, multiband HF antenna system: what it is and how it works.

by *Francesco Errante*

The multiband radio-electric transducer system disclosed hereby is based upon the natural *harmonic resonance*, a very well known physics phenomenon. This system, therefore, does not utilize traps or loading coils or other artifacts for band segmentation.

The radiators act, band by band, as mono-band full-size aerials by resonating on a different fraction or a multiple of the relevant signal wavelength, starting from 3/4 a wavelength at each set lowest operating band.

By virtue of this radiator design, it possible to get several, well distinct, sets of bands, each set giving 8 bands in the spectrum of the HF, each and all of them yelding a COMMON IMPEDANCE VALUE of 225 Ohm. **This particular common impedance value is the key factor in this system design**, while *Radiondistics' own patented virtual ground broadband impedance coupling and phasing network technology* has allowed this system to work seamlessly with an exceptional impedance matching throughout all its frequency ranges. **NO additional A.T.U. (antenna tuning unit) is needed and all its operating bands offer a generous width as this is an high impedance antenna system.**

Moreover, this system requires only a reasonable amount of space to be deployed, it is easy to install and does not require tedious adjustment of the length of its radiators, as all of its bands are well defined but NOT critical. If necessary, the radiator can even be bent along its axis (this will definitely affect its radiation pattern but not its original tuning) or installed as a top fed sloper or as a center-fed inverted-vee with any angle with respect to the ground. It also requires very little room clearance from the ground or any other surrounding obstacles. Thanks to the *Radiondistics' proprietary RF feeding system*, no de-tuning takes place on the aerials unless they are almost right down onto the ground. **For example, a 225 Ohm folded monopole arrangement, for all the I.T.U. amateur radio bands, from the 80 meter to the 10 meter, plus a lot more in between, would take about 26.5 meters of space on a streight line, while a center-fed dipole-like arrangement would need just twice as much**.

Accordingly, a range of feeding systems have been deviced for:

folded monopole antennas with Errante's capacitive earth grounding ;

center-fed dipole-like antennas. See Errante's 50/225+225 Ohm balun ;

folded monopole antennas with Errante's virtual grounding ;

folded monopoles in a turnstile antenna configuration.

RADIONDISTICS' products carry an unconditional lifetime guarantee.

Prices:

Errante's 50/225 Ohm capacitive earth grounded UN-UN,
Euro 250,00 +V.A.T. and shipping costs ;

Errante's 50/225+225 Ohm BalUn, as pictured above
Euro 500,00 +V.A.T. and shipping costs ;

Errante's 50/225 Ohm virtual ground feeding system,
Euro 700,00 +V.A.T. and shipping costs ;

Errante's 4x 225 Ohm broadband turnstile driver,
Euro 1650,00 +V.A.T. and shipping costs.

For an useful currency converter click here

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RADIONDISTICS' products carry an unconditional lifetime guarantee.

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34. ANTENNA BALUN : Errante's VIRTUAL GROUND BALUN for dipole & loop HF ANTENNAS

balun_errante.htm

RADIONDISTICS

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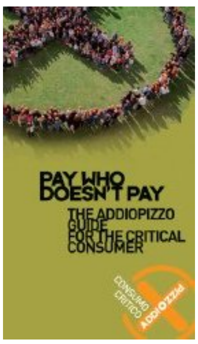
Go

Virtual ground node

Errante's CAPACITIVE EARTH GROUNDING SYSTEM

Balanced transmission lines in a progressive

Planning a trip to Sicily? �Then, make sure you won't be feeding the Mafia!



NEW **HF *Turnstile* antennas broadband driver, NOW available as standard! Find out more here.**

NEW **Errante's 2x225 Ohm virtual ground balun for the dipole-like center-fed folded monopoles multiband HF antenna system Find out more here.**

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ERRANTE's BALUN

Virtual ground balun for dipole & loop

A state-of-the-art high power broadband HF balun

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Patent n.0001347484 (ITAPA20030019) and
Patent n.0001347486 (ITAPA20030021)
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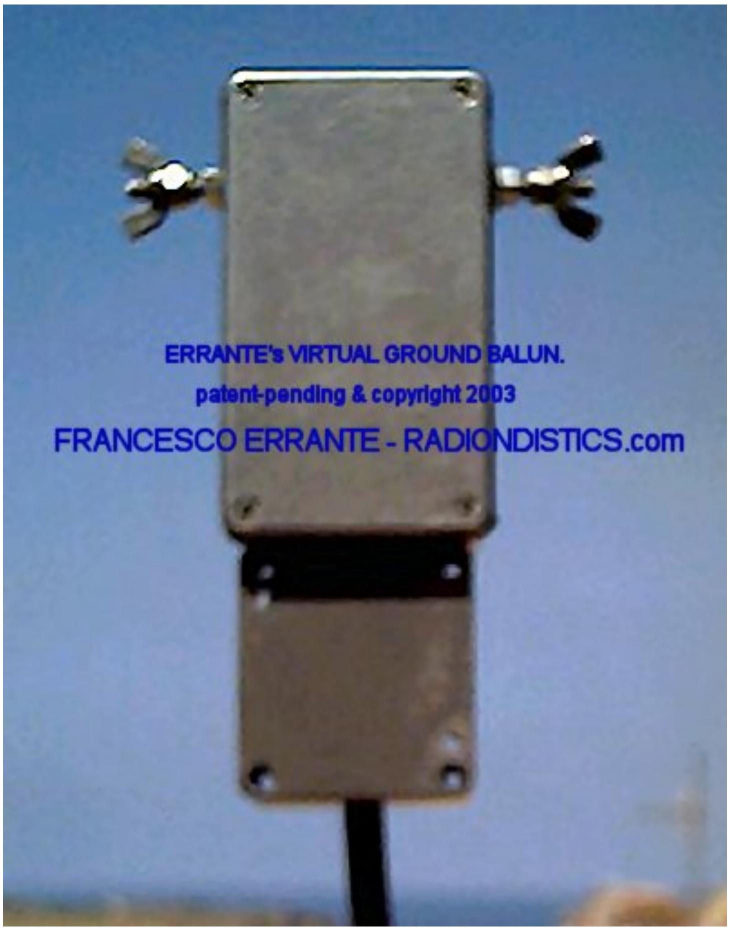

wave regime

Hertzian radiation: what it is and how it happens

Errante's 50/70 Ohm virtual ground balun
Freq. range: from 1 to 30 MHz
RF Power: 2KW CW

Click on the thumbnails to view larger pictures

   

N.B. The balun pictured above is available as standard for the following impedance values:

NEW! unbalanced 50 Ohm input to a 225+225 Ohm balanced output to suit the **Errante's 225 Ohm, multiband HF antenna system** ;

unbalanced 50 Ohm input to a 70 Ohm balanced output to suit an **� wavelength open dipole antenna**;

unbalanced 50 Ohm input to a 300 Ohm balanced output to suit an **� wavelength folded dipole antenna**;

unbalanced 50 Ohm input to a 50 Ohm balanced output to suit an **� wavelength *inverted Vee* dipole antenna**;

unbalanced 50 Ohm input to a 200 Ohm balanced output to suit a **full wavelength, single turn, Deltaloop antenna**.

Any other impedance values combination to be manufactured upon special order.

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ERRANTE's BALUN : what it is, how it works and how it came to be made.

by *Francesco Errante*

The virtual ground balun presented hereby is the practical application in the field of the short wave radio communications of what the Author has previously created for scientific purposes. It, infact, along with the virtual ground balun for doubling the � wavelength folded dipole antenna, descends from the **radio-electric circuitry for the suppression of anyone of the two branches of a � of wavelength open dipole**.

Errante's balun is, essentially, a pure electrical transformer which, in its most elementary layout, has a minimum of 3 tuned circuits: a primary circuit (primary winding LC network) and 2 secondary circuits (secondary windings LC networks).
Generally, the *Errante's* balun secondary circuits are identical to each other in order to feed symmetrical balanced antennas. When necessary, the secondary windings can be formed to have impedance values different between each other, while maintaining the phase angle difference of the balanced terminations to a constant �90 degree with respect to the virtual ground node. Moreover, as all the pure electrical transformers, *Errante's* balun can be made to have more than a couple of secondary windings.

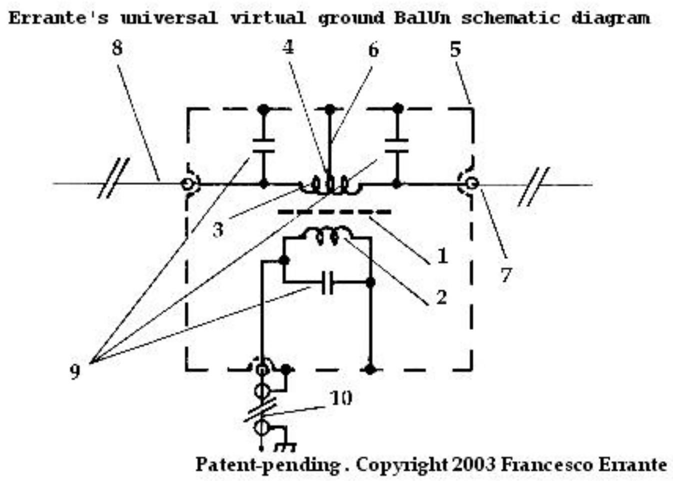
***Errante's* balun is characterized for NON allowing any impedance transformation to take place unless its terminations impedance values are matched (NON-fluctuating impedance).**

***Errante's* balun terminations impedance values are set by design, hence the impedance transformation ratio is a function of them and not viceversa, unlike it happens with any other balun arrangement.** See for example the *Guanella's* or the *Ruthroff's* transformers which are instead characterized by having a fluctuating terminations impedance values, as they are nothing else than an arrangement of auto-transformers, their input and output circuits are unseparable and, therefore, any change made to their primary circuits will inevitably affect their secondary circuits.
In practice, a balun having a fluctuating termination impedance value carries out its impedance transformation duty within a very ample range of impedance values, changing its input termination impedance according to the one of its load and viceversa. Therefore, supposed we have a *Guanella's* type balun with a transformation ratio of 4:1, it will show an input impedance of 75 Ohm if its load impedance is of 300 Ohm, or it will show an input impedance of 50 Ohm if its load impedance is of 200 Ohm. **This cannot happen if an *Errante's* type of balun is employed, because each termination impedance value is a fixed property of the relevant termination and does not vary even if there is a change in the values of the impedance of the circuits (antennas or transmission lines) connected to them.**
By limiting the line-load system impedance to the imposed values only, it is possible to inhibit both lines and loads (antennas) from giving origins to unwanted resonances, which are, generally, harmonic resonances, very often deriving from the presence of more than a load on the same

feed-point, which generates alternative paths for the RF. (for example: multiple dipoles o monopoles, fan antennas and so on). **This is something not to be underestimated, not only with regard to the transmitting phase but and above all, to the receiving phase as well. Employing a NON-fluctuating impedance balun, infact, introduces a very high rejection to interfering radio signals from sources outside the bands of the radio spectrum for which an antenna is ment. This translates in an excellent stability and selectivity of the transmission line or antenna systems which employ *Errante's* baluns.**

A virtual ground balun represents the ideal device for the correct feeding of dipole and loop antennas. The virtual ground balun carries out many functions at once allowing to:
1) match the RF source impedance to the chosen load while introducing neglectable insertion loss of power;
2) transform the RF signal fed to it into two identical portions with the correct phase angle difference for the natural feeding of open or folded dipoles and loop antennas;
3) obtain a virtual ground node for the correct polarization of its unbalanced port;
4) obtain a common ground path for the dissipation of the charges induced by the electrostatic phenomena;
5) obtain an high de-coupling between its two balanced ports.

The generic virtual ground balun's schematic diagram is shown here below.



The virtual ground balun is housed inside a weatherproof rugged metal enclosure having two separate compartments: one hosts the balun circuitry while the other one protects the coaxial connectors. The said metal enclosure not only ensures an high mechanical robustness and a total stability against the weather agents, it also works as the general electric conductor for all the references to the virtual ground node. Such an enclosure ensures also a full electrostatic shielding of the RF balun transformer circuitry.

Radiondistics claims the originality of the balun's unique electromechanical layout, as well as the balun radio-electric circuitry.

Operational advantages:

a. the virtual ground balun allows the feeding of dipole and loop antennas with much less loss than conventional baluns

b. the virtual ground balun allows an aerial to exhibit a much larger bandwidth compared to when the same antenna is fed by a conventional balun.

c. the virtual ground balun allows an antenna to exhibit a much stronger immunity to parasitic capacitive effects compared to when the same antenna is fed by conventional baluns.

d. the virtual ground balun allows an antenna to exhibit a strong resonance stability, regardless of the level and inclination of it above the ground.

e. the virtual ground balun allows an antenna to exhibit a much stronger immunity to out-of-band emissions compared to when the same aerial is fed by a conventional balun.

f. the virtual ground balun allows an antenna to exhibit a much stronger immunity to electrostatic charges and noises compared to when it is fed by a conventional balun.

g. the virtual ground balun, owing to the constructive and destructive interference phenomena taking place in the transformer allows the dipole antenna to exhibit a sharper radiation/reception pattern (off the beam axis signal rejection) compared to the same aerial when fed by a conventional balun.

h. the virtual ground balun allows a much faster antenna deployment time compared to conventional baluns.

i. the virtual ground balun allows an antenna to exhibit a much better compatibility with highly populated environments (EMC TVI) compared to when it is fed by a conventional baluns.

Applications:

a. SHORT WAVES FIXED & MOBILE RADIO STATION FOR CIVIL & MILITARY PURPOSES �

b. STEALTH, FAST & TACTICAL DEPLOYMENT PURPOSES �

c. SEARCH & RESCUE SERVICES�

d. SHIP to SHIP & SHIP to SHORE MARINE COMMUNICATIONS

e. DX-ing and metropolitan HAM RADIO STATIONS

f. ELEMENT in ARRAYS of AERIALS

g. RADIO-PHYSICS LABORATORIES

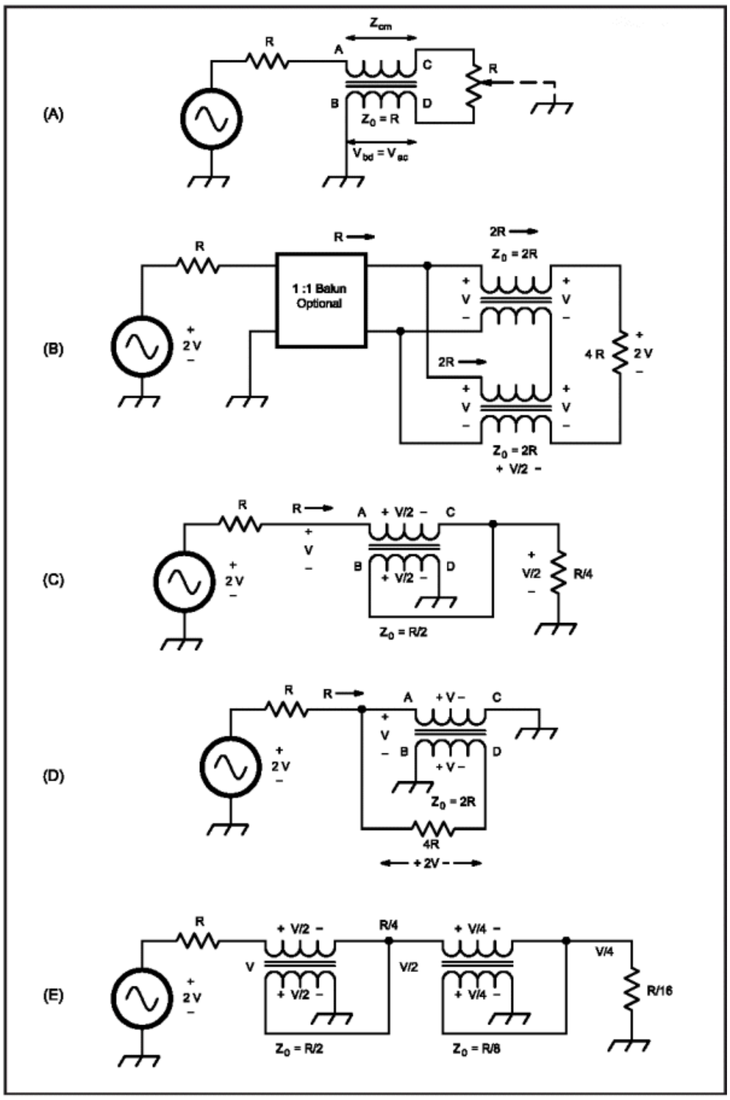
see also SWR-metered balun

�

Prices:
Euro 250,00 +V.A.T. and shipping costs.
N.B. The price of the new 50/225+225 is Euro 500,00 +V.A.T. and shipping costs.
For an useful currency converter click here

RADIONDISTICS' products carry an unconditional lifetime guarantee.

Schematic diagram of *Guanella's* and *Ruthroff's* baluns



At A, basic balun. At B, 1:4 Guanella transformer. At C, Ruthroff transformer, 4:1 unbalanced. At D, Ruthroff 1:4 balanced transformer. At E, Ruthroff 16:1 unbalanced transformer.

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35. HF Turnstile antenna broadband driver © 2026 Francesco Errante - Radiondistics.com

quadripole.htm

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The *ERRANTE's HF Turnstile antenna broadband driver*

NOW, also available in the MULTIBAND VERSION

An all-in-one solution for coupling, balancing and phasing the cross-dipole antenna for the HF spectrum.

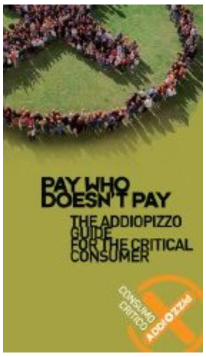
Virtual ground node

Errante's CAPACITIVE EARTH GROUNDING SYSTEM

Errante's virtual ground BalUn

ERRANTE's HF Turnstile antenna broadband driver.

Freq. range: 1 to 30 MHz
Input imp.: 50 Ohm, unbalanced
Output imp.: balanced 2 x 70 Ohm or 2 x 450 Ohm for the Errante's multiband system
Outputs phase shifts: 0, 90, 180, 270 degrees
SWR @ band-center: 1:1
Insertion loss: -0.01 dB
RF power: 4 KW CW
Sockets type: SO-239
Weight : 5 Kg ca.



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ERRANTE's HF Turnstile antenna broadband driver : what it is, how it works and what it does.

by *Francesco Errante*

The Errante's turnstile antenna driver, is a broadband radioelectric circuit based on our own original *virtual ground balun* patented design.
A *turnstile antenna*, which is also known as *cross-dipole or quadripole* is the top choice when a quasi-omnidirectional radiation pattern is requided in a horizontally polarized antenna system. � (see the 1937, George H. Brown's original patent application)
When it comes to point to point and broadcast radiocommunications through the ionospheric radio-propagation, such systems have the advantage of following the wanted signals, no matter what changes in the direction of propagation the sky makes.
Trans-ionospheric effects include refraction,

Balanced transmission lines in a progressive wave regime

Hertzian radiation: what it is and how it happens

amplitude and phase scintillation, signal delay and polarisation rotation. Such changes always affect the strength of the wanted signal during ionospheric radio propagation reception (fading, evanescence, QSB) and can easily reach a 20 dB loss. **A turnstile antenna system driven by this device of ours will recover that loss and on average all signals are 10 to 15 dB higher compared to what a single dipole would extract in the same conditions**.
Untill now, making a proper turnstile antenna for the short wave bands used to be a mammuth task and no previous system could allow multiband operation owing to their own delay line limitation. The device presented hereby is, instead, the ideal solutions to manage a cross dipole as you would do with a single one. It, infact, exhibits the same properties as our own virtual ground balun, while making easy to drive 2 identical dipoles or 2 identical fan dipoles in a *turnstile configuration*, thanks to its unique phasing system. By eliminating the need for a delay line, our system also eliminates all the loss of signal that inevitably comes with it, therefore, the quantity of energy tranferred to each of the 2 dipoles is precisely the same, this will translate in a truly symmetrical radiation patten.
Where fan-dipoles cannot be used through lack of space, multiband operations can also be achived by employing trapped dipole branches. **NOW, also available in the Errante's 225 Ohm, multiband monopoles version. (Errante's 225 Ohm, multiband HF antenna system)**

The turnstile antenna system, as presented hereby, it's the top choice for digital communications over the HF spectrum, such as the *e-mail over the HF services* and

the like.

Prices:
Euro 1250,00 +V.A.T. and shipping costs, for the
2x 70 Ohm output model.

Euro 1650,00 +V.A.T. and shipping costs for the
2x 450 Ohm output model.

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RADIONDISTICS' products carry an unconditional lifetime guarantee.

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All the concepts, methods, designs and devices presented on this web site
are the original novelty works of *FRANCESCO ERRANTE*.


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36. Frequency to wavelength converter - Wavelength to frequency converter - Energy calculator

frequency_wavelength_energy_radiation_calculator.htm

Energy and radiation wavelength

NB: this page is an appendix to the *Errante's* hertzian radiation theorem.

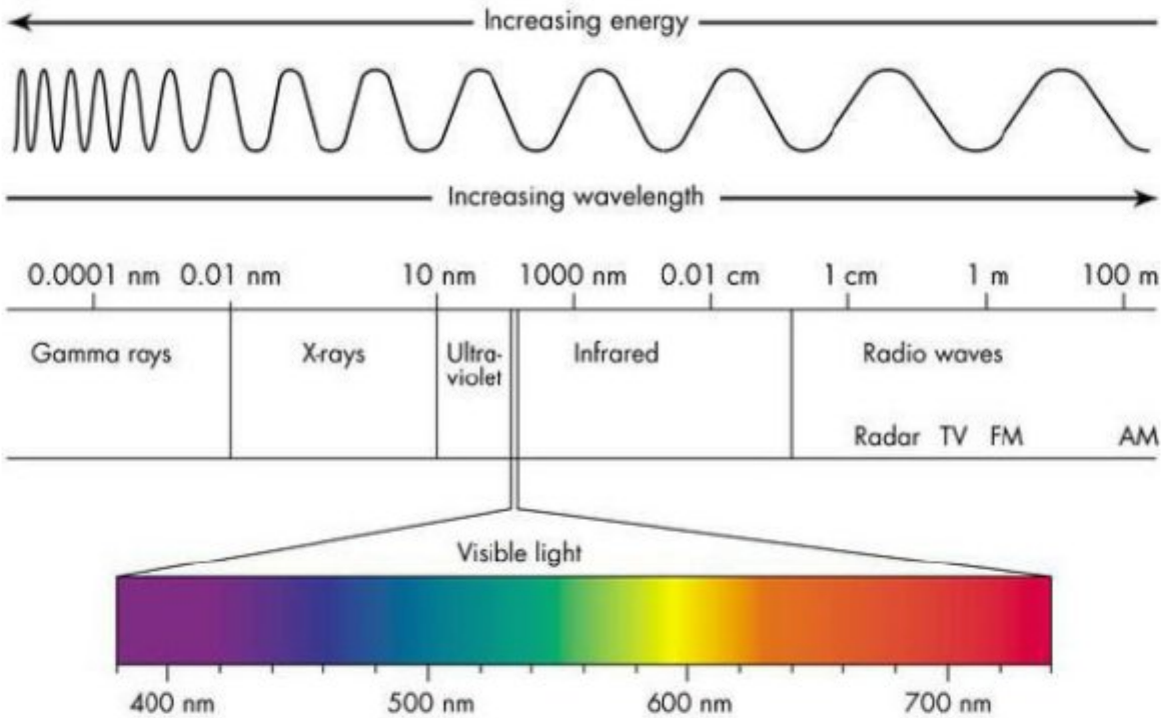
Understanding radiowaves.
The radio-electric transduction phenomenon originating the hertzian radiation, as d

Back to previous page

NEW Hertzian radiation as a convenient means for artificial lighting source.

NEW Hertzian radiation and the *Hallwachs-effect*, better known as the *photovoltai*

NEW Hertzian radiation studies deliver a major blow to the antimatter theory origin.



Frequency / Wavelength / Energy Calculator

Frequency and wavelength of radiations are related to one another through the spee

Equation: $f * \lambda = c$

Equation: $E = {}^{hc}/_{\lambda}$

where:

f = frequency in Hertz (Hz = ${}^{1\ \text{cycle}}/_{\text{sec}}$)

λ = wavelength in meters (m)

c = the speed of light (approximately equal to $3*10^8\ {}^{m}/_{s}$)

E = energy in electron Volts (eV)

h = Planck's constant ($6.626068 \times 10^{-34}\ {}^{m^2 kg}/_{s}$)

To convert wavelength to frequency enter the wavelength in microns (μm) and pres
GHz.

OR enter the frequency in gigahertz (GHz) and press "Calculate λ and E" to convert

Wavelength will be in µm.

OR enter the energy in eV and click "Calculate λ and F" and the values will appear i

Wavelength: (λ) 0 **[µm]**

Frequency: (f) 0 **[GHz]**

Calculate f and E

Calc

Radiation type:

R

eV to Watt per second converter

1 Watt-second (W · s) = 1 Joule (J)= 6.241 509 126 x 10^{18} electronVolts (

Energy (eV) **to**

6.241509126e+18

Reset!

Power to photon rate converter.

Power	0	W (kg·m²/s²)
Wavelength	0	m
Photons	Wait	
	per	s

Calculate! Reset!

Energy of a single photon at a given frequency

E = h·f

where:

f = frequency in Hertz (conversion table is at the bottom of this page)

E = energy in electron Volts (eV)

h = Planck's constant ($6.626068 \times 10^{-34}\ ^{m^2kg}/_s$)

f = frequency in Hertz (an equivalence calculator is at the bottom of this page)

Frequency (Hz)

6.626068e-34

Reset!

Power level variation by dB value

Power level (Watt) enter power value 1

Gain or attenuation (dB or -dB) enter dB or -dB value 0

Resulting power level

dBW = Watts

Calculate

Clear Values

Path loss in free space calculator

Select a distance unit:	distance in meters distance in km distance in million km distance in lightyears
Frequency:		Input frequency
Gain TX antenna:	0	Input gain TX anten
Gain RX antenna:	0	Input gain RX anten
Distance:	1	Calculate
Path loss:		Calculate

Reset!

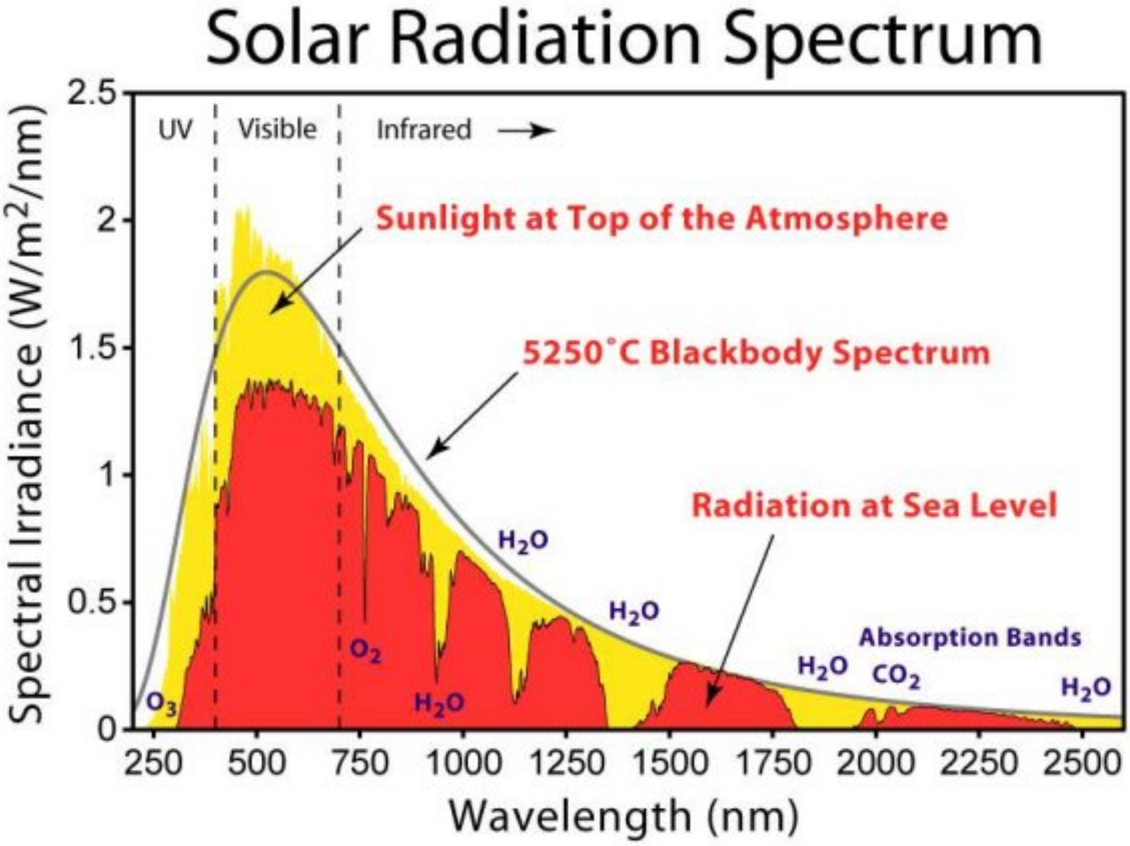
Wavelength to frequency and viceversa multiple output conve

Enter a valid value into a field below. Select a result in any which one fie

SI - Metric system prefixes

FACTOR	...or in full ...	or in words	SI PREFIX	SI SY

1,0E+24	1 000 000 000 000 000 000 000 000	septillion	yotta-	Y
1,0E+21	1 000 000 000 000 000 000 000	sextillion	zetta-	Z
1,0E+18	1 000 000 000 000 000 000	quintillion	exa-	E
1,0E+15	1 000 000 000 000 000	quadrillion	peta-	P
1,0E+12	1 000 000 000 000	trillion	tera-	T
1,0E+9	1 000 000 000	billion	giga-	G
1,0E+6	1 000 000	million	mega-	M
1,0E+3	1 000	thousand	kilo-	k
1,0E+2	100	hundred	hecto-	h
1,0E+1	10	ten	deca-	da
1,0E-1	0,1	tenth	deci-	d
1,0E-2	0,01	hundredth	centi-	c
1,0E-3	0,001	thousandth	milli-	m
1,0E-6	0,000 001	millionth	micro-	µ
1,0E-9	0,000 000 001	billionth	nano-	n
1,0E-12	0,000 000 000 001	trillionth	pico-	p
1,0E-15	0,000 000 000 000 001	quadrillionth	femto-	f
1,0E-18	0,000 000 000 000 000 001	quintillionth	atto-	a
1,0E-21	0,000 000 000 000 000 000 001	sextillionth	zepto-	z
1,0E-24	0,000 000 000 000 000 000 000 001	septillionth	yocto-	y


Solar Radiation Spectrum
UV
Visible
Infrared
Sunlight at Top of the Atmosphere
5250°C Blackbody Spectrum
Radiation at Sea Level
Absorption Bands
O_3
O_2
H_2O
CO_2
Spectral Irradiance ($W/m^2/nm$)
Wavelength (nm)
2.5
2
1.5
1
0.5
0
250
500
750
1000
1250
1500
1750
2000
2250
2500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without harming a single human being

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37. RIVELATORE PASSIVO DI RADIAZIONE HERTZIANA A FLUORESCENZA DA IONIZZAZIONE SECONDARIA - Radioluminescenza

detector_it.htm

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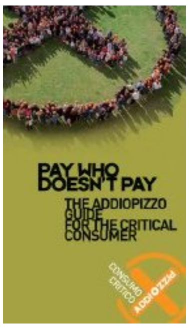

NEW La radiazione hertziana e l'*effetto Hallwachs*, meglio noto come *effetto fotoelettrico*.

NEW Gli studi sulla *radiazione hertziana* assestano un serio colpo alle origini della teoria dell'antimateria

La radiazione hertziana: cos'� e come avviene

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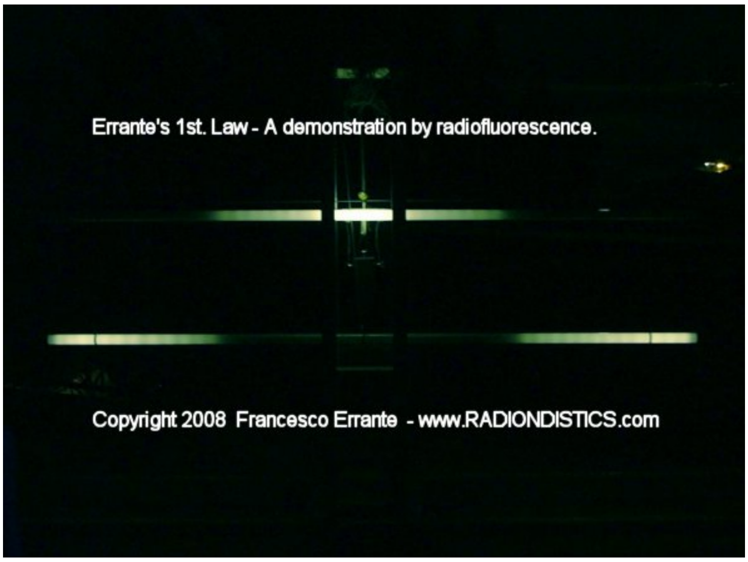
Il nodo di terra virtuale

Sistema di messa a terra capacitiva

RIVELATORE PASSIVO DI RADIAZIONE HERTZIANA A FLUORESCENZA DA IONIZZAZIONE SECONDARIA

La *RADIOLUMINESCENZA* nello spettro hertziano

Il POTERE IONIZZANTE DELLA *RADIAZIONE HERTZIANA* (Meglio nota come onde radio)



BREVETTO INDUSTIALE n.0001347487 (ITAPA20030022)
Visita www.uibm.gov.it ed immetti *Francesco Errante* nel campo di ricerca per vedere tutti i miei brevetti, o verifica direttamente sul sito dell'***European Patent Office*** QUI !

per RF

Le linee di trasmissione bilanciate in regime di onda progressiva

Apparecchi di illuminazione a R.F.

�

RIVELATORE PASSIVO DI RADIAZIONE HERTZIANA A FLUORESCENZA DA IONIZZAZIONE SECONDARIA

by *Francesco Errante*

Descrizione

La "*radiazione hertziana*" � uno dei modi meno in voga, ma pi� antichi per riferirsi alle emissioni nello spettro delle radio-onde. La presente invenzione concerne un rivelatore passivo a fluorescenza da ionizzazione secondaria, intrinsecamente non-invasivo, a bombardamento diretto, per radiazioni hertziane, capace di convertire la radiazione hertziana in radiazione luminosa nello spettro del visibile, consentendo l'apprezzamento visivo della distribuzione energetica sul corpo di un'antenna radio o su di una linea di trasmissione e nello spazio in loro prossimit� durante la sua trasmissione di segnali radioelettrici nello spazio. Il rivelatore passivo di radiazione hertziana qui presentato permette di essere posto a contatto dell'antenna od nella sua immediata prossimit� senza che lo stesso ne disturbi il suo comportamento, grazie al fatto che esso � radio-trasparente. Esso produce un'emissione luminosa nello spettro del visibile, dovuta a fluorescenza da ionizzazione secondaria provocata dalla radiazione hertziana sui gas presenti all'interno del tubo rivelatore stesso. La sua emissione luminosa cresce al crescere dell'intensit� del flusso della radiazione che lo investe.
Un rivelatore passivo di radiazione hertziana nel suo allestimento pi� elementare e pi� economico, consiste di un tubo fluorescente(1) ai vapori di mercurio, del tipo per illuminazione ambientale da 100W, della lunghezza di 2,4 metri ed un carrello su

ruote(2), costruito con materiale dielettrico che permette al tubo rivelatore(1) di essere mantenuto in posizione verticale e di poter scorrere lungo il corpo dell'antenna radio(3) sotto esame, posta orizzontalmente rispetto al suolo ed ad un livello tale da incontrare il tubo rivelatore, pressappoco in coincidenza del centro dell'altezza di quest'ultimo.
In genere, � sufficiente una potenza di poco superiore ai 40 Watt RF, perch� un radiatore risonante per onde corte come un dipolo aperto a � onda, inneschi la fluorescenza in tutto il tubo. Il tubo rivelatore una volta innescato permette, riducendo progressivamente la potenza fornita al radiatore sotto esame o scostando lo stesso dall'antenna, di apprezzare visivamente l'intensit� del campo prodotto dall'antenna in ogni suo punto e nello spazio in sua stretta prossimit�. Una volta innescato, il tubo rivelatore continuer� a mantenere la sua fluorescenza fino al cessare od al diminuire drasticamente della radiazione hertziana.
Riducendo progressivamente la potenza di alimentazione dell'antenna eccitatrice(3), si riduce corrispondentemente anche la lunghezza della regione di tubo rivelatore illuminata e la sua emissione luminosa si estingue non appena la potenza di eccitazione scende al di sotto di 0,1 Watt circa.
Con l'uso di pi� tubi rivelatori, � possibile creare un diagramma luminoso dinamico, bidimensionale o tridimensionale, dell'intensit� del campo irradiato dall'antenna, ma � consigliabile invece costruire il diagramma attraverso il montaggio di pi� fotogrammi dello stesso tubo su vari punti dell'antenna. Ci� � facilmente realizzabile facendo scorrere il tubo, una volta innescato, su vari punti dell'antenna in esame o dello spazio nell'immediata prossimit� dell'antenna in esame. Fino al raggiungimento della saturazione luminosa del

tubo, ad uguale intensit� della radiazione luminosa da esso emessa, corrisponde uguale intensit� della radiazione hertziana che ne sostiene l'eccitazione a quel livello. E'possibile, pertanto, stimare anche l'attenuazione della radiazione hertziana introdotta dallo spazio in stretta prossimit� dell'antenna.

E' stato cos� inventato un rivelatore passivo di radiazione hertziana a fluorescenza da ionizzazione secondaria che permette di compiere indagini visive dirette sul comportamento delle antenne radio e delle radiazioni da esse generate tramite lo sfruttamento del **potere ionizzante delle radio-onde, qui implicitamente dimostrato**.

Funzioni scientifiche

a. Il rivelatore passivo di radiazione hertziana a fluorescenza da ionizzazione secondaria qui descritto consente di dimostrare il potere ionizzante transitorio della radiazione hertziana.

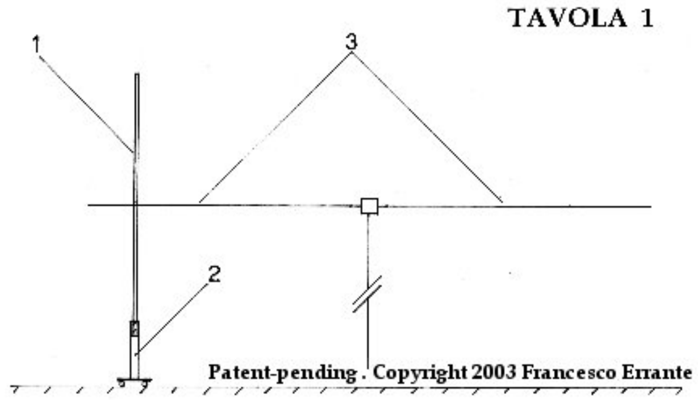
b. Il rivelatore passivo di radiazione hertziana a fluorescenza da ionizzazione secondaria qui descritto consente di apprezzare ad occhio nudo la distribuzione energetica della radiazione hertziana in un'antenna radio in fase di trasmissione.

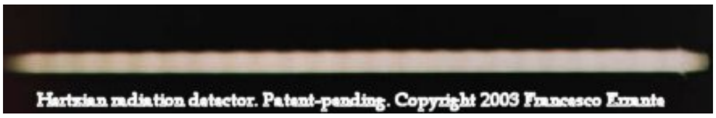
c. Il rivelatore passivo di radiazione hertziana a fluorescenza da ionizzazione secondaria qui descritto consente di apprezzare ad occhio nudo l'intensit� relativa e la distribuzione energetica di una radiazione hertziana nella regione di spazio immediatamente circostante un'antenna radio in fase di trasmissione.

d. Il rivelatore passivo di radiazione hertziana a

fluorescenza da ionizzazione secondaria qui descritto consente di vedere che � sempre la parte terminale di un radiatore, ovvero quella opposta al punto di alimentazione, ad essere responsabile per la gran parte della radiazione hertziana da esso emessa. La distribuzione energetica della radiazione all'atto della fuoriuscita dal radiatore NON � sinusoidale. E' ragionevole ipotizzare che il corpo del radiatore stesso funga da rampa per gran parte della sua lunghezza e che la radiazione avvenga soltanto alla sua estremita', ove le cariche elettriche non potendo pi� avanzare ritornano in dietro per scontrarsi con un nuovo fronte d'onda proveniente dal punto di alimentazione del radiatore stesso. Non � possibile ammettere che lo *scattering* avvenga soltanto perche' le cariche non possano pi� proseguire nel loro percorso una volta raggiunta la fine del conduttore e ci� perch�, altrimenti, non si avrebbe il fenomeno della risonanza. Ne deriva che la risonanza puo' concepirsi come un regime para-stazionario o controllato.
Leggi anche: La Radiazione Hertziana: un'osservazione radioscopica, di *Francesco Errante*

e. Il rivelatore passivo di radiazione hertziana a fluorescenza da ionizzazione secondaria qui descritto consente di dimostrare come i segnali radioelettrici impiegati nell'eccitazione per bombardamento diretto di tubi fluorescenti convenzionali per illuminazione, permettano agli stessi un'efficienza e versatilit� di molto superiore al metodo tradizionale di eccitazione per attraversamento galvanico da corrente alternata a bassa frequenza.





Vista parziale del tubo innescato a bassa potenza

Il rivelatore � totalmente radio-trasparente. Una volta innescato, esso diviene leggermente meno radio-trasparente. Questo, comunque, non ha effetti sul suo funzionamento, n� su quello del radiatore sotto esame. Nel caso peggiore, per esempio, nel caso di uno stretto accoppiamento parallelo completo tra radiatore e tubo rivelato apparir� un ROS di 1:1,6 a tubo completamente illuminato.

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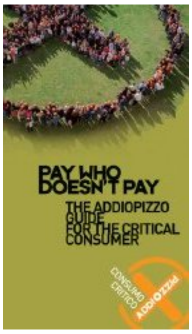
Richard P. Feynman

The Development of the Space-Time View of Quantum Electrodynamics

Nobel Lecture, December 11, 1965

We have a habit in writing articles published in scientific journals to make the work as finished as possible, to cover all the tracks, to not worry about the blind alleys or to describe how you had the wrong idea first, and so on. So there isn't any place to publish, in a dignified manner, what you actually did in order to get to do the work, although, there has been in these days, some interest in this kind of thing. Since winning the prize is a personal thing, I thought I could be excused in this particular situation, if I were to talk personally about my relationship to quantum electrodynamics, rather than to discuss the subject itself in a refined and finished fashion. Furthermore, since there are three people who have won the prize in physics, if they are all going to be talking about quantum electrodynamics itself, one might become bored with the subject. So, what I would like to tell you about today are the sequence of events, really the sequence of ideas, which occurred, and by which I finally came out the other end with an unsolved problem for which I ultimately received a prize.

I realize that a truly scientific paper would be of greater value, but such a paper I could publish in regular journals. So, I shall use this Nobel Lecture as an opportunity to do something of less value, but which I cannot do elsewhere. I ask your indulgence in another manner. I shall include details of anecdotes which are of no value either scientifically, nor for understanding the development of ideas. They are included only to make the lecture more entertaining.



I worked on this problem about eight years until the final publication in 1947. The beginning of the thing was at the Massachusetts Institute of Technology, when I was an undergraduate student reading about the known physics, learning slowly about all these things that people were worrying about, and realizing ultimately that the fundamental problem of the day was that the quantum theory of electricity and magnetism was not completely satisfactory. This I gathered from books like those of Heitler and Dirac. I was inspired by the remarks in these books; not by the parts in which everything was proved and demonstrated carefully and calculated, because I couldn't understand those very well. At the young age what I could understand were the remarks about the fact that this doesn't make any sense, and the last sentence of the book of Dirac I can still remember, "It seems that some essentially new physical ideas are here needed." So, I had this as a challenge and an inspiration. I also had a personal feeling, that since they didn't get a satisfactory answer to the problem I wanted to solve, I don't have to pay a lot of attention to what they did do.

I did gather from my readings, however, that two things were the source of the difficulties with the quantum electrodynamical theories. The first was an infinite energy of interaction of the electron with itself. And this difficulty existed even in the classical theory. The other difficulty came from some infinites which had to do with the infinite numbers of degrees of freedom in the field. As I understood it at the time (as nearly as I can remember) this was simply the difficulty that if you quantized the harmonic oscillators of the field (say in a box) each oscillator has a ground state energy of (½) and there is an infinite number of modes in a box of every increasing frequency w, and therefore there is an infinite energy in the box. I now realize that that wasn't a completely correct statement of the central problem; it can be removed simply by changing the zero from which energy is measured. At any rate, I believed that the difficulty arose somehow from a combination of the electron acting on itself and the infinite number of degrees of freedom of the field.

Well, it seemed to me quite evident that the idea that a particle acts on itself, that the electrical force acts on the same particle that generates it, is not a necessary one - it is a sort of a silly one, as a matter of fact. And, so I suggested to myself, that electrons cannot act on themselves, they can only act on other electrons. That means there is no field at all. You see, if all charges contribute to making a single common field, and if that common field acts back on all the charges, then each charge must act back on itself. Well, that was where the mistake was, there was no field. It was just that when you shook one charge, another would shake later. There was a direct interaction between charges, albeit with a delay. The law of force connecting the motion of one charge with another would just involve a delay. Shake this one, that one shakes later. The sun atom shakes; my eye electron shakes eight minutes later, because of a direct interaction across.

Now, this has the attractive feature that it solves both problems at once. First, I can say immediately, I don't let the electron act on itself, I just let this act on that, hence, no self-energy! Secondly, there is not an infinite number of degrees of freedom in the field. There is no field at all; or if you insist on thinking in terms of ideas like that of a field, this field is always completely determined by the action of the particles which produce it. You shake this particle, it shakes that one, but if you want to think in a field way, the field, if it's there, would be entirely determined by the matter which generates it, and therefore, the field does not have any *independent* degrees of freedom and the infinities from the degrees of freedom would then be removed. As a matter of fact, when we look out anywhere and see light, we can always "see" some matter as the source of the light. We don't just see light (except recently some radio reception has been found with no apparent material source).

You see then that my general plan was to first solve the classical problem, to get rid of the infinite self-energies in the classical theory, and to hope that when I made a quantum theory of it, everything would just be fine.

That was the beginning, and the idea seemed so obvious to me and so elegant that I fell deeply in love with it. And, like falling in love with a woman, it is only possible if you do not know much about her, so you cannot see her faults. The faults will become apparent later, but after the love is strong enough to hold you to her. So, I was held to this theory, in spite of all difficulties, by my youthful enthusiasm.

Then I went to graduate school and somewhere along the line I learned what was wrong with the idea that an electron does not act on itself. When you accelerate an electron it radiates energy and you have to do extra work to account for that energy. The extra force against which this work is done is called the force of radiation resistance. The origin of this extra force was identified in those days, following Lorentz, as the action of the electron itself. The first term of this action, of the electron on itself, gave a kind of inertia (not quite relativistically satisfactory). But that inertia-like term was infinite for a point-charge. Yet the next term in the sequence gave an energy loss rate, which for a point-charge agrees exactly with the rate you get by calculating how much energy is radiated. So, the force of radiation resistance, which is absolutely necessary for the conservation of energy would disappear if I said that a charge could not act on itself.

So, I learned in the interim when I went to graduate school the glaringly obvious fault of my own theory. But, I was still in love with the original theory, and was still thinking that with it lay the solution to the difficulties of quantum electrodynamics. So, I continued to try on and off to save it somehow. I must have some action develop on a given electron when I accelerate it to account for radiation resistance. But, if I let electrons only act on other electrons the only possible source for this action is another electron in the world. So, one day, when I was working for Professor Wheeler and could no longer solve the problem that he had given me, I thought about this again and I calculated the following. Suppose I have two charges - I shake the first charge, which I think of as a source and this makes the second one shake, but the second one shaking produces an effect back on the source. And so, I calculated how much that effect back on the first charge was, hoping it might add up the force of radiation resistance. It didn't come out right, of course, but I went to Professor Wheeler and told him my ideas. He said, - yes, but the answer you get for the problem with the two charges that you just mentioned will, unfortunately, depend upon the charge and the mass of the second charge and will vary inversely as the square of the distance R, between the charges, while the force of radiation resistance depends on none of these things. I thought, surely, he had computed it himself, but now having become a professor, I know that one can be wise enough to see immediately what some graduate student takes several weeks to develop. He also pointed out something that also bothered me, that if we had a situation with many charges all around the original source at roughly uniform density and if we added the effect of all the surrounding charges the inverse R square would be compensated by the R^2 in the volume element and we would get a result proportional to the thickness of the layer, which would go to infinity. That is, one would have an infinite total effect back at the source. And, finally he said to me, and you forgot something else, when you accelerate the first charge, the second acts later, and then the reaction back here at the source would be still later. In other words, the action occurs at the wrong time. I suddenly realized what a stupid fellow I am, for what I had described and calculated was just ordinary reflected light, not radiation reaction.

But, as I was stupid, so was Professor Wheeler that much more clever. For he then went on to give a lecture as though he had worked this all out before and was completely prepared, but he had not, he worked it out as he went along. First, he said, let us suppose that the return action by the charges in the absorber reaches the source by advanced waves as well as by the ordinary retarded waves of reflected light; so that the law of interaction acts backward in time, as well as forward in time. I was enough of a physicist at that time not to say, "Oh, no, how could that be?" For today all physicists know from studying Einstein and Bohr, that sometimes an idea which looks completely paradoxical at first, if analyzed to completion in all detail and in experimental situations, may, in fact, not be paradoxical. So, it did not bother me any more than it bothered Professor Wheeler to use advance waves for the back reaction -

a solution of Maxwell's equations, which previously had not been physically used.

Professor Wheeler used advanced waves to get the reaction back at the right time and then he suggested this: If there were lots of electrons in the absorber, there would be an index of refraction n, so, the retarded waves coming from the source would have their wave lengths slightly modified in going through the absorber. Now, if we shall assume that the advanced waves come back from the absorber without an index - why? I don't know, let's assume they come back without an index - then, there will be a gradual shifting in phase between the return and the original signal so that we would only have to figure that the contributions act as if they come from only a finite thickness, that of the first wave zone. (More specifically, up to that depth where the phase in the medium is shifted appreciably from what it would be in vacuum, a thickness proportional to $l/(n-1)$.) Now, the less the number of electrons in here, the less each contributes, but the thicker will be the layer that effectively contributes because with less electrons, the index differs less from 1. The higher the charges of these electrons, the more each contribute, but the thinner the effective layer, because the index would be higher. And when we estimated it, (calculated without being careful to keep the correct numerical factor) sure enough, it came out that the action back at the source was completely independent of the properties of the charges that were in the surrounding absorber. Further, it was of just the right character to represent radiation resistance, but we were unable to see if it was just exactly the right size. He sent me home with orders to figure out exactly how much advanced and how much retarded wave we need to get the thing to come out numerically right, and after that, figure out what happens to the advanced effects that you would expect if you put a test charge here close to the source? For if all charges generate advanced, as well as retarded effects, why would that test not be affected by the advanced waves from the source?

I found that you get the right answer if you use half-advanced and half-retarded as the field generated by each charge. That is, one is to use the solution of Maxwell's equation which is symmetrical in time and that the reason we got no advanced effects at a point close to the source in spite of the fact that the source was producing an advanced field is this. Suppose the source s surrounded by a spherical absorbing wall ten light seconds away, and that the test charge is one second to the right of the source. Then the source is as much as eleven seconds away from some parts of the wall and only nine seconds away from other parts. The source acting at time t=0 induces motions in the wall at time +10. Advanced effects from this can act on the test charge as early as eleven seconds earlier, or at t= -1. This is just at the time that the direct advanced waves from the source should reach the test charge, and it turns out the two effects are exactly equal and opposite and cancel out! At the later time +1 effects on the test charge from the source and from the walls are again equal, but this time are of the same sign and add to convert the half-retarded wave of the source to full retarded strength.

Thus, it became clear that there was the possibility that if we assume all actions are *via* half-advanced and half-retarded solutions of Maxwell's equations and assume that all sources are surrounded by material absorbing all the the light which is emitted, then we could account for radiation resistance as a direct action of the charges of the absorber acting back by advanced waves on the source.

Many months were devoted to checking all these points. I worked to show that everything is independent of the shape of the container, and so on, that the laws are exactly right, and that the advanced effects really cancel in every case. We always tried to increase the efficiency of our demonstrations, and to see with more and more clarity why it works. I won't bore you by going through the details of this. Because of our using advanced waves, we also had many apparent paradoxes, which we gradually reduced one by one, and saw that there was in fact no logical difficulty with the theory. It was perfectly satisfactory.

We also found that we could reformulate this thing in another way, and that is by a principle of least action. Since my original plan was to describe everything directly in terms of particle motions, it was my desire to represent this new theory without saying anything about fields. It turned out that we found a form for an action directly involving the motions of the charges only, which upon variation would give the equations of motion of these charges. The expression for this action A is

$$A = \Sigma_i m_i \int \left(\dot{X}^i_\mu \dot{X}^i_\mu\right)^{\frac{1}{2}} d\alpha_i + \frac{1}{2} \sum_{\substack{ij \\ i \neq j}} e_i e_j \int\int \delta(I_{ij}^2)\, \dot{X}^i_\mu(\alpha_i)\, \dot{X}^j_\mu(\alpha_j)\, d\alpha_i d\alpha_j \quad (1)$$

where

$$I_{ij}^2 = [X^i_\mu(\alpha_i) - X^j_\mu(\alpha_j)]\,[X^i_\mu(\alpha_i) - X^j_\mu(\alpha_j)]$$

where $\dot{X}^i_\mu(\alpha_i)$ is the four-vector position of the *i*th particle as a function of some parameter α_i, $\dot{X}^i_\mu(\alpha_i)$ is $dX^i_\mu(\alpha)/d\alpha_i$. The first term is the integral of proper time, the ordinary action of relativistic mechanics of free particles of mass m_i. (We sum in the usual way on the repeated index m.) The second term represents the electrical interaction of the charges. It is summed over each pair of charges (the factor ½ is to count each pair once, the term *i=j* is omitted to avoid self-action) .The interaction is a double integral over a delta function of the square of space-time interval I^2 between two points on the paths. Thus, interaction occurs only when this interval vanishes, that is, along light cones.

The fact that the interaction is exactly one-half advanced and half-retarded meant that we could write such a principle of least action, whereas interaction *via* retarded waves alone cannot be written in such a way.

So, all of classical electrodynamics was contained in this very simple form. It looked good, and therefore, it was undoubtedly true, at least to the beginner. It automatically gave half-advanced and half-retarded effects and it was without fields. By omitting the term in the sum when *i=j*, *I* omit self-interaction and no longer have any infinite self-energy. This then was the hoped-for solution to the problem of ridding classical electrodynamics of the infinities.

It turns out, of course, that you can reinstate fields if you wish to, but you have to keep track of the field produced by each particle separately. This is because to find the right field to act on a given particle, you must exclude the field that it creates itself. A single universal field to which all contribute will not do. This idea had been suggested earlier by Frenkel and so we called these Frenkel fields. This theory which allowed only particles to act on each other was equivalent to Frenkel's fields using half-advanced and half-retarded solutions.

There were several suggestions for interesting modifications of electrodynamics. We discussed lots of them, but I shall report on only one. It was to replace this delta function in the interaction by another function, say, $f(I^2_{ij})$, which is not infinitely sharp. Instead of having the action occur only when the interval between the two charges is exactly zero, we would replace the delta function of I^2 by a narrow peaked thing. Let's say that $f(Z)$ is large only near Z=0 width of order a^2. Interactions will now occur when T^2-R^2 is of order a^2 roughly where T is the time difference and R is the separation of the charges. This might look like it disagrees with experience, but if a is some small distance, like 10^{-13} cm, it says that the time delay T in action is roughly $\sqrt{R^2 \pm a^2}$ or approximately, - if R is much larger than a, $T=R\pm a^2/2R$. This means that the deviation of time T from the ideal theoretical time R of Maxwell, gets smaller and smaller, the further the pieces are apart. Therefore, all theories involving in analyzing generators, motors, etc., in fact, all of the tests of electrodynamics that were available in Maxwell's time, would be adequately satisfied if were 10^{-13} cm. If R is of the order of a centimeter this deviation in T is only 10^{-26} parts. So, it was possible, also, to change the theory in a simple manner and to still agree with all observations of classical electrodynamics. You have no clue of precisely what function to put in for f, but it was an interesting possibility to keep in mind when developing quantum electrodynamics.

It also occurred to us that if we did that (replace d by f) we could not reinstate the term i=j in the sum because this would now represent in a relativistically invariant fashion a finite action of a charge on itself. In fact, it was possible to prove that if we did do such a thing, the main effect of the self-action (for not too rapid accelerations) would be to produce a modification of the mass. In fact, there need be no mass m_i, term, all the mechanical mass could be electromagnetic self-action. So, if you would like, we could also have another theory with a still simpler expression for the action A. In expression (1) only the second term is kept, the sum extended over all i and j, and some function replaces d. Such a simple form could represent all of classical electrodynamics, which aside from gravitation is essentially all of classical physics.

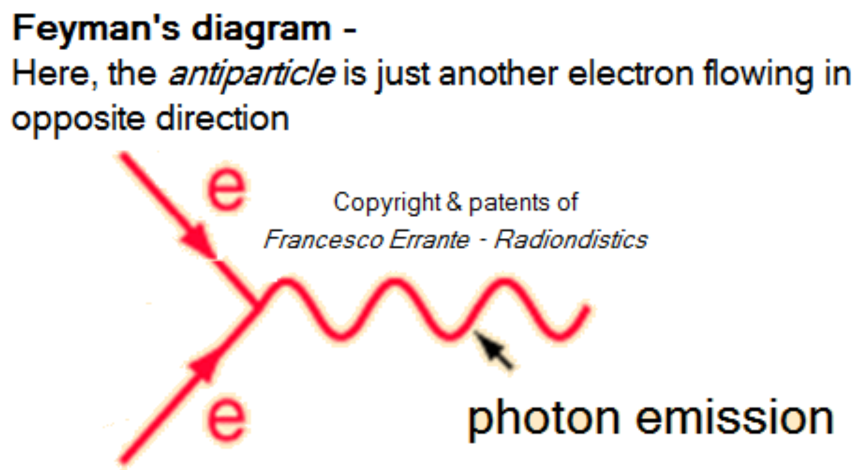
Although it may sound confusing, I am describing several different alternative theories at once. The important thing to note is that at this time we had all these in mind as different possibilities. There were several possible solutions of the difficulty of classical electrodynamics, any one of which might serve as a good starting point to the solution of the difficulties of quantum electrodynamics.

I would also like to emphasize that by this time I was becoming used to a physical point of view different from the more customary point of view. In the customary view, things are discussed as a function of time in very great detail. For example, you have the field at this moment, a differential equation gives you the field at the next moment and so on; a method, which I shall call the Hamilton method, the time differential method. We have, instead (in (1) say) a thing that describes the character of the path throughout all of space and time. The behavior of nature is determined by saying her whole spacetime path has a certain character. For an action like (1) the equations obtained by variation (of $X^i m (a_i)$) are no longer at all easy to get back into Hamiltonian form. If you wish to use as variables only the coordinates of particles, then you can talk about the property of the paths - but the path of one particle at a given time is affected by the path of another at a different time. If you try to describe, therefore, things differentially, telling what the present conditions of the particles are, and how these present conditions will affect the future you see, it is impossible with particles alone, because something the particle did in the past is going to affect the future.

Therefore, you need a lot of bookkeeping variables to keep track of what the particle did in the past. These are called field variables. You will, also, have to tell what the field is at this present moment, if you are to be able to see later what is going to happen. From the overall space-time view of the least action principle, the field disappears as nothing but bookkeeping variables insisted on by the Hamiltonian method.

As a by-product of this same view, **I received a telephone call one day at the graduate college at Princeton from *Professor (John Archibald) Wheeler*, in which he said, *"Feynman, I know why all electrons have the same charge and the same mass"* "Why?" *"Because, they are all the same electron!"* And, then he explained on the telephone, *"suppose that the world lines which we were ordinarily considering before in time and space - instead of only going up in time were a tremendous knot, and then, when we cut through the knot, by the plane***

corresponding to a fixed time, we would see many, many world lines and that would represent many electrons, except for one thing. If in one section this is an ordinary electron world line, in the section in which it reversed itself and is coming back from the future we have the wrong sign to the proper time - to the proper four velocities - and that's equivalent to changing the sign of the charge, and, therefore, that part of a path would act like a positron." "But, Professor", I said, "there aren't as many positrons as electrons." "Well, maybe they are hidden in the protons or something", he said. I did not take the idea that all the electrons were the same one from him as seriously as **I took the observation that positrons could simply be represented as electrons going from the future to the past** in a back section of their world lines. That, I stole!



***Errante*'s radio-electric transduction mechanism is fully consistent with *Prof. John Archibald Wheeler*'s own intuition.**

To summarize, when I was done with this, as a physicist I had gained two things. One, I knew many different ways of formulating classical electrodynamics, with many different mathematical forms. I got to know how to express the subject every which way. Second, I had a point of view - the overall space-time point of view - and a disrespect for the Hamiltonian method of describing physics.

I would like to interrupt here to make a remark. The fact that electrodynamics can be written in so many ways - the differential equations of Maxwell, various minimum principles with fields, minimum principles without fields, all different kinds of ways, was something I knew, but I have never understood. It always seems odd to me that the fundamental laws of physics, when discovered, can appear in so many different forms that are not apparently identical at first, but, with a little mathematical fiddling you can show the relationship. An example of that is the Schrödinger equation and the Heisenberg formulation of quantum mechanics. I don't know why this is - it remains a mystery, but it was something I learned from experience. There is always another way to say the same thing that doesn't look at all like the way you said it before. I don't know what the reason for this is. I think it is somehow a representation of the simplicity of nature. A thing like the inverse square law is just right to be represented by the solution of Poisson's equation, which, therefore, is a very different way to say the same thing that doesn't look at all like the way you said it before. I don't know what it means, that nature chooses these curious forms, but maybe that is a way of defining simplicity. Perhaps a thing is simple if you can describe it fully in several different ways without immediately knowing that you are describing the same thing.

I was now convinced that since we had solved the problem of classical electrodynamics (and completely in accordance with my program from M.I.T., only direct interaction between particles, in a way that made fields unnecessary) that everything was definitely going to be all right. I was convinced that all I had to do was make a quantum theory analogous to the classical one and everything would be solved.

So, the problem is only to make a quantum theory, which has as its classical analog, this expression (1). Now, there is no unique way to make a quantum theory from classical mechanics, although all the textbooks make believe there is. What they would tell you to do, was find the momentum variables and replace them by $(\hbar/i)(\partial/\partial x)$, but I couldn't find a momentum variable, as there wasn't any.

The character of quantum mechanics of the day was to write things in the famous Hamiltonian way - in the form of a differential equation, which described how the wave function changes from instant to instant, and in terms of an operator, H. If the classical physics could be reduced to a Hamiltonian form, everything was all right. Now, least action does not imply a Hamiltonian form if the action is a function of anything more than positions and velocities at the same moment. If the action is of the form of the integral of a function, (usually called the Lagrangian) of the velocities and positions at the same time

$$S = \int L(\dot{x}, x)\, dt$$

then you can start with the Lagrangian and then create a Hamiltonian and work out the quantum mechanics, more or less uniquely. But this thing (1) involves the key variables, positions, at two different times and therefore, it was not obvious what to do to make the quantum-mechanical analogue.

I tried - I would struggle in various ways. One of them was this; if I had harmonic oscillators interacting with a delay in time, I could work out what the normal modes were and guess that the quantum theory of the normal modes was the same as for simple oscillators and kind of work my way back in terms of the original variables. I succeeded in doing that, but I hoped then to generalize to other than a harmonic oscillator, but I learned to my regret something, which many people have learned. The harmonic oscillator is too simple; very often you can work out what it should do in quantum theory without getting much of a clue as to how to generalize your results to other systems.

So that didn't help me very much, but when I was struggling with this problem, I went to a beer party in the Nassau Tavern in Princeton. There was a gentleman, newly arrived from Europe (Herbert Jehle) who came and sat next to me. Europeans are much more serious than we are in America because they think that a good place to discuss intellectual matters is a beer party. So, he sat by me and asked, "what are you doing" and so on, and I said, "I'm drinking beer." Then I realized that he wanted to know what work I was doing and I told him I was struggling with this problem, and I simply turned to him and said, "listen, do you know any way of doing quantum mechanics, starting with action - where the action integral comes into the quantum mechanics?" "No", he said, "but Dirac has a paper in which the Lagrangian, at least, comes into quantum mechanics. I will show it to you tomorrow."

Next day we went to the Princeton Library, they have little rooms on the side to discuss things, and he showed me this paper. What Dirac said was the following: There is in quantum mechanics a very important quantity which carries the wave function from one time to another, besides the differential equation but equivalent to it, a kind of a kernal, which we might call $K(x', x)$, which carries the wave function j(x) known at time t, to the wave function j(x') at time, $t+e$ Dirac points out that this function K was *analogous* to the quantity in classical mechanics that you would calculate if you took the exponential of ie, multiplied by the Lagrangian $L(\dot{x},x)$ imagining that these two positions x,x' corresponded t and t+e. In other words,

$$K(x',x) \text{ is analogous to } e^{i\varepsilon L(\frac{x'-x}{\varepsilon},x)/\hbar}$$

Professor Jehle showed me this, I read it, he explained it to me, and I said, "what does he mean, they are analogous; what does that mean, *analogous*? What is the use of that?" He said, "you Americans! You always want to find a use for everything!" I said, that I thought that Dirac must mean that they were equal. "No", he explained, "he doesn't mean they are equal." "Well", I said, "let's see what happens if we make them equal."

So I simply put them equal, taking the simplest example where the Lagrangian is ½Mx^2 - $V(x)$ but soon found I had to put a constant of proportionality A in, suitably adjusted. When I substituted $Ae^{i\epsilon L/\hbar}$ for K to get

$$\psi(x', t+\varepsilon) = \int A \exp\left[\frac{i\varepsilon}{\hbar} L\left(\frac{x'-x}{\varepsilon}, x\right)\right] \psi(x, t)\, dx$$

and just calculated things out by Taylor series expansion, out came the Schrödinger equation. So, I turned to Professor Jehle, not really understanding, and said, "well, you see Professor Dirac meant that they were proportional." Professor Jehle's eyes were bugging out - he had taken out a little notebook and was rapidly copying it down from the blackboard, and said, "no, no, this is an important discovery. You Americans are always trying to find out how something can be used. That's a good way to discover things!" So, I thought I was finding out what Dirac meant, but, as a matter of fact, had made the discovery that what Dirac thought was analogous, was, in fact, equal. I had then, at least, the connection between the Lagrangian and quantum mechanics, but still with wave functions and infinitesimal times.

It must have been a day or so later when I was lying in bed thinking about these things, that I imagined what would happen if I wanted to calculate the wave function at a finite interval later.

I would put one of these factors e^{ieL} in here, and that would give me the wave functions the next moment, t+e and then I could substitute that back into (3) to get another factor of e^{ieL} and give me the wave function the next moment, t+2e and so on and so on. In that way I found myself thinking of a large number of integrals, one after the other in sequence. In the integrand was the product of the exponentials, which, of course, was the exponential of the sum of terms like eL. Now, L is the Lagrangian and e is like the time interval dt, so that if you took a sum of such terms, that's exactly like an integral. That's like Riemann's formula for the integral Ldt, you just take the value at each point and add them together. We are to take the limit as e-0, of course. Therefore, the connection between the wave function of one instant and the wave function of another instant a finite time later could be obtained by an infinite number of integrals, (because e goes to zero, of course) of exponential $^{(iS/\hbar)}$ where S is the action expression (2). At last, I had succeeded in representing quantum mechanics directly in terms of the action S.

This led later on to the idea of the amplitude for a path; that for each possible way that the particle can go from one point to another in space-time, there's an amplitude. That amplitude is e to the times the action for the path. Amplitudes from various paths superpose by addition. This then is another, a third way, of describing quantum mechanics, which looks quite different than that of Schrödinger or Heisenberg, but which is equivalent to them.

Now immediately after making a few checks on this thing, what I wanted to do, of course, was to substitute the action (1) for the other (2). The first trouble was that I could not get the thing to work with the relativistic case of spin one-half. However, although I could deal with the matter only nonrelativistically, I could deal with the light or the photon interactions perfectly well by just putting the interaction terms of (1) into any action, replacing the mass terms by the non-relativistic $(Mx^2/2)\mathrm{d}t$. When the action has a delay, as it now had, and involved more than one time, I had to lose the idea of a wave function. That is, I could no longer describe the program as; given the amplitude for all positions at a certain time to compute the amplitude at another time. However, that didn't cause very much trouble. It just meant developing a new idea. Instead of wave functions we could talk about this; that if a source of a certain kind emits a particle, and a detector is there to receive it, we can give the amplitude that the source will emit and the detector receive. We do this without specifying the exact instant that the source emits or the exact instant that any detector receives, without trying to specify the state of anything at any particular time in between, but by just finding the amplitude for the complete experiment. And, then we could discuss how that amplitude would change if you had a scattering sample in between, as you rotated and changed angles, and so on, without really having any wave functions.

It was also possible to discover what the old concepts of energy and momentum would mean with this generalized action. And, so I believed that I had a quantum theory of classical electrodynamics - or rather of this new classical electrodynamics described by action (1). I made a number of checks. If I took the Frenkel field point of view, which you remember was more differential, I could convert it directly to quantum mechanics in a more conventional way. The only problem was how to specify in quantum mechanics the classical boundary conditions to use only half-advanced and half-retarded solutions. By some ingenuity in defining what that meant, I found that the quantum mechanics with Frenkel fields, plus a special boundary condition, gave me back this action, (1) in the new form of quantum mechanics with a delay. So, various things indicated that there wasn't any doubt I had everything straightened out.

It was also easy to guess how to modify the electrodynamics, if anybody ever wanted to modify it. I just changed the delta to an *f*, just as I would for the classical case. So, it was very easy, a simple thing. To describe the old retarded theory without explicit mention of fields I would have to write probabilities, not just amplitudes. I would have to square my amplitudes and that would involve double path integrals in which there are two *S*'s and so forth. Yet, as I worked out many of these things and studied different forms and different boundary conditions. I got a kind of funny feeling that things weren't exactly right. I could not clearly identify the difficulty and in one of the short periods during which I imagined I had laid it to rest, I published a thesis and received my Ph.D.

During the war, I didn't have time to work on these things very extensively, but wandered about on buses and so forth, with little pieces of paper, and struggled to work on it and discovered indeed that there was something wrong, something terribly wrong. I found that if one generalized the action from the nice Langrangian forms (2) to these forms (1) then the quantities which I defined as energy, and so on, would be complex. The energy values of stationary states wouldn't be real and probabilities of events wouldn't add up to 100%. That is, if you took the probability that this would happen and that would happen - everything you could think of would happen, it would not add up to one.

Another problem on which I struggled very hard, was to represent relativistic electrons with this new quantum mechanics. I wanted to do a unique and different way - and not just by copying the operators of Dirac into some kind of an expression and using some kind of Dirac algebra instead of ordinary complex numbers. I was very much encouraged by the fact that in one space dimension, I did find a way of giving an amplitude to every path by limiting myself to paths, which only went back and forth at the speed of light. The amplitude was simple (*i*e) to a power equal to the number of velocity reversals where I have divided the time into steps and I am allowed to reverse velocity only at such a time. This gives (as approaches zero) Dirac's equation in two dimensions - one dimension of space and one of time $(\hbar=M=c=1)$.

Dirac's wave function has four components in four dimensions, but in this case, it has only two components and this rule for the amplitude of a path automatically generates the need for two components. Because if this is the formula for the amplitudes of path, it will not do you any good to know the total amplitude of all paths, which come into a given point to find the amplitude to reach the next point. This is because for the next time, if it came in from the right, there is no new factor *i*e if it goes out to the right, whereas, if it came in from the left there was a new factor ie. So, to continue this same information forward to the next moment, it was not sufficient information to know the total amplitude to arrive, but you had to know the amplitude to arrive from the right and the amplitude to arrive to the left, independently. If you did, however, you could then compute both of those again independently and thus you had to carry two amplitudes to form a differential equation (first order in time).

And, so I dreamed that if I were clever, I would find a formula for the amplitude of a path that was beautiful and simple for three dimensions of space and one of time, which would be equivalent to the Dirac equation, and for which the four components, matrices, and all those other mathematical funny things would come out as a simple consequence - I have never succeeded in that either. But, I did want to mention some of the unsuccessful things on which I spent almost as much effort, as on the things that did work.

To summarize the situation a few years after the way, I would say, I had much experience with quantum electrodynamics, at least in the knowledge of many different ways of formulating it, in terms of path integrals of actions and in other forms. One of the important by-products, for example, of much experience in these simple forms, was that it was easy to see how to combine together what was in those days called the longitudinal and transverse fields, and in general, to see clearly the relativistic invariance of the theory. Because of the need to do things differentially there had been, in the standard quantum electrodynamics, a complete split of the field into two parts, one of which is called the longitudinal part and the other mediated by the photons, or transverse waves. The longitudinal part was described by a Coulomb potential acting instantaneously in the Schrödinger equation, while the transverse part had entirely different description in terms of quantization of the transverse waves. This separation depended upon the relativistic tilt of your axes in spacetime. People moving at different velocities would separate the same field into longitudinal and transverse fields in a different way. Furthermore, the entire formulation of quantum mechanics insisting, as it did, on the wave function at a given time, was hard to analyze relativistically. Somebody else in a different coordinate system would calculate the succession of events in terms of wave functions on differently cut slices of space-time, and with a different separation of longitudinal and transverse parts. The Hamiltonian theory did not look relativistically invariant, although, of course, it was. One of the great advantages of the overall point of view, was that you could see the relativistic invariance right away - or as Schwinger would say - the covariance was manifest. I had the advantage, therefore, of having a manifestedly covariant form for quantum electrodynamics with suggestions for modifications and so on. I had the disadvantage that if I took it too seriously - I mean, if I took it seriously at all in this form, - I got into trouble with these complex energies and the failure of adding probabilities to one and so on. I was unsuccessfully struggling with that.

Then Lamb did his experiment, measuring the separation of the $2S½$ and $2P½$ levels of hydrogen, finding it to be about 1000 megacycles of frequency difference. Professor Bethe, with whom I was then associated at Cornell, is a man who has this characteristic: If there's a good experimental number you've got to figure it out from theory. So, he forced the quantum electrodynamics of the day to give him an answer to the separation of these two levels. He pointed out that the self-energy of an electron itself is infinite, so that the calculated energy of a bound electron should also come out infinite. But, when you calculated the separation of the two energy levels in terms of the corrected mass instead of the old mass, it would turn out, he thought, that the theory would give convergent finite answers. He made an estimate of the splitting that way and found out that it was still divergent, but he guessed that was probably due to the fact that he used an unrelativistic theory of the matter. Assuming it would be convergent if relativistically treated, he estimated he would get about a thousand megacycles for the Lamb-shift, and thus, made the most important discovery in the history of the theory of quantum electrodynamics. He worked this out on the train from Ithaca, New York to Schenectady and telephoned me excitedly from Schenectady to tell me the result, which I don't remember fully appreciating at the time.

Returning to Cornell, he gave a lecture on the subject, which I attended. He explained that it gets very confusing to figure out exactly which infinite term corresponds to what in trying to make the correction for the infinite change in mass. If there were any modifications whatever, he said, even though not physically correct, (that is not necessarily the way nature actually works) but any modification whatever at high frequencies, which would make this correction finite, then there would be no problem at all to figuring out how to keep track of everything. You just calculate the finite mass correction Dm to the electron mass m_o, substitute the numerical values of m_o+Dm for m in the results for any other problem and all these ambiguities would be resolved. If, in addition, this method were relativistically invariant, then we would be absolutely sure how to do it without destroying relativistically invariant.

After the lecture, I went up to him and told him, "I can do that for you, I'll bring it in for you tomorrow." I guess I knew every way to modify quantum electrodynamics known to man, at the time. So, I went in next day, and explained what would correspond to the modification of the delta-function to *f* and asked him to explain to me how you calculate the self-energy of an electron, for instance, so we can figure out if it's finite.

I want you to see an interesting point. I did not take the advice of Professor Jehle to find out how it was useful. I never used all that machinery which I had cooked up to solve a single relativistic problem. I hadn't even calculated the self-energy of an electron up to that moment, and was studying the difficulties with the conservation of probability, and so on, without actually doing anything, except discussing the general properties of the theory.

But now I went to Professor Bethe, who explained to me on the blackboard, as we worked together, how to calculate the self-energy of an electron. Up to that time when you did the integrals they had been logarithmically divergent. I told him how to make the relativistically invariant modifications that I thought would make everything all right. We set up the integral which then diverged at the sixth power of the frequency instead of logarithmically!

So, I went back to my room and worried about this thing and went around in circles trying to figure out what was wrong because I was sure physically everything had to come out finite, I couldn't understand how it came out infinite. I became more and more interested and finally realized I had to learn how to make a calculation. So, ultimately, I taught myself how to calculate the self-energy of an electron working my patient way through the terrible confusion of those days of negative energy states and holes and longitudinal contributions and so on. When I finally found out how to do it and did it with the modifications I wanted to suggest, it turned out that it was nicely convergent and finite, just as I had expected. Professor Bethe and I have never been able to discover what we did wrong on that blackboard two months before, but apparently we just went off somewhere and we have never been able to figure out where. It turned out, that what I had proposed, if we had carried it out without making a mistake would have been all right and would have given a finite correction. Anyway, it forced me to go back over all this and to convince myself physically that nothing can go wrong. At any rate, the correction to mass was now finite, proportional to $\ln(ma/\hbar)$ where *a* is the width of that function *f* which was substituted for d. If you wanted an unmodified electrodynamics, you would have to take *a* equal to zero, getting an infinite mass correction. But, that wasn't the point. Keeping a finite, I simply followed the program outlined by Professor Bethe and showed how to calculate all the various things, the scatterings of electrons from atoms without radiation, the shifts of levels and so forth, calculating everything in terms of the experimental mass, and noting that the results as Bethe suggested, were not sensitive to *a* in this form and even had a definite limit as *a*g0.

The rest of my work was simply to improve the techniques then available for calculations, making diagrams to help analyze perturbation theory quicker. Most of this was first worked out by guessing - you see, I didn't have the relativistic theory of matter. For example, it seemed to me obvious that the velocities in non-relativistic formulas have to be replaced by Dirac's matrix a or in the more relativistic forms by the operators γ_μ. I just took my guesses from the forms that I had worked out using path integrals for nonrelativistic matter, but relativistic light. It was easy to develop rules of what to substitute to get the relativistic case. I was very surprised to discover that it was not known at that time, that every one of the formulas that had been worked out so patiently by separating longitudinal and transverse waves could be obtained from the formula for the transverse waves alone, if instead of summing over only the two perpendicular polarization directions you would sum over all four possible directions of polarization. It was so obvious from the action (1) that I thought it was general knowledge and would do it all the time. I would get into arguments with people, because I didn't realize they didn't know that; but, it turned out that all their patient work with the longitudinal waves was always equivalent to just extending the sum on the two transverse directions of polarization over all four directions. This was one of the amusing advantages of the method. In addition, I included diagrams for the various terms of the perturbation series, improved notations to be used, worked out easy ways to evaluate integrals, which occurred in these problems, and so on, and made a kind of handbook on how to do quantum electrodynamics.

But one step of importance that was physically new was involved with the negative energy sea of Dirac, which caused me so much logical difficulty. I got so confused that I remembered Wheeler's old idea about the positron being, maybe, the electron going backward in time. Therefore, in the time dependent perturbation theory that was usual for getting self-energy, I simply supposed that for a while we could go backward in the time, and looked at what terms I got by running the time variables backward. They were the same as the terms that other people got when they did the problem a more complicated way, using holes in the sea, except, possibly, for some signs. These, I, at first, determined empirically by inventing and trying some rules.

I have tried to explain that all the improvements of relativistic theory were at first more or less straightforward, semi-empirical shenanigans. Each time I would discover something, however, I would go back and I would check it so many ways, compare it to every problem that had been done previously in electrodynamics (and later, in weak coupling meson theory) to see if it would always agree, and so on, until I was absolutely convinced of the truth of the various rules and regulations which I concocted to simplify all the work.

During this time, people had been developing meson theory, a subject I had not studied in any detail. I became interested in the possible application of my methods to perturbation calculations in meson theory. But, what was meson theory? All I knew was that meson theory was something analogous to electrodynamics, except that particles corresponding to the photon had a mass. It was easy to guess the d-function in (1), which was a solution of d'Alembertian equals zero, was to be changed to the corresponding solution of d'Alembertian equals m^2. Next, there were different kind of mesons - the one in closest analogy to photons, coupled *via* $\gamma_\mu\gamma_\mu$, are called vector mesons - there were also scalar mesons. Well, maybe that corresponds to putting unity in place of the γ_μ, I would here then speak of "pseudo vector coupling" and I would guess what that probably was. I didn't have the knowledge to understand the way these were defined in the conventional papers because they were expressed at that time in terms of creation and annihilation operators, and so on, which, I had not successfully learned. I remember that when someone had started to teach me about creation and annihilation operators, that this operator creates an electron, I said, "how do you create an electron? It disagrees with the conservation of charge", and in that way, I blocked my mind from learning a very practical scheme of calculation. Therefore, I had to find as many opportunities as possible to test whether I guessed right as to what the various theories were.

One day a dispute arose at a Physical Society meeting as to the correctness of a calculation by Slotnick of the interaction of an electron with a neutron using pseudo scalar theory with pseudo vector coupling and also, pseudo scalar theory with pseudo scalar coupling. He had found that the answers were not the same, in fact, by one theory, the result was divergent, although convergent with the other. Some people believed that the two theories must give the same answer for the problem. This was a welcome opportunity to test my guesses as to whether I really did understand what these two couplings were. So, I went home, and during the evening I worked out the electron neutron scattering for the pseudo scalar and pseudo vector coupling, saw they were not equal and subtracted them, and worked out the difference in detail. The next day at the meeting, I saw Slotnick and said, "Slotnick, I worked it out last night, I wanted to see if I got the same answers you do. I got a different answer for each coupling - but, I would like to check in detail with you because I want to make sure of my methods." And, he said, "what do you mean you worked it out last night, it took me six months!" And, when we compared the answers he looked at mine and he asked, "what is that Q in there, that variable Q?" (I had expressions like $(\tan^{-1}Q)/Q$ etc.). I said, "that's the momentum transferred by the electron, the electron deflected by different angles." "Oh", he said, "no, I only have the limiting value as Q approaches zero; the forward scattering." Well, it was easy enough to just substitute Q equals zero in my form and I then got the same answers as he did. But, it took him six months to do the case of zero momentum transfer, whereas, during one evening I had done the finite and arbitrary momentum transfer. That was a thrilling moment for me, like receiving the Nobel Prize, because that convinced me, at last, I did have some kind of method and technique and understood how to do something that other people did not know how to do. That was my moment of triumph in which I realized I really had succeeded in working out something worthwhile.

At this stage, I was urged to publish this because everybody said it looks like an easy way to make calculations, and wanted to know how to do it. I had to publish it, missing two things; one was proof of every statement in a mathematically conventional sense. Often, even in a physicist's sense, I did not have a demonstration of how to get all of these rules and equations from conventional electrodynamics. But, I did know from experience, from fooling around, that everything was, in fact, equivalent to the regular electrodynamics and had partial proofs of many pieces, although, I never really sat down, like Euclid did for the geometers of Greece, and made sure that you could get it all from a single simple set of axioms. As a result, the work was criticized, I don't know whether favorably or unfavorably, and the "method" was called the "intuitive method". For those who do not realize it, however, I should like to emphasize that there is a lot of work involved in using this "intuitive method" successfully. Because no simple clear proof of the formula or idea presents itself, it is necessary to do an unusually great amount of checking and rechecking for consistency and correctness in terms of what is known, by comparing to other analogous examples, limiting cases, etc. In the face of the lack of direct mathematical demonstration, one must be careful and thorough to make sure of the point, and one should make a perpetual attempt to demonstrate as much of the formula as possible. Nevertheless, a very great deal more truth can become known than can be proven.

It must be clearly understood that in all this work, I was representing the conventional electrodynamics with retarded interaction, and not my half-advanced and half-retarded theory corresponding to (1). I merely use (1) to guess at forms. And, one of the forms I guessed at corresponded to changing d to a function *f* of width a^2, so that I could calculate finite results for all of the problems. This brings me to the second thing that was missing when I published the paper, an unresolved difficulty. With d replaced by *f* the calculations would give results which were not "unitary", that is, for which the sum of the probabilities of all alternatives was not unity. The deviation from unity was very small, in practice, if *a* was very small. In the limit that I took *a* very tiny, it might not make any difference. And, so the process of the renormalization could be made, you could calculate everything in terms of the experimental mass and then take the limit and the apparent difficulty that the unitary is violated temporarily seems to disappear. I was unable to demonstrate that, as a matter of fact, it does.

It is lucky that I did not wait to straighten out that point, for as far as I know, nobody has yet been able to resolve this question. Experience with meson theories with stronger couplings and with strongly coupled vector photons, although not proving anything, convinces me that if the coupling were stronger, or if you went to a higher order (137th order of perturbation theory for electrodynamics), this difficulty would remain in the limit and there would be real trouble. That is, I believe there is really no satisfactory quantum electrodynamics, but I'm not sure. And, I believe, that one of the reasons for the slowness of present-day progress in understanding the *strong* interactions is that there isn't any relativistic theoretical model, from which you can really calculate everything. Although, it is usually said, that the difficulty lies in the fact that strong interactions are too hard to calculate, I believe, it is really because strong interactions in field theory have no solution, have no sense they're either infinite, or, if you try to modify them, the modification destroys the unitarity. I don't think we have a completely satisfactory relativistic quantum-mechanical model, even one that doesn't agree with nature, but, at least, agrees with the logic that the sum of probability of all alternatives has to be 100%. Therefore, I think that the renormalization theory is simply a way to sweep the difficulties of the divergences of electrodynamics under the rug. I am, of course, not sure of that.

This completes the story of the development of the space-time view of quantum electrodynamics. I wonder if anything can be learned from it. I doubt it. It is most striking that most of the ideas developed in the course of this research were not ultimately used in the final result. For example, the half-advanced and half-retarded potential was not finally used, the action expression (1) was not used, the idea that charges do not act on themselves was abandoned. The path-integral formulation of quantum mechanics was useful for guessing at final expressions and at formulating the general theory of electrodynamics in new ways -

although, strictly it was not absolutely necessary. The same goes for the idea of the positron being a backward moving electron, it was very convenient, but not strictly necessary for the theory because it is exactly equivalent to the negative energy sea point of view.

We are struck by the very large number of different physical viewpoints and widely different mathematical formulations that are all equivalent to one another. The method used here, of reasoning in physical terms, therefore, appears to be extremely inefficient. On looking back over the work, I can only feel a kind of regret for the enormous amount of physical reasoning and mathematically re-expression which ends by merely re-expressing what was previously known, although in a form which is much more efficient for the calculation of specific problems. Would it not have been much easier to simply work entirely in the mathematical framework to elaborate a more efficient expression? This would certainly seem to be the case, but it must be remarked that although the problem actually solved was only such a reformulation, the problem originally tackled was the (possibly still unsolved) problem of avoidance of the infinities of the usual theory. Therefore, a new theory was sought, not just a modification of the old. Although the quest was unsuccessful, we should look at the question of the value of physical ideas in developing a *new* theory.

Many different physical ideas can describe the same physical reality. Thus, classical electrodynamics can be described by a field view, or an action at a distance view, etc. Originally, Maxwell filled space with idler wheels, and Faraday with fields lines, but somehow the Maxwell equations themselves are pristine and independent of the elaboration of words attempting a physical description. The only true physical description is that describing the experimental meaning of the quantities in the equation - or better, the way the equations are to be used in describing experimental observations. This being the case perhaps the best way to proceed is to try to guess equations, and disregard physical models or descriptions. For example, McCullough guessed the correct equations for light propagation in a crystal long before his colleagues using elastic models could make head or tail of the phenomena, or again, Dirac obtained his equation for the description of the electron by an almost purely mathematical proposition. A simple physical view by which all the contents of this equation can be seen is still lacking.

Therefore, I think equation guessing might be the best method to proceed to obtain the laws for the part of physics which is presently unknown. Yet, when I was much younger, I tried this equation guessing and I have seen many students try this, but it is very easy to go off in wildly incorrect and impossible directions. I think the problem is not to find the *best* or most efficient method to proceed to a discovery, but to find any method at all. Physical reasoning does help some people to generate suggestions as to how the unknown may be related to the known. Theories of the known, which are described by different physical ideas may be equivalent in all their predictions and are hence scientifically indistinguishable. However, they are not psychologically identical when trying to move from that base into the unknown. For different views suggest different kinds of modifications which might be made and hence are not equivalent in the hypotheses one generates from them in ones attempt to understand what is not yet understood. I, therefore, think that a good theoretical physicist today might find it useful to have a wide range of physical viewpoints and mathematical expressions of the same theory (for example, of quantum electrodynamics) available to him. This may be asking too much of one man. Then new students should as a class have this. If every individual student follows the same current fashion in expressing and thinking about electrodynamics or field theory, then the variety of hypotheses being generated to understand strong interactions, say, is limited. Perhaps rightly so, for possibly the chance is high that the truth lies in the fashionable direction. But, on the off-chance that it is in another direction - a direction obvious from an unfashionable view of field theory - who will find it? Only someone who has sacrificed himself by teaching himself quantum electrodynamics from a peculiar and unusual point of view; one that he may have to invent for himself. I say sacrificed himself because he most likely will get nothing from it, because the truth may lie in another direction, perhaps even the fashionable one.

But, if my own experience is any guide, the sacrifice is really not great because if the peculiar viewpoint taken is truly experimentally equivalent to the usual in the realm of the known there is always a range of applications and problems in this realm for which the special viewpoint gives one a special power and clarity of thought, which is valuable in itself. Furthermore, in the search for new laws, you always have the psychological excitement of feeling that possible nobody has yet thought of the crazy possibility you are looking at right now.

So what happened to the old theory that I fell in love with as a youth? Well, I would say it's become an old lady, that has very little attractive left in her and the young today will not have their hearts pound anymore when they look at her. But, we can say the best we can for any old woman, that she has been a very good mother and she has given birth to some very good children. And, I thank the Swedish Academy of Sciences for complimenting one of them. Thank you.

From *Nobel Lectures, Physics 1963-1970*, Elsevier Publishing Company, Amsterdam, 1972

39. RADIATING SISTEMS WHERE ELEMENTS INTERFERE WHITH EACH OTHER FIELDS - SELF-ANTAGONIST SYSTEMS

self_antagonist.htm

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The self-antagonist (radiating) systems.

Examples of auto-lesionism! The G5RV is one of those.
by *Francesco Errante*

NEVER use a balanced antenna system when the lack of space compels to operate with radiators (radio-electric transducers) having their physical length considerably shorter than the wavelength of the radio signals they are ment to be running on. That's because, by doing so, it will give origin to *self-antagonist* systems and as far as the antenna's gain is concerned, almost nothing else is more

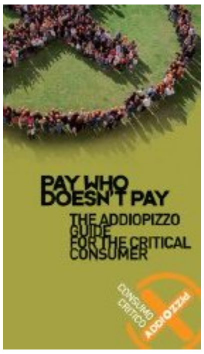
Radio antennas are: radio-electric transducers

�

Hertzian radiation: what it is and how it happens

Balanced transmission

counterproductive than this.
So, what is a *self-antagonist* system ? It is a system where its radiators are fed in a counterphase arrangement (balanced systems) while being so close to each other that they can be considered as two almost conciding point sources (e.g. short center-fed dipoles). Although, those systems can radiate, their fields tend to cancel each other's effects out. This is because each of the system radiators generate its own field, which in turn, at any given point in the space, will induce an electro motive force equally strong to the one induced by its counterpart but with an opposite phase.



An example of such systems is the so-called *G5RV*.

Needlessly to say, that even an array of active and properly resonant full-size elements or the elements of a *Yagi-Uda* beam antenna if deliberately mis-spaced would result in self-antagonist arrangements.

To understand what happens it is useful to know the *theory of the point sources*.
By applying the theory of the point sources to two short monopoles, in a center-fed dipole-like counterphase arrangement, therefore ensuring that they are themselves counterphased with each other, it becomes clear that the fields generated by their hertzian radiation must have, point by point, an angular distance which is much less than 180 degrees. Accordingly, a rule of thumb can be extracted as follows: "**the shorter the monopoles, the smaller the phase difference between the fields generated by them and the grater is their tendency to**

lines in a progressive wave regime

Balanced virtual ground node generator

Errante's 225 Ohm multiband HF antenna system

cancel each other out". This phenomenon is called "*destructive wave interference*".

All the concepts, methods, designs and devices presented on this web site are the original novelty works of *FRANCESCO ERRANTE*.

�

3D wave simulator

A simulation of the interference between 2 point source's field, almost coinciding with each other and fed in a counterphase arrangement, is easily obtainable by running the java applet associated to this page, if properly configured, placing the "phase difference" slide in the middle of its range (180�) and shifting the "source separation" slide towards left (minimum). Observe how the effects of the two field will cancel each other out.

NB: The java applet, to which this page refers to, opens automatically in a new "pop-up" window.

The video below shows the dynamic diagram and terminates with its rotation in the space

It is a simulator that shows waves in three dimensions. It can show point sources, line sources, plane waves and interference between sources.

When the applet starts up you will see red and green waves emanating from two sources in the center of a cubic box. The wave color indicates the fields' magnitude. The green areas are negative and the red areas are positive.

To get started with the applet, just go through the items in the Setup menu in the upper right.

Rotate the box with the mouse to view it from different angles.

NB: To start the applet afresh, simply reload this web page.

Sorry, you need a Java-enabled browser to see the simulation.

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40. © 2003 Francesco Errante

`mismatching_simulator.htm`

This experiment requires a **Java**-enabled Web Browser.

41. TRANSMISSION LINE CALCULATOR © 2000 Michael A. Lee and Kevin E. Schmidt

tlc.htm

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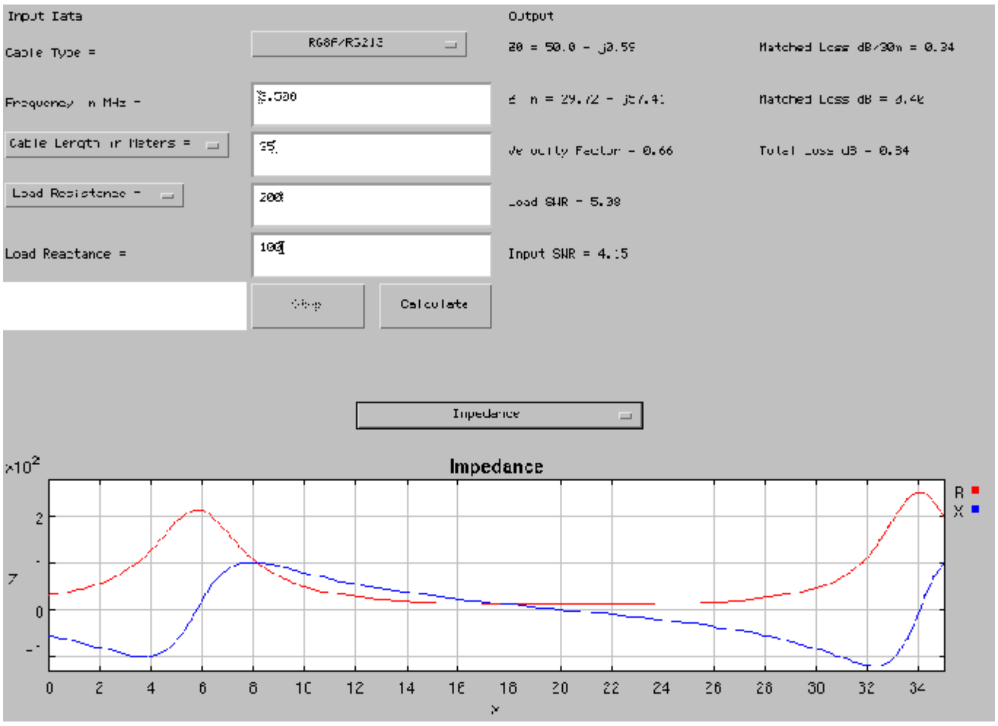
A transmission line calculator

NB: this page is an appendix to the *Errante*'s apparatus for the physics of the balanced transmission lines for radio-frequency signals.

Back to previous page

To use the *Transmission Line Calculator* applet below, change the values in the input fields, and click calculate.
Most of the fields will be self explanatory but further explanation is provided below.

Your browser can't run Java Applets: Here's a picture of the calculator instead.



To interrupt a calculation in progress, click stop.

The cable type is a menu of common cable types that can be used. In addition **two user defined entries allow you to enter custom cable types** as described below.

You need to enter the frequency in the first text field. The label "Cable Length in Feet" can be changed to "Cable Length in Meters" which will change all the length units from feet to meters. You then enter the length in the cable length field. Similarly, the "Load Resistance" label can be changed to "Input Resistance" so that measured impedances at the input end of a cable can be converted to antenna impedances. Enter the resistance and reactance (positive for inductive, negative for capacitive) in those fields.

When your input is set, press calculate. If there is an input error a message will be displayed in the field to the left of the stop button. The output side only changes when you press calculate, and will display the characteristic impedance Zo of the line, the impedance at the other end of the line, the velocity factor, the SWR at each end of the line, the matched loss of the line, and the total loss with your load impedance.

The bottom part of the applet is a plot. The x axis is the position along the line; x=0 is the input. The default shows the power, current and voltage on the line all on one plot. To make them all appear on a single plot, I take the current to have magnitude one at the load, I divide the voltage by the characteristic resistance of the line, and I divide the power by its value at the load. At the top of the plot is a choice menu that allows you to also select the power in watts, voltage magnitude in volts, or current magnitude in amps with 1500 watts input to the line. You can also plot the impedance in ohms seen along the line. The red curve is the resistance and the blue one is the reactance. The plotting is implemented using a stripped down version of *ptplot* by Edward A. Lee and Christopher Hylands, copyright University of California. One of the nice features of this package is that **you can zoom in on a portion of the plot** by holding down the left mouse button on that portion and move down and to the right to highlight the region with a little square. You can do this multiple times too. If you hold down the left mouse button and move up and to the left, you can unzoom. The easiest way to get back to the original is to click calculate again.

Two user defined cable types are provided. For *user 1*, you enter the resistive part of the characteristic impedance, the velocity factor and the attenuation in dB/100ft at some frequency. If you want to extrapolate to other frequencies, the exponent would be exactly 0.5 if the loss mechanism were purely conductor losses, and that should be good enough for a first guess. Otherwise, the loss at a new frequency will be the ratio of the new frequency to this frequency raised to the exponent power times this attenuation. The reactive part of the characteristic impedance is calculated assuming only conductor losses.

User 2 does not scale the results with frequency. Instead you input the characteristic impedance both resistive and reactive parts, the attenuation and the velocity factor. These characteristics are assumed to not change with frequency.

Click here to jump back to the Errante's apparatus for the physics of the balanced transmission lines page

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42. Reginald Fessenden and Guglielmo Marconi, the history of radio.

fessenden_marconi_radio_differences.htm

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NB: this page is an appendix to the *Marconi's* Nobel lecture page

Back to previous page

Fessenden and Marconi: their differing technologies and transatlantic experiments during the first decade of this century

by John S. Belrose
International Conference on 100 Years of Radio -- 5-7 September 1995

1. Introduction

Many scientists and engineers have contributed to the early development of electromagnetic theory, the invention of wireless signaling by radio, and the development of antennas needed to transmit and receive the signals. These include, Henry, Edison, Thomson, Tesla, Dolbear, Stone-Stone, Fessenden, Alexanderson, de Forest and Armstrong in the United States; Hertz, Braun and Slaby in Germany; Faraday, Maxwell, Heaviside, Crookes, Fitzgerald, Lodge, Jackson, Marconi and Fleming in the UK; Branly in France; Popov in the USSR; Lorenz and Poulsen in Denmark; Lorentz in Holland; and Righi in Italy. The inventor of wireless telegraphy, that is messages as distinct from signals, is Italian-born Guglielmo Marconi, working in England; and the inventor of wireless telephony is Canadian-born Reginald Aubrey Fessenden, working in the United States.

According to Marconi, he was an amateur in radio: in fact this was far from the truth. He foresaw the business side of wireless telegraphy. He was aware, however, of his own limitations as a scientist and engineer, and so he enlisted (in 1900) the help of university professor John Ambrose Fleming, as scientific advisor to the Marconi Company; and he chose engineers of notably high caliber, R.N. Vyvyan and others, to form the team with which he surrounded himself. Marconi's systems were based on spark technology, and he persevered with spark until about 1912. He saw no need for voice transmission. He felt that the Morse code was adequate for communication between ships and across oceans. He was a pragmatist and uninterested in scientific inquiry in a field where commercial viability was unknown. He, among others, did not foresee the development of the radio and broadcasting industry.

For these reasons Marconi left the early experimentation with wireless telephony to others, Reginald Fessenden and Lee de Forest.

Fessenden was a radio scientist and an engineer, but he did not confine his expertise to one discipline. He worked with equal facility in the chemical, electrical, radio, metallurgical and mechanical fields. He recognized that continuous wave transmission was required for speech and continued the work of Nikola Tesla, John Stone-Stone, and Elihu Thomson on this subject. Fessenden also felt that he could transmit and receive Morse code better by the continuous wave method than with the spark apparatus that Marconi was using.

This paper overviews the differing technologies of Fessenden and Marconi at the turn of the century, and their endeavours to achieve transatlantic wireless communications.

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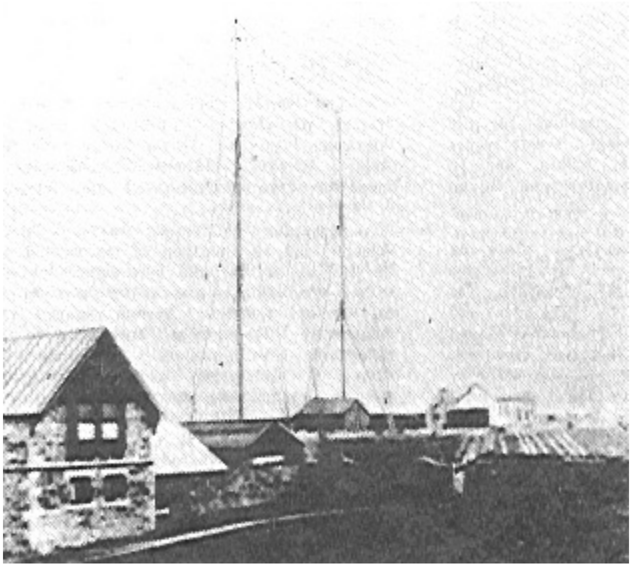

Figure 1: Marconi's antenna system at Poldhu, Cornwall, December 1901. (John Belrose)

2. Transatlantic wireless communications began at LF

Heinrich Hertz's classical experiments were conducted in his laboratory using a small end-loaded dipole driven by an induction coil and a spark gap for his transmitter. His receiver was a small loop, and detection was by induced sparking. Since the frequency generated by a spark transmitter is determined by the resonant mean frequency of the antenna system, his experiments in 1887 were conducted at VHF/UHF (60 to 500 Mhz) -- the corresponding wavelengths (5.0 to 0.6 metre) being practical for indoor experiments.

Marconi started experimenting with Hertz's apparatus in 1894. He was fascinated by the idea that by means of Hertzian waves it might be possible to send telegraph signals, without wires, far enough for such a system to have commercial value. By 1896 he achieved a transmission distance of 2.5 kilometres, by using an earth and an elevated aerial at both transmitter and receiver (nowadays called a Marconi antenna). His first permanent station established a link between the Isle of Wight and Bournemouth, England, some 22 km away (in 1897). He established communications across the Channel in 1899. By now he must have been using frequencies in the low HF band, since his aerial systems were much larger.

In 1900 he decided to try and achieve transatlantic communications. The required aerial size, and so the signalling frequency, at best could only be projected by extrapolation from values successful over a range of much shorter distances. The aerial at Poldhu, Cornwall in December 1901 (see Fig. 1), more by circumstance than design (to be discussed), radiated signals in the MF band (about 850 kHz).

Marconi kept building larger antenna systems, larger since he was striving for greater transmission distance and improved signal reception, which lowered the operating frequency. At Poldhu the frequency of his station in October 1902 was 272 kHz. His initial station at Table Head, Glace Bay, NS in December 1902 was a massive structure comprising 400 wires suspended from four 61 metre wooden towers, with down leads brought together in an inverted cone at the point of entry into the building. The frequency was 182 kHz. By 1904 his English antenna had become a pyramidal monopole with umbrella wires, and the frequency was 70 kHz. In 1905 his Canadian antenna, moved to Marconi Towers, Glace Bay was a capacitive top-loaded structure, with 200 horizontal radial wires each 305 metres long, at a height of 55 metres, and the frequency was 82 kHz. � By late in 1907 he was using a frequency of 45 kHz.

Fessenden's early experiments using spark transmitters were probably conducted at a frequency in the lower part of the HF band, since initially he was testing over short links of a few kilometres using 50-metre masts to support wire aerials. His belief was that radio transmission should be by way of continuous waves (CW), not the damped-wave or whip-and-lash type of transmission provided by spark-gap transmitters. The only way he knew to generate true CW was by a high-frequency alternator, and in the period 1890-1905 10 kHz was the highest frequency achieved using an HF alternator. But the efficiency of practical aerial systems was very poor at such a low frequency. So he strove to increase the speed and frequency of his HF alternator. In the meantime he invented the synchronous rotary-spark-gap transmitter. His transatlantic experiments in 1906 were conducted using such a transmitter and 420-foot umbrella top loaded antennas at Brant Rock, MA and Machrihanish, Scotland, tuned to a frequency of about 80 kHz.

3. Marconi and Fessenden Their Differing Technologies

Marconi, those working with him, and most experimenters in the new field of wireless communications at the turn of the century, were unanimous in their view that a spark was essential for wireless, and he actively pursued this technology from the beginning (in 1895) until about 1912.

Fessenden was a proponent of the continuous wave (CW) method of wireless transmission. Somewhat alone in this direction in 1900-1906, his CW patents had little impact on the users of radio technology. The golden age for spark was from 1900 to 1915; dominated by Marconi, who fought to quell any divergence from that mode. The fact that the damped wave-coherer system could never be developed into a practical operative telegraph system and that the sustained oscillation method should be used was perceived by Fessenden in 1898 [see Electrical World, July 29, August 12, September 16, 1899 and Proceedings American Institute of Electrical Engineers, November, 1899, p. 635 and November 20, 1906, p. 7311. In 1900-1902 only two methods were available for generating CW: 1) the HF alternator; and 2) the oscillating arc.

Plain Aerial Apparatus

Marconi's early experiments employed plain aerial apparatus, and placed the spark gap directly across the terminals of his vertical wire aerial-ground antenna. His receiver employed a similar set-up, with a coherer type of detector. The transmitter/receiver systems were untuned, excepting by the natural amplitude-frequency response of the aerials. Unbeknownst to him his transmitter and receiver were in effect "tuned" to different frequencies. The oscillating damped wave on the transmitting aerial, which was in effect "connected" to ground through the low resistance of the conducting spark, was in effect "tuned" to the fundamental quarter wave resonant response of the aerial. His receiver however, awaiting reception of the spark signal, would in effect be tuned to the half-wave resonant frequency of the wire aerial -- since the coherer prior to the reception of the RF impulse-like signal would present a high impedance between the aerial and ground.

This problem was solved by using a closed tuned circuit for the receiver; and for the transmitter by using the circuit arrangement devised by Braun, in which the oscillatory circuit (discharge capacitor and spark gap) was placed in a separate primary circuit transformer-coupled to the antenna system. This latter arrangement also lengthened the duration of the damped wave signal, since when the spark ceased, the oscillation in the antenna circuit continued, damped only by its natural L-C-R response.

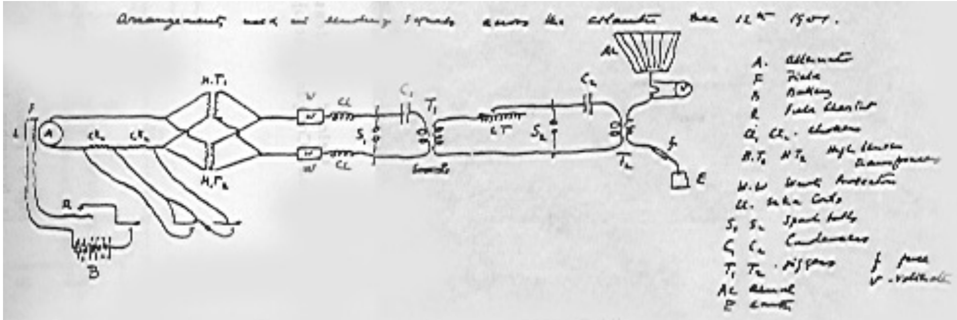

Figure 2: The circuit diagram of the December 1901 Poldhu transmitter in J.A. Fleming's handwriting. (Bondyopadhyay)

Transmitter Technology

The Poldhu transmitter was a curious two-stage circuit, in which a first-stage spark at some attainable lower voltage provided the energy for the second stage in tandem (Fig. 2), to spark at a specified higher voltage. While this voltage multiplication system was innovative in the field of wireless at the time, it carried with it many problems, and the inefficiencies of two spark stages.

Marconi clearly realized that to achieve high power from a spark transmitter it was necessary to charge the condenser to a very high voltage (voltages of up to 150 kV were spoken about and may have been realized); and that a very large discharge capacitance was needed, since the stored energy in the condenser was understood (Energy equals 1/2 CV^2). But he carried the latter requirement to an extreme.

The power capability of the Poldhu AC generator (25 amperes at 1500 volts) in 1901 was quite insufficient to recharge the condenser every period. It seems like several periods of the supply generator (operating at 36 Hz) were required to bring the condenser voltage to gap break-down potential. Fleming's estimates of the spark rate lie between wide limits. Thackeray [1992] has estimated that the spark rate for the primary circuit was 7.5 to 12 sparks/sec at most; and the spark rate for the secondary circuit might have been as low as two or three sparks/sec. After that time there was clearly a redesign to a single-stage transmitter that sparked directly from the power transformer; and Fleming began to develop rotating dischargers in an attempt to achieve rapid quenching of the spark.

It is perhaps ironic that the low spark rate was compromised by Marconi himself, when in Newfoundland he put a telephone receiver to his ear to listen for the dot transmissions from Poldhu. At the low spark rate he employed all he would hear would be a click, not distinguishable from an atmospheric. But recall that Marconi's early experience was with coherer-type detectors, which worked best when the spark rate was low.

Figure 3: Clifden, Ireland condenser under construction. (Marconi Company)

Before leaving our discussion about Marconi's methodology, let me comment on some of the physical arrangements for his stations. The discharge capacitor for his Clifden and Marconi Towers, Glace Bay stations consisted of thousands of steel plates hanging from floor to ceiling, which filled the wings of the building, and this room was subsequently called the "condenser building" (see Fig. 3). The power supply was a 15 kV DC generator (three 5 kV generators in series) driven by a steam engine. Note the power source was DC. Standby batteries (6000, 2 volt, 30 AH batteries in series) at both stations may well have been the largest battery the world has ever seen. The heart of his Clifden/Marconi Towers stations was a whirling five foot spark discharge disk, with studs on its perimeter. Each time a stud passed between two electrodes, a 15 kV spark jumped the gaps. The regular spark rate was about 350 sparks/sec. The awesome size of the station and the din of the transmitter must have been something to behold. The power consumed by these stations was in the range of 100 to 300 kW, and the spark was a display of raw power. It is said that the awesome din of the transmitter could be heard several kilometres away.

Fessenden's technology and circuit arrangements were very different. He tried all the various methods of generating wireless signals in the early days, by spark, by arc and by the high frequency alternator. It is likely that he would have used the HF alternator from the outset, see for example his patent No. 706,787 filed 29 May 1901; excepting that a suitable HF alternator, generating frequencies above about 10 kHz was not available until 1906. There is no fundamental reason that long distance wireless communications could not have begun at VLF, except for the practical realization of efficient antenna systems for such a low frequency.

Fessenden's work was dominated by his interest in transmitting words without wires. By 1903 and 1904 fairly satisfactory speech had been transmitted by the arc method, but the news of Marconi's attempts to achieve transatlantic wireless telegraphy transmission had caught the attention of the world. Since the development of his HF alternator was taking longer than anticipated, Fessenden set his mind to make a more CW-like spark transmitter. This led to the development of the synchronous rotary-spark-gap transmitter. An AC generator was used, driven by a steam engine, which as well as providing the energy for the spark transmitter, was directly coupled to a rotating spark gap so that sparks occurred at precise points on the input wave, viz. at waveform maximum for best efficiency. The spark was between fixed terminals on the stator and terminals on the rotor, which was in effect a spoked wheel, rotating in synchronism with the AC generator.

As the speed of the wheel and the AC frequency both depend on the speed of the generator, the number of times/sec at which the condenser voltage reaches a peak value and the number of opportunities it has for discharging can be made equal, and the positions of the stator terminals can be arranged so that these conditions occur simultaneously. Another advantage was realized, since in effect a rotary gap was a kind of a mechanically quenched spark-gap transmitter. The oscillations in the primary circuit ceased after a few oscillations, when the rotating gap opened. The quenched gap was more efficient and certainly less noisy than the unquenched gap. With a synchronous spark-discharger phased to fire on both positive and negative peaks of a 3-phase waveform, precisely at waveform peak, a 125 Hz generator could produce a spark rate of 750 times a second. These rotating gaps produced clear almost musical signals, very distinctive and easily distinguished from any other signal at the time. It was not true CW but it came as close as possible to that, and the musical tone could be easily read through noise and interference from other transmitters.

Fessenden's Brant Rock and Machrihanish stations employed a rotary gap 1.8 metres in diameter at the rotor. Its rotor had 50 electrodes (poles) and its stator had four. It was driven by a 35 kVA alternator, powered by a steam engine.

The synchronous rotary gap spark discharger should not be confused with the asynchronous rotary gap that was in more general use at the time (e.g. by Marconi ship-borne equipment, and radio amateurs in general used asynchronous rotary gaps). Here the speed of rotation of the wheel is entirely independent of the speed of the generator, and while it was possible to realize several sparks during one cycle of the generator, the sparks occur at different points on the cycle.� The conditions are not exactly repeated each time as in the case of the synchronous spark, because the charging current from the generator is charging up the condenser during different parts of its own cycle of variation, and hence neither the voltage to which it is charged, nor the breakdown voltage is constant. Not only is it possible to miss a spark altogether, but the interval between sparks is not absolutely constant. In addition, the energy stored in the condenser and the proportion radiated in the separate wave trains is variable. The result is that the note heard at the receiving station is impure.

By the summer of 1906 many of the difficulties had been overcome and the Alexanderson HF alternator developed by GE for Fessenden giving 50 kHz was installed at Brant Rock. Various improvements were made by Fessenden and his assistants, and by the fall of 1906 the alternator was working regularly at 75 kHz with an output of one half a kilowatt. This was the beginning of pure CW transmission, c.f. Alexanderson [1919].

Continuous waves was the method of generation Fessenden had long sought, since he wanted to transmit words without wires. He inserted a carbon microphone in series with the lead from his alternator to the antenna, and he had an amplitude modulated transmitter. But more on that later.

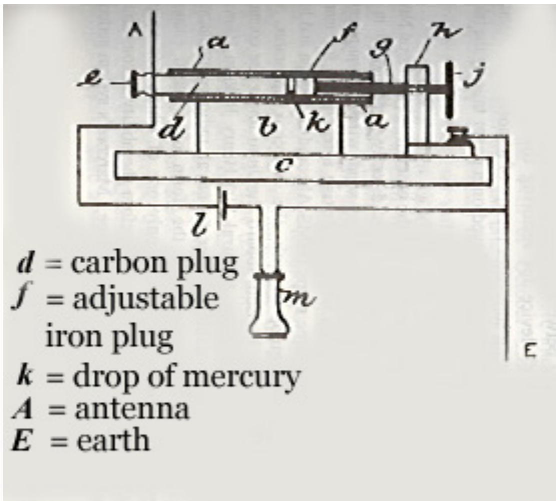

Figure 4: The Italian Navy coherer as patented by Marconi in September 1901 (Brit. Pat. 18 105). (Phillips)

Receiver Technology

The ability to receive wireless signals at the turn of the century was very poor, for several reasons: 1) Initially the receiver was untuned or if tuned the selectivity was poor; 2) there was no means to amplify the signal; and 3) a sensitive detector had yet to be invented.

The early experiments employed a device called a coherer. The coherer as we have noted was a device which normally exhibited a high resistance, but when subject to a voltage above a given threshold there was a marked decrease in this resistance. The change in resistance could be detected by means of a secondary relay circuit, or by listening to the current change with a telephone earpierce. The filings coherer was a bistable device. It needed an electrical voltage to effect one transition, and a physical shock (a tapper) to return it to its initial state. The sensitivity of the device was poor; the action of the receiver depended upon a voltage rise and so was independent of the energy of the signal; it did not discriminate between impulses of different character, viz. between signals and atmospherics; the selectivity of the receiver was a function of the state of the coherer; and it could not be used as a detector for continuous waves.

For his transatlantic experiment in 1901 Marconi had two types of receivers, and three types of coherers. One was a tuned receiver, which he referred to as a "syntonic receiver", that is a receiver tuned to the frequency of the transmitter. The second earlier receiver was untuned. The three types of coherers that he used were: one containing loose carbon filings; another designed by Marconi containing a mixture of carbon dust and cobalt dust; and thirdly the Italian Navy coherer (see Fig. 4) containing a globule of mercury between a carbon plug and a moveable iron plug. This latter device, when critically adjusted or more or less by luck, acted like a crude form of a rectifier, but its performance was poor and unpredictable [Phillips, 1993]. Later, in 1902, he devised a form of current operated receiver, called a magnetic detector, which greatly enhanced his receiver sensitivity. This detector was used by Marconi until it was replaced by the vacuum tube in 1913.

When Marconi designed the receiver he intended to use for the first transatlantic HF experiment, he designed it so it could be tuned, and so respond selectively to signals of different frequencies -- his famous four sevens patent of 1900. This idea was however not his own, as was the case for many of Marconi's "inventions", but was devised by Oliver Lodge, who in 1897 had filed four patents. Two dealt with improvements to coherers, and two to "tuning" or "syntony" [Austin, 1994].

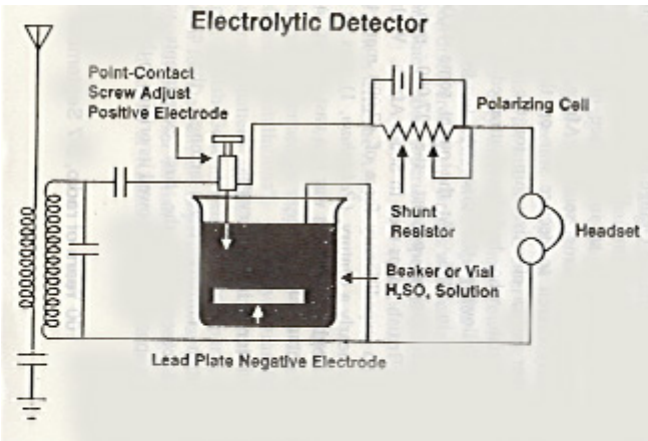

Figure 5: An early version of Fessenden's receiver which employed an electrolytic rectifier (his barretter), a battery and earphones. (John Belrose)

Fessenden was convinced that a successful detector for reception of wireless signals must be constantly receptive, instead of requiring resetting as was the characteristic of the coherer. Although his experiments with wireless receivers began when he was a Professor of Electrical Engineering in the Western University of Pennyslvania, in 1896/97; it was not until 1901/02 that he discovered the electrolytic detector. In 1902 and 1903 he patented the first practical detector [US Patents 706,744 and 727,331] -- which he called a barretter, a name coined from the French word for exchanger. This name implies the exchange of AC for DC, i.e. the device behaved like a rectifier, c.f. [Pickworth, 1994]. In Fig. 5 we sketch one of Fessenden's early radio receivers using this detector; which was the standard of sensitivity for many years until it was replaced by the vacuum tube some ten years later.

Fessenden's barretter detector was however useless for reception of unmodulated CW. All that one would hear would be the clicks as the Morse telegraph key was closed and opened. However, very early, some 11 years ahead of its time, Fessenden's fertile mind had already devised a solution. He invented the methodology (and the word) for combining two frequencies to derive their sum and difference frequencies, viz. the heterodyne method of detecting continuous waves (US Pat. No. 706,740 dated 12 August 1902; and Nos. 1,050,441 and 1,050,728 dated 14 January, 1913). But it was not before 1912-1914, when the triode's versatility to be an oscillator, or RF source, was established, that the heterodyne receiver became a practical method for detecting CW. Today, heterodyning is fundamental to the technology of radio communications.

Fleming in 1904 invented the valve diode (known as a Fleming valve). The patent which covered its use as a detector of Hertzian waves became the property of the Marconi Company, and eventually, but not until after WWI, Fleming valves were put into operation in Marconi stations.

Meanwhile de Forest, who was following the footsteps of Fessenden, experimenting with the electric arc as a CW wireless transmitter for telephony needed a good detector. The electrolytic detector that he had been using was judged by The Courts in 1906 to be an infringement on Fessenden's patent. As a result he had to change all of his stations to use the silicon detector, which had been patented in 1906 by H.H. Dunwoody, an officer in his company. Because of this incident, de Forest resigned from the company in November 1906. De Forest started looking for a better valve detector. He made some Fleming valves, and, in a moment of inspiration, he added a third element, a control element shaped like a grid-iron, called a grid. De Forest patented his audion, the first three-element valve in 1907. Although the audion was more sensitive as a detector than the Fleming valve he was prevented from using it for commercial purposes, by a lawsuit launched by the Marconi Company (claiming infringement in spite of the fact that it was a different valve). De Forest did not understand how the valve operated, and it remained for Langmuir, Armstrong, and van der Pol to discover its full possibilities. The time interval between Fessenden's heterodyne receiver (1902) and Armstrong's "feedback" receiver or regenerative receiver (1913) is the 11 years mentioned above. Armstrong's superheterodyne receiver was not invented until 1918.

4. The First Transatlantic Experiment

On 12 December 1901 signals from a high power spark transmitter located at Poldhu were reported to have been received by Marconi and his assistant George Kemp, at a receiving station on Signal Hill, near St. John's, Newfoundland. The signals had traveled a distance of 3500 kilometres. Even at the time of the experiment there were those who said, indeed there are some who still say, that he misled himself and the world into believing that atmospheric noise crackling was in fact the Morse code letter 'S'.

A little later, in February 1902, when Marconi returned to England on the SS Philadelphia, using a tuned ship-borne antenna, he received signals using his filings coherer from the same sender up to distances of 1120 km by day and 2500 km by night. Even these distances are rather remarkable considering the receiving apparatus he used.

We discuss here in detail that first transatlantic experiment.

The Poldhu Station

Marconi's ambition at the turn of the century to demonstrate long-distance wireless communication, and develop a profitable long-distance wireless telegraph service, led to his pragmatic proposal in 1900 to send a wireless signal across the Atlantic. He conceived a plan to erect two super-stations, one on each side of the Atlantic, for two-way wireless communications, to bridge the two continents together in direct opposition to the cable company (Anglo-American Telegraph Company). For the eastern terminal, he leased land overlooking Poldhu cove in southwestern Cornwall, England. For the western terminal the sand dunes on the northern end of Cape Cod, MA at South Wellfleet, was chosen.

The aerial systems comprised 20 masts, each 61 metres high, arranged in a circle 61 m in diameter. The ring of masts supported a conical aerial system of 400 wires, each insulated at the top and connected at the bottom, thus forming an inverted cone. Vyvyan [1933], the Marconi engineer who worked on the 1901 experiment, when shown the plan, did not think the design sound. Each mast was stayed to the next one, and only to ground in a radial direction, to and away from the centre of the mast system. He was overruled, construction went ahead, and both aerial systems were completed in early 1901.

However, before testing could begin catastrophe struck, the Poldhu aerial collapsed in a storm on 17 September, and the South Wellfleet aerial suffered the same fate on 26 November, 1901.

At Poldhu Marconi quickly erected two masts and put up an aerial of 54 wires, spaced 1 metre apart, and suspended from a triadic stay stretched between these masts at a height of 45.7 m. The aerial wires were arranged fan shaped, presumably insulated at the top, as was his conical wire aerial, and connected together at the lower end, see Fig. 1. This photograph has been published and republished, and clearly one can see only 12 wires -- but the view generally held is that the aerial system as described above by Vyvyan [1933) is right, that is there were 54 wires, and the photograph has been retouched.

The antenna was driven by the curious two stage spark transmitter, previously discussed. There were many problems in getting it to work at the high power levels desired [see Thackeray, 1992]. Our principal concern here is the frequency generated by the Poldhu station. The oscillation frequency is determined by the natural resonant response of the antenna system, which includes the inductance of the secondary of the antenna transformer T_2, since in effect the antenna system is a base-loaded monopole (see Fig. 2).

The primary of this transformer consisted of 2 7/20 wires in parallel, the secondary consisted of 7 or 9 wires of 7/20 wire in series. Fleming's sketch indicates 9 wires; Entwisle [1922] said there were 7 wires. The inductance values for this transformer have long been debated, since the original transformer is lost, there are no drawings, and reports about them differ [Thackeray, 1992]. G. Garratt made a copy of L_s (the secondary of this transformer) and measured its inductance to be 6×10^{-6} H [see Ratcliffe 1974].

While the inductance L_s changes the resonant frequency, the exact value does not change the conclusion reached in our study. For L_s equal to 6×10^{-6} H, we have modelled Marconi's Poldhu antenna, assuming the fan comprised 12 wires. According to the antenna analysis code MININEC, the resonant frequency of the antenna system was 850 kHz.

A number of scientists and engineers interested in the actual frequency or frequencies radiated by this first high power transmitter at Poldhu have discussed the possibility that the aerial transformer was overcoupled, resulting in a double-humped frequency/amplitude response. We do know that Fleming tuned the primary oscillatory circuit by varying the discharge capacitor C_2 to maximize the aerial current. Since our best estimate for the component values (C_2 = 0.037 muF and $L_p = 8 \times 10^{-7}$ H) would result in a resonant frequency of 925 kHz, it seems logical to conclude that the overall system response would result in a single peak centred on the resonant frequency of the aerial system, viz. about 850 kHz.

Historians have also speculated that the transmitter might also have radiated a high-frequency signal as well, since an HF signal would have been more suitable for transatlantic communications (to be discussed), see for example Ratcliffe [1974]. If Marconi had used a thin wire transmitting antenna at Poldhu, this antenna would indeed have radiated efficiently at odd harmonics of the fundamental resonant frequency. But for our model the antenna is inductive for all frequencies greater then the fundamental resonant frequency response of the antenna system. One must conclude therefore that the Poldhu spark-transmitter system radiated efficiently only on the fundamental oscillation frequency of the tuned antenna system -- about 850 kHz.

Marconi himself has been evasive concerning the frequency of his Poldhu transmitter. Fleming in a lecture he gave in 1903 said that the wavelength was a 1000 feet or more, say, one-fifth to one-quarter of a mile (820 kHz is the generally quoted frequency). Marconi remained silent on this wavelength, but in 1908 in a lecture to the Royal Institution he quotes the wavelength as 1200 feet, see Bondyopadhyay [1993].

Reception on Signal Hill

For his transatlantic experiment, Marconi decided to set up receiving equipment in Newfoundland. In December 1901 he set sail for St. John's, with a small stock of kites and balloons to keep a single wire aloft in stormy weather.

A site was chosen on Signal Hill, and apparatus was set up in an abandoned military hospital. A cable was sent to Poldhu, requesting that the Morse letter " S " be transmitted continuously from 3:00 to 7:00 PM local time.

On 12th December, 1901, under strong wind conditions, a kite was launched with a 155 m long wire. The wind carried it away. A second kite was launched with a 152.4 m wire attached. The kite bobbed and weaved in the sky, making it difficult for Marconi to adjust his new syntonic receiver which employed the Italian Navy coherer. "Difficult" I will accept, but how he determined the frequency of tuning for his receiver is a mystery to me. Whatever, because of this difficulty, Marconi decided to use his older untuned receiver. History has assumed that he substituted the metal filings coherer previously used with this receiver for the newly acquired Italian Navy coherer, but Marconi never really said he did [see Phillips, 1993]. He referred only to the use of three types of coherers.

Despite the crude equipment employed, and in our view the impossibility of hearing the signal, Marconi and his assistant George Kemp convinced themselves that they could hear on occasion the rhythm of three clicks more or less buried in the static, and clicks they would be if heard at all, because of the low spark rate. Marconi wrote in his laboratory notebook: Sigs at 12:30, 1:10 and 2:20 (local time). This notebook is in the Marconi Company archives and is the only proof today that the signal was received.

The Enigma

Today we know that signals (depending on frequency used) can indeed travel across the Atlantic, and far beyond. But in 1901, anyone who believed that they could, and did, believed so as an act of faith based on the integrity of one man -- Marconi.

If 850 kHz was indeed the frequency used, the tests took place at the worst time of day, because the entire path would have been daylight, and the daytime skywave would be heavily attenuated, even though it was a winter day, in sunspot minimum period, and there were no magnetic storms at the time, or for ten days before. The day-time absorption of an ionospherically-reflected signal is a maximum in the LF/MF band. Ratcliffe [1974] has deduced that, from a knowledge only of propagation conditions, reception on Signal Hill is consistent with the observed limiting ranges of reception on the ship only if the untuned landbased receiver was 10-100 times more sensitive than the tuned receiver on the ship.

It is therefore difficult to believe that signals could have been heard on Signal Hill, since the receiving equipment after all consisted of a long-wire antenna, coupled to an untuned receiver which had no means of amplification whatsoever, and the type of detector used was less sensitive and its performance unpredictable compared with Fessenden's barretter detector, or the galena crystal detector which evolved a few years later.

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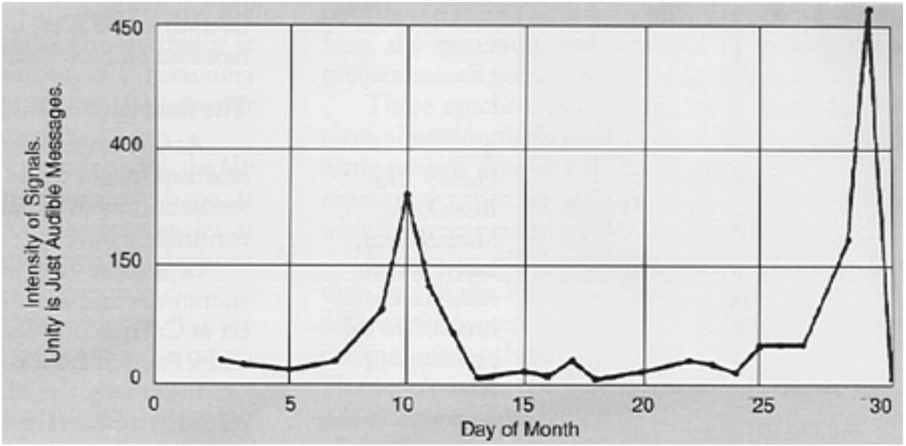

Figure 6: Curve showing the variaton of the intensity of transatlantic signals for the month of January, 1906. Unity corresponds to a just audible message. Such a curve is certainly one of the first reocrdings of LF propagation data. (Fessenden)

5. The First Radio Propagation Experiments

There is no evidence that Marconi made any serious attempt to systematically investigate the charactericistics of the HF, MF and LF portions of the radio frequency spectrum when he began the downward frequency trend, in his struggle to achieve transatlantic wireless communications. He did not do this until 20 years later, in the early 1920's, when attracted to the HF band by amateur radio operators. The radio amateurs had been banished to the then-believed useless frequencies higher than 1500 kHz.

The first record showing qualitatively the variation of the intensity of transatlantic messages transmitted between Brant Rock, MA and Machrihanish, Scotland, at night, during the month of January 1906 is reproduced in Fig. 6 [Fessenden, 1908]. Nothing at all was received that month during daytime.

It was found (measurements made during 1906) that absorption at a given instant was a function of direction as well as distance, since on a given night the signals received by stations in one direction would be greatly weakened, while there would be less weakening of the signals received by stations lying in another direction; and a few hours or minutes later the reverse would be the case. It was also found that variations of absorption on transatlantic signals appeared to have a quite definite relation with variations of the geomagnetic field, i.e., the greater the absorption the greater the magnetic variation [Fessenden, 1908].

Experiments were made between Brant Rock and the West Indies, a distance of 2735 km, during the spring and summer of 1907. Frequencies in the band 50 kHz to 200 kHz were used. It was found that the absorption at 200 kHz was very much greater than at 80 kHz, and that messages could be successfully received over this path in daytime at the latter frequency. Antenna radiation efficiency was an important factor for frequencies less than 80 kHz. No messages were received in daytime with the higher frequency.

The fact that these experiments were made during summer, that the receiving station was in the Tropics (high noise levels), and the fact that the distance, 2735 km was practically the same as between Ireland and Newfoundland was reported by Fessenden [1907]. After publication of the above results, Marconi, in early October, 1907 abandoned his previously used frequencies, and immediately succeeded in operating between Glace Bay and Clifden, a distance of more than 3000 km, the frequency being about 70 kHz. The same messages were received at Brant Rock, MA, a distance of nearly 4825 km. A little later Marconi moved to an even lower frequency, 45 kHz.

6. Verifiable Transatlantic Radio Communications

The first East-West transatlantic radio transmission was made during October 1902 from Poldhu, Cornwall to the Italian cruiser Carlos Alberto anchored in the harbour of Sydney, NS with Marconi aboard. The frequency employed was about 272 kHz. This successful transmission was considered an experimental prerequisite to the start-up of the permanent land based wireless Marconi station under construction at Glace Bay, NS.

The first West-East transatlantic radio transmission was recorded on 5 December 1902 between Glace Bay and Poldhu. The frequency was about 182 kHz.

The first Canada/UK transatlantic radio message (as opposed to hearing the signal) was sent from Glace Bay to Poldhu on 15 December 1902. It was a press message from a London Times correspondent at Glace Bay to his home office.

The first USA/UK transatlantic radio message received at Poldhu from the Marconi station at South Wellfleet, MA was from President Roosevelt to King Edward VII, on 18 January, 1903.

History has recorded that the above messages were successfully transmitted, but how well these messages were received is a matter of conjecture. In 1902-c.1912, both the Clifden and Glace Bay stations were using "disc discharger" transmitters, and a form of current operated receiver (Marconi's magnetic detector). It is clear that Marconi was still struggling in 1908 to achieve reliable transatlantic radio communications. It is interesting to read a letter written on 19 March, 1909 to Hon. Chauncey M. Depew, US Senate, Washington, DC, signed by five members of The Junior Wireless Club (now The Radio Club of America). The thrust of the letter was to comment on a proposed bill before the Senate, that would in effect restrict the use of the air waves by radio amateurs, because of presumed malicious interference caused by radio amateurs. I quote from a part of that letter, which can be found in the Seventy-Fifth Anniversary Diamond Jubilee Year Book of The Radio Club of America, 1984:

> "At the Narragansett Bay there were certain Naval tests made about two years ago, and the various so-called Wireless Companies wanted to get the first news to the newspapers of these tests, so as to boom their companies' stocks, and to say the news was received first through their company, and when some of them found they were unable to cut out interference between themselves, in order to prevent other Wireless Companies from getting the news first they sent a lot of fake messages of confused dashes.
>
> "Only a few of the so-called Wireless Companies have efficient methods of cutting out interference, and these are the companies that are now crying for the most protection.
>
> "You probably have heard of the tests made last year between Glace Bay, NS and Clifden, Ireland, when the National (Electric) Signaling Company (Fessenden's Brant Rock

> station) picked up the messages, which Marconi, on the test, was unable to deliver between his own stations, from both Glace Bay and Clifden, Ireland, in spite of the fact that the Marconi Company kept up a constant interference of dash, dash dash, from their Cape Cod Station for 48 hours without interruption, but the National (Electric) Signaling Company paid no attention to such interference and picked up all the messages, which Marconi was unable to exchange between their own stations, and all these messages were handed over to Lord Northcutt at the Hotel St. Regis."

Marconi himself, in his 1909 Nobel Prize address said: "What often happens in pioneer work repeated itself in the case of radiotelegraphy. The anticipated obstacles or difficulties were purely imaginary or else easily surmountable, but in their place unexpected barriers presented themselves, and recent work has been directed to the solutions of problems that were neither expected or anticipated when long distances were first attempted". Certainly after Marconi's first transatlantic radio experiment in 1901, he found that the realization of reliable transatlantic radio communications was more distant (for him) than he realized at the time.

The first two-way transatlantic radio telegraphy transmission took place on 10 January 1906, between Fessenden's stations at Brant Rock, MA and Machrihanish Scotland. Repeatedly regular exchange of messages across the Atlantic Ocean took place on most days during winter, spring and into early summer. The frequencies used were in the 80-100 kHz band. The reliability and the quality of signal reception (signal-to-noise ratio) for the Fessenden system must have been very much better than anything Marconi could achieve at this point in time. Fessenden was using his synchronous rotary-spark transmitters at both ends, and tuned receivers with his barretter detector. The signals were superior to other signals used at the time, which by comparison were rough and ragged. His antenna system was an umbrella top-loaded radiator 128 metres high. The Marconi antennas were multi-wire conical structures, or wire antennas with extensive top loading, 61 metres high. Since the radiation efficiency of electrically short antennas varies (approximately) as the height of the antenna red, Fessenden's antenna systems were probably four times more efficient than Marconi's.

Radio telephony

At the turn of the century Fessenden was using a spark transmitter, employing a Wehnelt interrupter operating a Ruhmkorff induction coil. In 1899 he noted, when the key was held down for a long dash, that the peculiar wailing sound of the Wehnelt interrupter could be clearly heard in the receiving telephone. He must have had a detector of some sort that was working for him, even at this early stage in the development of wireless. This suggested to him that by using a spark rate well above voice band (10,000 sparks/sec), wireless telephony could be achieved; and this he did transmitting speech over a distance of 1.5 km on 23 December 1900, between 15 metre masts on Cob Island, MD [Belrose, 1994a; 1994b].

In autumn of 1906 Fessenden had his HF alternator working adequately on frequencies up to about 100 kHz. About midnight in November, 1906 Mr. Stein at Fessenden's Brant Rock station was telling the operator at a nearby test station at Plymouth, MA how to run the HF alternator. It was usual for these two operators to use speech over this short distance. However his voice was heard by Mr. Armor at Machrihanish, Scotland with such clarity that there was no doubt about the speaker, and the station log books confirmed the report.

Fessenden's equipment was working exceptionally well in the early hours of that morning, and (remarkable for that time) the echo of the telegraphy signals from the Scotland station could clearly be heard one fifth of a second later, having travelled the long way around the earth.

The Machrihanish tower crashed to the ground on 5 December, 1906 during a severe winter storm. The station was never rebuilt, and so Fessendeil's transatlantic experiments came to an abrupt end.

Fessenden's greatest success took place on Christmas Eve 1906, when he and his colleagues presented the world's first wireless broadcast. The transmission included a speech by Fessenden and selected music for Christmas. Fessenden played Handel's Largo on the violin. That first broadcast, from his transmitter at Brant Rock, MA was heard by radio operators on board US Navy and United Fruit Company ships equipped with Fessenden's wireless receivers at various distances over the South and North Atlantic, and in the West Indies. The wireless broadcast was repeated on New Year's Eve. The transmitter was an HF alternator, in which one terminal was connected to ground, the other terminal to the tuned antenna, and a carbon microphone was inserted in the antenna lead.

Recall that Fleming, in the first edition of his book on Electromagnetic Waves published in 1906, stated that an abrupt impulse was a necessary condition for wireless transmission, and that high frequency currents, even of sufficient frequency could not produce radiation. The highest frequency of HF alternators prior to the summer of 1906 was about 10 kHz. This belief, and an earlier belief that the terminals of an antenna had to be bridged by a spark, show how wrong some of the early "experts" were.

Continuing in the same vein, Fessenden in his 1908 paper restated his long held view: "The coherer is well adapted for working with damped waves, but the coherer-damped wave method can never be developed into a practical telegraph system. It is a question whether the invention of the coherer has not been on the whole a misfortune as tending to lead development of the art astray into impracticable and futile lines, and thereby retarding the development of a really practical system".

7. Concluding Remarks

There are those that say that Marconi's greatest triumph (the mother of all experiments) was when he succeeded in 1901 in passing signals across the Atlantic. There are those that say that he misled himself and the world into believing that atmospheric noise crackling was in fact the Morse code letter 'S'. Whether Marconi heard the three faint dots or not is really unimportant. His claim "sparked" a controversy among contemporary scientists and engineers about the experiment that continues today.

Certainly engineers and scientists of the present day are unanimous in admiring the bold and imaginative way in which Marconi attempted to take one spectacular step forward, to extend the range of wireless communications from one or two hundred kilometres to the 3500 kilometre distance across the Atlantic ocean.

The world has acclaimed Marconi as the "father of wireless", although some say that Alexander Popov and Oliver Lodge were first in the field. History has accredited Marconi with the invention of an early form of radio telegraphy.

Fessenden's continuous waves, a new type of detector, and, his invention of the method as well as the coining of the word heterodyne, did not by any means constitute a satisfactory wireless telegraphy or wireless telephony system, judged by today's standards. They were, however, the first real departure from Marconi's damped-wave-coherer system for telegraphy which other experimenters were merely imitating or modifying. They were the first pioneering steps toward radio communications and radio broadcasting.

Today, heterodyning is fundamental to the technology of radio communications. Some historians consider that Fessenden's heterodyne principle is his greatest contribution to radio science. Edwin Howard Armstrong's super-heterodyne receiver is based on the heterodyne principle. Except for method improvement Armstrong's superheterodyne receiver remains the standard radio receiving method today.

Fessenden, a genius, and a mathematician was the inventor of radio as we know it today.

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43. GENERATORE di NODO di TERRA VIRTUALE per linee bilanciate © 2008 Francesco Errante

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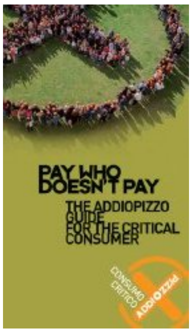

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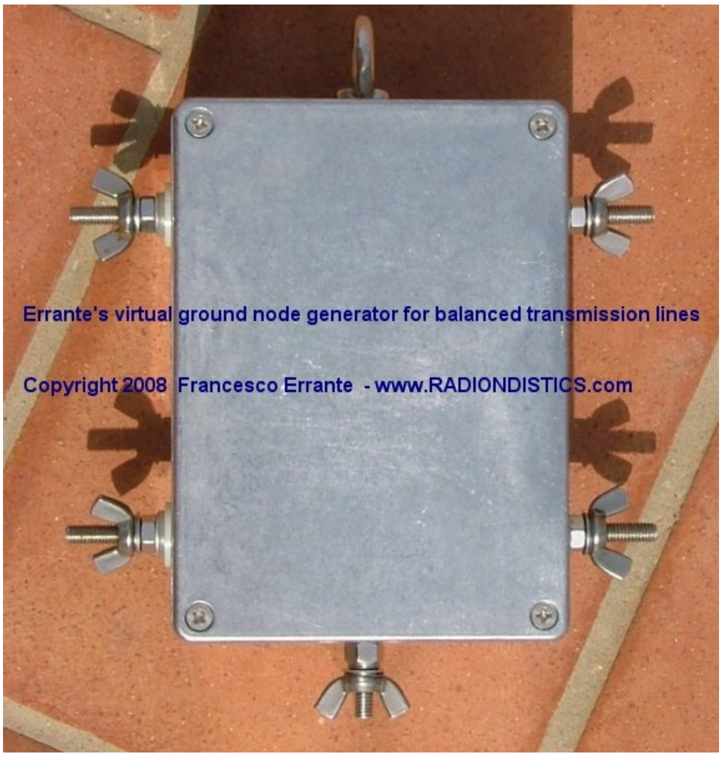

Il generatore di nodo di terra virtuale per linee bilanciate .

Freq. di lavoro: da 1 a 30 MHz
Imp. d'ingresso: 300 Ohm
Imp. d'uscita: 300 Ohm
R.O.S.di ingresso/uscita: ï¿½ 1:1 entro la banda
Perdita d'inserzione: -0.01 dB
Potenza RF: 3 KW cw
Connettori tipo: bulloni M6
Peso : 1 Kg ca.

ï¿½

Generatore di nodo di terra virtuale per linee bilanciate sottoposto al controllo

regime
di onda
progressiva

La radiazione
hertziana:
cos'ï¿½ e come
avviene

della linearitï¿½ della sua risposta in frequenza

ï¿½

Il generatore di nodo di terra virtuale per linee bilanciate: cos'ï¿½, come funziona e cosa fa.

Generatore di nodo di terra virtuale per linee bilanciate

di *Francesco Errante*

Il lettore che non abbia giï¿½ familiarizzato col meccanismo di generazione del nodo di terra virtuale ï¿½ vivamente pregato di leggere: *l'antenna HF monopolare con terra virtuale* ed anche il circuito soppressore con terra virtuale prima di procedere nella lettura di questa pagina.

Il generatore di nodo di terra virtuale per linee bilanciate e sistemi di antenna bilanciati (B.V.G.N.G. per brevitï¿½) ï¿½ un circuito radioelettrico basato sullo stesso principio di funzionamento del generatore di nodo di terra virtuale per linee sbilanciate ed ï¿½ realizzato mediante l'impiego di un trasformatore ad accoppiamento di flusso per RF, con ingresso bilanciato su 300 Ohm con presa centrale ed uscita anch'essa bilanciata su 300 Ohm con presa centrale (Bal-Bal 1:1). Le prese centrali, sedi di due nodi di terra virtuale sono collegate fra di loro ed allo chassis, assicurando cosi una totale continuitï¿½ galvanica fra tutti i rami del dispositivo.

Questo dispositivo permette di disaccoppiare il

dipolo ripiegato a mezz'onda (impedenza 300 Ohm) dalla linea bilanciata che lo alimenta (impedenza 300 Ohm), allo scopo di imporre un netto confine per i segnali a RF tra linea e dipolo, in modo tale che il dipolo esibisca la sua reale banda di lavoro e la linea bifilare non interferisca con la stessa.
In mancanza di un netto confine elettrico tra linea e radiatore, infatti, la linea di trasmissione si rende parte del circuito del radiatore, modificandone inevitabilmente la lunghezza elettrica e dando essa stessa origine a radiazione hertziana, con spiacevoli conseguenze come: perdita di segnale, induzione di interferenze sugli impianti circostanti e soprattutto l'insorgenza di apparenti fenomeni di risonanza della linea stessa.
Le linee bilanciate sono, invece, a-periodiche, come dimostrato qui.

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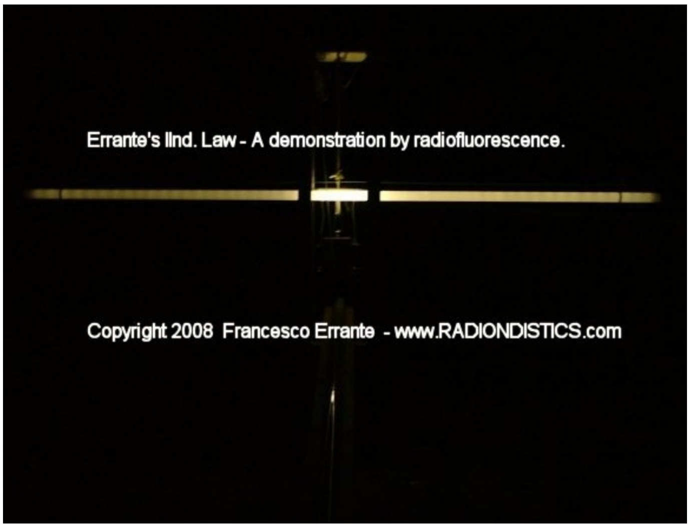

Dipolo ripiegato disaccoppiato dalla linea mediante BVGNG - Rilevamento del campo generato dal dipolo ripiegato e verifica dell'assenza di campo generato dalla linea bilanciata.

Linea bilanciata a 300 Ohm disaccoppiata dal dipolo ripiegato mediante BVGNG e sottoposta a misura radioscopica ad alta potenza. Verifica della IIa Legge di Errante

Linea bilanciata a 300 Ohm disaccoppiata dal dipolo ripiegato mediante BVGNG e sottoposta a misura radioscopica ad alta potenza. Verifica della IIa Legge di Errante

Questi dispositivi sono realizzabili per qualsiasi rapporto di trasformazione. Per esempio: per alimentare correttamente un dipolo aperto a mezz'onda con una linea bifilare a 300 Ohm ï¿½ necessario un Bal-Bal con ingresso a 300 Ohm ed uscita a 70 Ohm.

Il B.V.G.N.G. puï¿½, inoltre, generare un nodo di terra virtuale in un qualsivoglia punto di una data linea di trasmissione bilanciata.

Importanza scientifica:

Il BVGNG dimostra che le linee bilanciate, anche quando siano terminate su carichi reattivi, se opportunamente disaccoppiate dai radiatori, hanno sempre un comportamento a-periodico.
Vedasi: **comportamento delle linee di trasmissione bilanciate per segnali elettrici a radio frequenza in regime di onda progressiva**

Vantaggi operativi:

a) Sicura a-periodicitï¿½ della linea di trasmissione bilanciata;

b) inibizione dell'insorgenza di radiazione hertziana sulla linea stessa

c) reiezione totale delle correnti in fase indotte su di una linea di trasmissione bilanciata da una qualsiasi fonte esterna di RF;

d) diminuzione delle perdite di segnale.

ï¿½

Prezzo:
Euro 300,00 +IVA e spese di spedizione

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44. GENERATORE di NODO di TERRA VIRTUALE in linea © 2007 Francesco Errante

vgng_it.htm

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NEW SISTEMA CAPACITIVO DI TERRA PER RF

Vedi anche: Generatore di nodo di terra virtuale per linee di trasmissione bilanciate

�

Il *GENERATORE di NODO di TERRA VIRTUALE* in-line "**V.G.N.G.**", per brevit�

Il nodo di terra virtuale

�

Sistema di messa a terra capacitiva per RF

�

Le linee di trasmissione

Il generatore di nodo di terra virtuale in-linea.

Freq. di lavoro: da 1 a 30 MHz
Imp. d'ingresso: 50 Ohm
Imp. d'uscita: 50 Ohm
R.O.S di ingresso/uscita: 1:1 entro la banda
Perdita d'inserzione: -0.01 dB
Potenza RF: 3 KW cw
Connettori tipo: SO-239
Peso : 1 Kg ca.

�

Il generatore di nodo di terra virtuale in-linea: cos'�, come funziona e cosa fa.

Generatore universale di nodo di terra virtuale in-linea

di *Francesco Errante*

Il generatore di nodo di terra virtuale in-linea di *Errante* e' un circuito radio-elettrico basato sullo stesso principio di funzionamento e schema del progetto originale dell'antenna monopolare con terra virtuale
Il lettore che non abbia gi� familiarizzato col meccanismo di generazione del nodo di terra virtuale e' vivamente pregato di leggere: *l'antenna HF monopolare con terra virtuale* ed anche il circuito soppressore con terra virtuale prima di procedere nella lettura di questa pagina.

Il generatore di nodo di terra virtuale in-linea (in-line V.G.N.G. per brevit�) e' una rete radio-elettrica simmetrica che converte un segnale RF sbilanciato in due segnali bilanciati identici e li riconverte nuovamente in un segnale sbilanciato non-invertito.

bilanciate
in
regime
di onda
progressiva

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La radiazione
hertziana:
cos'� e come
avviene

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Nel far ci� la rete genera nel suo centro un nodo di terra virtuale. Una tale rete pu� efficacemente essere descritta come "Un-Bal-Bal-Un".
In pratica questo dispositivo, come presentato qui, ha il suo ingresso e la sua uscita entrambi terminati su un valore di impedenza pari a 50 Ohm ed usa connettori del tipo SO-239 per per adattarsi al pi� consueto standard di impedenza di linea co-assiale di trasmissione ed e' ottimizzato per operazioni nello spettro delle HF da 1 a 30 MHz. All'interno della sua banda di lavoro, questo dispositivo si dimostra "trasparente" tra la fonte RF ed il carico [SWR = 1:1] ed esibisce una bassissima perdita d'inserzione [-0.01 dB]. Comunque, esso pu� essere realizzato per qualsivoglia valore d'impedenza d'ingresso e di uscita, nonch� per qualsivoglia spettro di frequenze.

Il V.G.N.G. in-linea pu� generare un nodo di terra virtuale in un qualsivoglia punto di una data linea co-assiale in un sistema a linea di trasmissione sbilanciata. Questa �, infatti, la sua funzione. � anche disponibile la versione per linee bilanciate.

Applicazioni principali:

> **1)** Pu� essere impiegato per stabilizzare un intero sistema di RTX su linea sbilanciata, indipendentemente dalla sua complessit�. Esso fornisce un affidabile e centralizzato sistema di messa a terra per RF.
> Come molti lettori hanno gi� avuto modo di constatare, l'ottenimento di una buona messa a terra per RF � un'ardua impresa e non ha niente a che fare con le

soluzioni adottabili per la messa a terra convenzionale;
N.B. Per ragioni di sicurezza, ove richiesto, tutti gli apparecchi elettrici devono essere messi a terra nel modo convenzionale e lo chassis del VGNG in-linea pu� anch'esso essere collegato al dispersore di sicurezza. Gli chassis di tutti gli apparati vanno collegati ai terminali di massa di cui il VGNG in-linea dispone. Tutti i collegamenti da realizzarsi attraverso il percorso pi� breve.

2) Pu� essere impiegato per fornire un modo affidabile per corto-circuitare le antenne che non offrono protezioni antistatiche, senza doverle modificare o doverle raggiungere sul tetto. Grazie al fatto che il VGNG in-linea esibisce un solido corto circuito alla corrente continua su entrambi le sue porte RF, tutte le cariche elettrostatiche raccolte dall'antenna e/o conduttore esterno del cavo coassiale saranno instradate via percorso di terra comune verso il dispersore di terra di sicurezza dell'edificio;

3) Pu� essere usato per stabilizzare le linee co-assiali esposte a radiazione hertziana provenienti da altre fonti vicine;

4) Pu� essere impiegato per stabilizzare carichi reattivi nel loro punto di alimentazione.

�

Importanza scientifica:

Il VGNG in-linea permette a laboratori di fisica e RF di disporre di un affidabile, veloce ed economico sistema di messa a terra per RF da impiegarsi come preciso riferimento per misure e collaudi.

Vantaggi operativi:

a. Il VGNG in-linea � facile e veloce da installare;

b. Il VGNG in-linea aiuta a migliorare la compatibilit� E.M.C. all'interno di ambienti ad alta densit� di apparecchiature radio-elettriche.

�

F.A.Q.:

Domanda: Utilizzo un'antenna HF con terra virtuale, devo ugualmente impiegare un VGNG in-linea ?
RISPOSTA: NO ! Il tuo sistema di antenna con terra virtuale fornisce di gi� tutta la necessaria messa a terra per la R.F. Comunque, usando un'antenna differente pu� sorgere la necessit� di dover utilizzare un VGNG in-linea.

Domanda: In quale punto della mia linea di trasmissione dovrei inserire il VGNG?
RISPOSTA Laddove il sistema RTX su linea sbilanciata comprende soltanto il TX/RTX-SWRmeter-coaxial-antenna, il VGNG in-linea

deve essere collegato direttamente al connettore d'antenna del TX/RTX.
N.B. Se si usa un'adattatore d'impedenza d'antenna, il VGNG in-linea deve sempre precederlo. Ovvero, v� sempre collegato direttamente all'ingresso dell'adattatore d'impedenza.
Se si usa un amplificatore lineare di potenza, il VGNG in-linea deve essere sempre posto all'uscita dell'amplificatore.
Laddove il sistema comprende il TX/RTX, l'amplificatore lineare di potenza ed anche un adattatore d'impedenza d'antenna, il VGNG in-linea deve essere posto tra l'amplificatore e l'adattatore d'impedenza d'antenna.
Se sulla linea � presente un filtro, ad esempio un filtro anti-TVI, il filtro deve sempre trovarsi tra l'uscita dell'amplificatore lineare di potenza ed il VGNG in-linea. Inoltre, va' notato come il filtraggio migliori ed il sistema si giover� di un addizionale attenuazione delle emissioni fuori banda dovuta alla tipica rejezione dei segnali fuori banda propria del VGNG in-linea.
N.B. Il VGNG in-linea deve essere inserito nella linea di trasmissione usando un breve ma comodo tratto di cavo RG-213, o migliore, debitamente intestato con connettori Amphenol tipo PL-259. Ovunque sia possibile, preferire l'uso di giunti doppio maschio al posto di collegamenti con cavo co-assiale.

Prezzo:
Euro 160,00 +IVA e spese di spedizione

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45. ATTREZZATURE DIDATTICHE - PRODOTTI SCIENTIFICI - RADIAZIONI TRASDUTTORI RADIOELETTRICI

apparato_errante.htm

RADIONDISTICS

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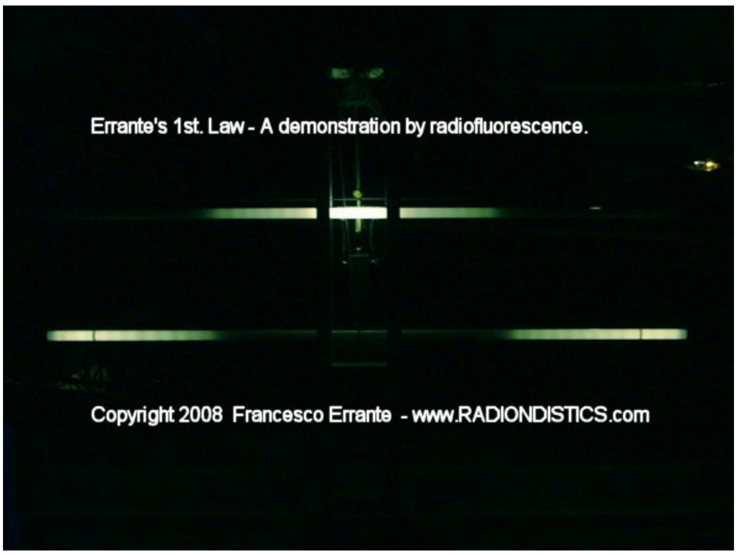
Apparato di Errante
per la fisica della radiazione hertziana.

Questo apparato permette di eseguire numerosi esperimenti e misure in materia di fisica dei segnali e trasduttori radio-elettrici.



Il nodo di terra virtuale

Generatore di nodo di terra virtuale per linee bilanciate

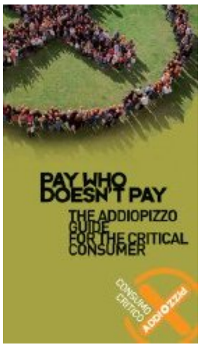
La radiazione hertziana:

Premessa

Sebbene sia trascorso oltre un secolo dal giorno in cui Heinrich Hertz realizzo' la prima trasmissione radio artificiale, la comprensione dei fenomeni radio-elettrici (piï¿½ comunemente intesi come fenomeni elettromagnetici) e' stata limitata alla sola teoria. Cio' perche' l'analisi dei fenomeni radio-elettrici aveva, finora, fatto affidamento esclusivamente sullo strumento matematico. Quest'ultimo pero', pur essendo di grande ausilio alla fisica, non puo' sostituirsi all'osservazione strumentale dei fenomeni naturali. L'osservazione strumentale e' essenzialmente legata al progresso tecnologico ma necessita soprattutto dell'inventiva del ricercatore scevro da preconcetti.

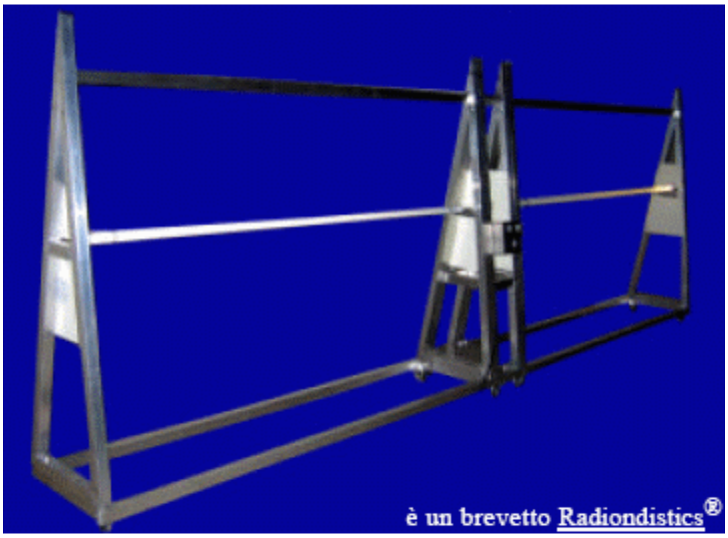
cos'ï¿½ e come avviene

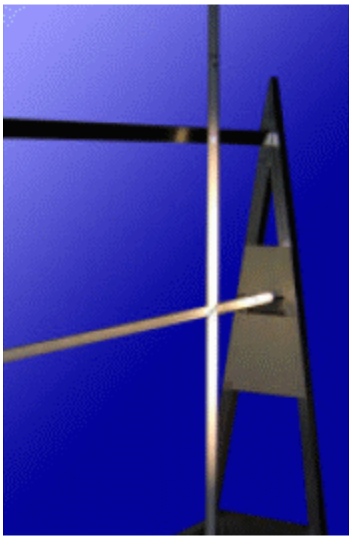
Rivelatore di radiazione hertziana mediante radio luminescenza

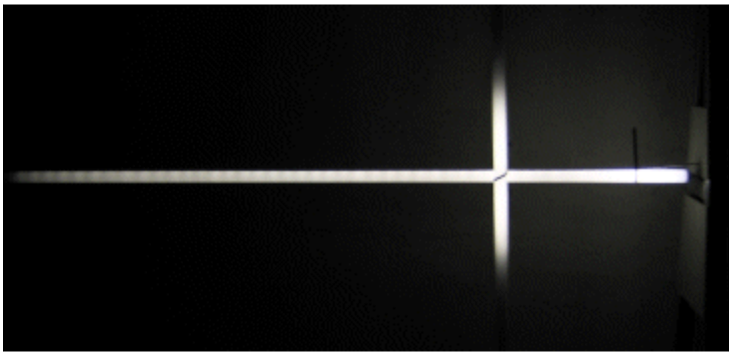


L'apparato qui presentato consente di riprodurre passo passo, i principali esperimenti, ideati e realizzati dall'autore per compiere indagini sulla fisica dei segnali e dei trasduttori radio-elettrici (antenne radio), nonchï¿½ per le indagini sulla radiazione hertziana (le radio-onde). mediante i quali lo stesso ha scoperto l'effettivo meccanismo della radiazione hertziana (trasduzione radio-elettrica). Tutti i dispositivi radio-elettrici essenziali utilizzati in questo apparato sono coperti da brevetto industriale.

Questo apparato consente di effettuare numerosi esperimenti e misure e permette di comprendere il meccanismo che origina la radiazione hertziana su un conduttore e la sua propagazione nello spazio, mediante l'uso contemporaneo di tubi rivelatori e strumenti elettrici, mettendo l'allievo in condizione di vedere ad occhio nudo la distribuzione energetica della radiazione hertziana su entrambi i rami di un antenna a dipolo e nelle sue immediate vicinanze, nonche 'di poter variare e misurare la potenza del segnale iniettato alla'ntenna e misurare i rapporti stazionari durante la fase di trasduzione radio-elettrica allo scopo di poter univocamente correlare cause ad effetti. (Proporzionalitï¿½ tra potenza del segnale radio-elettrico ed intensitï¿½ del campo generato; attenuazione introdotta dallo spazio etc...)







- **I^a esperienza di laboratorio:**

Verifica sperimentale della sopprimibilita' di uno qualsiasi dei due rami del dipolo aperto.

Descrizione:

questo esperimento permette di dimostrare che una qualsiasi antenna a dipolo non e' un'*antenna elementare*, bensi' essa e' una *cortina elementare* formata da due antenne indipendenti (i due rami del dipolo stesso). Esso si basa sul ns. circuito radio-elettrico per la soppressione di uno qualsiasi dei rami di un dipolo aperto a 1/2 onda (vedasi ns. soppressore) e permette l'estromissione alternata dal meccanismo di trasduzione radioelettrica, di entrambi i rami del dipolo e contemporaneamente permette di effettuare sia la misura del rapporto onde stazionarie (ROS) agli ingressi di entrambi i rami del dipolo che all'ingresso del sistema di simmetrizzazione (vedasi ns. balun rosmetrico con terra virtuale).

- **II^{a} esperienza di laboratorio:**

Verifica sperimentale sperimentale dello sdoppiamento elettrico del dipolo ripiegato.

Descrizione:

questo esperimento permette di dimostrare che una qualsiasi antenna a dipolo ripiegato non e' un'*antenna elementare*, bensi' essa e' una *cortina elementare* formata da due antenne indipendenti ma coincidenti. Esso si basa sul ns. balun con terra virtuale per la sdoppiamento del filamento rami di un dipolo aperto a 1/2 onda (vedasi ns. sdoppiatore) e permette l'estromissione alternata dal meccanismo di trasduzione radioelettrica, di entrambi i rami sdoppiati del dipolo ripegato e contemporaneamente permette di effettuare sia la misura del rapporto onde stazionarie (ROS) agli ingressi di entrambi i rami del dipolo che all'ingresso del sistema di simmetrizzazione (vedasi ns. balun rosmetrico con terra virtuale).

- **IIIª esperienza di laboratorio:**

  Verifica sperimentale del potere ionizzante della radiazione hertziana.

  Descrizione:

  Questo esperimento consente di verificare il potere ionizzante della radiazione hertziana (radio-onde) tramite l'impiego di un rivelatore passivo di radiazione hertziana a fluorescenza da ionizzazione secondaria (Vedasi ns. rivelatore).

- **IVª esperienza di laboratorio:**

  Verifica sperimentale del meccanismo della radiazione hertziana.
Iª e IIª legge di Errante

  Descrizione:

  Questo esperimento consente di verificare come un qualsiasi segnale elettrico a radiofrequenza od un qualsiasi impulso elettrico a radiofrequenza, iniettato su un qualsiasi conduttore elettrico di qualsiasi forma, dia sempre origine a radiazione hertziana con inizio sul lato opposto a quello da dove viene iniettato. Parimenti, consente di verificare come un qualsiasi segnale elettrico a radiofrequenza od un qualsiasi impulso elettrico a radiofrequenza, iniettato su un qualsiasi conduttore elettrico di qualsiasi forma, debitamente terminato su un carico anti-induttivo di impedenza pari a quella della sua fonte, non dia mai origine a radiazione hertziana. (Vedasi La radiazione hertziana - osservazione radioscopica di *Francesco Errante*)

- **Vª esperienza di laboratorio:**

  Verifica sperimentale della direttivitï¿½ del dipolo. (Guadagno e reiezione)

  Descrizione:

  Questo esperimento consente di verificare come la f.e.m. indotta su un dipolo varii al variare del suo orientamento rispetto ad una fonte remota di radiazione hertziana.

- **VIª esperienza di laboratorio:**

  Accoppiamento di due dipoli paralleli nello spazio. (Radiazione hertziana ed induzione di f.e.m.)

  Descrizione:

  Questo esperimento consente di verificare, macroscopicamente, il trasferimento di energia nello spazio mediante radiazione hertziana.

E', adesso, commercialmente disponibile il ns. apparato per la fisica delle linee di trasmissione bilanciate per segnali elettrici a radio frequenza. Leggi !

RADIONDISTICS e' un *OEM* (Original Equipment Manufacturer) e produce dispositivi e sistemi radio-elettrici e radio-elettronici interamente basati sulla propria tecnologia brevettata. Oltre ai prodotti standard qui mostrati, Radiondistics puo' progettare, produrre e fornire articoli, materiali, attrezzature e sistemi di misura completi per qualsiasi esigenza del Cliente.

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46. BALUN ROSMETRICO CON TERRA VIRTUALE PER DIPOLO APERTO © 2003 Francesco Errante

balun_rosmetrico.htm

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Planning a trip to Sicily? Then, make sure you won't be feeding the Mafia!

BALUN ROS-metrico

di *Francesco Errante*

Il *balun rosmetrico* � un modello di utilita' basato sul circuito del soppressore del ramo di dipolo. Esso � dotato di tre rosmetri incorporati per il monitoraggio di ogni sua porta. Inoltre il balun rosmetrico incorpora due commutatori coassiali e due carichi fittizii per la soppressione alternata di entrambi i rami del dipolo sotto esame.

Il *balun rosmetrico* � il simmetrizzatore ideale per gli esperimenti con il rivelatore di radiazione hertziana a fluorescenza da ionizzazione secondaria.

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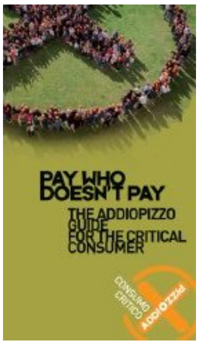
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47. BALUN CON TERRA VIRTUALE PER LO SDOPPIAMENTO DEL DIPOLO RIPIEGATO A ½ ONDA

balun_300_it.htm

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NEW **Pilota a larga banda per antenne a dipoli incrociati, note anche come antenne *turnstile* o quardipoli.**

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BALUN di ERRANTE

Il nodo di terra virtuale

Generatore di nodo di terra virtuale per disaccoppiare una linea di trasmissione bilanciata dal

BalUn con terra virtuale per dipoli ripegati & loops

A state-of-the-art high power broadband HF balun

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BALUN CON TERRA VIRTUALE PER LO SDOPPIAMENTO DEL DIPOLO RIPIEGATO A ½ ONDA

by *Francesco Errante*

Dipolo ripiegato a 1/2 onda: un'antenna... anzi due! (BIS)

N.B. Questa pagina tratta di una speciale applicazione a fini scientifici del balun di *Errante* con terra virtuale. Per una panoramica su tutti i baluns di *Errante* con terra virtuale per l'impiego nelle radio comunicazioni in HF e per ulteriori approfondimenti teorici, si prenda visione della pagina successiva.

Funzioni scientifiche

a. Il balun qui descritto consente lo sdoppiamento fisico ed elettrico del filamento del dipolo ripiegato a ½ onda.

b. Il balun qui descritto consente di dimostrare che il filamento di un dipolo ripiegato a ½ onda (avente

suo carico

Sistema di messa a terra capacitiva per RF

Le linee di trasmissione bilanciate in regime di onda progressiva

La radiazione hertziana: cos'è e come avviene

un'impedenza caratteristica di 300 Ohm) si comporta come se fosse costituito da due filamenti distinti ma identici e coincidenti alimentati in controfase ed aventi, ognuno, un'impedenza caratteristica di 150 Ohm, riferita al nodo di terra virtuale.

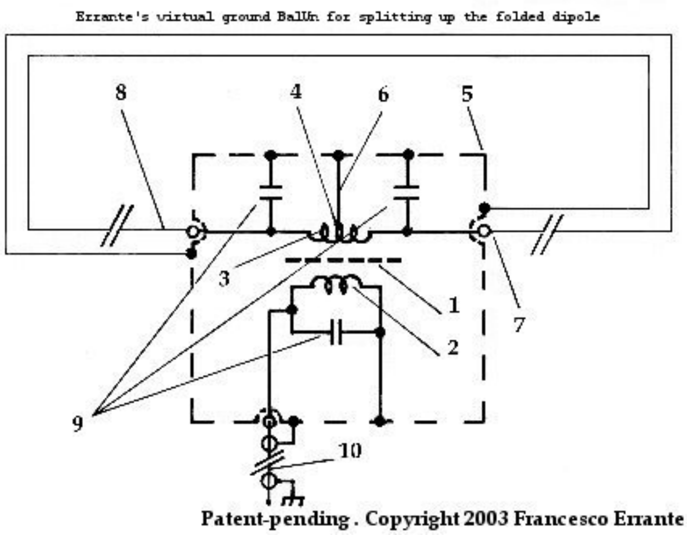
c. Il balun qui descritto consente di dimostrare che il filamento del dipolo ripiegato a ½ onda e, generalizzando, anche il filamento di qualsiasi altro loop risonante, di qualsiasi forma, sia esso ad unica spira che a spire multiple, quando viene correttamente alimentato con un segnale radio-elettrico sinusoidale di idonea lunghezza d'onda, esso è interamente percorso, da capo a capo, da correnti che per un semiperiodo hanno verso orario e per il successivo semiperiodo hanno verso antiorario o viceversa.

Descrizione

L'Autore, mediante l'uso di un particolare circuito radio-elettrico per la soppressione di uno qualsiasi dei due rami di un dipolo aperto a ½ onda, ha precedentemente dimostrato la possibilità di alimentare separatamente ed indipendentemente i due rami di un dipolo aperto a ½ onda, nella stessa occasione è stato introdotto il concetto ed il metodo per la generazione di un nodo di terra virtuale per la separazione delle correnti di ramo.
Il dipolo ripiegato a ½ onda è costituito da un unico conduttore e non consente l'utilizzazione indipendente delle correnti che lo attraversano nei due versi.
La presente invenzione permette lo sdoppiamento del filamento del dipolo ripiegato a ½ d'onda per consentire di dimostrare che lo stesso filamento si comporta come se fosse costituito da due filamenti

identici e coincidenti ma ognuno di essi attivo soltanto per la durata di un semiperiodo. Tale risultato è stato ottenuto mediante l'impiego di un circuito radioelettrico a costanti concentrate, basato su di un trasformatore elettrico(1) a larga banda per se-gnali a radiofrequenza, avvolto su nucleo di ferrite di forma binoculare, avente un avvolgimento primario(2) con impedenza di valore pari a quello della linea di trasmissione adottata(10) ed un avvolgimento secondario(3) con presa centrale(4) ed impedenza pari a 150 +150 Ohm riferiti al nodo di terra virtuale. Il nodo di terra è propagato a tutto il circuito mediante la gabbia di scher-maggio elettrostatico(5) connessa alla presa centrale del trasformatore(1) mediante un breve collegamen-to elettrico(6). L'ottimizzazione del punto di lavoro del trasformatore è ottenuta mediante compensazione con l'utilizzo di capacitá elettriche ad alto isolamento(9). Il trasformatore balun oggetto di questa invenzione permette lo sdoppiamento del filamento originario del dipolo ripiegato a ½ onda, in due filamenti gemelli(7, 8) fornendo loro due segnali identici ma in controfase tra di loro ed aventi ognuno un' impedenza pari a 150 Ohm sbilanciati e riferiti al nodo di terra virtuale. Le due spire(7, 8) elettricamente indipendenti ma concentriche rispetto al loro punto di alimentazione, possono ora assumere la posizione di parallele tra di loro, consentendo il loro funzionamento concorrente sulla stessa direttrice oppure possono essere disposte l'una perpendicolare all'altra rispetto alla loro dimensione maggiore, permettendo al dipolo di operare con un diagramma di radiazione quasi-omnidirezionale.

E' stato così inventato un *balun con terra virtuale* che consente lo sdoppiamento del filamento del dipolo ripiegato a ½ onda e permette di dimostrare che esso in realta' lavora come se fossero due filamenti identici e coincidenti ma alimentati in controfase.



Peculiaritá elettriche ed operative

1. Il balun qui descritto è caratterizzato dal fatto di avere una presa centrale sul suo avvolgimento secondario per la generazione di un nodo di terra virtuale che costituisce il riferimento elettrico a potenziale zero di tutto il circuito e ne permette una precisa stabilizzazione elettrica.

2. Il balun qui descritto è caratterizzato dal fatto che l'avvolgimento secondario(4) del trasformatore, durante la trasmissione di segnali radio-elettrici nello spazio, provvede ad erogare due segnali radioelettrici identici ma in controfase tra di loro, ovvero, ognuno dei segnali presenta una differenza dell'angolo di fase rispetto al nodo di terra virtuale pari a ±90 gradi.

3. Il balun qui descritto è caratterizzato dal fatto che, quando le due spire del dipolo ad esso collegate sono orientate nella medesima direzione, durante la ricezione di segnali radioelettrici provenienti dallo spazio da direzioni fuori orientamento rispetto al dipolo stesso, il trasformatore(1) nel provvedere a combinare qualsiasi segnale bilanciato captato dai due rami del dipolo, introduce una reiezione elettrica che aumenta col diminuire della differenza dell'angolo di fase tra i due segnali ad esso inviati dai rami del dipolo stesso. Ciò permette di esaltare la già nota proprietà di *reiezione di fianco* del dipolo, che si traduce in una maggiore attenuazione dei segnali indesiderati provenienti da direzioni al di fuori dell'orientamento del dipolo stesso.

4. Il balun qui è caratterizzato dal fatto che permette di variare il diagramma di radiazione di un dipolo ripiegato a ½ onda, consentendo d'irradiare lungo un'unica direttrice oppure di operare con un diagramma di radiazione quasi-omnidirezionale (**arrangiamento del tipo *turnstile***).

5. Il balun qui descritto è caratterizzato dal fatto che la larghezza di banda del dipolo ripiegato a ½ onda ad esso connesso è sensibilmente maggiore rispetto a quando lo stesso dipolo è utilizzato con altri balun già noti.

6. Il balun qui descritto è caratterizzato dal fatto che la suscettibilità alle influenze da parte di capacità parassite (suolo, muri etc.) di ognuno dei due rami del dipolo stesso, è notevolmente ridotta rispetto a quando il loro adattamento alla linea di trasmissione sbilanciata è fatto con altri sistemi di adattamento già noti (BALUN), consentendo di trasmettere con notevoli potenze anche in stretta prossimità del suolo, di muri od ostacoli vari e con qualsiasi inclinazione rispetto agli stessi, senza che si verifichino apprezzabili variazioni d'impedenza (disadattamenti) all'ingresso del trasformatore(1).

7. Il balun qui descritto è caratterizzato dal fatto che tutte le porte del trasformatore(1) presentano un cortocircuito alla corrente continua e sono tutte collegate alla terra fisica tramite il conduttore esterno della linea di trasmissione coassiale(3), consentendo lo scaricamento delle cariche elettrostatiche via terra comune.

8. Il balun di *Errante* si caratterizza per essere un trasformatore elettrico per RF ad *impedenza NON fluttuante*. I valori d'impedenza delle terminazioni del balun di *Errante* sono determinate per costruzione, ne deriva che il rapporto di trasformazione e' una sua funzione e NON viceversa, diversamente da quanto avviene con tutti gli altri arrangiamenti bal-un.

N.B. Questa pagina tratta di una speciale applicazione a fini scientifici del balun di *Errante* con terra virtuale. Per una panoramica su tutti i baluns di *Errante* con terra virtuale per l'impiego nelle radio comunicazioni in HF e per ulteriori approfondimenti teorici, si prenda visione della pagina successiva.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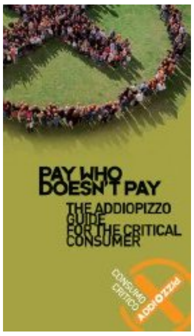

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49. Sistema d'antenna MULTIBANDA per le HF con radiatore a 225 Ohm © 2009 Francesco Errante - www.Radiondistics.com

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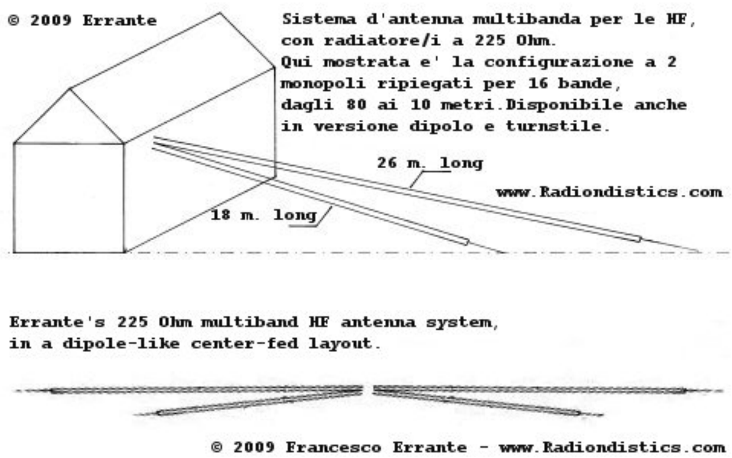
Sistema d'antenna MULTIBANDA per le HF, con radiatore a 225 Ohm.



Implementazione completa delle proprietá multibanda del radiatore elementare a 225 Ohm, nella configurazione a monopolo ripiegato, dipolo ed antenna turnstile.

Tutte le bande radio amatoriali per tutte le Regioni I.T.U. dagli 80 ai 10 metri e molte

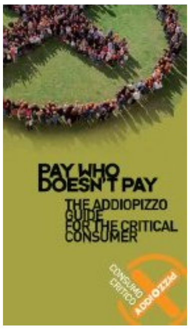
Sistema di messa a terra capacitiva per RF

Il nodo di terra virtuale

Il *radiatore elementare*

altre ancora, in 16 ampii segmenti, con l'uso di due soli radiatori!

NB. Allo scopo di evitare malintesi, si fa notare che il presente progetto NON ha niente a che spartire con il *monopolo ripiegato, elettricamente corto* (<< ¼ lambda) noto anche come "*folded unipole*", che è un radiatore a bassissima impedenza, con perdite intrinseche.



Fai click sulle immagini per ingrandirle

BalUn
con terra
virtuale

La radiazione
hertziana:
cos'è
e come
avviene

Balun con terra virtuale per il sistema d'antenna HF multibanda con monopoli ripiegati, da 225 Ohm, in configurazione dipolo.
Si notino i due isolatori pivottanti, realizzati in Nylon resitente ai raggi ultravioletti - nella fattispecie lunghi 42 cm l'uno - per l'ancoraggio di due monopoli ripiegati per lato. Una soluzione ideale per l'installazione dei rami a qualsiasi angolo dai 120 circa ai 180 gradi e nuovamente in sù a 120 gradi. Da "V" invertita a -> dipolo aperto e sù anche -> a "V".
Gli isolatori sono liberi di ruotare attorno

all'asse costituito dai perni M8, che sono anche i terminali del balun stesso ed i capicorda sono serrati indipendentemente dagli isolatori.

Sistema d'antenna MULTIBANDA per le HF con radiatore a 225 Ohm: cos'è e come funziona.

by *Francesco Errante*

Il sistema di antenna multibanda qui presentato si basa sulla *risonanza armonica*, un fenomeno fisico ben noto. Questo sistema, pertanto, NON utilizza trappole, nè cariche, nè altri artificii per la segmentazione delle bande.

I radiatori, banda per banda, si comportano da mono-banda full-size, risonando su una differente frazione o multiplo della lunghezza d'onda del segnale desiderato, ad iniziare dai 3/4 della lunghezza d'onda della banda piú bassa di ogni serie di risonanze.

Per virtú di ció, è possibile ottenere numerosi e ben distinti gruppi di bande di frequenza, ognuno con 8 utili bande nello spettro delle HF, ognuna delle quali esibisce un VALORE COMUNE DI IMPEDENZA, pari a 225 Ohm. **Questo particolare valore d'impedenza comune è la chiave del progetto di questo sistema**, mentre la tecnologia per l'adattamento e messa in fase delle reti RF a larga banda propria di *Radiondistics* ha permesso a questo sistema di lavorare in modo fluido, con un eccezionale adattamento d'impedenza su tutte le sue bande. **NESSUN "*accordatore d'antenna*" è necessario e tutte le gamme operative**

offrono una ampia larghezza di banda, propria dei sistemi ad alta impedenza.

Inoltre, questo sistema, per il suo ingombro, specie nella versione a monopoli, non necessita di spazii eccessivi per la sua posa, e' facile da installare e NON richiede tediosi aggiustamenti della lunghezza dei suoi radiatori per il suo accordo. Le sue bande di lavoro, infatti, risultano ben definite ma NON sono critiche. Se necessario, i suoi radiatori si prestano ad essere ulteriormente piegati rispetto al proprio asse (anche se ció determina una variazione del diagramma di radiazione del sistema il suo accordo rimane immutato). I radiatori possono essere installati in tutti i modi (sloper, v-invertita, L etc.) e con qualsiasi angolo rispetto al suolo. Il sistema mantiene i suoi accordi anche in prossimitá della terra o altri ostacoli. Grazie alla tecnologia di alimentazione a RF, propria di *Radiondistics*, nessun de-tuning avviene sui radiatori, ammenocchè essi non vengano ammainati al suolo. **Per esempio, un arrangiamento da 2 monopoli ripiegati a 16 bande (come mostrato sopra), per tutte le bande radio amatoriali dagli 80 ai 10 metri e molte altre ancora per tutte le Regioni I.T.U., occupa circa 26 metri di spazio, se disposto su una linea retta, mentre una configurazione a dipolo alimentata a centro necessita semplicemente uno spazio doppio**.

Visti i vantagi di questo sistema multibanda, è stata sviluppata ed e' ora disponibile una completa linea di sistemi di alimentazione per:

monopoli ripiegati con sistema di terra capacitivo ;

configurazioni a dipolo. Vedi 50/225+225 Ohm virtual ground balun ;

monopoli ripiegati sistema di terra virtuale ;

monopoli ripiegati in configurazione turnstile.

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Prices:

Errante's 50/225 Ohm capacitive earth grounded UN-UN,
Euro 250,00 +V.A.T. and shipping costs ;

Errante's 50/225+225 Ohm BalUn, as pictured above
Euro 500,00 +V.A.T. and shipping costs ;

Errante's 50/225 Ohm virtual ground feeding system,
Euro 700,00 +V.A.T. and shipping costs ;

Errante's 4x 225 Ohm broadband turnstile driver,
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50. BALUN UNIVERSALE CON TERRA VIRTUALE PER DIPOLO © 2003 Francesco Errante

balun_errante_it.htm

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NEW **Pilota a larga banda per antenne a dipoli incrociati, note anche come antenne *turnstile* o quardipoli.**

NEW **Balun con terra virtuale per il sistema d'antenna HF multibanda con monopoli ripiegati, da 225 Ohm, in configurazione dipolo**

�

BALUN di ERRANTE

BalUn con terra virtuale per dipoli & loops

�

Planning a trip to Sicily? Then, make sure you won't be feeding the Mafia!

Il nodo di terra virtuale

Sistema di messa a terra capacitiva per RF

Le linee di trasmissione bilanciate in

A state-of-the-art high power broadband HF balun

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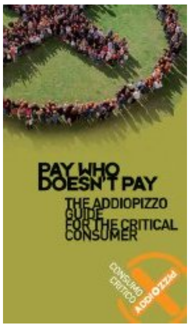
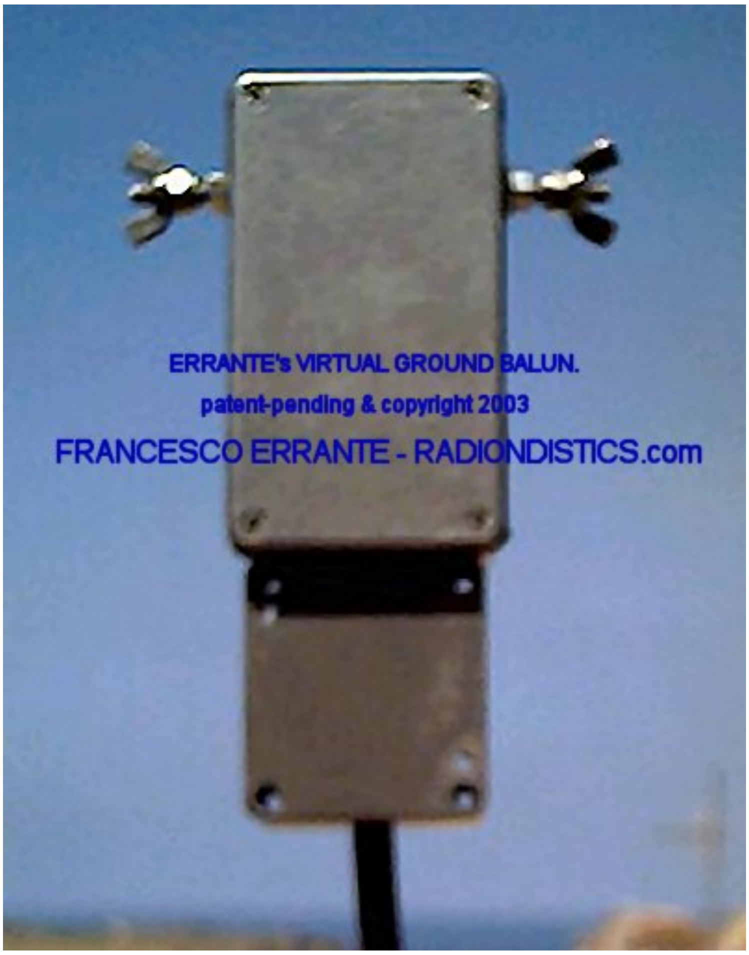
Visita www.uibm.gov.it ed immetti *Francesco Errante* nel campo di ricerca per vedere tutti i miei brevetti, o verifica direttamente sul sito dell'***European Patent Office*** QUI e QUI !





regime
di onda
progressiva

La radiazione
hertziana:
cos'� e come
avviene

Immagini del BALUN con terra virtuale, 50/70 Ohm
Banda di lavoro del BALUN: da 1 a 30 MHz
Potenza RF: 2KW CW

Fai click sulle immagini per ingrandirle




N.B. L'esemplare qui raffigurato � disponibile in allestimenti normalizzati per:

NEW! Dipolo multibanda 225+225 Ohm - Ingresso 50 Ohm sbilanciati uscita 225+225 Ohm bilanciati **Sistema d'antenna MULTIBANDA a 225 Ohm** ;

Dipolo aperto a � onda - Ingresso 50 Ohm sbilanciati, uscita 70 Ohm bilanciati;

Dipolo ripiegato a � onda - Ingresso 50 Ohm sbilanciati, uscita 300 Ohm bilanciati;

Dipolo a *V invertita* da � onda - Ingresso 50 Ohm sbilanciati, uscita 50 Ohm bilanciati;

Deltaloop onda intera a singola spira - Ingresso 50 Ohm sbilanciati, uscita 200 Ohm bilanciati.

Differenti valori d'impedenza o differenti terminazioni sono disponibili solo su richiesta.

�

Il Balun con terra virtuale per dipoli & loops : cos'�, come funziona e come si � pervenuti alla sua realizzazione.

di *Francesco Errante*

Il Balun con terra virtuale, qui presentato, � l'applicazione pratica nel campo delle radio comunicazioni in onde corte di quanto precedentemente realizzato dell'Autore per applicazioni scientifiche. Esso, come il suo omologo il circuito radioelettrico per lo sdoppiamento del dipolo ripiegato a � onda, deriva, infatti, dal **circuito radioelettrico per la soppressione di uno qualsiasi dei due rami di un dipolo aperto a � onda** .

Il balun di *Errante* e' essenzialmente un trasformatore elettrico puro che nella sua versione pi� elementare ha un minimo di 3 circuiti accordati: il circuito primario (rete LC dell'avvolgimento primario) e 2 circuiti secondari (reti LC degli avvolgimenti secondari). Generalmente, i circuiti secondari del balun di *Errante* sono identici, per essere idonei ad alimentare antenne bilanciate simmetriche. All'occorrenza, essi possono essere anche formati per valori d'impedenza tra di loro differenti, mantenendo, pero', costante l'angolo di sfasamento delle terminazioni bilanciate a �90 gradi rispetto al nodo di terra virtuale. Inoltre, come tutti i trasformatori elettrici puri, il balun di *Errante* puo' essere costruito con pi� di una coppia di secondari.

Il balun di *Errante* si caratterizza per NON permettere alcuna trasformazione

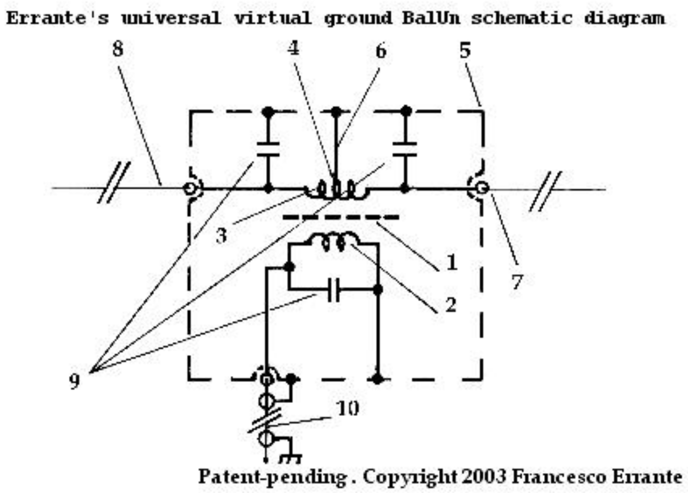
d'impedenza fintantocche' non si verifichi l'adattamento d'impedenza delle proprie terminazioni (impedenza NON-fluttuante). I valori d'impedenza delle terminazioni del balun di *Errante* sono determinate per costruzione, ne deriva che il rapporto di trasformazione e' una sua funzione e NON viceversa, diversamente da quanto avviene con tutti gli altri arrangiamenti bal-un. Vedasi ad esempio il trasformatore di *Guanella* od il trasformatore di *Ruthroff* che invece sono caratterizzati dall'avere un rapporto di trasformazione fluttuante. Cio' perche' essi non sono altro che arragiamenti di auto-trasformatori e quindi i loro circuiti d'ingresso e di uscita sono inscindibili, con la conseguenza che ogni variazione apportata ai loro circuiti primarii si riflette sui loro circuiti secondari.

In pratica, un balun che permette la fluttuazione dei valori d'impedenza delle proprie terminazioni effettua la trasformazione d'impedenza all'interno di una vasta escursione di valori d'impedenza, modificando l'impedenza della sua terminazione d'ingresso in funzione di quella del carico e viceversa. Quindi, supponendo di disporre di un balun del tipo *Guanella* con rapporto di trasformazione 4:1, si avra' al suo ingresso un'impedenza di 75 Ohm se il carico e' da 300 Ohm, oppure un'impedenza di 50 Ohm se il carico e di 200 Ohm e cosi' via. **Questo non puo' invece avvenire con l'impiego di un balun del tipo *Errante*, perche' i valori d'impedenza di ognuna delle sue terminazioni sono proprieta' fisse delle stesse e non variano al variare dei valori di impedenza dei circuiti (antenne o linee) ad essi collegati. Limitando le impedenze dei sistemi linea-**

carico ai soli valori imposti, consente di inibire sia le linee che i carichi (antenne) dal prestarsi a risonanze indesiderate, generalmente frutto di risonanze armoniche, molto spesso derivanti dalla presenza di pi� carichi sullo stesso punto di alimentazione che generano percorsi alternativi per la RF. (vedasi dipoli o monopoli multipli, antenne a ventaglio e cosi via). **Questo e' un particolare da non sottovalutare, non solo in fase di trasmissione ma anche e soprattutto in fase di ricezione. L'impiego di un balun ad impedenze NON fluttuanti, infatti, introduce un'altissima reiezione ai segnali radio interferenti provenienti da fonti al di fuori delle bande di spettro radio per cui un'antenna e' destinata. Da qui ne deriva un'eccellente stabilita' e selettivita' dei sistemi di linea e d'antenna che impiegano i balun di *Errante***

Un balun con terra virtuale rappresenta il dispositivo ideale per la corretta alimentazione di dipoli e loops. Esso provvede contemporaneamente ad:
1) adattare l'impedenza della sorgente RF al carico, mediante trasformazione a perdite irrisorie;
2) ottenere due porzioni identiche del segnale RF ad esso fornito e dotate del corretto sfasamento tra di esse per la naturale alimentazione dei dipoli aperti o ripiegati (ivi incluse le antenne del tipo *deltaloop*);
3) ottenere un nodo di terra virtuale per la corretta polarizzazione della sua porta sbilanciata;
4) ottenere un percorso di terra comune per la dissipazione delle tenzioni indotte dai fenomeni elettrostatici;
5) ottenere un elevato disaccoppiamento tra i due rami bilanciati.

Lo schema elettrico del balun con terra virtuale � riportato qu� appresso.



Il balun con terra virtuale � alloggiato all'interno di un contenitore metallico a doppio corpo dotato di un vano a tenuta stagna per la circuiteria del balun e di un secondo vano per la protezione dei connettori della linea coassiale. Detto contenitore metallico oltrecch� assicurare un'elevata robustezza meccanica ed una totale stabilit� nei confronti degli agenti atmosferici assolve anche alla funzione di conduttore elettrico generale per il riferimento al nodo di terra virtuale ed assicura un completo schermaggio elettrostatico della circuiteria del balun stesso.

Radiondistics, parimenti, rivendica l'originalita' delle soluzioni elettromeccaniche adottate.

Vantaggi operativi:

a. Il balun con terra virtuale permette all'antenna di operare con perdite ridotte rispetto alla stessa antenna alimentata con balun convenzionali.

b. Il balun con terra virtuale permette all'antenna di esibire una larghezza di banda caratteristica pi� ampia rispetto alla stessa antenna alimentata con balun convenzionali.

c. Il balun con terra virtuale permette all'antenna di esibire un'elevata immunit� agli effetti delle capacit� parassite.

d. Il balun con terra virtuale permette all'antenna di esibire un'alta stabilit� del punto di risonanza indipendentemente dalla sua inclinazione e distanza dal suolo.

e. Il balun con terra virtuale permette all'antenna di esibire un maggiore reiezione ai segnali fuori-banda, rispetto alla stessa alimentata con balun convenzionali.

f. Il balun con terra virtuale permette all'antenna di esibire un maggiore immunita' alle cariche ed ai rumori di natura elettrostatica rispetto alla stessa alimentata con balun convenzionali.

g. Il balun con terra virtuale, grazie ai fenomeni di interferenza costruttiva e distruttiva che hanno luogo nel trasformatore, permette all'antenna a dipolo di esibire un diagramma di radiazione/ ricezione (reiezione dei segnali fuori dall'orientamento dell'asse del fascio) pi� netto e definito rispetto alla stessa alimentata con balun convenzionali.

h. Il balun con terra virtuale permette all'antenna di essere installata in tempi molto pi� rapidi di quanto richiesto dalle stessa se alimentata attraverso un balun convenzionale.

i. Il balun con terra virtuale migliora la compatibilita' ambientale (EMC & TVI) delle antenne previamente utilizzate con baluns convenzionali.

Applicazioni:

a. STAZIONI RADIO TERRESTRI FISSE E MOBILI PER IMPIEGHI CIVILI E MILITARI.

b. INSTALLAZIONI RAPIDE, SEGRETE e STRATEGICHE. �

c. SERVIZI DI LOCALIZZAZIONE e SALVATAGGIO.

d. RADIO COMUNICAZIONI NAVALI (SHIP to SHIP & SHIP to SHORE).

e. IMPIEGHI RADIOAMATORIALI DXing ed installazioni su propieta' condominiali.

f. ELEMENTI DI CORTINE d'ANTENNE OMOGENEE

g. LABORATORI di FISICA delle RADIO ONDE.
Vedi anche: Balun rosmetrico

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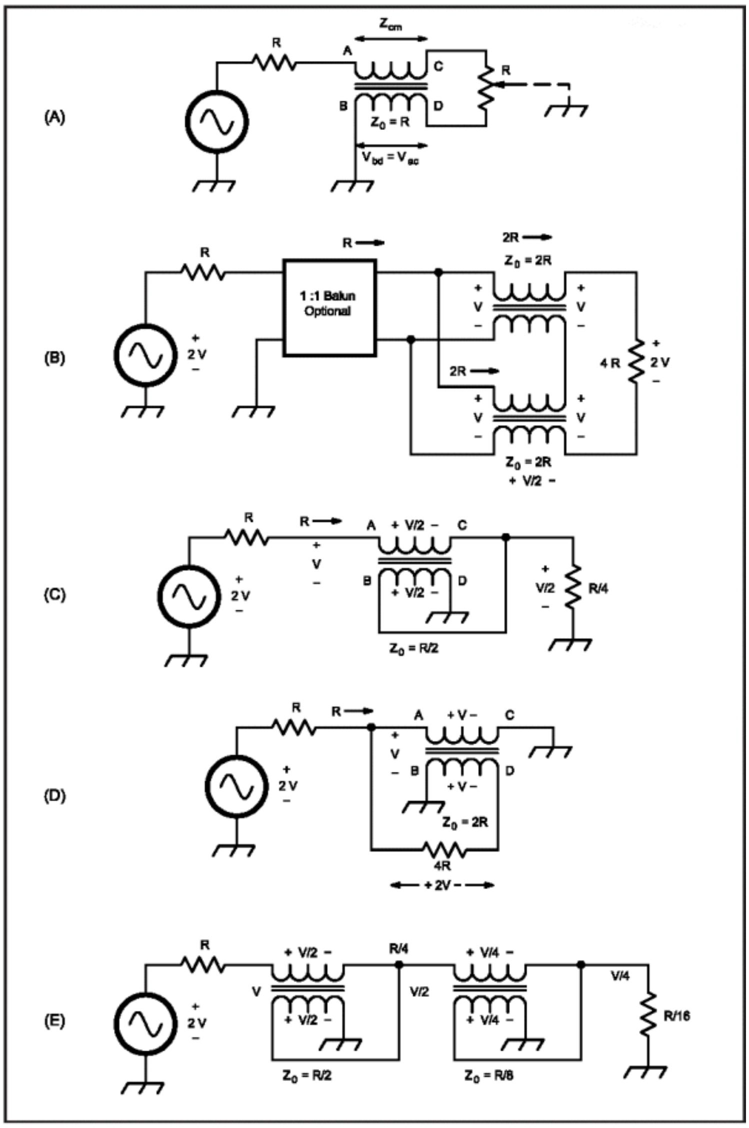
Prezzo al pubblico:
Euro 250,00 +IVA e spese di spedizione.
N.B. Il prezzo della versione 50/225+225 � di Euro 500,00 +IVA e spese di spedizione.

RADIONDISTICS garantisce a vita i propri prodotti.

Schemi elettrici del balun di *Guanella* e quello di *Ruthroff*



At A, basic balun. At B, 1:4 Guanella transformer. At C, Ruthroff transformer, 4:1 unbalanced. At D, Ruthroff 1:4 balanced transformer. At E, Ruthroff 16:1 unbalanced transformer.

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Over 25 years of direct action for animals
without harming a single human being

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51. Antenna TURNSTILE HF, dipolo a croce, quadripolo - Pilota a larga banda © 2008 Francesco Errante - Radiondistics.com

quadripolo.htm

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Planning a trip to Sicily? Then, make sure you won't be feeding the Mafia!

Pilota a larga banda per antenne HF turnstile

ORA, anche disponibile nella VERSIONE MULTIBANDA

La soluzione tutto-in-uno per accoppiare, bilanciare e fasare il dipolo a croce per lo spettro delle HF

Il nodo di terra virtuale

Sistema di messa a terra capacitiva per RF

BalUn con terra virtuale

ERRANTE's HF Turnstile antenna broadband driver.

Freq. range: 1 to 30 MHz
Input imp.: 50 Ohm, unbalanced
Output imp.: balanced 2 x 70 Ohm or 2 x 450 Ohm for the Errante's multiband system
Outputs phase shifts: 0, 90, 180, 270 degrees
SWR @ band-center: 1:1
Insertion loss: -0.01 dB
RF power: 4 KW CW
Sockets type: SO-239
Weight : 5 Kg ca.

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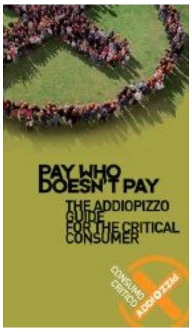

Pilota a larga banda per antenne HF turnstile : cos'�, come funziona e cosa fa.

di *Francesco Errante*

Il pilota per antenne HF turnstile, � un circuito radioelettrico fondato sul ns. progetto originale e brevettato del *balun con terra virtuale*.
L'antenna *turnstile*, nota anche come *dipolo a croce* o *quadripolo* � la scelta primaria laddove � richiesto un diagramma di radiazione quasi-omnidirezionale in un sistema d'antenna a polarizzazione orizzontale.� (documenti del brevetto di George H. Brown, anno 1937)
Nelle radio comunicazioni da punto a punto o per diffusione via radio propagazione ionosferica tali sistemi hanno il vantaggio di seguire i segnali desiderati, indipendentemente dai cambiamenti di direzione di provenienza del segnale introdotti dal cielo.
I fenomeni trans-ionosferici includono la rifrazione, la variazione di ampiezza e di fase, nonch� ritardi e

Le linee di trasmissione bilanciate in regime di onda progressiva

La radiazione hertziana: cos'� e come avviene

rotazioni di polarizzazione dei segnali radio. Tali variazioni influiscono sempre sulla forza del segnale desiderato durante la ricezione tramite radio propagazione ionosferica (evanescenza, QSB) e possono facilmente introdurre perdite che raggiungono i 20 dB. **Un sistema di antenna del tipo *turnstile* pilotato con questo ns. dispositivo recupera quelle perdite ed in media tutti i segnali sono dai 10 ai 15 dB pi� forti se comparati ai segnali estratti da un singolo dipolo nelle medesime condizioni.** .
Finora, la realizzazione di un'antenna turnstile per le bande delle onde corte � stata un'impresa da titani e nessun sistema precedente permette operazioni multibanda, a causa delle limitazioni proprie delle loro linee di ritardo.
Il dispositivo qui presentato �, invece, la soluzione ideale per utilizzare due dipoli incrociati con la stessa semplicit� richiesta da un singolo dipolo. Esso, infatti, esibisce le stesse propriet� del ns. balun con terra virtuale ma permette di pilotare 2 identici dipoli o 2 identici dipoli a ventaglio in *configurazione turnstile*, grazie al suo peculiare sistema di fasamento. Con l'eliminazione della linea di ritardo, il ns. sistema ha eliminato anche la perdita di segnale inevitabilmente introdotta dalla stessa. Ne consegue che la quantit� di energia traferita ad ognuno dei 2 dipoli � identica, questo si traduce in un diagramma di irradiazione realmente simmetrico.
Laddove i dipoli a ventaglio non possano venire impiegati per mancanza di spazio, le operazioni multibanda sono sempre possibili con l'impiego di rami trappolati. **ORA, disponibile anche nella versione per monopoli multibanda da 225 Ohm (Antenna multibanda di Errante a 225 Ohm)**

Il sistema d'antenna *turnstile*, qui presentato, e' la scelta primaria per le comunicazioni digitali sullo spettro delle HF, come ad esempio i servizi di *e-mail via HF* etc.

Prezzo:
Euro 1250,00 +IVA e spese di spedizione, per il modello 2x 70 Ohm

Euro 1650,00 +IVA e spese di spedizione, per il modello 2x 450 Ohm

RADIONDISTICS garantisce a vita i propri prodotti.

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ANIMAL
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Over 25 years of direct action for animals
without harming a single human being

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52. SISTEMI RADIANTI AUTO-ANTAGONISTI - SISTEMI RADIANTI IMPROPRII - Radiondistics.com

sistemi_autoantagonisti.htm

Se hai un interesse scientifico in materia di fisica della radio, visiona questo sito internet nell'ordine consigliato!

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I sistemi (radianti) auto-antagonisti.

Il massimo per gli auto-lesionisti! La G5RV ne e' un esempio.
di *Francesco Errante*

Rice-trasmettere con antenne (trasduttori radio-elettrici) relativamente molto corte rispetto alla lunghezza d'onda di lavoro e' svantagioso, ma lo e' ancor di pi� se lo si fa con sistemi d'antenna bilanciati. Cio', perche' cosi' facendo si da' origine a sistemi radianti *auto-antagonisti* ed ai fini del guadagno d'antenna, pressocche' null'altro e' pi� controproducente.
Cosa sono i sistemi radianti auto-antagonisti? Sono sistemi nei quali i radiatori sono alimentati in opposizione di fase tra di loro (bilanciati) mentre si trovano ad una distanza tra di

Le antenne radio sono: trasduttori radio-elettrici

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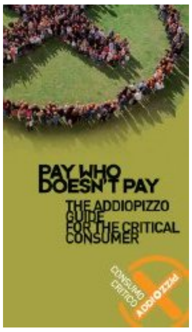
La radiazione hertziana: cos'� e come avviene

Le linee di trasmissione

loro cosi breve da poter essere considerati come due sorgenti coincidenti (vedi dipoli corti alimentati al centro). Sebbene tali sistemi irradino, i campi da loro prodotti tendono ad annullarsi. Questo a causa del fatto che ognuno dei radiatori del sistema genera un suo proprio campo, che a sua volta induce, in qualsiasi punto dello spazio, una forza elettromotrice di uguale intensita' a quella dell'altro ma con fase opposta.

Un esempio di sistema auto-antagonista e' la cosiddetta "*G5RV*"

Si noti come anche una cortina di elementi attivi *full-size* e risonanti o gli elementi di un'antenna direttiva del tipo *Yagi-Uda* se deliberatamente maldistanziati finiscano per costituire dei sistemi auto-antagonisti.



Per comprendere cosa avviene bisogna conoscere la *teoria delle sorgenti puntiformi*. Applicando la teoria delle sorgenti puntiformi a due monopoli corti in configurazione dipolo alimentato al centro, che essendo, per costruzione, alimentati da due segnali in controfase, sono anch'essi in controfase nel tempo, si comprende come i campi prodotti, punto per punto, dalla loro radiazione hertziana, abbiano tra di loro una distanza angolare inferiore - e di molto - ai 180 gradi. Quindi vale la seguente regola: "**pi� sono corti i monopoli del *dipolo elettricamente corto*, minore e' la distanza angolare tra i campi da loro prodotti e maggiore e' la loro tendenza ad elidersi**". Questo fenomeno e' chiamato "*interferenza distruttiva*".

Tutti i concetti, metodi, disegni e prodotti qui presentati sono originali e sono

bilanciate in regime di onda progressiva

Generatore di nodo di terra virtuale per linee bilanciate

Sistema d'antenna multibanda con radiatori a 225 Ohm

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esclusivamente frutto del lavoro di ricerca scientifica in materia di fisica delle radio onde, condotto da FRANCESCO ERRANTE.

�

Simulatore d'onde a rappresentazione tridimensionale

Una simulazione dell'interferenza tra i campi di due sorgenti puntiformi, quasi coincidenti ed alimentati in controfase, e' facilmente ottenibile dall'applet java concatenato a questa pagina, se debitamente configurato, ponendo il cursore "phase difference" a centro (180�) della sua corsa e facendo scorrere il cursore "source separation" verso sinistra (minimo). Si noti il progressivo annullamento degli effetti dei due campi.

NB: L'applet java a cui questa pagina fa riferimento si apre automaticamente in una nuova finestra "pop-up".

Il video qui in basso mostra il diagramma dinamico e termina con la sua rotazione nello spazio

It is a simulator that shows waves in three dimensions. It can show point sources, line sources, plane waves and interference between sources.

When the applet starts up you will see red and green waves emanating from two sources in the center of a cubic box. The wave color indicates the fields' magnitude. The green areas are negative and the red areas are positive.

To get started with the applet, just go through the items in the Setup menu in the upper right.

Ruotare il cubo col mouse per vedere la simulazione da un angolo differente.

NB: Per rilanciare nuovamente l'applet, si aggiorni questa pagina web.

Sorry, you need a Java-enabled browser to see the simulation.

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53. RF LIGHTING DEVICES, RFLD, HELUs, RF powered fluorescent tubes, high efficiency light sources, RF-LIGHTING.com

rf_lighting_devices/index.html

RF LIGHTING DEVICES

Welcome to RF-Lighting.com - The official ***RF Lighting Devices***' web site.

RF Lighting Devices is the world's sole designer and manufacturer of artificial lighting sources based on the direct bombardment of fluorescent tubes by progressive radio waves.

A fluorescente tube (Rapid T12, 2,4 m. long) excited through secondary ionization by radio waves bombardment.

Several means are known for obtaining light emission out of fluorescent tubes and lamps. Some are more convenient than others and however similar their outcome may look they only share part of the phenomena involved.

Ever since 1970, when John M. Anderson disclosed his own invention regarding (US Patent #3,500,118) an *"electrodeless gaseous electric discharge device utilizing ferrite cores"*, there has been a proliferation of similar devices, culminating in what they are now known as *"induction lamps"* and some manufacturers even call them "RF lighting devices" .

According to the internationally recognised classification, the RF lighting devices are those *devices capable of producing light by using RF energy to stimulate gases contained inside a lamp* and not the ones having just circuitry dealing with radio frequency electric signals. Therefore, it must be stressed that none of them can technically be defined as being radio-frequency lighting devices and by no means they should be confused with our products or viceversa.

The underlying principle behind the functioning of true RF lighting devices is the ionizing power of the radio waves.

The ionizing power of the radio waves and the radioluminescence effect exploited in the investigation of antennas and RF transmission lines behaviour.

RF Lighting devices employes *Radiondistics'* proprietary patented technology stemming from *Francesco Errante'* s brainchild.

Find out more on the science and technology behind the *radio-electric transduction light source*, here!

Until the year 2003, the radioluminescence effect in the shortwave spectrum of the radio waves (HF) was *RADIONDISTICS'* most tightly kept secret, this because the exploitation of this phenomenon in connection with readily available linear fluorescent tubes was and still is the only means to visually investigate the *radio-electric transduction* mechanism and the behaviour of radio waves being radiated from radio antennas. (see: A secondary ionization fluorescence hertzian radiation passive detector)

The radioluminescence stimulated by radio waves direct bombardment of fluorescent tubes is much more than just a means for detecting radio waves emission. Infact, **the ionizing power of the radio waves in the shortwave spectrum (HF) can advantageously be exploited to energize any fluorescent tube to provide artificial light at a fraction of the energy that traditional electric discharge methods would require**, including those systems powered through an electronic ballast or even modern *"magnetic induction lamps"* (aka: "induction lamps" for short).

Fluorescent lamps are the most efficient white light sources, way above any LED device developed to date.

However, due to the poor energy conversion efficiency afforded by both the galvanic and the inductive method of ignition, the overall efficiency drops drastically.
So, it must be made clear that it is not the fluorescent tube it-self the one responsable for the bulk of its energy consumption but the devices, the circuitry or the method employed to energize it.

Luminaires based on fluorescent neon tubes as luminous bodies, for decades, have maintained their primacy on energy efficiency, luminous efficiency and color rendering

over the rest of the electric power artificial lighting sources available.
With the advent of new technologies, the polyvalent primacy of the fluorescent tubes has been attacked on several fronts, not only on the energy efficiency one, but also on the aesthetic and the miniaturization fronts. To date, however, no new source of artificial lighting beats them in terms of color rendering. For this important reason, the industry has invested considerable resources to catch up with other rivaling technologies in terms of energy performance and their size, with notable successes also with regard to the ability to control their brightness - something impossible to be obtained with old tubes and old electrical ballast.
However, the fluorescent tubes are penalized by the fact that the electrical phenomenon used to give rise to their ignition, known as arc discharge (galvanic crossing of the gases to work of electric current), it is inherently expensive and requires an amount of energy which is not technically possible to reduce and consequently the power consumption values of the conventional fluorescent tubes luminaires cannot be improved any further.
The fluorescent tubes, however, lend themselves advantageously to be triggered and maintained switched on by means of other forms of excitation energy that allow it to exhibit a light efficiency far higher than that when the arc discharge is employed. Leaving aside the radioluminescence by permanent radioactive sources, they are:

1) the one, improperly, known as *"magnetic induction";*
2) the *direct bombardment by radio waves.*

The *"induction lamps",* although still undermarketed are available for lighting small and large surfaces, (eg: Philips QL Lamp, now QL Company and ICETRON Osram-Sylvania) with their own limitations in terms of energy efficiency and luminous bodies size, while the luminaires based on the direct bombardment by radio waves have only recently appeared in the consumer lighting market.

Lighting is a major energy consumer. For many buildings, such as commercial and public sites, lighting uses up more energy than anything else.

The need for artificial lighting is a large and rapidly growing source of energy demand and greenhouse gas emissions. At the same time the savings potential of lighting energy is high. New energy efficient lighting technologies are coming onto the market. Currently, more than 33 billion lamps operate worldwide, consuming more than 2650 TWh of energy annually, which is 19% of the global electricity consumption.
It is estimated that energy consumption for commercial lighting accounts for something in region of 25% to 40% of their entire power requirement and substantial savings can be made in this area by designing and installing the right units and controls based on the

most efficient equipment available.
Substitution of standard or obsolete lighting units by high efficiency lighting units can typically generate savings of between 20% and 70%.

Today, fluorescent lamps light over 80% of the building floor space.

In conventional discharge lamps, the energy required to supply the light source is provided by direct or low frequency alternating electric currents, which require the presence of electrodes within the lamp to trigger and maintain the discharge. The presence of electrodes places severe restrictions on lamp design and is a major cause of failure, hence limiting lamp life. Fluorescent lamps operate with thermionic cathodes; that is to say, an electron emitting material (such as barium oxide) is coated onto the electrode. The inevitable evaporation of emitter material during life causes unsightly darkening of the end of the lamp and results eventually in lamp failure. Conventional fluorescent lamps have rated life of around 5,000-20,000 hrs, while electrodeless fluorescent lamps are rated at 100,000 hrs.
The prospect of developing "electrodeless" lamps has challenged the lighting industry for more than a century, since *Tesla* demonstrated in 1891 that light could be produced in the presence of a high frequency electric field. The introduction of lamps based on the electrodeless principle for general purpose lighting has been hampered by the lack of readily available, compact, cheap and reliable driving electronics. However, in the last decade, progress in semiconductor electronics and power switching technology have made this approach commercially feasible.

The luminaires based on the bombardment by progressive regime radio waves designed and produced by the *RF Lighting Devices* allow, with safety and reliability, to achieve energy efficiency (luminous efficiency) far superior to any other artificial lighting system known so far, including those LED and allow fluorescent tubes to maintain the same characteristics of color rendering as shown when traditionally excited by arc.
Moreover, the radio waves direct bombardment system does not introduce any limitation to the physical size of luminous bodies that can be used.

Our products are most suitable for large indoor areas where a daylight-like and high luminance lighting is desirable.

True *RF lighting devices* exhibit several advantages:

• Huge energy saving: >50% compared to other energizing methods;
• Employ readily available cheap off-the-shelf conventional fluorescent tubes;
• Ease of retrofitting in linear fluorescent tube fixtures existing arrangment;
• Reduced number of units required compared to round induction lamps;
• No need for high rased bay for covering large areas;
• Fully electrode-less;
• No extra safety concerns. Any existing safety device for fluorescent tubes, such as protecting sleeves or shatterproof srhrinkwrap can be easily adopted. Industry standard colour filters can continue to be used too.
• No electric noise emissions (ample spectrum radio frequency interferences is a major EMC problem caused by the electronic ballasts, see also spectrum analisys of high energy efficiency fluorescent lights, the induction lamps, the plasma lighting systems and even the LED based energy efficient lighting);
• Virtually, no heat generation (resulting in even less energy consumption, longer tube life expectancy and low lumen depreciation);
• No thermal inertia (making them suitable for special light effects, eg.: flashing and sweeping light bars);
• Immediate lamp starting;
• No flickering nor strobing;
• Dimmerable down to >100 mW RF output (ultrafast dynamic dimmering);
• Reasonable cost of the equipment and installation.

***RF Lighiting Devices'* equipment are designed and constructed in accordance with good engineering practice and state-of-the-art technology and have sufficient shielding and filtering to provide adequate suppression of all radiated and/or conducted unwanted emission, according to the highest *interference-causing equipment* industrial standard.**

RF interference rf noise chemiluminescence phosphore phisical principles induction lamps plasma lighting microwave lamps lighting devices energizing a plasma arc without using filaments or electrodes gas discharge tube fluorescent light bulbs and tubes irradiance disadvantages: Bulky design for large area lighting, the discharge tube is large compared with HID lamps. New and Old technology: it is new: it is still expensive to buy the lamps. It is old: most companies that make the lamps are using 20 year old ballast technology copied from OSRAM and Philips. The ballasts have a high failure rate. The technology is under commercialized. Radio interference is a major problem to be worked out. The lamps are limited in use due to this issue. An induction lamp is a fluorescent lamp where the electric discharge is induced by a magnetic field, rather than an electric field as in a fluorescent lamp, and therefore does not have any electrodes. Induction lamps produce light by exciting the same phosphors found in conventional fluorescent lamps. The radio frequency (RF) power supply sends an electric current to an induction coil, generating an electromagnetic field. This field excites the mercury in the gas fill, causing the mercury to emit UV energy. The UV energy strikes and excites the phosphor coating on the inside of the glass bulb, producing light. Electrodeless lamps have efficacies similar to those of CFLs or HID lamps of comparable light output. Electrodeless lamps use rare-earth phosphors, giving them color properties similar to those of higher-end fluorescent lamps. Because the lamp has no electrodes that usually cause lamp failure, the life of this system is limited by the induction coil. Induction lamps are rated at 100,000 h of life. Because of this long life, and the good color rendition, induction technology is coming into use for areas where maintenance to change the lamp is expensive, such as high ceilings in commercial and industrial buildings, atria, tunnels, roadway sign lighting, etc. Induction lamps are electronic devices, and like all electronic devices they may generate electromagnetic interference (EMI) if unwanted electromagnetic signals, which can travel through wiring or radiate through the air, interfere with desirable signals from other devices. Shielding of the system to protect people and equipment from these emissions is important. Manufacturers must comply with national regulations on EMI to sell products in any country. This Interference-Causing Equipment Standard ICES-005, Issue 3, for Radio Frequency Lighting Devices (RFLD) Electrical lighting and similar equipment EN 55015 CISPR 15 AS/NZS CISPR 15 EMC phenomenon of emitted disturbance associated with: Conducted (continuous and intermittent) radio frequency disturbance Radiated radio frequency disturbance Test procedures and requirements associated with the EMC phenomenon Electrodeless lamps For lighting, the relevant standard is CISPR 15 (4th edition, September 1992), although there may be local differences from country to country. In this standard, lighting devices are allowed to operate in the ISM (industrial, scientific and medical) bands. For induction lamps there is a band 14 kHz wide at 13.56 MHz that has been used, with the advantage that there is also an ISM band at twice the frequency, so that the first harmonic does not have to be particularly well screened. The allowed level of emission is sufficiently high for practical lamps, but the small bandwidth means that accurate frequency control is necessary, which increases circuit costs significantly. Another band is available for non-communications use at 2400-2500 MHz, centred on 2450 MHz. This is the band that is used for microwave ovens, but is also used for light sources (see section 11.5). CISPR 15 also permits a relaxed level for the 'magnetic field induced current' between 2.2 and 3 MHz, a region of poor broadcast reception which lies between the medium and short wave radio bands. It is practically impossible to provide screening of the magnetic field of electrodeless lamps to the level required outside this band by CISPR 15. Although the emission allowed between 2.2 and 3 MHz, is still quite low, it has proved practical to use this band for the operation of electrodeless lamps. It is fortunate that commercially useful lamps operate at close to optimum in this frequency range. Clearly the CISPR 15 standard has a dominant influence on the choice of frequencies for electrodeless lamps. Another source of interference is terminal disturbance voltage (TDV), sometimes called conducted interference. RF currents may flow out of the lamp and into cables and other conductors, from which they may radiate and interfere with broadcast reception. In many electronically ballasted lamps the main source of TDV is called differential mode (DM) noise. This results in switching transients causing current to flow out of, say, the live lead, through the cabling and back into the neutral. Filtering must be provided at the mains input end of the circuit to reduce the DM currents to acceptable levels. In induc tioo lamps another source, called common mode (CM) noise, can be dominant. The lamp coil is coupled capacitatively to the ground plane. Driven by the coil voltage, current can flow through this capacitance to ground and back through both the supply wires to the lamp. CM noise is also measured in the CISPR 15 TDV test. A number of means for reducing CM TDV are available, of which the simplest is to enclose the lamp in a Faraday cage, which may be either a fitting or a transparent conducting coating on the lamp Any RF power which escapes from electrodeless lamps must be small enough to be safe. At present there are no standards for human exposure to EM radiation. However, there are many national recommended level" of exposure which are broadly similar lo each other. In the UK for example, the National Radiological Protection Board publishes Restrictions on human exposure to static and time varying electromt:{gneticfields and radiation. Electrodeless lamps need to comply with national recommendations.The induction lamp is also referred to as the 'electrodeless lamp'. It relies upon both magnetic and fluorescent principles for its operation. The constructional features of the lamp Energy transfer using magnetism (following the electrical transformer principle) is employed with the low pressure mercury filling in the lamp acting as a secondary coil of the transformer. A high frequency alternating electrical current in the primary winding is supplied from an external source.The current induced in the mercury vapour gives rise to emission of ultraviolet radiation in a similar manner to that in a conventional fluorescent lamp and the phosphor coating on the inside of the lamp envelope converts this UV radiation into visible light. The lamp life is typically 60000 h to 100000 h, which makes the use of such lamps in relatively inaccessible areas particularly beneficial. The useful life of conventional fluorescent lamps is primarily limited by the life of the cathodes. A secondary issue is phosphor life, which is partly dependent on the presence of impurities some of which come from the cathodes. It can therefore be surmised that if it was possible to make an electrodeless lamp, the result would be a lamp with exceptionally long life. In a normal fluorescent lamp a longtitudinal electric field between the two electrodes drives the discharge. The induction lamp is a device whereby an electric field is induced into an ionized vapor, without the need for electrodes. A way by which this can be done is illustrated in Figure 5.1. Here a ferrite coil is carrying a high frequency alternating current. Its magnetic field is continually varying, but we know that if a conductor is placed in a varying magnetic field, a voltage is induced in it. The conductor itself then produces an electric field. If the conductor is itself a plasma, then the field is set up within it. Figure 5.1 shows that the electric field is at right angles to the magnetic field. If the field is strong enough and is within the mercury vapor, then the result is the production of UV radiation, just as it is in a fluorescent lamp. Inductive discharges were known about in the 19th century, and an induction lamp was patented as early as 1907. However, the realization of a practical induction lamp had to wait until the arrival of compact power electronics. The design of induction lamps is a juggling act between getting the physics of the discharge right, achieving a good power conversion efficiency, and meeting EMC requirements. Lamp manufacturers have approached the problem

54. Apparecchi di illuminazione ad altissima efficienza luminosa a radio onde - RF-LIGHTING.com

`rf_lighting_devices/index_it.htm`

RF-Lighting.com	**RF Lighting Devices** a *Radiondistics'* spin-off	

RF LIGHTING DEVICES

Home | Retrospettiva | Brevetti | Prodotti | Contattaci

Benvenuti su RF-Lighting.com - Sito Internet ufficiale de ***RF Lighting Devices - Italy***

RF Lighting Devices è l'unico progettista e produttore, al mondo, di fonti di illuminazione artificiali basate sul bombardamento diretto di tubi fluorescenti, mediante onde radio in regime progressivo.

Tubo fluorescente (T12 rapido da m.2,4) eccitato tramite ionizzazione secondaria da bombardamento da radio onde

Several means are known for obtaining light emission out of fluorescent tubes and lamps. Some are more convenient than others and however similar their outcome may look they only share part of the phenomena involved.

Ever since 1970, when John M. Anderson disclosed his own invention regarding (US Patent #3,500,118) an *"electrodeless gaseous electric discharge device utilizing ferrite cores"*, there has been a proliferation of similar devices, culminating in what they are now known as *"induction lamps"* and some manufacturers even call them "RF lighting devices" .

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The underlying principle behind the functioning of true RF lighting devices is the ionizing power of the radio waves.

The ionizing power of the radio waves and the radioluminescence effect exploited in the investigation of antennas and RF transmission lines behaviour.

RF Lighting devices employes *Radiondistics'* proprietary patented technology stemming from *Francesco Errante'* s brainchild.

Find out more on the science and technology behind the *radio-electric transduction light source*, here!

Until the year 2003, the radioluminescence effect in the shortwave spectrum of the radio waves (HF) was *RADIONDISTICS'* most tightly kept secret, this because the exploitation of this phenomenon in connection with readily available linear fluorescent tubes was and still is the only means to visually investigate the *radio-electric* *transduction* mechanism and the behaviour of radio waves being radiated from radio antennas. (see: A secondary ionization fluorescence hertzian radiation passive detector)

The radioluminescence stimulated by radio waves direct bombardment of fluorescent tubes is much more than just a means for detecting radio waves emission. Infact, **the ionizing power of the radio waves in the shortwave spectrum (HF) can advantageously be exploited to energize any fluorescent tube to provide artificial light at a fraction of the energy that traditional electric discharge methods would require**, including those systems powered through an electronic ballast or even modern *"magnetic induction lamps"* (aka: "induction lamps" for short).

I tubi fluorescenti sono le fonti di luce bianca più efficienti, di gran lunga superiori a qualsiasi dispositivo a LED finora sviluppato.

Tuttavia, a causa della bassa efficienza nella conversione energetica esibita dai metodi di accensione generalmente impiegati, quali quello galvanico e quello induttivo, l'efficienza generale degli impianti diminuisce drasticamente.
Pertanto, va chiarito che non sono i tubi fluorescenti i responsabili della gran parte del loro consumo energetico, ma i dispositivi, i circuiti o i metodi adottati per la loro eccitazione.

Le luminarie basate sui tubi fluorescenti al neon come corpi luminosi hanno mantenuto per decenni il primato del rendimento energetico, dell'efficienza luminosa e della resa cromatica rispetto al resto delle fonti d'illuminazione artificiale elettriche disponibili.
Con l'avvento delle nuove tecnologie il primato plurimo dei tubi fluorescenti è stato attaccato su diversi fronti, non solo su quello del rendimento energetico, ma anche sul fronte dell'estetica e della miniaturizzazione.
A tutt'oggi, però, nessuna nuova fonte di illuminazione artificiale li batte in termini di resa cromatica. Per questo importante motivo, l'industria ha investito notevoli risorse per far riguadagnare terreno ai tubi fluorescenti sul piano del rendimento energetico e delle loro dimenzioni, con notevoli successi anche per quanto riguarda la possibilità di controllo della loro luminosità - cosa questa impossibile da ottenersi coi vecchi tubi e vecchi circuiti elettrici d'innesco.
Tuttavia, i tubi fluorescenti rimangono penalizzati dal fatto che il fenomeno elettrico impiegato per dare origine alla loro accensione, noto come scarica ad arco (attraversamento galvanico dei gas ad opera di corrente elettrica), è intrinsecamente oneroso e necessita di una quantità di energia sotto la quale non è tecnicamente possibile scendere e di conseguenza i valori di assorbimento elettrico delle luminarie convenzionali a tubi fluorescenti non sono più sostanzialmente migliorabili.
I tubi fluorescenti, però, si prestano vantaggiosamente ad essere innescati e mantenuti accesi per mezzo di altre forme di eccitazione energetica che gli permettono di esibire un'efficienza luminosa di gran lunga superiore rispetto

a quella ove la scarica ad arco è impiegata. Tralasciando la radioluminescenza da fonti radioattive permanenti, essi sono:

1) quella impropriamente nota come *"induzione magnetica"* ;
2) quella per *bombardamento diretto da radio onde.*

Le *"lampade ad induzione"*, sebbene con una diffusione minoritaria, sono già presenti sul mercato dell'illuminotecnica per ampie superfici, (ad esempio: Philips QL Lamp, adesso QL Company and ICETRON Osram-Sylvania) pur se con le loro limitazioni in termini di rendimento energetico e dimenzioni dei corpi luminosi, mentre, le luminarie a bombardamento diretto da radio onde a regime progressivo sono appena agli albori della loro commercializzate nel mercato dell'illuminotecnica di ampio consumo.

Lighting is a major energy consumer. For many buildings, such as commercial and public sites, lighting uses up more energy than anything else.

The need for artificial lighting is a large and rapidly growing source of energy demand and greenhouse gas emissions. At the same time the savings potential of lighting energy is high. New energy efficient lighting technologies are coming onto the market. Currently, more than 33 billion lamps operate worldwide, consuming more than 2650 TWh of energy annually, which is 19% of the global electricity consumption.
It is estimated that energy consumption for commercial lighting accounts for something in region of 25% to 40% of their entire power requirement and substantial savings can be made in this area by designing and installing the right units and controls based on the most efficient equipment available.
Substitution of standard or obsolete lighting units by high efficiency lighting units can typically generate savings of between 20% and 70%.

Today, fluorescent lamps light over 80% of the building floor space.

In conventional discharge lamps, the energy required to supply the light source is provided by direct or low frequency alternating electric currents, which require the presence of electrodes within the lamp to trigger and maintain the discharge. The presence of electrodes places severe restrictions on lamp design and is a major cause of failure, hence limiting lamp life. Fluorescent lamps operate with thermionic cathodes; that is to say, an electron emitting material (such as barium oxide) is coated onto the electrode. The inevitable evaporation of emitter material during life causes unsightly darkening of the end of the lamp and results eventually in lamp failure. Conventional fluorescent lamps have rated life of around 5,000-20,000 hrs, while electrodeless fluorescent lamps are

rated at 100,000 hrs.
The prospect of developing "electrodeless" lamps has challenged the lighting industry for more than a century, since *Tesla* demonstrated in 1891 that light could be produced in the presence of a high frequency electric field. The introduction of lamps based on the electrodeless principle for general purpose lighting has been hampered by the lack of readily available, compact, cheap and reliable driving electronics. However, in the last decade, progress in semiconductor electronics and power switching technology have made this approach commercially feasible.

Le luminarie a bombardamento da radio onde a regime progressivo ideate e realizzate dalla *RF Lighting Devices* permettono, in tutta sicurezza ed affidabilità, di ottenere un rendimento energetico (efficienza luminosa) di gran lunga superiore a qualsiasi altro sistema d'illuminazione artificiale finora noto, inclusi quelli a L.E.D. e permettono di mantenere le medesime caratteristiche di resa cromatica proprie dei tubi fluorescenti, offerte dagli stessi quando eccitati tradizionalmente ad arco. Inoltre, il sistema di bombardamento diretto a radio onde non introduce alcuna limitazione alle dimenzioni fisiche dei corpi luminosi utilizzabili.

Our products are most suitable for large indoor areas where a daylight-like and high luminance lighting is desirable.

True *RF lighting devices* exhibit several advantages:

- Huge energy saving: >50% compared to other energizing methods;
- Employ readily available cheap off-the-shelf conventional fluorescent tubes;
- Ease of retrofitting in linear fluorescent tube fixtures existing arrangment;
- Reduced number of units required compared to round induction lamps;
- No need for high rased bay for covering large areas;
- Fully electrode-less;
- No extra safety concerns. Any existing safety device for fluorescent tubes, such as protecting sleeves or shatterproof srhrinkwrap can be easily adopted. Industry standard colour filters can continue to be used too.
- No electric noise emissions (ample spectrum radio frequency interferences is a major EMC problem caused by the electronic ballasts, see also spectrum analisys of high energy efficiency fluorescent lights, the induction lamps, the plasma lighting systems and even the LED based energy efficient lighting);
- Virtually, no heat generation (resulting in even less energy consumption, longer

tube life expectancy and low lumen depreciation);
• No thermal inertia (making them suitable for special light effects, eg.: flashing and sweeping light bars);
• Immediate lamp starting;
• No flickering nor strobing;
• Dimmerable down to >100 mW RF output (ultrafast dynamic dimmering);
• Reasonable cost of the equipment and installation.

***RF Lighiting Devices'* equipment are designed and constructed in accordance with good engineering practice and state-of-the-art technology and have sufficient shielding and filtering to provide adequate suppression of all radiated and/or conducted unwanted emission, according to the highest *interference-causing equipment* industrial standard.**

RF lampade principi fisici di induzione RF interferenze chemiluminescenza Phosphore forno a microonde al plasma illuminazione lampade ad arco al plasma dispositivi energizzanti senza utilizzare filamenti o elettrodi lampadine fluorescenti tubo a scarica di gas e tubi di irraggiamento Svantaggi: disegno ingombranti per la grande illuminazione della zona, il tubo di scarico è grande rispetto Con le lampade HID. La nuova tecnologia e Old: è nuovo: è ancora costoso per comprare le lampade. E vecchio: la maggior parte delle aziende sono che fanno le lampade con 20 anni di tecnologia di zavorra Copiato da OSRAM e Philips. I reattori hanno un alto tasso di fallimento. La tecnologia è commercializzato sotto. interferenze radio è un grave problema per essere elaborato. Le lampade sono limitato in uso a causa di questo problema. Una lampada induzione è una lampada fluorescente Dove scarica elettrica è indotta da un campo magnetico, un campo elettrico invece che come in un lampada fluorescente, e quindi non ha alcun elettrodi. lampade a induzione producono luce. eccitando gli stessi fosfori trovati nelle lampade fluorescenti convenzionali. La radiofrequenza (RF) alimentatore invia una corrente elettrica ad una bobina di induzione, generando un campo elettromagnetico. Questo campo eccita il mercurio nel gas di riempimento, causando mercurio per emettere UVenergy. Il UVenergy colpisce ed eccita il rivestimento fosforo all'interno del il bulbo di vetro, la produzione di luce. Avere lampade electrodeless simili a quelle di efficacies CFL o lampade HID di analoga emissione luminosa. lampade electrodeless utilizzano fosfori di terre rare. dando loro proprietà colore simile a quelle delle lampade fluorescenti di fascia alta. Poiché la Quella lampada non ha elettrodi causano il fallimento della lampada Di solito, la vita del sistema ESTA è limitato dalla bobina di induzione. lampade a induzione sono stimati a 100.000 ore di vita. A causa del lungo ESTA la vita, e la buona resa cromatica, la tecnologia ad induzione è in arrivo in uso per le zone dove per cambiare la manutenzione della lampada è costoso. come i soffitti alti in commerciale e Edifici industriali, atri, gallerie, illuminazione del segno della carreggiata, ecc lampade a induzione sono dispositivi elettronici, e, come tutti i dispositivi elettronici generano elettromagnetico Possono interferenze (EMI) se i segnali elettromagnetici indesiderati, che può viaggiare attraverso il cablaggio o irradiare attraverso l'aria, interferire con i segnali provenienti da altri dispositivi desiderabili.

schermatura il sistema per proteggere le persone e le attrezzature da queste emissioni è importante. Produttori Devono essere conformi alle normative nazionali in materia di EMI di vendere i prodotti in qualsiasi paese questo Apparecchiature che causano interferenze standard ICES-005, Issue 3, per Radio Frequency Lighting Devices (RFId) Attrezzature di illuminazione elettrica e simili EN 55015 CISPR 15 AS / NZS CISPR 15

EMC fenomeno di disturbo emesso associati: Condotto (continuo e intermittente) a radiofrequenza irradiati a radiofrequenza test di disturbo Procedure e requisiti associati a EMC phenomenon Electrodeless Per lampade illuminazione, lo standard è CISPR 15 rilevante (4a edizione, Settembre 1992), anche se ci Può essere differenze locali da paese a paese. In questo standard, i dispositivi di illuminazione sono ha permesso di operare nel ISM (, scientifiche e mediche industriali) bande. per l'induzione Lampade c'è un ampia banda 14 kHz a 13.56 MHz che è stato utilizzato, con il vantaggio che Inoltre vi è una banda ISM al doppio della frequenza, in modo che la prima armonica che non ha

Particolarmente di essere proiettati. Il livello consentito di emissioni è sufficientemente elevato per lampade pratico, ma la piccola larghezza di banda significa che è il controllo della frequenza preciso. Necessario, che aumenta in modo significativo i costi dei circuiti. Un altra band è disponibile per i non-comunicazione usano a 2400-2500 MHz, centrata sulla 2450 MHz. Questa è la band che viene utilizzato per i forni a microonde, ma è utilizzato per sorgenti luminose passato semplice (vedi paragrafo 11.5). Anche CISPR 15

Consente un livello rilassato per il campo magnetico indotto corrente tra 2,2 e 3 MHz, per regione di cattiva ricezione di trasmissioni che si trova tra la media e onde corte radio bande. Praticamente Il è impossibile fornire schermatura del campo magnetico il lampade electrodeless al livello richiesto di fuori di questa banda per CISPR 15., anche se Tra 2,2 e permesso di emissione 3 MHz. È ancora abbastanza basso è risultato pratico usare questa banda per il funzionamento di lampade senza elettrodi. È una fortuna che commercialmente lampade utili funzionano a vicino al ottimale nella gamma di frequenza ESTA. Chiaramente il CISPR 15 Standard hanno un influenza dominante sulla scelta delle frequenze per le lampade senza elettrodi. Un altra fonte di interferenza è terminale di tensione di disturbo (VDT), a volte chiamato disturbi condotti. correnti RF flusso può fuori dalla lampada e in fili e altre conduttori, da cui possono irradiare e interferire con la ricezione di trasmissione. in molti zavorrato elettronicamente lampade la principale fonte di TDV si chiama modo differenziale (OM) rumore. Ciò si traduce nel passaggio transitori di corrente ha causato a flow fuori, per esempio, il live piombo, attraverso il cablaggio e nuovamente dentro il neutro. Filtraggio deve essere fornita al

alimentazione estremità di ingresso del circuito per ridurre le correnti DM a livelli accettabili. In Induc tion altre lampade di origine, chiamati modo comune (CM) rumore, può essere dominante. La bobina e lampada capacitivamente accoppiato al piano di massa. Spinto dalla tensione della bobina, la corrente può fluire Attraverso questa capacità a terra e indietro attraverso i fili di alimentazione sia alla lampada. CM

Anche il rumore viene misurato in test CJSPR 15 TDV. Un certo numero di mezzi per ridurre CM TDV sono disponibili, di cui la più semplice è quella di racchiudere la lampada in una gabbia di Faraday, che possono essere O un raccordo o di un rivestimento conduttore trasparente sulla lampada Eventuali Quale perdite di potenza RF

dalle lampade electrodeless devono essere abbastanza piccolo per essere sicuri. Al momento non ci sono norme per l'esposizione umana alle radiazioni EM. Tuttavia, ci sono molti livello raccomandato nazionale Quali esposizione sono sostanzialmente uguali tra di loro. Nel Regno Unito, per esempio, il Consiglio nazionale per la protezione radiologica Pubblica Restrizioni esposizione umana a statica e il tempo campi elettromagnetici variabili e radiazioni lampade electrodeless necessità di rispettare

reccomandazioni. la lampada ad induzione chiamata anche come il electrodeless

lamp. Esso si basa su entrambi i principi magnetiche e fluorescenti per il suo funzionamento. il caratteristiche costruttive del trasferimento Lampada di energia utilizzando il magnetismo (in seguito alla elettrica principio del trasformatore) è impiegato con il ripieno di mercurio a bassa pressione nel lampada agisce come bobina secondaria del trasformatore. Un alta frequenza alternata elettrica avvolgimento corrente del primario è alimentato da una corrente source. The esterna indotta Fornisce l'aumento vapori di mercurio ad emissione di radiazioni ultraviolette in un modo simile a

Che in una lampada fluorescente convenzionale e il rivestimento fosforo all'interno della lampada. Converte busta esta radiazione UV in luce visibile. La durata della lampada è 60000 Tipicamente h a 100000 h, il che rende l'uso di tali lampade in zone impervie relativamente Particolarmente vantaggioso. La vita utile delle lampade fluorescenti è principalmente limitata dalla durata dei catodi. Un problema secondario è la vita di fosforo, che è in parte dipendente dalla presenza di impurità alcuni dei quali mangia dai catodi. Si può Quindi essere se è stato ipotizzato che possibile fare una lampada senza elettrodi, il risultato sarebbe una lampada con lunghissima durata. In lampada fluorescente normale longtitudinal Tra i due campo elettrico aziona i elettrodi di scarica. La lampada di induzione è un Dispositivo cui un campo elettrico viene indotta in un vapore ionizzato senza la necessità di elettrodi. Un modo per esta che può essere fatto è illustrata nella Figura 5.1. Qui di ferrite coil sta trasportando una corrente alternata ad alta frequenza. Il suo campo magnetico è continuamente variabile; ma sappiamo che se un pilota si trova in un campo magnetico variabile, una tensione è indotta in esso. Il conducente stesso produce un campo elettrico poi. Se il conducente è plasma stesso, allora un campo e il set-up Da esso. La figura 5.1 mostra l'elettrico che il campo è perpendicolare al campo magnetico. Se il campo è abbastanza forte ed è Da il vapore di mercurio Poi il risultato è la produzione di radiazione UV, così come è in un lampada fluorescente. scariche induttivo Acerca erano conosciuti nel 19 ° secolo, ed un lampada di induzione è stato brevettato già nel 1907. Tuttavia, la realizzazione di una pratica

Abbiamo dovuto aspettare lampada di induzione Fino all'arrivo di elettronica di potenza compatti. La progettazione di lampade a induzione è un atto di giocoleria tra ottenere la fisica del diritto di scarico, Il raggiungimento di una buona efficienza di conversione di potenza, e soddisfare i requisiti di compatibilità elettromagnetica. lampada Sono produttori affrontato il problema

55. RF LIGHTING DEVICES, RFLD, HELUs, RF powered fluorescent tubes, high efficiency light sources, RF-LIGHTING.com

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RF Lighting Devices

Home

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N.B. If your mail server's IP address has no reverse DNS record, our server will reject your mail. That's because more than 99% of the e-mail we receive from hosts with no reverse DNS record is spam, so we have no choice but to reject e-mails from these sources.

56. Radio frequency noise from high energy efficiency fluorescent lights ballasts

rf_lighting_devices/radio_interference_from_high_efficiency_fluorescent_lights.htm

RF LIGHTING DEVICES

Radiated RF noise from electronic ballasts. The video herebelow shows a spectrum analyzer sweeping in the frequency range between 10 Hz to 1 MHz.

Radio frequency noise from high energy efficiency fluorescent lights ballast. Courtesy of *MegawattKS*

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